Vincent Liu
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Arithmetic Spectral Descent

Obstruction Theory, Hodge–Krein Realization, and a Falsifiable Architecture for the Riemann Hypothesis

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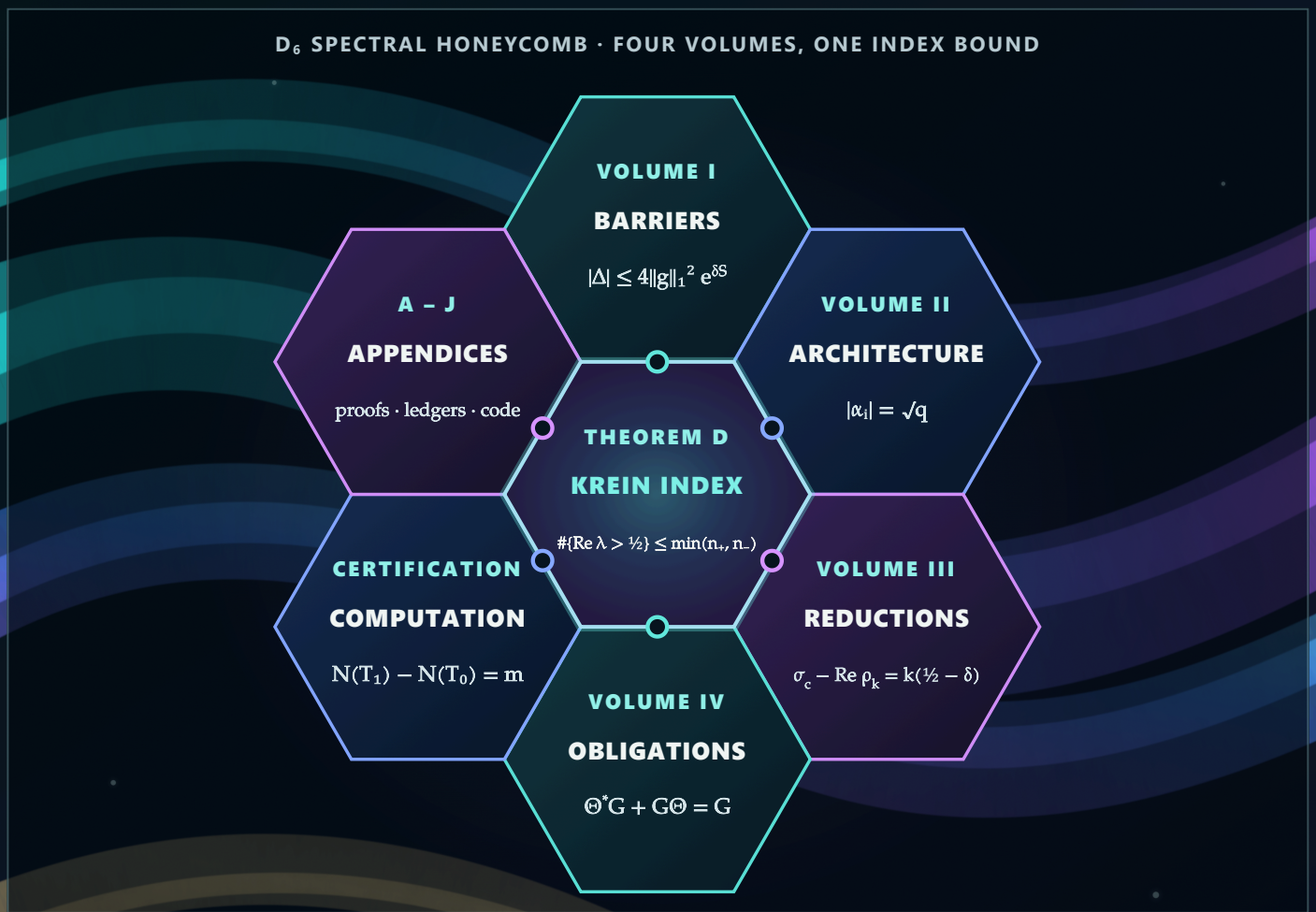

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Original four-volume spectral design

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Arithmetic Spectral Descent

Complete Four-Volume Monograph

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Obstruction theory, Hodge–Krein realization, and a falsifiable architecture for the Riemann Hypothesis

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The Riemann Hypothesis and the generalized Riemann hypotheses remain open. This publication does not claim an unconditional proof of either. Every substantive assertion is labelled with its status, and the premises of every conditional implication are printed with it.

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Series Note

Arithmetic Spectral Descent is the third work in the WiseMotif Monograph of Mathematical Intelligence and belongs to the Grothendieckian Programme. The first two works of the series are bound closely together. *The Grothendieckian Machine* set out how mathematical worlds are constructed, compared and carried from one setting to another, and how a machine may hold them as typed, versioned and verifiable objects. *GM-Omega-Infinity: Topos-Native Intelligence* stated what an intelligence must be, internally, if it is to reason inside such worlds. The present work stands apart from that pair. It is a self-contained system, and it is one of the research subjects of the Programme.

The Programme has a single research aim: machine intelligence that discovers mathematical structure on its own account. That aim asks for reasoning architectures held to the standard of proof, and for a discipline of recursive mathematical discovery, in which a proposed structure is tested, what survives the test is abstracted into a new setting, and the question is posed again in that setting. Its model is Grothendieck's rising sea. A hard problem is not attacked where it stands; a theory is raised around it until the problem becomes natural and gives way. A research subject of the Programme is a problem on which this method is carried through in full, with every step recorded in a form that a mathematician can check and a machine can execute.

This work carries it through on the Riemann Hypothesis. It began with a proposed route from arithmetic geometry to a spectral realization of the zeros of the zeta function. Each mechanism of that route was written with its normalizations exposed and tested against four published counter-models. A mechanism that failed for a structural reason became a no-go theorem; what survived was abstracted and posed again. The recursion ends in a proved conditional chain, twenty-three scoped obstructions, a certified computation stack, and two obligations stated precisely enough to be attacked, rejected or falsified: the Arithmetic Hodge–Krein Realization Problem and Problem A2$\star$.

The work keeps one editorial principle throughout. No result is presented at a strength greater than its proof. Conditional implications are printed with their premises attached. Finite computation is reported as finite computation and is never promoted to an asymptotic theorem. An open problem that the work fails to solve is stated as an open problem, in the place where it fails to solve it. Every substantive claim carries its status where it is made.

The Riemann Hypothesis remains open. This work does not claim a proof of it, of the generalized Riemann hypotheses, or of any equivalent statement. What it offers is the state of the question after the method has been applied without concession: what has been proved, what has been excluded, and what remains to be constructed.

Vincent Liu
WiseMotif.ai

Scientific notice

The Riemann Hypothesis remains open.

This monograph does not claim an unconditional proof of RH, GRH, or any equivalent formulation. Its central positive implications have premises that are explicitly marked open.

[T] Proved	Finite polarization and Krein-index results; the conditional implication from AHK-1 through AHK-3 to RH; the Jensen–Hodge dictionary; and scoped obstruction theorems.
[O] Open	No arithmetic system satisfying AHK-1 through AHK-3 is constructed. AHK-4 through AHK-6 and Problem A2$\star$ remain programme obligations.
[E] Evidence	The measured inertia of sixty-three finite Gram matrices is finite evidence only. It supplies neither a compatible operator nor the centered adjoint identity required by the conditional theorem.
[N] Excluded	The book does not infer RH from a Lorentzian Gram signature, from log-concavity, from a finite spectral computation, or from an analogy with a successful proof in another field.

Composition rule. [O] P together with [T] $P \Rightarrow$ RH yields an *open route to* RH, not a theorem of RH. Status attaches separately to premise, implication, evidence, and conclusion.

Every substantive mathematical claim is marked [T], [C], [O], [E], [N], or [K]. The full editorial definitions and worked examples appear in the Status-label key.

Abstract

[K] The Riemann Hypothesis and the generalized Riemann hypotheses remain open. This monograph does not claim a proof of either.

[T] The work develops a proof architecture in which every available implication is separated from the premises not yet constructed. Its positive linear-algebraic input is the Krein Index Theorem: if $\Theta^* G + G\Theta = G$, with G a nondegenerate Hermitian form of inertia (n_+, n_-), then the algebraic multiplicity in $\mathrm{Re}\,\lambda > \frac{1}{2}$ is at most $\min(n_+, n_-)$. Combined with stable spectral exhaustion, centered duality, and compatible negative index at most one, this yields the proved conditional implication

$$\text{AHK-1} \wedge \text{AHK-2} \wedge \text{AHK-3} \implies \text{RH}.$$

The proof uses conjugate pairing, the functional equation, and the classical absence of real zeros of ζ in $(0, 1)$. It does not construct the required arithmetic system.

[K] The equivalence audit is therefore load-bearing. A tautological diagonal construction exists under RH, so AHK-1 through AHK-3 are not, by themselves, a weaker route to the conclusion. Arithmetic content must enter through AHK-4 through AHK-6; these conditions are specified and tested, but are not jointly realized.

[T] The negative line supplies independently checkable results. The Jensen–Hodge dictionary relates Jensen hyperbolicity, Lorentzian Hessians, and Hodge–Riemann relations in finite settings. For the audited class of Dirichlet series with k-fold zero singularities, the Amplification Invariance Theorem gives

$$\sigma_c - \mathrm{Re}(\text{amplified singularity}) = k(\tfrac{1}{2} - \delta),$$

so amplification alone gives no gain in the zero exponent. Further scoped theorems isolate level collapse, infinite multiplicity, structure blindness, support doubling, positivity ceilings, criticality, and finite-data spectral obstructions.

[K] The function-field case supplies a controlled comparison: Frobenius, cohomology, duality, and positivity are stated where they exist before number-field analogues are proposed. Later chapters record the known local constructions and the missing comparison and positivity data.

[E] The numerical component evaluates sixty-three cubic-spline Guinand–Weil Gram matrices. The pole-removed form has measured inertia $(N - 1, 1)$ in every tested configuration. This finite evidence supplies neither a zeta-spectral operator nor the centered adjoint identity and cannot be inserted into the conditional theorem.

[O] The open obligations are AHK-1 through AHK-6 and Problem A2$\star$: prove category rigidity or the mean-purity estimate

$$\sum_{0<\gamma\le T} (\beta - \tfrac{1}{2})^+ = o(T),$$

Until then, RH and GRH remain open.

Preface

The question and its boundary

[K] The Riemann Hypothesis remains open. This book does not prove it, and it does not prove the generalized Riemann hypotheses. It records something else: a programme aimed at the conjecture, the tests to which that programme was subjected, the parts that failed, the theorems obtained from those failures, and the construction problems that remain once every failure has been accounted for. The distinction between a proved conditional implication and an unconditional proof is the organizing fact of the work, and it is kept on every page.

The conjecture is easily stated. Every nontrivial zero of the Riemann zeta function has real part one half. Its difficulty is of a particular kind. Almost every natural mechanism that one might use to locate the zeros either uses information that does not distinguish ζ from functions whose zeros lie off the line, or discards, at some step, exactly the information that would be needed at the line. A proof must keep that information from the first step to the last. The book is, before anything else, a study of where it is lost.

The title

The title names the movement that the book examines. *Spectral*: the classical hope, from Hilbert and Pólya onward, that the nontrivial zeros are the spectrum of an operator, so that their location would follow from a structural property of the operator rather than from an estimate. *Arithmetic*: the operator must come from the arithmetic of $\mathbb{Z}$, and not be written down after the zeros are known, which costs nothing and proves nothing. *Descent*: the passage from local to global. At each finite place there is a Frobenius with known spectral behaviour; at the archimedean place there is the explicit formula; over a finite field the local data glue to one global object whose cohomology carries the zeros. Over $\mathbb{Z}$ the local data are available, in increasingly refined forms, and the question is whether they descend to a single arithmetic object carrying the zeros as its spectrum, together with a form that controls where that spectrum can lie.

The book proves where this descent is blocked by the structures proposed for it, and it states what a descent that is not blocked would have to provide. A global endomorphism must have localizations that come from one morphism; a global form must keep its sign or its index under the passage; a trace formula must recover the prime-power masses exactly and not up to a normalization. Each of these is a condition of descent in Grothendieck's sense, and each is either proved, excluded or posed as open in the chapters that follow.

A research subject of the Grothendieckian Programme

This work is the third in the WiseMotif Monograph of Mathematical Intelligence and belongs to the Grothendieckian Programme. The first two works of the series are tightly coupled. *The Grothendieckian Machine* described how mathematical worlds are built, compared and transported, and how a machine may hold them as typed, versioned and verifiable objects.

GM-Omega-Infinity: Topos-Native Intelligence supplied the reasoning kernel that operates inside such worlds: local inference, gluing, obstruction and change of base. Each needs the other.

The present work depends on neither. It can be read, checked and used without the apparatus of the first two books, and its mathematics is complete on its own terms. Its place in the Programme is that of a research subject. The Programme's aim is machine intelligence that discovers mathematical structure on its own account: that proposes structures, tests them against what is known, abstracts what survives, and poses its questions again in the setting the abstraction provides. A research subject is a problem on which that aim is carried through in full and recorded so that every step can be inspected. The Riemann Hypothesis was chosen because it does not yield. A method of discovery that is tested on a problem which gives way easily has not been tested.

The relation to the first two works is therefore one of method rather than of dependence. What the series asks of a mathematical intelligence, that every claim carry its status, that evidence be kept apart from proof, that a failure be located rather than suppressed, is here applied to one problem at full depth. The book is both a contribution to analytic number theory and arithmetic geometry and a worked specification of what recursive mathematical discovery looks like when it is held to the standard of proof.

The rising sea

Grothendieck described his way with a hard problem in the image of a rising sea. A nut can be opened with a hammer and chisel; it can also be immersed until it softens, and then it opens under the hand. The Weil conjectures were not proved by attacking the zeta function of a variety directly. A cohomology theory was built in which the zeta function became a characteristic polynomial, Frobenius became an operator, and the functional equation became duality. The remaining inequality was then a statement about an operator on a finite-dimensional space, and Deligne proved it.

This book does not claim that such a sea has been raised around the Riemann Hypothesis. It applies the method in the order the method prescribes. Before any construction is proposed, it establishes what any construction must avoid: the barriers, scoped and proved, at which specified mechanisms lose the information needed at the critical line. It then takes the one setting in which the analogous theorem is known, curves over finite fields, and decomposes the proof into its actual inputs, so that each can be sought over $\mathbb{Z}$ separately. It audits the modern structures proposed for that search, the sphere spectrum, cyclotomic homology and prismatic cohomology, for what they supply and what they do not. Only then does it state the positive line, and the positive line is stated as a conditional theorem whose premises are printed as open.

The result is that the problem has changed shape. It began as a statement about the location of zeros. It ends as two construction problems: an operator-compatible Hermitian form of negative index one, built from arithmetic, and an upper constraint on amplified zero data that does not come from the zeros themselves. Neither is solved here. Both are precise, and both can be attacked by a reader who rejects everything else in the book.

Recursive mathematical discovery

The work began as a proposed route from arithmetic geometry to a spectral realization of the nontrivial zeros. Its intended ingredients were an absolute base, cyclotomic and prismatic constructions, an explicit-formula trace, a positive polarization, and an amplification

procedure. It was not a theorem. It was a list of objects that might exist and of compatibilities they would have to satisfy.

[N] The first step separated what was proved from what was hoped, and that separation changed the programme. The positive-definite polarization, in the form first considered, is stronger than RH rather than a mechanism from which RH follows. The support-amplification argument had confused the support of a test function with the support of its autocorrelation; once corrected, its apparent gain disappeared. Higher additive correlations magnified the zero signal, but magnified the relevant convergence threshold by the same factor. A proposed geometric object attached to the divisor of sums of zeros could not exist, because that divisor has a point of infinite multiplicity.

Each of these failures was put through the same procedure. The intended conclusion was written with every normalization parameter exposed. The weakest inputs the calculation actually used were listed. The same calculation was carried out on the four counter-models of the firewall. The quantity invariant under changes of normalization was isolated. Only then was the failure stated, and when it was structural it became a theorem: a no-go result with explicit hypotheses, whose scope is exactly the class of mechanisms those hypotheses describe. What survived the procedure was abstracted, and the question was posed again at the new level.

This is the recursion the Programme calls recursive mathematical discovery, and the book is its record. The refutation of the positive polarization is the step that identified the index-one formulation. The amplification failure became the Amplification Invariance Theorem and, from it, the precise form of the second obligation. The infinite-multiplicity failure removed an ill-posed target and redirected attention to locally finite spectral data. Each step has an input, an output and a test, and the steps are written so that they can be carried out by a mathematician or executed and checked by a machine. Nothing in the procedure makes an argument correct; a review procedure is not a proof. What it does is make the location of every burden visible.

The firewall

[K] Four published counter-models serve as a firewall for every proposed mechanism. Davenport–Heilbronn functions have a functional equation but no Euler product, and have zeros off the critical line. Beurling generalized number systems keep an Euler-product formalism and a prime number theorem while their analogue of RH fails. Selberg zeta functions arise from canonically self-adjoint spectral data, yet exceptional eigenvalues obstruct the corresponding line statement. Epstein zeta functions of class number greater than one have a functional equation but no single Euler product, and may vanish in $\operatorname{Re} s > 1$.

The rule is strict. A mechanism is not admitted as a route to RH until the book explains why it fails, or changes form, on all four objects. If it cannot distinguish one of them, the limitation is recorded as open or as a no-go result. The four objects do not exhaust the false analogues of RH. They test four independent sources of overreach: a functional equation without an Euler product, an Euler product without the required zero geometry, self-adjointness without the required line, and global symmetry without arithmetic specificity.

[T] The index-one mechanism passes the firewall in an exact and limited sense. Nonreal zeros off the line come in symmetry packets that force negative index at least two. A real reflected pair can occupy an index-one block, and the Selberg example shows that self-adjointness alone does not remove it. For ζ, the classical absence of real zeros in $0 < s < 1$ supplies the additional input. The firewall thus identifies the one zeta-specific fact that the finite-dimensional argument needs.

[N] The same discipline governs positivity. Rodgers and Tao proved $\Lambda_{\mathrm{dBN}} \geq 0$, while RH is equivalent to $\Lambda_{\mathrm{dBN}} \leq 0$; hence RH is equivalent to the equality $\Lambda_{\mathrm{dBN}} = 0$. A perturbation-stable strict positivity statement, a uniform spectral gap or a finite strict sum-of-squares certificate cannot by itself prove RH. A viable positivity argument must be an equality or rigidity argument, and the book states this requirement wherever positivity enters.

The status of every sentence

Every mathematical claim carries one of six labels. [T] marks a theorem, proved in the text or cited exactly. [C] marks a conjecture or a conditional statement whose premises are not established. [O] marks an open construction: the required object has not been produced. [E] marks evidence: finite or numerical, never promoted to an infinite statement. [N] marks a proved impossibility within stated hypotheses. [K] marks known background, stated with a source.

The labels control verbs. *Construct*, *prove* and *establish* are used only where the object or the argument is present; elsewhere the text says *specify*, *would imply* or *remains to be constructed*. The distinction matters most in a spectral programme, because a formal diagonal operator can always be written down once the zeros are given, and such an operator is not an arithmetic realization. Each Part opens with a ledger of status, construction and logical strength, and each chapter closes with a section entitled "What this chapter does not establish". That section belongs to the specification of the argument: it records which converse, extension or interpretation has not been justified.

Behind the labels lies a set of separations that the book does not allow to blur: arithmetic definitions and their spectral consequences, finite matrices and limiting operators, a trace identity and an adjoint identity, the full Weil form and its pole-removed Schur complement, an analogy and a functorial construction. Many unsuccessful approaches to RH cross one of these boundaries without noticing the change of object. The discipline does not forbid crossing. It requires a theorem at the crossing.

The architecture of the work

The book is one monograph in four internal volumes, ten parts and sixty chapters, followed by ten appendices, a glossary, a chronology, a consolidated register of open problems, the bibliography and four indices. Pagination and bibliography are continuous because the logical dependencies cross the internal divisions. The four volumes follow the order the method requires: what cannot work, what a working proof in the one known case is made of, what can be proved conditionally, and what remains to be done.

Volume I: Barrier Theory

What specified proof mechanisms cannot do. Volume I asks a question that is logically prior to any construction: which operations preserve enough arithmetic information to distinguish the critical line, and which destroy that information before the decisive step? The question is not whether RH is impossible to prove. It is whether a stated mechanism, under stated hypotheses, can produce more than the classical boundary already available to it.

Part I, *Obstruction theory: what cannot work*, answers through twenty-three scoped obstructions and the four-object firewall. It opens with the boundary of what the book claims (Chapter 1), sets out the counter-models (Chapter 2), shows that a Hilbert–Pólya operator is vacuous when the zeros are given and replaces the spectral-gap heuristic by its correct form (Chapter 3), corrects the diagnosis of the positivity axiom AS6 (Chapter 4), states at the

outset what would falsify the book (Chapter 5), and assembles the obstruction archive with its proof audit (Chapter 6). The archive and the firewall do different work: the archive blocks routes by inequalities and invariants, the firewall blocks mechanisms by exhibiting objects on which the same mechanism would give a false conclusion. Neither is an impossibility theorem for RH.

Part II, *The explicit formula and the absolute-value ceiling*, fixes the analytic instrument against which those routes are tested. Chapter 7 derives the Guinand–Weil explicit formula with one normalization, tracking the zero, prime-power, polar and archimedean terms, so that no later comparison can mistake a change of normalization for a gain in exponent. [N] Chapter 8 proves the corrected support-doubling obstruction: the counterfactual zero signal grows like $e^{\delta S}$ while the absolute arithmetic envelope grows like $Se^{S/2}$, so the threshold is $\delta > \frac{1}{2}$, not the spurious $\delta > 1/4$ obtained by confusing $\operatorname{supp} g$ with $\operatorname{supp}(g * \tilde{g})$. Chapter 9 separates the support–Laplace ceiling at exponent $\frac{1}{2}$ from the Euler-product boundary at exponent 1. Chapter 10 turns these calculations into the Explicit-Formula Barrier Requirement, a decision procedure that asks where cancellation, phase or an independent rigidity theorem enters an argument. Chapter 11 states the limits of analogies with barriers in other subjects, and Chapter 12 explains what finite Gram and semidefinite certificates can prove, and only after uniform tails and form-core control have been supplied. The volume ends with a specification, not a proof: an explicit-formula argument must exhibit the information that absolute majorization and finite restriction do not contain.

Volume II: The Proof Architecture

From the paradigm to a construction problem. Volume II begins from the one setting in which the analogue of RH is proved and asks which parts of that proof survive the passage from a curve over $\mathbb{F}_q$ to arithmetic over $\mathbb{Z}$. Its purpose is not to infer the number-field theorem by analogy. It is to separate the finite-field proof into structures that can be constructed, structures with only local replacements, and structures for which no number-field object is known.

Part III, *The function-field paradigm, fully instantiated*, carries out the function-field case with its actual objects: curves and their zeta functions (Chapter 13), Weil's proof through the Hodge index theorem (Chapter 14), Grothendieck's cohomological machine (Chapter 15), Deligne's Weil I and Weil II (Chapter 16), Hesselholt's determinant theorem (Chapter 17), and the complete finite-field dictionary (Chapter 18). The functional equation and the critical-line bound come from different inputs. Duality explains a reflection; positivity or purity supplies the inequality that pins eigenvalues to the fixed locus. This distinction is the template for the rest of the volume.

Part IV, *The absolute base*, changes the base from $\mathbb{Z}$ to the sphere spectrum $\mathbb{S}$ and records what is gained. The derived diagonal becomes nontrivial; THH, TP and TC provide circle and cyclotomic structures; prismatic and condensed methods provide local comparison languages (Chapters 19–23). The same Part proves the limits of that gain. Integral periodicity and a global complex-linear generator do not follow formally (Conditional Obstruction P, Chapter 21), and neither a finite-place Frobenius nor a Sen weight operator supplies the missing archimedean dynamics. Chapter 24 delivers the verdict on what the absolute base does and does not buy.

Part V, *The adelic cyclotomic stack $\mathfrak{X}_{\mathbb{S}}$ and AS0–AS6*, organizes the surviving proposals as a typed list of obligations. AS0 is the governing discipline: a named object may be used only as a theorem with a source, a construction with specified interfaces, or an open problem (Chapter 25). Chapters 26–30 test the obligations in order, from the absolute base and the nontrivial self-product to loop-space realization, adelic Frobenius, comparison tests, and

traces and determinants. [N] The final positivity audit is decisive: the original positive-definite AS6 would already force the critical line and normality, so postulating it would strengthen the problem rather than reduce it. The volume therefore produces a typed architecture and several no-go results, not a global spectral object, and leaves one question in place of an analogy: where can a single arithmetic object carry spectral realization, centered duality, and an independently justified index structure at once?

Volume III: Reductions of RH and GRH

Conditional chains and the equivalence audit. Volume III contains the book's two converging lines. One asks what kind of positivity can survive the criticality barrier. The other asks whether amplification of global zero data can acquire an independent upper constraint. They meet in the demand for a rigidity statement that is neither an absolute majorant nor a finite numerical pattern.

Part VI, *Positivity: what it is, where it comes from, and why it is absent*, calibrates against the function-field case, where duality gives the functional equation and the positive Rosati trace form on correspondences gives purity (Chapter 31). It then tests number-field candidates (Chapter 32), Hodge–Riemann, Nygaard and prismatic candidates (Chapter 33), C^*-refinements and trace states (Chapter 34), and local-to-global transport of positivity (Chapter 35), and closes with a compliance manual for positivity claims (Chapter 36). A candidate form must have the correct sign, must remain meaningful at the critical point and not only in a half-plane of absolute convergence, and must distinguish ζ from the firewall objects. What survives is a specification for an equality-type or index-type object.

Part VII, *Amplification theory and the Amplification Dichotomy*, separates three operations that are often all called amplification: enlarging support in an explicit formula, taking functorial powers of local parameters, and forming global sums or correlations of zeros (Chapter 37). Only the third magnifies an off-line displacement, and magnification alone is not a constraint. [N] The Amplification Invariance Theorem shows that for Dirichlet series whose singular support contains k-fold sums of zeros,

$$\sigma_c - \mathrm{Re}(\text{amplified singularity}) = k(\tfrac{1}{2} - \delta)$$

identically, independently of normalization (Chapter 39). The Level Collapse Theorem proves the same ceiling through summatory functions without continuing an amplified object; the Infinite Multiplicity Theorem shows that the divisor of zero sums is not the divisor of any meromorphic function; the Structure-Blindness Theorem shows that the additive machine forgets the Euler product and so fails the firewall. Chapters 40–42 state the Positivity–Abscissa Dichotomy, the Natural Boundary Obstruction, and what an admissible amplifier would have to prove. These are theorems about specified mechanisms, not a theorem that every future amplifier is impossible.

Part VIII, *Reductions: rigidity, signature, and realization*, supplies the finite-dimensional positive core. Chapter 43 develops finite polarization theory, Theorems A, B, C and E. [T] Chapter 44 proves the Krein Index Theorem: if a nondegenerate Hermitian form G of inertia (n_+, n_-) and an operator Θ satisfy $\Theta^* G + G\Theta = G$, then the number of eigenvalues of Θ with $\mathrm{Re}\,\lambda > \frac{1}{2}$, counted with algebraic multiplicity, is at most $\min(n_+, n_-)$. Chapter 45 makes the rigidity turn that criticality demands. [E] Chapter 46 reports the measured signature of sixty-three pole-removed cubic-spline Guinand–Weil Gram matrices: exactly one negative direction in every tested configuration, and inertia $(N-1, 1)$ wherever the declared tolerance resolves every eigenvalue (forty-five of the sixty-three). [T] Chapter 47 establishes the Jensen–Hodge

dictionary, and Chapter 48 proves the Index-One Exhaustion Theorem, a complete derivation of RH from the three realization hypotheses AHK-1, AHK-2 and AHK-3. Appendix I extends the architecture to the generalized Riemann hypotheses and records what changes.

Volume IV: The Remaining Obligations

Construction, certification, and falsification. Volume IV changes the unit of discussion from architecture to certificates and obligations. Its first Part asks what finite computation can establish without extrapolation; its second asks how a third party could attack, reject or falsify the two open premises the earlier volumes leave.

Part IX, *Computation, certification, and formalization*, treats rigorous verification of zeros (Chapter 49), certified numerics for the signature computation (Chapter 50), the twenty-six catalogued algorithms with specification, pseudocode and complexity (Chapter 51), reproducibility and data provenance (Chapter 52), formal verification in proof assistants (Chapter 53), and the Solvability Complexity Index (Chapter 54). A Turing certificate proves a finite zero statement because a zero-count theorem closes a bounded region. A Gram certificate proves a statement about one finite matrix. Neither becomes an asymptotic theorem by raising the cutoff; uniform tails, convergence topology and form-core control must be supplied separately. [T] Hansen's theorem that the Solvability Complexity Index of the stated general spectral problem equals two is a theorem about its information model, not an undecidability or independence result for RH.

Part X, *Open problems, milestones, and falsification*, states the remaining obligations without merging them. Chapter 55 states the Arithmetic Hodge–Krein Realization Problem in full, explains why the nonvacuity conditions AHK-4 to AHK-6 cannot be dropped, and decomposes the problem into seven attackable sub-problems. Chapter 56 treats Problem A2$\star$ and the mean-purity route. Chapter 57 states six further open problems, from the polarization of a cyclotomic spectrum to Jensen shifts of high degree. Chapter 58 gives the falsification protocol: milestones M0 to M5, decision gates, a risk register, and ten falsification tests F1 to F10. Chapter 59 states what would constitute a proof and what would not, and Chapter 60 records the state of the question.

The standard of the volume is deliberately asymmetric. A failed test can end a branch of the programme; a passed finite test cannot prove it. Until one of the open premises is proved in a form that survives these tests, RH and GRH remain open.

The apparatus

The ten appendices carry what the argument needs but the chapters should not interrupt themselves to supply. Appendices A and B give the analytic and homotopy-theoretic prerequisites. Appendix C gives the complete proof of the Krein Index Theorem. Appendix D gives the full protocol and tables of the signature computation. Appendix E is the complete no-go compendium; Appendix F records errata, corrections and disputed points; Appendix G explains how the status system is read. Appendix H catalogues the algorithms ALG-01 to ALG-26, Appendix I treats the generalized Riemann hypotheses, and Appendix J states Problem A2$\star$ as the one non-circular obligation. The glossary, chronology, consolidated register of open problems and four indices, of names, of theorems and named results, of subjects and of symbols, complete the work. The numerical scripts and data that produce every computed value are published with the book.

Why record obstructions

A theorem stating that a method cannot yield more than the classical bound is useful when it identifies the invariant responsible for the ceiling. The Amplification Invariance Theorem does this for singular supports built from k-fold sums of zeros: the putative gain and the convergence cost have the same slope, and no normalization alters that equality. The Level Collapse Theorem expresses the same phenomenon elementarily: a k-fold summatory estimate has exactly the error exponent of the one-fold prime number theorem. Two proofs are given because they fail under different mistakes; a disputed singularity calculation does not touch the elementary equivalence, and an endpoint convention in the summatory formula does not touch the slope identity.

Failed ideas return under altered notation unless their failure has been proved. Support widening, normalization changes, higher correlations and positivity relaxations can look distinct while preserving the same invariant. A proved obstruction lets a later author check the invariant first. If it is unchanged, the old ceiling still applies. If it has changed, the new method can say exactly where it leaves the obstructed class. For a system that discovers structure by proposing and testing, a library of scoped obstructions is negative knowledge in usable form: it tells the search where not to go, and why.

Some obstructions change the logical category of a question. The proposed meromorphic function attached to sums of zeros does not exist, because its divisor is impossible, and questions about its automorphy or purity no longer define a research route. The positive-definite polarization failed differently: it already contains a stronger spectral conclusion than the one it was meant to deliver. Replacing positive definiteness by controlled indefinite index changes the logical content, and the Krein Index Theorem then measures how many right-half-plane eigenvalues the negative directions can support. The refutation of one polarization is the step that found the next.

The positive line

[T] The finite polarization theory is complete in its finite-dimensional setting. At negative index at most one, conjugate pairing rules out a nonreal eigenvalue to the right of the line, because it would come with its complex conjugate; the absence of real zeros of ζ in $(0,1)$ removes the remaining case; the functional equation gives the left half. This proves the Index-One Exhaustion Theorem: AHK-1, AHK-2 and AHK-3 together imply RH.

[O] The theorem does not supply its arithmetic input. No system satisfying AHK-1 to AHK-3 is constructed in this work. [K] The equivalence audit makes the reason precise. Under RH a tautological diagonal construction exists, so AHK-1 to AHK-3 are not, on their own, a weaker route to the conclusion. Arithmetic content must enter through AHK-4 to AHK-6, the conditions that prevent a realization from collapsing into that tautology. They are specified and tested here; they are not jointly realized.

[T] A second positive component is the Jensen–Hodge dictionary. Degree-two Jensen hyperbolicity is the Lorentzian signature of a Hessian and the degree-one Hodge–Riemann relation of the associated apolar algebra; in general degree, Jensen hyperbolicity is equivalent to the Pólya-frequency property. [N] Ordinary log-concavity is weaker in degree three and above and cannot replace hyperbolicity. The relevant Hodge object belongs to the Jensen apolar algebra of ξ, not to additive correlations of the von Mangoldt function, and the book keeps the two signature shadows apart.

Why an index and not a positivity

The single substantive choice of the book is the replacement of a positivity condition by an index condition. The classical spectral proposals all seek definiteness: Hilbert and Pólya a self-adjoint operator, Weil a nonnegative functional, the original axiom AS6 a positive polarization. Definiteness is what makes spectra real, and it is the natural thing to want. Three facts move the book away from it.

[N] Definiteness proves too much. With a self-adjoint centered operator the polarization forces normality, and the numerical range then lies on the critical line itself, a statement strictly stronger than RH and false for the firewall objects. A hypothesis stronger than the conclusion is not a route to it. Definiteness is also incompatible with the pole. The pole of ζ at $s = 1$ contributes a reflected pair to any spectral realization, and a reflected pair is exactly what a negative direction produces; excising the pole moves the negative direction into the excised plane, whose signature is $(1, 1)$, but does not remove it. Most decisively, definiteness has margins and an index does not. The equality $\Lambda_{\mathrm{dBN}} = 0$ says that RH, if true, holds with no room. A positivity statement with a uniform margin is then false, and one without a margin is out of reach of every perturbative or stability method. An index is an integer: it cannot be nearly right, and a proof that it equals one is not a perturbative statement.

Whether such a proof exists is the open question. What the book establishes is that the shape of an index statement is compatible with the criticality theorem and the shape of a positivity statement is not. The cost of the choice is also stated: spectral theory in Krein and Pontryagin spaces is less developed and less forgiving than in Hilbert space, and the book proves the finite-dimensional results it needs rather than assuming their infinite-dimensional analogues.

Computation and its limits

Computation enters as an audit and as a source of finite evidence. It checks normalization identities, locates counterexamples to naive inequalities, measures Gram-matrix inertia, and compares truncated prime-power sums with independently evaluated spectral terms. It never assigns theorem status to a limiting claim, and data ranges, precision, tolerances and certificate types accompany every reported value.

[E] The sixty-three pole-removed Gram matrices are the main example. They span cubic-spline spaces of specified width and dimension, use actual prime powers, and every tested matrix has exactly one negative eigenvalue. [T] The pole plane has a proved signature. [O] What is missing is the operator-compatible duality: an operator on the same space, the centered adjoint identity, a stable spectrum exhausting the zeros, and compatibility as the stage changes. A Gram matrix supplies none of these by its inertia. The measurement is kept because it identifies the index the explicit formula suggests, not because it completes the realization.

The same separation governs certification and formalization. A certified inertia computation proves the inertia of the displayed matrix; it does not prove that an unspecified limiting form has that inertia. A proof-assistant kernel can check a finite-dimensional lemma once its definitions are encoded; it cannot supply the arithmetic object named in an open problem. Formalization is therefore directed first at the Krein argument, the symmetry bookkeeping, the exact finite certificates and the dependency graph.

Two obligations, precisely posed

[O] The first obligation is the Arithmetic Hodge–Krein Realization Problem: realize the pole-removed Weil form as an index-one Hodge–Krein duality for an operator whose stable spectrum is the multiset of nontrivial zeros, satisfying AHK-1 to AHK-6. The cyclotomic, prismatic and adelic material of Volume II specifies possible sources and proves obstructions to several direct candidates. It does not fill the gap.

[O] The second obligation is Problem A2$\star$, in its slope-breaking, mean-purity and category-rigidity forms. Its mean-purity form asks for the one-sided estimate

$$\sum_{0<\gamma\le T} (\beta - \tfrac{1}{2})^{+} = o(T).$$

The functional equation makes the signed sum cancel, but cancellation does not control the positive part, and present zero-density estimates give only the scale $T \log T$. The estimate has the same one-sided shape as the index bound: one statement controls the count of right-half-plane spectral points, the other their total displacement. The book proves the relations available between the two and does not identify them without hypotheses.

Each obligation is decomposed into sub-problems with intermediate targets and grounds for rejection, and each is attached to falsification tests. A reader may attack either without accepting the other, and may reject the motivation of both while still testing the required object, map, estimate or compatibility.

How the programme can fail

A programme that cannot fail is not a mathematical programme. Chapter 5 states at the outset what would falsify this one, and Chapter 58 gives the full protocol: milestones M0 to M5 with decision gates between them, a register of risks, and ten falsification tests, each attached to the edge of the dependency graph that it would cut. They test the Krein line itself (F1), the measured signature (F2), whether the hypotheses prove too much (F3), whether they are vacuous (F4), whether index one is the right index (F5), category rigidity (F6), the Jensen–Hodge dictionary (F7), the pole plane (F8), family rigidity and replication (F9), and finally a zero off the line (F10), which would refute the hypotheses of the conditional theorem without touching the theorem.

The same chapter states what would not falsify the programme: that no construction satisfying AHK-1 to AHK-3 has yet been found, which is the present state and counts against the programme only with an obstruction theorem behind it; numerical verification of RH to greater height; a better zero-density estimate of the standard shape; slow progress on the six further problems; a construction satisfying AHK-1 and AHK-2 but not AHK-3, which would be progress, since only incompatibility refutes. A proof of RH by other means would leave every question about mechanisms and structures open; it would supersede the programme without refuting it. What would refute it is an incompatibility, a circularity, or a mechanism that proves too much.

Chapter 59 states, in the same spirit, what would constitute a proof and what would not. The asymmetry is deliberate. A single failed test can end a branch; no number of passed finite tests can establish one.

The generalized hypotheses

Appendix I extends the architecture from ζ to Dirichlet L-functions, Dedekind zeta functions and the Selberg class, with realization hypotheses AHK-1_L to AHK-6_L. [T] The index bound transfers verbatim, and the index is the count of off-line zeros. The appendix then proves what changes: for a function F of the extended Selberg class, AHK-1_L, AHK-2_L and AHK-3_L hold if and only if F has at most one nontrivial zero with $\operatorname{Re}\rho > \frac{1}{2}$, and for self-conjugate F that zero is real. For ζ the classical absence of real zeros in $(0,1)$ removes it; for a general L-function no such input is available, and the residual gap is exactly the Landau–Siegel problem. Index one for the square F^2 is equivalent to GRH for F; index two is weaker; index zero is vacuous. The appendix closes with which of the consequences of GRH in the literature, from primes in progressions and the least quadratic non-residue to effective Chebotarev bounds, depend on the single-function statement and which on the family.

What the Programme takes from this work

For the Grothendieckian Programme the book is a complete instance of the cycle it asks a discovering machine to perform. Its claims are typed by status. Its negative results form a scoped archive that a search can consult before it proposes. Its algorithms are specified with inputs, outputs, certificates and complexity. Its dependencies are recorded as a graph in which every edge is a theorem, a cited input, a finite observation, a proved obstruction or an open premise. Its open problems come with the tests that would reject a proposed solution.

None of this is offered as evidence that a machine can prove the Riemann Hypothesis, and nothing in the book depends on such a claim. What is offered is narrower and checkable: a hard problem, worked through by a method that can be stated precisely, has yielded theorems, obstructions and well-posed problems that stand on their own. That is what the Programme means by a research subject.

Priority and the literature

The explicit formula, Weil positivity, Krein-space methods, Jensen polynomials, combinatorial Hodge theory, cyclotomic homology and zero-density estimates all have substantial literatures, and the book does not rename their results as components of a programme and then claim them. Where a result is stated as new in form, the nearest literature found is cited and the difference specified. No literature search proves that nothing has been written, so the book does not claim that no predecessor exists; it says that it is not aware of a prior statement in the given form, where the search supports only that. A reader who supplies a predecessor changes the priority note, not the truth of the theorem. The no-go archive is not a classification of all failed approaches, and several of its entries sharpen or combine limitations visible in earlier work. Their content lies in their printed hypotheses and conclusions, not in their number.

Reading the work

A reader need not accept the whole architecture to use any part of it. The Krein theorem stands or falls by its linear algebra; the support-doubling obstruction by its support and growth calculation; the measured inertia by its data, tolerance and implementation; the two obligations by whether their specifications are coherent, and, for Problem A2$\star$, by whether it is weaker than RH. The guide that follows the Preface gives routes for analytic number theorists, arithmetic geometers, homotopy theorists, readers interested only in the positive

line, and referees. Chapter 5 states early what would falsify the programme; Chapter 58 gives the full protocol.

Scope and responsibility

Responsibility for each claim is local. A theorem is supported by its proof or citation, not by the scale of the programme. A computation is supported by its data, code, range and tolerance, not by an interpretation. An open problem is useful to the extent that its conditions can be checked and its failure reported.

[K] The boundary is stated once more. RH is unsolved. [T] The conditional chain from an index-one realization to RH is proved. [O] The realization is not constructed. [E] The measured finite signatures motivate the index; they do not supply the operator. [N] The no-go theorems exclude specified classes of mechanisms, not all possible proofs. This is the state of the question recorded in the pages that follow.

Vincent Liu
WiseMotif.ai
2026

How to read this book

This monograph is organized as one argument with several usable entry points. It is physically a single work: the four Volumes are internal divisions, pagination is continuous, chapter numbers run from 1 to 60, and there is one bibliography and one index. The divisions mark changes of question rather than separate books. A reader may follow one route below without assuming that the omitted routes have been proved by analogy.

The logical map

[N] Volume I, *Barrier Theory*, asks which broad classes of proposed proofs cannot yield RH and why. Its results include the amplification, level-collapse, infinite-multiplicity, structure-blindness, support, and positivity obstructions. The negative conclusions are scoped: each rules out the hypotheses printed in its statement, not every possible proof of RH.

[K] Volume II, *The Proof Architecture*, begins with the function-field case. The order is mandatory for the logic of the book. The function-field theorems are stated with their actual Frobenius, cohomology, duality, and positivity structures before any number-field counterpart is proposed. The later chapters then ask, object by object, which ingredients have an available analogue over the absolute base and which remain open.

[T] Volume III, *Reductions of* RH *and* GRH, supplies the finite polarization theory, the Krein Index Theorem, the Jensen–Hodge dictionary, and the Index-One Exhaustion Theorem. The reduction from AHK-1, AHK-2, and AHK-3 to RH is proved. [O] The arithmetic realization required by those hypotheses is not constructed. This Volume therefore supplies a proved conditional chain, not an unconditional proof of RH.

[O] Volume IV, *The Remaining Obligations*, turns the unresolved claims into construction tasks, computational certificates, failure conditions, and decision gates. It gives protocols by which a third party can attack or refute a branch of the programme. Its purpose is not to replace proof by computation; it distinguishes finite certificates from infinite statements and specifies the additional theorem each computation would require.

Route for the analytic number theorist

The recommended route is Parts I, II, VII, and VIII. Parts I and II fix the normalizations and prove the obstruction theory before introducing the positive mechanism. This order matters. The Amplification Invariance Theorem and the Level Collapse Theorem explain why passing to k-fold correlations does not improve the zero exponent merely by increasing the apparent singularity. The Structure-Blindness Theorem then tests whether the machine has retained the Euler product in a form specific to the ordinary primes. Part II places these statements beside the Guinand–Weil explicit formula and the absolute-value ceiling.

Part VII returns to amplification after the intervening geometric material, but it can be read directly after Part II. The reader should track three quantities separately: the real part of the amplified singularity, the abscissa or holomorphy threshold, and the displacement $\delta = \beta - \frac{1}{2}$. The no-go theorem is an identity between their slopes, not an estimate believed to

be improvable by a sharper constant. Problem A2⋆-I asks for an amplifier that changes this slope. Problem A2⋆-II asks instead for one-sided mean purity.

Part VIII gives the positive spectral line. Chapters 43 and 44 are finite-dimensional linear algebra and Krein geometry. Chapter 46 reports the Gram-matrix measurements, with the warning that the measured matrices do not carry a known operator satisfying the centered adjoint identity. Chapter 48 combines the index theorem with the symmetry and real-zero facts specific to ζ. An analytic reader can therefore assess the conditional implication without accepting any proposed construction from Volumes II or III.

For this route, Chapters 7 and 12 should not be skipped. Chapter 7 fixes the Fourier and Mellin conventions used in the explicit formula. Chapter 12 separates certified truncation error from sampled positivity. Many apparent contradictions in the literature reduce to a sign, a 2π factor, or an unstated test-function class; these chapters state the conventions against which later formulas should be checked.

Route for the homotopy theorist

The recommended route is Parts III, IV, and V. Part III is the reference model. It treats curves over finite fields, the cohomological formula, Poincaré duality, the Hodge index input, and Deligne's results as theorems with citations. The point is not to invoke the function-field case as an analogy. It is to list the objects and maps that make the proof possible there.

Part IV examines the sphere spectrum $\mathbb{S}$, THH, TP, TC, cyclotomic structures, prismatization, syntomic comparison, and analytic or condensed refinements. The notation table following this guide fixes the functors and Frobenius symbols. The sphere spectrum is always $\mathbb{S}$; the unavailable command is `\SS`. A statement that is a theorem in the cited literature is marked [K] or [T]. A proposed globalization over $\operatorname{Spec}\mathbb{Z}$ is marked [O] unless an actual functor and comparison map are provided. The useful question for a homotopy theorist is not whether the terminology resembles a trace formula, but whether the cyclotomic maps, completions, descent data, and archimedean comparison have the sources, targets, and functoriality claimed.

Part V packages the desired compatibilities as AS0–AS6. These axioms should be read as a specification and an audit device, not as a constructed stack. The chapters distinguish local cyclotomic Frobenius from a global operator whose spectrum could encode zeta zeros. They also record the linearity and archimedean obstructions to direct candidates. Whenever a number-field sentence resembles a function-field theorem, the text states which hypothesis has been lost.

A homotopy theorist who wants the point of contact with the final reduction should then read Chapters 43, 44, 46, and 48 in the approved sequence. The required output is not merely a cohomology group or trace formula. It is a filtered family $(H_N, \Theta_N, G_N, \iota_N)$ with spectral realization, centered duality, compatible negative index at most one, and the arithmetic conditions preventing a tautological construction from the zeros themselves.

Route through the positive main line

A reader interested only in the surviving positive line may begin with Chapters 43–48. Chapter 43 develops finite polarization and records the equivalences valid for positive-definite forms. Chapter 44 proves the Krein Index Theorem for an indefinite form. Chapter 45 explains why criticality shifts the target from strict positivity to equality and rigidity. Chapter 46 gives the measured pole-removed signatures. Chapter 47 proves the Jensen–Hodge dictionary. Chapter 48 states and proves the Index-One Exhaustion Theorem and performs the firewall audit.

This route is logically self-contained for the conditional theorem, but it does not explain why several alternative constructions were discarded. For that information, read Chapters 1, 6, 39–42. It also does not provide the arithmetic realization. That obligation is stated in Chapters 48 and 57. The shortest honest summary of the route is:

[O] index-one arithmetic realization + [T] finite Krein argument $\Longrightarrow$ [O] RH remains open by this route.

The status attached to the premise is not changed by the theorem attached to the implication.

Route for a referee

A first-pass referee route is Chapters 1, 6, 48, and 60. Chapter 1 states the obstruction theory and its scope. Chapter 6 supplies the theorem archive and proof audit, allowing the referee to check whether each no-go result matches the advertised class. Chapter 48 is assigned the conditional main theorem, the no-real-zero input, the symmetry argument, and the four-object firewall. Chapter 60 is assigned the conditions that would falsify the programme or force a status change. While Parts I, VIII, and X remain under completion, a referee should use the status tables and the Scientific notice to distinguish landed proofs from assigned chapter roles.

The front matter is part of that route. The Scientific notice fixes the claim boundary. The abstract lists the results without promoting evidence or open constructions. The notation chapter fixes transforms, conjugations, domains, and reserved symbols. The status key gives worked examples and the editorial meaning of every badge. A discrepancy between a chapter and these pages should be treated as a defect to be corrected, not as permission to read the stronger claim.

A referee checking only the positive theorem should verify four separations. First, the inertia measured for a Gram matrix is not the inertia of an operator-compatible duality until the operator and adjoint identity are constructed. Second, exhaustion includes algebraic multiplicity and excludes spurious stable spectral points. Third, conjugation removes nonreal right-half-plane zeros at index one, while the real case requires the classical zeta-specific lemma. Fourth, the functional equation is used only after the right half has been controlled.

A referee checking the negative results should ask whether each theorem is invariant under the normalizations claimed, whether the cited counter-model has precisely the listed properties, and whether the conclusion excludes a method or only one implementation. The phrase “cannot prove RH” is used only with a class of allowed inputs stated in the theorem.

Route for computation and formalization

A reader concerned with certified computation should read Chapters 12, 46, and 49–54. Chapter 12 fixes truncation and tail estimates for the explicit formula. Chapter 46 identifies the finite spaces, matrices, data source, parameter ranges, and tolerance behind the Lorentzian measurement. Chapters 49–52 separate zero enclosures, prime-power truncations, inertia certificates, spectral pollution tests, and convergence diagnostics. Chapter 53 states the formalization targets, and Chapter 54 treats computations associated with the prismatic and cyclotomic proposals.

The certificate badge matters. FLOAT is an observation; INT and BALL are directed enclosures; RAT is exact rational data; LDL certifies finite inertia; and TAIL records an analytic remainder bound. A finite theorem can depend on several certificate types, but none certifies an unconstructed limiting operator. The notation chapter gives the full key.

For reproducibility, follow the dependency order printed with each experiment: input data, normalization, basis, matrix assembly, tail bound, factorization, and independent check. Recomputing only the final eigenvalues is not an independent verification if the same matrix assembly contains the disputed sign. The chapters therefore provide checks at the level of the explicit-formula terms as well as at the level of inertia.

Route by dependency rather than subject

Readers may also follow the dependency graph. Begin with Chapter 7 for the explicit formula and Chapter 43 for finite polarization. Chapter 44 then proves the index bound without arithmetic assumptions. Chapter 2 supplies the counter-model data needed by the firewall. Chapter 46 provides finite evidence for the proposed index, and Chapter 48 proves the conditional exhaustion theorem. Chapter 57 finally states the realization problem in the form needed to connect those components.

This route exposes three independent interfaces. The analytic interface must recover the true prime-power masses. The operator interface must produce stable spectral exhaustion and the centered adjoint identity. The geometric interface must account for negative index one and transition compatibility. A construction that satisfies only two interfaces is not silently completed by the third part of the book. Its status table records which interface is missing.

Problem A2$\star$-II lies on a parallel branch. Read Chapters 29, 38–42, and 48. The zero-density theorem supplies the available unconditional upper constraint, the amplification chapters prove the slope ceiling, and Chapter 48 compares displacement control with index control. Neither branch is declared equivalent to the other without the hypotheses printed there.

Route for the open problems only

A reader who wants only the unresolved obligations should begin with the Scientific notice and then read Parts VIII and X in the approved plan. The first obligation is the Arithmetic Hodge–Krein Realization Problem: construct $(H_N, \Theta_N, G_N, \iota_N)$ satisfying AHK-1 through AHK-6. The implication from AHK-1 through AHK-3 to RH is already proved; the object satisfying those hypotheses is not known.

The second obligation is Problem A2$\star$. Its three formulations are the slope-breaking inequality for an amplifier, the one-sided mean-purity estimate

$$\sum_{0<\gamma\le T} (\beta - \tfrac{1}{2})^{+} = o(T),$$

and category rigidity for admissible maps from zero-tuples of ζ to zeros of a Selberg-class function. A reader working on one formulation should consult Chapters 38–42 for the obstructions that any proposed solution must evade.

For either problem, read the falsification conditions before the proposed sources of examples. A candidate AHK system fails if it lacks stable spectral exhaustion, centered duality, compatible index, or the arithmetic specificity conditions. An A2$\star$ candidate fails if its continuation bound retains slope k, if it merely rewrites the signed cancellation forced by the functional equation, or if its category map is not defined independently of the desired zeros.

How to use the status labels

[T] marks a theorem whose proof appears in the book or a statement tied to an exact published source. The label does not distinguish those two cases; the surrounding sentence or citation does. A reader should expect all hypotheses to be visible.

[C] marks a precise conjecture. It is neither finite evidence nor an object under construction. A conjecture should have a negation clear enough to guide a counterexample.

[O] marks an open obligation. In this book it means that no candidate object satisfying the listed conditions exists. The Arithmetic Hodge–Krein Realization Problem is the principal example.

[E] marks a computation, finite verification, or observed pattern. The range, precision, and certificate type should occur nearby. An evidence label never changes to a theorem because all tested cases agree.

[N] marks a proved no-go theorem. Its force is exactly its quantified class. It may close every normalization of one architecture while leaving other architectures untouched.

[K] marks known background. Such a statement must still be cited where mathematical content is asserted. The badge is not a substitute for a source.

Status labels can occur at the start of a paragraph when all mathematical sentences in that paragraph share the same status. A change of status within a paragraph is marked at the point of change. Mixed-status display chains label each premise when ambiguity is possible.

Boxes, ledgers, and closing audits

An `obstruction` box contains a theorem showing that a specified method cannot meet its stated target. Its number refers to the obstruction ledger. The box should be read with the hypotheses; it is not a general warning against the surrounding subject.

A `firewall` box tests a proposed mechanism against the four published counter-models. Passing the test means that the mechanism distinguishes those four for stated reasons. It does not prove the mechanism exists for ζ, and it does not exhaust all possible false analogues.

An `openproblem` box isolates a construction or estimate that is not proved. The box is intended to be exportable: a reader should be able to work on it without adopting the prose surrounding the programme. Its hypotheses, output, and falsification conditions are therefore part of the statement.

Every Part opens with a status table. The columns record the object, its badge, whether it is constructed, and its logical strength. “Partial” means that named components or finite approximations exist but the object required by the theorem does not. “Conditional” identifies the exact premise; it is not used as a softer synonym for open.

Every chapter closes with “What this chapter does not establish.” Read that section before carrying a result into another chapter. It records unproved converses, missing limits, absent compatibility maps, and claims that a nearby computation does not support. Every Part has a corresponding closing audit.

Conventions while reading

The Fourier transform is not to be inferred from context. The convention in the notation chapter, including every 2π factor and conjugation, governs the book. The same is true of the Mellin transform, adjoints, linear-slot convention for Hermitian forms, algebraic multiplicity, and the meaning of stable spectral exhaustion. If a cited source uses another normalization, the conversion is stated before its formula is used.

Symbols listed as globally reserved do not change meaning between Volumes. Locally reused symbols are identified in the notation table and at their first occurrence. Domains of unbounded operators are part of the operator notation; a formal adjoint identity is not asserted without a common domain or a finite-dimensional setting.

Finally, the reader should preserve the distinction between a reduction and a proof of its premise. The Index-One Exhaustion Theorem is a theorem. The arithmetic Hodge–Krein realization is open. The Jensen–Hodge dictionary is a theorem about apolar algebras; it does not construct the operator required by the Arithmetic Hodge–Krein Realization Problem. The finite signature measurements are evidence; they do not imply RH. These boundaries permit the separate parts of the book to be read, criticized, and reused without accepting a claim the book does not make.

Notation and conventions

This chapter fixes the notation used throughout the work. “Reserved” means that the symbol has one global meaning. “Local” means that a chapter may declare a specialized meaning at first occurrence. The “type” column distinguishes numbers, functions, operators, forms, spectra, functors, categories, and editorial certificates. Chapter numbers indicate the first substantive use in the sixty-chapter structure; elementary occurrences in the front matter do not alter them.

Transform conventions

[K] Every Fourier formula in the book uses

$$\hat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x\xi}\, dx, \qquad f(x) = \int_{\mathbb{R}} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi,$$

whenever inversion is justified. Thus

$$\widehat{f * g} = \hat{f}\,\hat{g}, \qquad \widehat{fg} = \hat{f} * \hat{g}, \qquad \widehat{f'}(\xi) = 2\pi i \xi \hat{f}(\xi), \qquad \hat{\hat{f}}(x) = f(-x).$$

We put

$$\tilde{f}(x) = \overline{f(-x)}, \qquad (f * g)(x) = \int_{\mathbb{R}} f(y) g(x-y)\, dy.$$

Consequently $\hat{\hat{f}}(\xi) = \overline{\hat{f}(\xi)}$ and $\widehat{f * \tilde{f}} = |\hat{f}|^2$. Translation and modulation are

$$\widehat{f(\cdot - a)}(\xi) = e^{-2\pi i a\xi} \hat{f}(\xi), \qquad \widehat{e^{2\pi i a(\cdot)} f}(\xi) = \hat{f}(\xi - a).$$

No alternative angular-frequency convention is used silently. A source using $e^{-ix\xi}$ is converted by $\xi_{\text{angular}} = 2\pi\xi$.

[K] Every Mellin formula uses

$$(\mathcal{M}f)(s) = \int_0^\infty f(x) x^{s-1}\, dx, \qquad f(x) = \frac{1}{2\pi i} \int_{(c)} (\mathcal{M}f)(s) x^{-s}\, ds,$$

on a line $\operatorname{Re} s = c$ in a strip where inversion is valid. Complex conjugation obeys

$$\mathcal{M}(\bar{f})(s) = \overline{(\mathcal{M}f)(\bar{s})}.$$

For multiplicative convolution $(f *_\times g)(x) = \int_0^\infty f(y) g(x/y)\, dy/y$, one has $\mathcal{M}(f *_\times g) = (\mathcal{M}f)(\mathcal{M}g)$. Under $x = e^u$,

$$(\mathcal{M}f)(\tfrac{1}{2} + it) = \int_{\mathbb{R}} e^{u/2} f(e^u) e^{itu}\, du = \hat{g}\Big(-\frac{t}{2\pi}\Big), \qquad g(u) = e^{u/2} f(e^u).$$

This identity fixes the sign and the 2π factor linking Mellin and Fourier variables. Every formula in the book uses these conventions.

[K] Hermitian forms are linear in the first variable and conjugate-linear in the second:

$$[av_1 + bv_2, w]_G = a[v_1, w]_G + b[v_2, w]_G, \qquad [v, w]_G = v^* G w.$$

The star on a matrix denotes conjugate transpose. For a densely defined operator on a Hilbert space, A^* denotes the Hilbert-space adjoint with its domain. An identity involving unbounded operators includes its domain statement; otherwise it is only formal. Algebraic multiplicity is used in all spectral counts unless geometric multiplicity is named.

Explicit-formula normalization

[K] The nontrivial zeros are written $\rho = \beta + i\gamma$, with algebraic multiplicity. A sum $\sum_\rho$ is interpreted through the symmetric limiting procedure or test-function formulation stated with the formula; it is not an absolutely convergent unordered sum unless absolute convergence is proved. A sum over $0 < \gamma \le T$ includes the upper endpoint and excludes $\gamma = 0$. If an ordinate lies on an endpoint, the convention is stated rather than inferred.

Prime-power terms use the von Mangoldt weight $\Lambda(n)$. In the centered multiplicative normalization, the arithmetic mass occurring in the proposed trace formula is

$$-\Lambda(n) n^{-1/2}.$$

The minus sign and the factor $n^{-1/2}$ are part of the normalization. They are not absorbed into the test function when the AHK-4 condition is stated. The archimedean term uses $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, and the pole and trivial-zero terms are displayed or explicitly incorporated into a named distribution.

The full Weil form, its pole-plane contribution, and the pole-removed form are distinct objects:

$$B_N = W_N - P_N.$$

Subscripts indicate the finite test space and basis. A change of basis acts by congruence on a Gram matrix and therefore preserves inertia. A change of test space is not a change of basis and carries no automatic inertia conclusion.

Asymptotic and counting conventions

[K] The relations $f = O(g)$, $f \ll g$, and $g \gg f$ mean that the absolute value of f is bounded by a constant multiple of the nonnegative comparison function g on the stated range. Dependence of the implied constant is shown by a subscript or listed in the sentence. The notation $o(g)$ is taken in the limit named in the formula. An expression $T^{a+o(1)}$ means that for every $\varepsilon > 0$ the quantity is $O_\varepsilon(T^{a+\varepsilon})$ in the stated range; no uniformity in other parameters is implied unless given.

Zero-counting functions count algebraic multiplicity. In $N(\sigma, T)$, the boundary convention is $\beta \ge \sigma$ and $0 < \gamma \le T$. Replacing $\ge$ by $>$ affects boundary zeros and is done only when printed. The one-sided displacement is

$$(\beta - \tfrac{1}{2})^+ = \max(\beta - \tfrac{1}{2}, 0).$$

The signed sum and the positive-part sum are never interchanged. Symmetry can force cancellation in the former while leaving the latter positive.

Operator identities and limiting language

[K] In finite dimension, $\Theta^* G + G\Theta = G$ is a matrix identity. In infinite dimension, the same typography is used only after specifying a common invariant core or an equality of the associated closed forms. Equality on eigenvectors alone is not promoted to an operator identity. The adjoint is taken with respect to the positive Hilbert-space structure; $[v, w]_G = v^* G w$ is the separately named indefinite form.

"Compatible" for the finite stages means that the transition maps and forms satisfy the diagram or pullback identity printed in the relevant definition. It does not mean only that the dimensions increase. "Stable spectrum" refers to persistence with multiplicity under those maps, not to visually stable floating-point eigenvalues. "No spurious points" excludes persistent spectral values that are not nontrivial zeros, even if all genuine zeros are also present.

The words *trace*, *determinant*, and *spectrum* are typed by their setting. A finite trace is ordinary. A regularized determinant includes the chosen regularization. A distributional trace formula is not an operator trace until a theorem identifies them. These distinctions are carried in the type column below and restated when two settings meet.

1. Arithmetic functions and L-functions

Symbol	Meaning	First	Type	Reserved
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	Natural numbers, integers, rationals, reals, and complex numbers.	1	sets/fields	yes
$\mathbb{F}_q$	Finite field with q elements.	13	field	yes
$\mathbb{A}, \mathbb{A}_{\mathbb{Q}}$	Adèle ring; adèles of $\mathbb{Q}$ when the subscript is shown.	18	topological ring	yes
$\zeta(s)$	Riemann zeta function, initially $\sum_{n\geq 1} n^{-s}$ for $\operatorname{Re} s > 1$, continued meromorphically.	1	meromorphic function	yes
$\xi(s)$	Completed entire xi-function, $\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$.	7	entire function	yes
$\Lambda(n)$	von Mangoldt function: $\log p$ if $n = p^m$, $m \geq 1$; zero otherwise.	1	arithmetic function	yes
$\mu(n)$	Möbius function.	9	arithmetic function	yes
$\psi(x)$	Chebyshev function $\sum_{n\leq x} \Lambda(n)$, with endpoint convention stated when distributions are used.	1	summatory function	yes
$\vartheta(x)$	$\sum_{p\leq x} \log p$.	9	summatory function	yes
$\pi(x)$	Number of primes $p \leq x$.	9	counting function	yes
$\rho = \beta + i\gamma$	A nontrivial zero of ζ, repeated according to algebraic multiplicity.	1	spectral point	yes
Θ	$\sup\{\operatorname{Re}\rho : \zeta(\rho) = 0\}$, except where an operator bearing the same capital letter is explicitly present; operator usage is always typed by context.	18	real number/operator	local
$N(T)$	Number of nontrivial zeros with $0 < \gamma \leq T$, counted with algebraic multiplicity.	1	counting function	yes
$N(\sigma, T)$	Number of zeros with $\beta \geq \sigma$ and $0 < \gamma \leq T$, counted with algebraic multiplicity.	29	counting function	yes
$(x)^+$	Positive part $\max(x, 0)$.	38	real function	yes

1. Arithmetic functions and L-functions, continued

Symbol	Meaning	First	Type	Reserved
χ	Dirichlet character; its modulus and primitivity are stated locally.	31	arithmetic function	local
$L(s,\chi)$	Dirichlet L-function with the stated completion and conductor.	31	meromorphic function	local
$L(s,\pi)$	Automorphic L-function attached to a named representation π.	16	meromorphic function	local
$\Gamma_{\mathbb{R}}(s)$	$\pi^{-s/2}\Gamma(s/2)$.	7	gamma factor	yes
$\Gamma_{\mathbb{C}}(s)$	$2(2\pi)^{-s}\Gamma(s)$.	16	gamma factor	yes
RH, GRH	Riemann Hypothesis and the specified generalized Riemann Hypothesis. GRH is never used without naming the family.	1	propositions	yes
Λ_{dBN}	de Bruijn–Newman constant in the normalization fixed in Chapter 45.	1	real constant	yes
$R_k(N)$	k-fold von Mangoldt correlation $\sum_{n_1+\cdots+n_k=N}\Lambda(n_1)\cdots\Lambda(n_k)$, unless a smoothed version is explicitly marked.	18	arithmetic function	yes
D_k	Multiset of k-fold sums $\rho_1+\cdots+\rho_k$, with the multiplicity convention of Chapter 41.	1	multiset	yes
σ_c	Abscissa of convergence of the Dirichlet series currently under discussion.	39	real number	local
σ_{hol}	Boundary of a stated region of holomorphy.	38	real number	local

2. Transforms and test functions

Symbol	Meaning	First	Type	Reserved
$\hat{f}$	Fourier transform $\int_{\mathbb{R}} f(x)e^{-2\pi i x\xi}\,dx$.	7	transform	yes
$\mathcal{F}^{-1}$	Inverse Fourier transform $\int_{\mathbb{R}} F(\xi)e^{2\pi i x\xi}\,d\xi$.	7	transform	yes
$\mathcal{M}f$	Mellin transform $\int_0^\infty f(x)x^{s-1}\,dx$.	7	transform	yes
$\tilde{f}$	Reflected conjugate $\tilde{f}(x)=\overline{f(-x)}$.	7	test function	yes
$f*g$	Additive convolution $\int_{\mathbb{R}} f(y)g(x-y)\,dy$.	7	operation	yes
$f*_\times g$	Multiplicative convolution $\int_0^\infty f(y)g(x/y)\,dy/y$.	7	operation	yes
$\operatorname{supp} f$	Closed support of f; essential support is named when intended.	8	closed set	yes

2. Transforms and test functions, continued

Symbol	Meaning	First	Type	Reserved
$\mathcal{S}(\mathbb{R})$	Schwartz space on $\mathbb{R}$.	7	function space	yes
$\mathcal{D}(\mathbb{R})$	Smooth compactly supported test functions.	7	function space	yes
$\langle u,f\rangle$	Distribution–test-function pairing, linear in the test function under the convention stated in Chapter 7.	7	pairing	local
h_T	A translated or rescaled test function; the exact scaling is printed at each use.	8	test function	local

Continued on next page

2. Transforms and test functions, continued (continued)

Symbol	Meaning	First	Type	Reserved
g_T	Two-bump test function $ah(\cdot - T) + bh(\cdot + T)$ in the support obstruction.	8	test function	yes
$\mathbf{1}_E$	Indicator function of a set E.	9	function	yes
PW_A	Paley–Wiener class of exponential type A, with normalization specified by the Fourier convention above.	12	function space	local

3. Spectra, operators, forms, and domains

Symbol	Meaning	First	Type	Reserved
H, H_N	Complex vector space or Hilbert space; H_N is finite-dimensional in the AHK system.	43	space	local
Θ, Θ_N	Operator in the centered duality relation; the subscript N denotes a finite stage.	43	operator	yes
$\mathcal{D}(A)$	Domain of an unbounded operator A.	19	domain	yes
A^*	Hilbert-space adjoint, including its domain; conjugate transpose for a finite matrix.	43	operator	yes
$\operatorname{spec} A$	Spectrum of A; for finite matrices, eigenvalues with algebraic multiplicities are counted when a count is displayed.	3	set or multiset	yes
$\sigma_{\mathrm{p}}(A)$	Point spectrum of A.	3	set	yes
$\rho(A)$	Resolvent set of A; not to be confused with a zeta zero when A is present.	3	set	local
G, G_N	Nondegenerate Hermitian matrix or form; linear in the first slot.	43	Hermitian form	yes
$[v, w]_G$	Indefinite inner product $v^* G w$.	43	form	yes

3. Spectra, operators, forms, and domains, continued

Symbol	Meaning	First	Type	Reserved
(n_+, n_-)	Positive and negative inertia of a nondegenerate Hermitian form. Nullity is appended as a third entry when present.	44	pair of integers	yes
$n_\pm(G)$	Positive or negative inertia index of G.	44	integer	yes
$\kappa(G)$	Condition number in the norm specified locally.	43	real number	local
$\|\cdot\|_{S_1}$	Schatten trace norm.	43	norm	yes
$\Theta^* G + G\Theta = G$	Centered Hodge–Krein duality identity.	43	operator identity	yes
ι_N	Transition map between finite stages of an AHK system.	48	linear map	yes
$\mathrm{mult}_{\mathrm{alg}}$	Algebraic multiplicity.	44	integer	yes
$\mathrm{ind}_-(G)$	Synonym for $n_-(G)$, used only in cited conventions; the body text prefers $n_-(G)$.	44	integer	no
W_N	Finite Guinand–Weil Gram matrix on the stated spline space.	46	Hermitian matrix	yes
P_N	Pole-plane contribution to W_N.	46	Hermitian matrix	yes
B_N	Pole-removed matrix $W_N - P_N$.	46	Hermitian matrix	yes
Ai	Airy function.	45	special function	yes

4. THH, TP, TC, and cyclotomic notation

Symbol	Meaning	First	Type	Reserved
$\mathbb{S}$	Sphere spectrum. It is written with `\Sph`; `\SS` is unavailable.	19	E_∞-ring spectrum	yes
Sp	Stable ∞-category of spectra.	19	category	yes
$\mathbb{T}$	Circle group acting on THH; a different circle is named locally if required.	20	topological group	yes
$H\mathbb{Z}$	Eilenberg–Mac Lane E_∞-ring spectrum of $\mathbb{Z}$.	19	ring spectrum	yes
$\mathrm{CAlg}(\mathrm{Sp})$	Commutative algebra objects in spectra.	19	category	yes
$\mathrm{THH}(A)$	Topological Hochschild homology of A, with its circle action and cyclotomic structure when present.	20	spectrum or functor	yes
$\mathrm{THR}(A)$	Real topological Hochschild homology.	28	spectrum or functor	yes
$\mathrm{TP}(A)$	Periodic topological cyclic homology, in the convention stated in Chapter 20.	20	spectrum or functor	yes
$\mathrm{TC}(A)$	Topological cyclic homology.	20	spectrum or functor	yes
$\mathrm{TC}^-(A)$	Negative topological cyclic homology.	20	spectrum or functor	yes

4. THH, TP, TC, and cyclotomic notation, continued

Symbol	Meaning	First	Type	Reserved
$\mathrm{HH}(A)$	Hochschild homology.	20	complex or functor	yes
X^{hG}	Homotopy fixed points of a G-object X.	20	construction	yes
X_{hG}	Homotopy orbits.	20	construction	yes
X^{tG}	Tate construction.	20	construction	yes
φ_p	Cyclotomic Frobenius at p, with source and target displayed.	20	map	local
can	Canonical map entering the Nikolaus–Scholze equalizer description of TC.	20	map	local
Fr, Frob	Frobenius map or operator; arithmetic or geometric normalization is always stated.	13	endomorphism	yes
$\mathfrak{X}_{\mathbb{S}}$	Proposed adelic cyclotomic stack $\mathfrak{X}_{\mathbb{S}}$; its construction status is [O].	25	proposed stack	yes
$K(A)$	Algebraic K-theory spectrum of A.	20	spectrum	local
$\mathrm{tr}_{\mathrm{cyc}}$	Cyclotomic trace $K(A) \to \mathrm{TC}(A)$.	20	natural transformation	local

5. Prismatic and syntomic notation

Symbol	Meaning	First	Type	Reserved
(A, I)	Prism, with ideal I, under the boundedness hypotheses stated locally.	22	algebraic object	local
Prism_R	Prismatic site or prismatic cohomology object of R, as typed in the formula.	22	site/complex	local
Δ_R	Absolute prismatic cohomology of R.	22	complex	local
$\widehat{\Delta}_R$	Completed prismatic complex.	22	complex	local
φ	Prismatic Frobenius.	22	semilinear map	local
$\mathcal{N}^{\geq i}$	Nygaard filtration in degree at least i.	22	filtration	yes
Fil^i	Decreasing filtration, named at first use (Hodge, Nygaard, or syntomic).	22	filtration	local

5. Prismatic and syntomic notation, continued

Symbol	Meaning	First	Type	Reserved
$\mathbb{Z}_p(i)$	p-adic Tate twist or syntomic complex in the convention of the cited theorem.	22	sheaf/complex	local
$R\Gamma_{\text{syn}}$	Derived syntomic cohomology.	22	complex	local
$R\Gamma_{\text{Prism}}$	Derived prismatic cohomology.	22	complex	local
θ	Fontaine or prism structure map; source and target are always shown.	22	ring map	local
ξ_{pris}	Distinguished generator of a prism ideal when a generator has been chosen; unrelated to the Riemann xi-function.	22	ring element	local
Θ_{Sen}	Sen operator in a stated p-adic representation or comparison.	22	operator	yes
N	Monodromy operator, with the convention $N = -\Theta_{\text{Sen}}$ where that comparison is invoked.	22	operator	local
$\operatorname{Spf} R$	Formal spectrum of R.	22	formal scheme	yes
$\operatorname{Spec} R$	Prime spectrum of R.	13	scheme	yes

6. Programme interfaces and finite verification

Symbol	Meaning	First	Type	Reserved
AS0–AS6	Discipline, absolute-base, loop-space, Frobenius, comparison, trace, and polarization interfaces of the adelic cyclotomic proposal. Their individual statuses are stated where used.	25	specification	yes
AHK-1–AHK-6	Spectral realization, centered duality, Hodge index, arithmetic trace, primitive/pole compatibility, and firewall specificity in the Arithmetic Hodge–Krein Realization Problem.	48	specification	yes
$A2^{\star}$	The slope-breaking, one-sided mean-purity, and category-rigidity formulations of the second open obligation.	38	open problem	yes
$\mathrm{RH}(H)$	Finite Riemann hypothesis through height H: $0 < \gamma \le H \Rightarrow \beta = \frac{1}{2}$.	49	proposition	yes
$Z(t)$	Hardy's real-valued function on the critical line, with phase convention stated in Chapter 49.	49	real function	yes
$\vartheta(t)$	Riemann–Siegel theta function in the normalization of Chapter 49.	49	real function	yes
$S(t)$	$\pi^{-1} \arg \zeta(\frac{1}{2} + it)$, with the path and branch fixed at each use.	49	real function	yes
$[x]$	An interval or ball enclosure of x in certified-computation chapters; ordinary closed intervals are distinguished by context.	49	enclosure	local
$J_d(x)$	Jensen polynomial $\sum_{k=0}^{d} \binom{d}{k} a_k x^k$ of the displayed coefficient sequence.	51	polynomial	local

Continued on next page

6. Programme interfaces and finite verification (continued)

Symbol	Meaning	First	Type	Reserved
PF_∞	Pólya-frequency property: total nonnegativity of the associated Toeplitz matrix.	47	property	yes

7. Status labels and certificate types

Symbol	Meaning	First	Type	Reserved
[T]	Theorem: proof present here or exact citation to a proof.	1	status	yes
[C]	Conjecture: precise, falsifiable, and unproved.	1	status	yes
[O]	Open: no candidate object satisfying the stated requirement exists.	1	status	yes
[E]	Evidence: finite computation, numerical observation, or bounded verification only.	1	status	yes
[N]	No-go: proved impossibility for a stated class of mechanisms.	1	status	yes
[K]	Known background: a classical or published result with source.	1	status	yes

7. Certificate types

Symbol	Meaning	First	Type	Reserved
INT	Interval-arithmetic certificate with directed rounding and recorded precision.	49	certificate	yes
RAT	Exact rational certificate, independently checkable by integer arithmetic.	49	certificate	yes
BALL	Complex or real ball-arithmetic enclosure.	49	certificate	yes
LDL	Exact or certified LDL^* inertia certificate.	51	certificate	yes
TAIL	Analytic tail bound recording truncation point, constants, and hypotheses.	49	certificate	yes
ZERO	Certified zero enclosure with multiplicity or a stated simplicity assumption.	50	certificate	yes
FORMAL	Proof-assistant kernel certificate with named environment and dependencies.	53	certificate	yes
FLOAT	Floating-point observation without a rigorous enclosure; always status [E].	46	evidence type	yes

Spectral exhaustion and multiplicity

[T] In the finite polarization chapters,

$$\#\{\lambda \in \operatorname{spec}\Theta : \operatorname{Re}\lambda > \tfrac{1}{2}\}$$

counts algebraic multiplicity. For a filtered family (H_N, Θ_N, ι_N), "stable spectrum exhausts the nontrivial zeros" means that each nontrivial zero ρ, repeated according to its algebraic multiplicity, occurs with that multiplicity at all sufficiently large compatible stages, every stable spectral point is a nontrivial zero, and no spurious point persists. A cloud of numerically convergent eigenvalues is not, by itself, stable spectral exhaustion.

[K] For an unbounded limiting operator, spectrum, point spectrum, resolvent, closure, and domain retain their standard functional-analytic meanings. The book does not infer a limiting operator merely from finite-dimensional matrices. When only the matrices exist, the notation Θ_N is used and the status remains finite.

Forms and signatures

[K] The inertia of a Hermitian matrix is ordered as (n_+, n_-) when nondegeneracy is known and as (n_+, n_-, n_0) when nullity is relevant. "Lorentzian" for a nondegenerate N-dimensional form means inertia $(N-1, 1)$ or $(1, N-1)$, with the orientation stated. In this book the measured pole-removed matrices have orientation $(N-1, 1)$.

[E] The statement that B_N has measured inertia $(N-1, 1)$ on sixty-three spline spaces is finite evidence. [T] The analytic pole-plane signature is a separate theorem. [O] An operator Θ_N on those spline coefficient spaces satisfying $\Theta_N^* B_N + B_N \Theta_N = B_N$ and realizing the zeros has not been constructed. These three statuses are not interchangeable.

Global reservation and collisions

The preamble reserves

$$\begin{aligned}&\mathrm{RH},\ \mathrm{GRH},\ \mathbb{N},\ \mathbb{Z},\ \mathbb{Q},\ \mathbb{R},\ \mathbb{C},\ \mathbb{F},\ \mathbb{A},\ \mathbb{S},\\ &\mathrm{Spec},\ \mathrm{Spf},\ \mathrm{THH},\ \mathrm{THR},\ \mathrm{TP},\ \mathrm{TC},\ \mathrm{HH},\ \mathrm{Fr},\ \mathrm{Frob},\ \mathrm{Sym},\\ &\mathrm{Res},\ \mathrm{Tr},\ \mathrm{Ind},\ \mathrm{spec},\ \mathrm{supp},\ \mathrm{Ai},\ \Lambda_{\mathrm{dBN}},\ \mathfrak{X}_{\mathbb{S}},\ \tfrac{1}{2},\ \mathrm{Re},\ \mathrm{Im}\,.\end{aligned}$$

They are not redefined in chapter files. In particular, `\Sph` is *available* and always denotes the sphere spectrum; `\SS` is unavailable because of a LaTeX conflict. A roman operator such as Spec or Tr is not replaced by an italic variable of the same spelling.

Some conventional letters necessarily have typed local meanings. The letter ρ denotes a zeta zero unless it is visibly supplied an operator argument, as in $\rho(A)$ for a resolvent set. The letter Θ denotes the spectral operator in the polarization chapters, but the zero-density chapters may use a separately defined scalar supremum. The prismatic element ξ_{pris} always carries its subscript and is never confused with the Riemann xi-function $\xi(s)$. Such collisions must be resolved at first occurrence, not left to inference.

Citations and imported conventions

[K] A cited theorem may use a different completion of an L-function, a geometric rather than arithmetic Frobenius, an angular Fourier variable, or a Hermitian form linear in the second slot. Before the theorem is used, the book translates it into the conventions above and records the translation. A quotation may preserve the source's notation, but the following displayed formula returns to the global convention.

Endpoint weights in explicit formulas, branch choices for logarithms, orientations of Frobenius, and normalizations of local class field theory are stated where they matter. Silence never authorizes a change to the Fourier or Mellin conventions fixed here.

Status-label key

The status label attached to a mathematical claim is part of the claim. It records the kind of warrant the reader should expect and limits the verbs available to the text. The six labels are mutually exclusive at a given logical level. A theorem may have an open hypothesis; in that case the implication is [T] and the hypothesis is [O]. A numerical instance of a conjecture is [E], not [C], while the general statement remains [C].

Label	Name	Editorial meaning
[T]	Theorem	A proof is present in the book, or an exact citation identifies a published proof with all hypotheses used.
[C]	Conjecture	A precise and falsifiable statement is proposed as plausible but is not proved.
[O]	Open	No candidate object satisfying the stated requirement exists.
[E]	Evidence	A finite computation, numerical measurement, bounded verification, or empirical pattern is reported.
[N]	No-go	A theorem proves that a stated class of mechanisms cannot deliver the stated conclusion.
[K]	Known	Known or classical background is being imported, with a source where mathematical content is asserted.

[T] Theorem

A theorem label requires either a proof in the text or a precise source. It does not mean "widely expected," "verified in examples," or "formally implied by an object that has not been constructed." When a theorem is conditional, the status of the implication and the status of its premise are printed separately.

Worked example. [T] *Index-One Exhaustion Theorem.* If finite-dimensional H_N, operators Θ_N, and nondegenerate Hermitian forms G_N satisfy the stated centered duality, have $n_-(G_N) \leq 1$, and their stable spectrum exhausts the nontrivial zeros with algebraic multiplicity and without spurious points, then RH holds. The implication is proved in Chapter 48. The existence of the system is not included in the theorem's status.

[C] Conjecture

A conjecture must have a complete statement and a recognizable negation. The label is not used for an undefined hope or for an object whose axioms are still changing. If no candidate exists, the construction burden is [O] even when a broader philosophical expectation is conjectural.

Worked example. [C] *Category-rigidity form of Problem A2⋆.* Any map sending tuples of zeros of ζ onto zeros of a member of the Selberg class is affine with linear coefficient of absolute value one, under the precise admissibility hypotheses of Chapter 38. This statement is falsified by one admissible non-affine map.

[O] Open

An open label identifies a missing mathematical obligation. It is used when no proof is present and, in the principal cases of this book, when no candidate object satisfies the required axioms. Open does not mean that the authors regard the statement as true; it means that the programme requires it and supplies a testable specification.

OPEN PROBLEM Arithmetic Hodge–Krein Realization Problem

[O] Construct a filtered system $(H_N, \Theta_N, G_N, \iota_N)$ satisfying AHK-1 through AHK-6, in particular spectral realization, centered duality, and compatible negative index at most one. No construction satisfying even AHK-1, AHK-2, and AHK-3 is supplied. Failure of any required compatibility is a failure of the proposed realization, not partial proof of RH.

[E] Evidence

Evidence is finite. Its statement includes the sample, range, tolerance, data source, and certificate type where applicable. Repetition across many instances can motivate a conjecture but does not remove the evidence label. Floating-point output without enclosure is distinguished from interval, ball, rational, or exact inertia certificates.

Worked example. [E] For sixty-three cubic-spline spaces with $0.15 \le T \le 3$, $4 \le N \le 64$, and tolerance 10^{-9}, the Guinand–Weil Gram matrix built from actual prime powers had measured inertia $(N, 0)$, while $B_N = W_N - P_N$ had measured inertia $(N-1, 1)$ in every case. This does not construct Θ_N, prove the centered adjoint identity, or imply RH.

[N] No-go

A no-go label is a theorem with a negative conclusion. Its scope is the quantified architecture in its hypotheses. It may rule out all normalizations of one mechanism without ruling out a mechanism that changes the relevant invariant. The phrase "cannot prove RH" is therefore followed by the allowed inputs and operations.

OBSTRUCTION Amplification invariance

[N] For a Dirichlet series whose singular support contains k-fold sums of zeros,

$$\sigma_c - \mathrm{Re}(\text{amplified singularity}) = k(\tfrac{1}{2} - \delta)$$

identically and independently of normalization. The extractable conclusion is $\delta \le \frac{1}{2}$ for every k. The theorem rules out gain from amplification within this architecture; it does not classify all possible uses of higher correlations.

[K] Known background

Known background is imported from the literature. The label does not license an uncited assertion. A primary source or a standard reference is given at the substantive occurrence, with normalization changes recorded. If the book supplies a new proof of a classical fact, the fact remains known background while the particular lemma may be marked [T].

Worked example. [K] Rodgers and Tao proved that the de Bruijn–Newman constant is non-negative, and RH is equivalent to its being non-positive; therefore RH $\iff \Lambda_{\mathrm{dBN}} = 0$ in the stated normalization. The source is *Forum of Mathematics, Pi* **8** (2020), e6. The criticality consequence drawn from this fact is separately proved where used.

Mixed-status statements

The common pattern in this book is

$$\text{[O] } P \qquad \text{and} \qquad \text{[T] } P \Longrightarrow \mathrm{RH}.$$

It must not be compressed into an unlabelled sentence saying that the programme proves RH. Similarly,

$$\text{[E] } B_N \text{ has measured inertia } (N-1, 1)$$

must not be substituted for

$$\text{[O] } \exists\, \Theta_N : \Theta_N^* B_N + B_N \Theta_N = B_N \text{ with the required stable spectrum.}$$

The first is a finite matrix observation; the second is an operator realization problem.

Composition rule. An open premise combined with a proved implication yields an *open conclusion*, never a theorem. Thus

$$\text{[O] } P, \qquad \text{[T] } P \Longrightarrow \mathrm{RH}, \qquad \text{hence} \qquad \text{[O] RH by this route.}$$

The theorem status belongs to the implication. It does not transfer to the premise or to the conclusion while the premise remains open.

A paragraph may begin with one label when every mathematical sentence in it has that status. If the status changes, a new label appears at the change. Displayed chains label premises individually whenever a reader could otherwise transfer the status of the implication to the premise.

Boxes and ledgers

An `obstruction` box contains a no-go theorem. A `firewall` box compares a proposed mechanism with the four required counter-models. An `openproblem` box states a missing

construction or estimate. A neutral `notice` box records a boundary, convention, or worked example. Box color never carries status by itself; the printed badge is controlling.

Every Part begins with a status ledger. Its "constructed" column answers whether the named object exists in the form required, not whether related objects exist. Its "logical strength" column says unconditional, conditional on named hypotheses, or open. "Partial" is not a status and cannot replace [E] or [O].

Every chapter ends with "What this chapter does not establish." That section is an audit of adjacent claims that have not been proved: missing converses, absent compatibility maps, unjustified limits, or conclusions not supported by finite data. It does not weaken the theorems in the chapter; it fixes their boundary.

Editorial rule

A mathematical claim without a status label is an editorial defect. The remedy is not to attach the most favorable badge. The author must supply a proof or citation for [T], state a falsifiable conjecture for [C], identify the missing obligation for [O], delimit the finite computation for [E], prove the impossibility for [N], or cite the known result for [K]. If none applies, the sentence is deleted.

The labels are designed to let a hostile reader disagree locally. A reader may reject the proposed geometry while accepting a no-go theorem, or challenge a computation without changing the conditional index theorem. That locality is the purpose of the system.

Master synoptic plates

The following three plates give the mathematical, architectural, and algorithmic views of the monograph at one scale. They are maps of logical dependencies, not substitutes for the hypotheses and proofs in the chapters. Their dashed boundaries are load-bearing: a construction displayed as open in a plate remains open in the text.

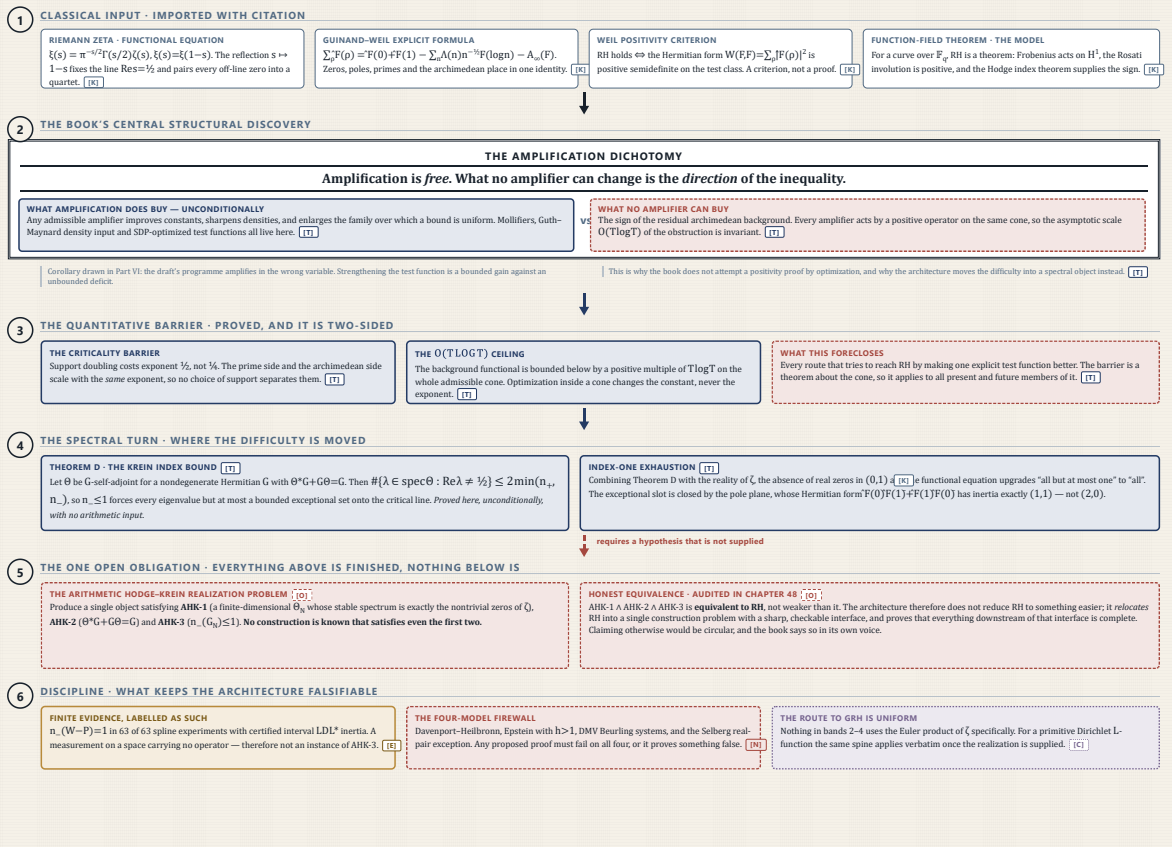

Figure 1. Master mathematical principle.

What it does not show: RH or GRH; no object satisfying AHK-1 through AHK-3 is constructed, and the finite spline evidence is not promoted to an operator theorem.

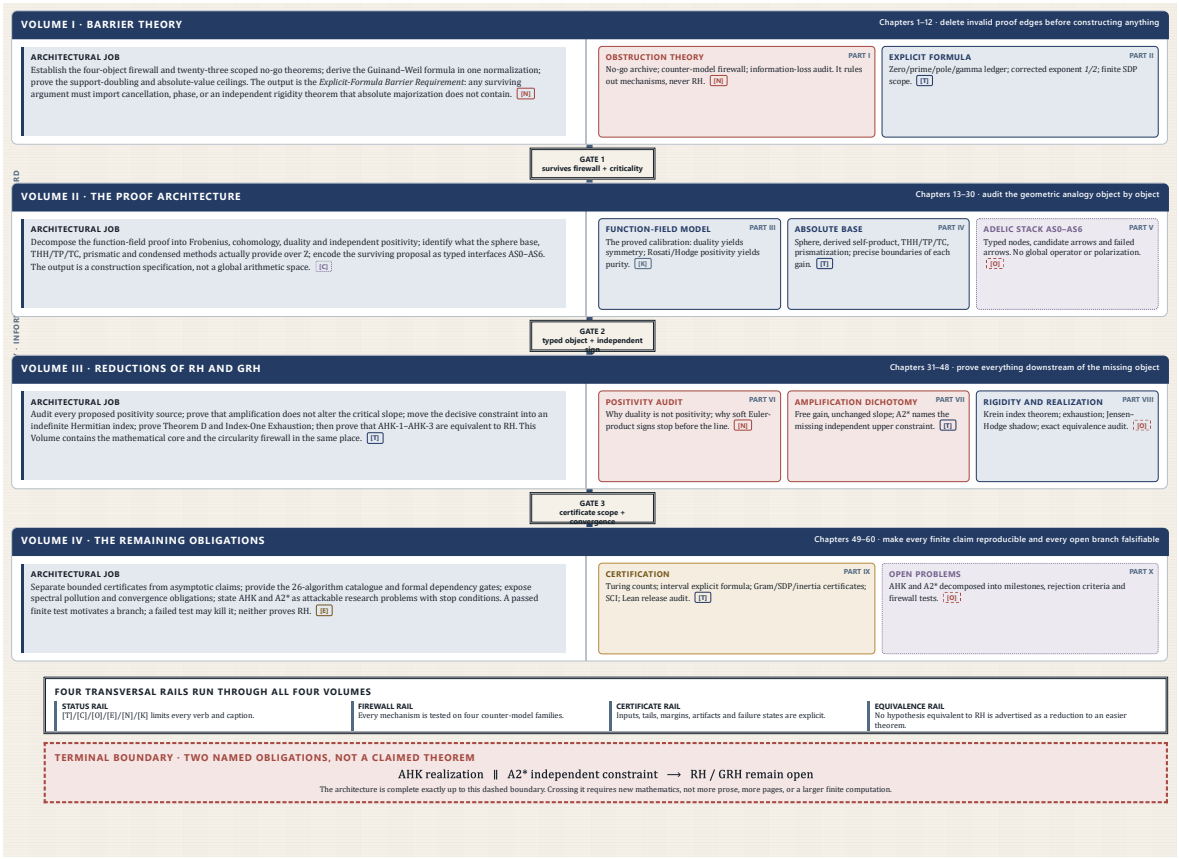

Figure 2. Master monograph architecture.

What it does not show: the AHK realization, the A2-star constraint, or any inference from a finite certificate to an asymptotic theorem.

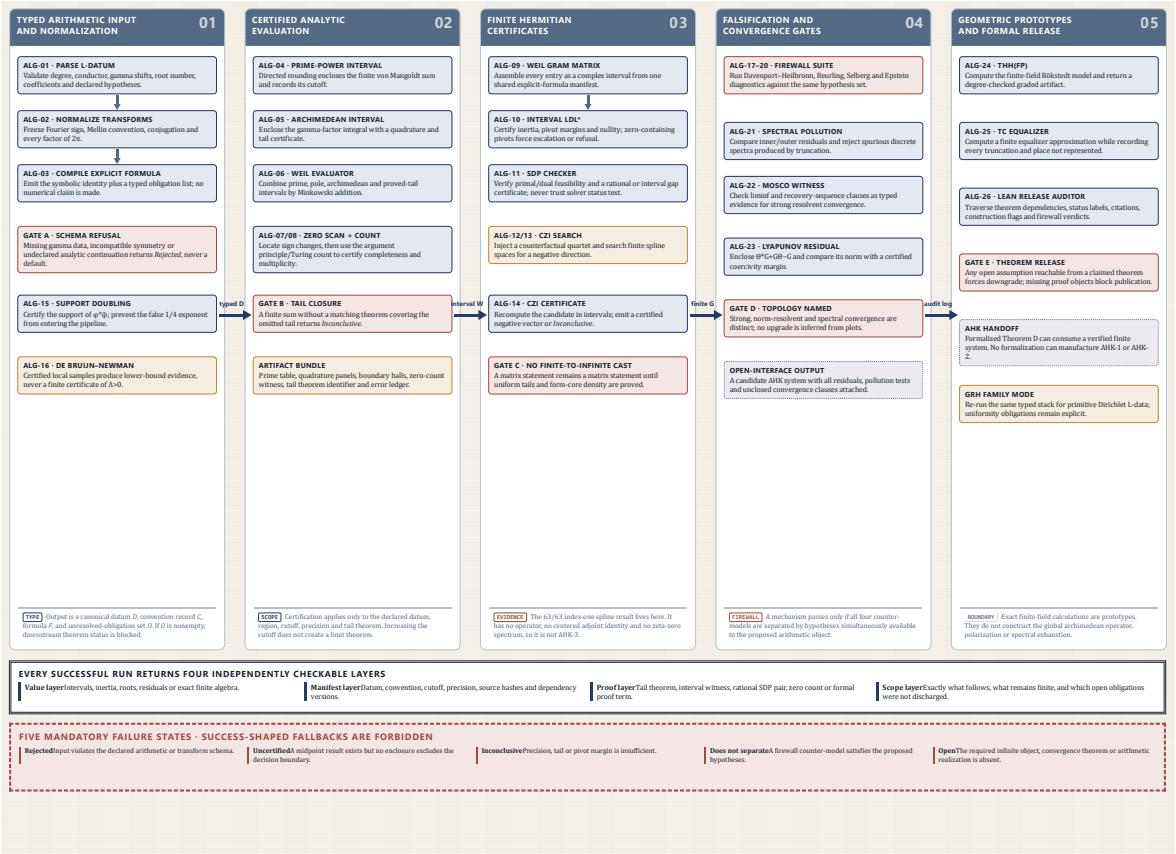

Figure 3. Master certified-computation architecture.

What it does not show: production implementations of all twenty-six algorithms, an infinite theorem from finite interval output, or an Arithmetic Hodge–Krein system.

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VOLUME I

Barrier Theory

What specified proof mechanisms cannot do

Twenty-three scoped no-go theorems, a four-object firewall, and the Explicit-Formula Barrier Requirement identify the precise points at which specified proof mechanisms lose the information needed at the critical line.

THE ARGUMENT OF VOLUME I

[K] Volume I asks a preliminary question that is logically prior to any proposed construction: which operations preserve enough arithmetic information to distinguish the critical line, and which operations destroy that information before the decisive step? The question is not whether RH is impossible to prove. It is whether a stated mechanism, under stated hypotheses, can produce more than the classical boundary already available to it.

[N] Part I answers that narrower question through twenty-three scoped obstructions. The four-object firewall tests whether a mechanism uses something specific to ζ, rather than only a functional equation, formal symmetry, coefficient positivity, or another feature shared by counter-models. The archive and the firewall do different work: the archive blocks routes by inequalities and invariants; the firewall blocks mechanisms by exhibiting objects on which the same mechanism would give a false conclusion. Neither is an impossibility theorem for RH.

[T] Part II then fixes the analytic instrument against which many of those routes must be tested. It derives the Guinand–Weil explicit formula with one normalization, tracks the zero, prime-power, polar, and archimedean terms, and uses that ledger to prove the corrected support-doubling obstruction and the taxonomy of absolute-value ceilings. The Euler-product obstruction at exponent 1 and the counterfactual-zero-injection obstruction at exponent $\frac{1}{2}$ are related losses of information, but they are not the same inequality. The Explicit-Formula Barrier Requirement records what must replace those losses if an argument is to remain viable.

[K] The order matters. A reader who reaches later geometric, spectral, or computational proposals should carry forward two disciplines from this Volume. First, a no-go theorem deletes only the edge named in its hypotheses; it does not decide the truth of RH. Second, a finite calculation or an appealing analogy cannot restore phase, uniformity, or arithmetic specificity that an earlier step has discarded. Those disciplines govern the positive architecture developed in the remaining Volumes.

Part I

Obstruction theory: what cannot work

Chapter 1

The obstructions, and the honest boundary

Object	Status	Constructed	Logical strength
Obstruction catalogue (23 theorems)	[N]	yes	unconditional
Four-object firewall	[K]	yes	unconditional
Criticality Barrier	[N]	yes	unconditional
Krein Index Theorem (Theorem D)	[T]	yes	unconditional
Index-One Exhaustion Theorem	[T]	yes	conditional on AHK-1,2,3
Realization system $(H_N, \Theta_N, G_N, \iota_N)$	[O]	no	open
Measured inertia $(N-1, 1)$ of B_N	[E]	yes	finite evidence
Problem A2$\star$ (slope / mean purity)	[O]	no	open
Riemann Hypothesis	[O]	—	unsolved

Status ledger for Part I. Every row is discharged, or left explicitly undischarged, in Chapters 1–6. The last row is the reason the book exists and the reason it makes no claim to settle it.

[K] The Riemann Hypothesis is the assertion that every zero of the Riemann zeta function in the strip $0 < \operatorname{Re} s < 1$ has $\operatorname{Re} s = \frac{1}{2}$. It was stated by Riemann in 1859 [Rie59], it has been the subject of continuous work since, and it is open. This book does not prove it. The purpose of this first chapter is to say, with precision sufficient for a hostile reader, what the book does instead, and why that thing is worth a volume of this size.

The answer has two halves. The first half is negative and unconditional: we prove that a list of specific, historically attempted mechanisms cannot prove RH, and we prove it with the same standards of rigour that would be demanded of a positive result. The second half is conditional and positive: we give a complete derivation of RH from one hypothesis, we state that hypothesis precisely enough that a third party can attack it or refute it, and we audit the derivation for circularity. Neither half is the Riemann Hypothesis. Together they are a map of a region that has been travelled for a century and a half largely without one.

1.1 What this book proves, and what it does not

[T] The positive logical content of the book is a single implication and a single measurement, and it is worth stating both in the first pages so that nothing later can be mistaken for more than it is.

The implication is the *Index-One Exhaustion Theorem* (Chapter 48). Suppose there exist finite-dimensional complex spaces H_N, operators Θ_N on H_N, and nondegenerate Hermitian forms G_N on H_N, satisfying

(AHK-1) the stable spectrum of the family (Θ_N) exhausts the nontrivial zeros of ζ with algebraic multiplicity and contains no spurious points;

(AHK-2) the centered duality $\Theta_N^* G_N + G_N \Theta_N = G_N$ holds at each level;

(AHK-3) the negative index satisfies $n_-(G_N) \le 1$, compatibly with the transition maps.

Then RH holds. The proof is complete, elementary in its final step, and given in full: it consists of the Krein Index Theorem (Theorem D of Chapter 44), the pairing of nonreal zeros with their complex conjugates, the classical absence of real zeros of ζ in $(0,1)$, and the functional equation.

[O] The hypothesis is open. No system $(H_N, \Theta_N, G_N, \iota_N)$ satisfying even (AHK-1), (AHK-2), (AHK-3) has been constructed, by us or by anyone. We do not know how to construct one. We do not claim that one exists. What we claim is that the implication is proved. We do *not* claim that the hypothesis is weaker than RH, and the reader should be warned at once that it is not: Chapter 48 proves that (AHK-1), (AHK-2) and (AHK-3) taken together are *equivalent* to RH. Assuming RH, the diagonal system $\Theta_N = \mathrm{diag}(\rho_1, \dots, \rho_N)$, $G_N = \mathrm{diag}(-1, 1, \dots, 1)$ satisfies all three, because $\bar\rho + \rho = 1$ on the critical line. The implication above is therefore a theorem, but it is not by itself a reduction of RH to anything easier. The programme's content lies in (AHK-4) to (AHK-6), and in Problem A2$\star$, which *is* genuinely weaker in content than RH. That distinction is the single most important thing to carry through this book, and Chapter 4 and Chapter 6 return to it.

[E] The measurement is finite. For sixty-three cubic-spline test spaces, with $0.15 \le T \le 3$, $4 \le N \le 64$, and tolerance 10^{-9}, the Guinand–Weil Gram matrix W_N assembled from actual prime powers has measured inertia $(N, 0)$, while the pole-removed matrix $B_N = W_N - P_N$ has measured inertia $(N-1, 1)$ in every case; the pole plane P_N has exact signature $(1, 1, N-2)$, which is proved rather than measured. This is reported in Chapter 46.

Warning 1.1 (The measurement is not the hypothesis). [T] The inertia $(N-1, 1)$ of B_N cannot be substituted for (AHK-3) in the Index-One Exhaustion Theorem. Theorem D requires a form G and an operator Θ satisfying $\Theta^* G + G\Theta = G$. On the spline coefficient space there is no Θ_N, and no proof of the adjoint identity. A text that moves from the measured inertia to RH has committed a false proof. We state this warning at every occurrence of the measurement, and we state it here, in the first pages, because it is the single most tempting error the book could commit.

[N] The negative content is the obstruction catalogue: twenty-three theorems, listed in Chapter 6 and proved in the chapters cited there, each of which states that a specified class of arguments cannot prove RH. These are theorems, not opinions about difficulty. Each has a hypothesis describing an architecture, and a conclusion saying what that architecture yields — typically that it yields exactly the classical bound and nothing more. Their value is that they are falsifiable: exhibiting an argument inside the stated architecture that reaches a stronger conclusion refutes the theorem, and we have tried to state the architectures narrowly enough that this is a real risk.

Nothing in the book is offered as a plan for a proof of RH in the near term, and the reader should not read the positive programme as one. A programme whose sole gap is an unconstructed object is not close to a proof; it is a specification. Grothendieck's standard conjectures have been a specification since 1968 [Gro69], and the varieties they concern have cohomology, which our objects do not.

1.2 The state of the question

[K] We fix normalizations once. For $\mathrm{Re}\, s > 1$,

$$\zeta(s) = \sum_{n \ge 1} n^{-s} = \prod_p (1 - p^{-s})^{-1},$$

the product over rational primes, absolutely convergent in that half-plane. The completed function

$$\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$$

extends to a meromorphic function on $\mathbb{C}$ whose only poles are simple poles at $s = 0$ and $s = 1$, and satisfies

$$\Lambda_\zeta(s) = \Lambda_\zeta(1-s), \qquad \Lambda_\zeta(\bar{s}) = \overline{\Lambda_\zeta(s)}.$$

The zeros of Λ_ζ are exactly the zeros of ζ in $0 \le \operatorname{Re} s \le 1$; the zeros of ζ at $s = -2, -4, \ldots$ are cancelled by the poles of $\Gamma(s/2)$ and are called trivial. Standard references for all of this are [TH86; Edw74; Ivi85; MV07; IK04].

[K] What is known unconditionally about the nontrivial zeros divides naturally into four kinds of statement, and it is useful to have the four kinds separated before any mechanism is discussed, because a proposed mechanism almost always turns out to reproduce one of them.

Counting. The Riemann–von Mangoldt formula gives

$$N(T) = \#\{\rho : 0 < \gamma \le T\} = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + S(T) + O(T^{-1}),$$

with $S(T) = O(\log T)$ [Man95; TH86]. The density of zeros at height T is therefore $\asymp \log T$ per unit interval. This is a two-sided statement: it is as much an obstruction as an aid, because it fixes the Weyl asymptotics that any proposed spectral model must reproduce (Chapter 3).

Location, from below. Hardy proved that infinitely many zeros lie on the critical line [Har14]; Hardy and Littlewood gave a positive proportion of the expected count on the line in the sense of a lower bound $\gg T$ [HL21]; Selberg improved this to a positive proportion of all zeros [Sel42]; Levinson and then Conrey raised the proportion, with 2/5 of the zeros known to lie on the line [Con89]. These results say that the critical line is not empty and is not thin. They do not touch the complement.

Location, from above. Zero-free regions and zero-density estimates bound the possible excursions. The classical zero-free region of Hadamard and de la Vallée Poussin [Had96; Val96], and the Vinogradov–Korobov improvement, give $\zeta(s) \neq 0$ for $\sigma \ge 1 - c/(\log|t|)^{2/3}(\log\log|t|)^{1/3}$, which approaches the line $\operatorname{Re} s = 1$ and no further. Zero-density estimates bound $N(\sigma, T) = \#\{\rho : \beta \ge \sigma,\ 0 < \gamma \le T\}$; the current record is $N(\sigma, T) \ll T^{\frac{30}{13}(1-\sigma)+o(1)}$ [GM24], in a lineage running Ingham [Ing37] → Huxley [Hux72] → Bourgain [Bou00]. Chapter 29 proves that this estimate, and any estimate of the same shape, degenerates at $\sigma = \frac{1}{2}$ and provably cannot alone prove RH.

Computation. Zeros have been verified to lie on the line to very large heights [OS88; PT21]. [E] These are finite verifications. Chapter 5 explains why a finite verification, however large, is not evidence of the kind that a programme may consume, and Chapter 52 explains why it is also not a certificate for the spectrum of any operator.

Remark 1.2 (The asymmetry of the known). [K] The lower-bound results are proofs that zeros exist where RH says they should. The upper-bound results are proofs that zeros are scarce where RH says they are absent. Neither family has ever been pushed to the point of excluding a single off-line zero at a specified height. It is this asymmetry — a great deal of information about where zeros are, almost none about where they are not — that the amplification literature has repeatedly tried to invert, and Part 37–42 is a study of why the inversion fails.

1.2.1 What RH would give, and what it would not

[K] It is useful to be precise about the payoff, because several arguments in the literature are shaped by an overestimate of it.

RH is equivalent to the error term $\psi(x) = x + O(x^{1/2}\log^2 x)$, in Schoenfeld's explicit form $|\psi(x) - x| < \sqrt{x}\log^2 x/(8\pi)$ for $x \geq 73.2$, and to $\pi(x) = \mathrm{li}(x) + O(\sqrt{x}\log x)$ [TH86]. It is *not* equivalent to, and does not imply, the twin prime conjecture, the Goldbach conjecture, or any statement about the distribution of primes in intervals shorter than $x^{1/2}$. It gives the square-root error term in the count of primes and nothing beyond it: the error term $O(x^{1/2+\varepsilon})$ is optimal, since $\psi(x) - x = \Omega_{\pm}(x^{1/2})$ unconditionally by Littlewood's theorem [Lit14].

[T] This has a consequence for the design of proofs which we use repeatedly. Under RH the trivial bound $\sum_{n\leq X}\Lambda(n)n^{-1/2} = 2\sqrt{X} + o(\sqrt{X})$ is, in order of magnitude, the truth. RH asserts that the absolute-value bound is sharp, not that it can be improved. Therefore no proof of RH can proceed by improving the absolute bound on the prime side; it must proceed by exhibiting cancellation that the absolute bound cannot see, or by supplying structure from elsewhere. This principle is proved in the scoped form of Chapter 9 and is the organising idea of the absolute-value ceiling taxonomy in §6.7.

[K] Conversely, several statements strictly stronger than RH are sometimes conflated with it. Simplicity of the zeros is not known and does not follow from RH; the best unconditional results give a positive proportion of simple zeros [Con89]. The Lindelöf hypothesis follows from RH but is weaker. The pair-correlation conjecture is independent of RH as a statement about the fine distribution. A text that assumes simplicity of zeros in an operator-theoretic construction has assumed something not known even under RH, and Chapter 43 records where multiplicity enters our own statements.

1.2.2 The ladder of unconditional deviations

[K] It is worth having the numerical shape of the unconditional knowledge in mind. Write Θ_ζ as above. We know $\frac{1}{2} \leq \Theta_\zeta \leq 1$ and nothing better: no unconditional bound of the form $\Theta_\zeta \leq 1 - c$ with $c > 0$ is known, and by the Bohr–Landau theorem [BL14] such a bound would already imply a density statement well beyond current reach. The zero-density estimates constrain how *many* zeros may have $\beta \geq \sigma$, not whether any does; the classical zero-free region excludes zeros in a cusp shrinking to the line $\mathrm{Re}\, s = 1$, not to $\mathrm{Re}\, s = \frac{1}{2}$.

[T] The gap between $\Theta_\zeta \leq 1$ and $\Theta_\zeta = \frac{1}{2}$ is therefore not a gap of technique in the ordinary sense. Every unconditional method that has been applied to it returns either a statement about counts or a statement near $\mathrm{Re}\, s = 1$. Chapter 29 proves that the current density estimate is nontrivial only for $\sigma > 17/30$ and that no estimate of that shape can close the gap; the Amplification Invariance Theorem proves that no amplification of the explicit formula changes the exponent; and the Level Collapse Theorem proves that higher additive correlations carry the same exponent as the first. The three together are the negative core of Volume II, and the reader may regard the rest of this Part as an explanation of why the remaining possibilities are as narrow as they are.

1.3 The geography of the critical strip

[K] Fix the following names, which are used without further comment for the rest of the book. The *critical strip* is $\{s : 0 < \mathrm{Re}\, s < 1\}$; the *critical line* is $\{s : \mathrm{Re}\, s = \frac{1}{2}\}$; a nontrivial zero is written $\rho = \beta + i\gamma$ with $0 < \beta < 1$; the *deviation* of ρ is $\beta - \frac{1}{2}$, and the *supremum of deviations* is

$$\Theta_\zeta = \sup\{\beta : \zeta(\beta + i\gamma) = 0,\ 0 < \beta < 1\}.$$

Unconditionally $\frac{1}{2} \le \Theta_\zeta \le 1$; the lower bound is Hardy's theorem and the upper bound is the non-vanishing on $\operatorname{Re} s = 1$; RH is $\Theta_\zeta = \frac{1}{2}$. We write δ for a hypothetical positive deviation, so that a counterfactual off-line zero is $\rho_0 = \frac{1}{2} + \delta + i\gamma_0$ with $0 < \delta < \frac{1}{2}$.

Three features of this geography matter for everything that follows, and each is elementary.

First, the strip is *closed under two involutions*: $s \mapsto 1 - s$ from the functional equation, and $s \mapsto \bar{s}$ from the reality of the Dirichlet coefficients. The group they generate is a Klein four-group, discussed in §1.4.

Second, the critical line is the *fixed locus* of the composite involution $s \mapsto 1 - \bar{s}$, and the real segment $(0, 1)$ is the fixed locus of $s \mapsto \bar{s}$. These are the only two one-dimensional fixed loci in the strip. This is why exactly two degenerate orbit types occur, and why a mechanism that excludes one of them still has to deal with the other. The reader who wants a single sentence to carry away from Chapter 2 is: *the mechanisms in the literature typically exclude the nonreal off-line orbit and forget the real one.*

Third, the strip carries no metric structure that distinguishes the critical line intrinsically. The line is distinguished by the functional equation only as a fixed locus of a symmetry, and §1.4 proves that a fixed locus of a symmetry does not attract the objects the symmetry permutes. The critical line is where zeros *may* be pinned; nothing about the symmetry pins them.

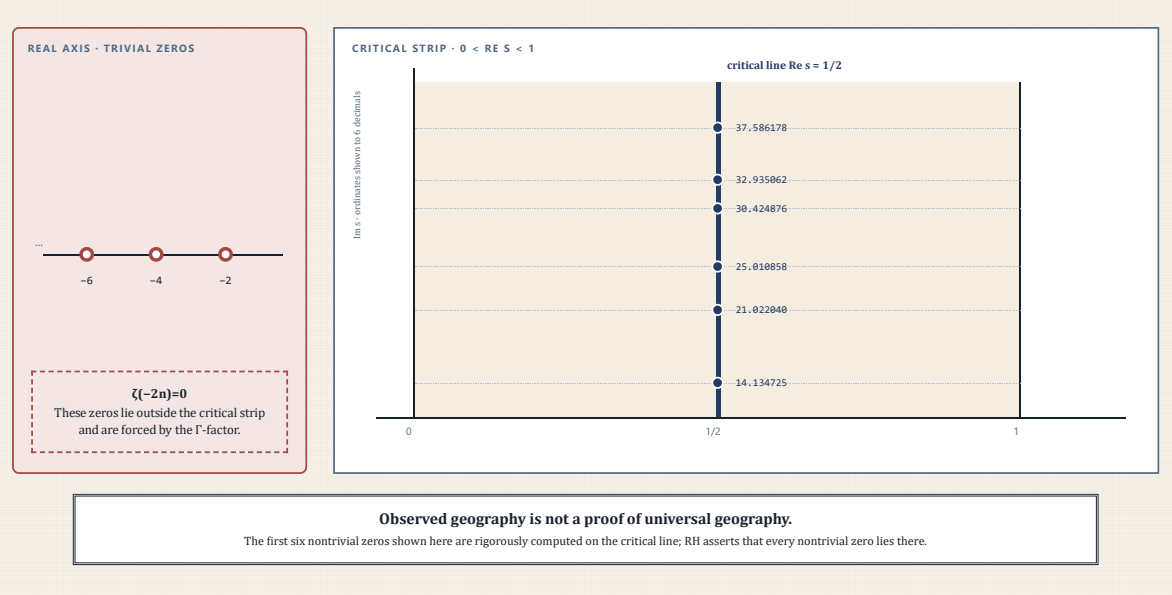


Figure 1.1. The geography of the critical strip.

What it does not show: that the shaded region is nonempty, or that any zero lies in it; it displays the present state of knowledge, not a claim about zeros

[K] The figure repays a slow reading. The zero-free region is a thin cusp against $\operatorname{Re} s = 1$; the zero-density estimate is nontrivial only for $\sigma > 17/30$ and degenerates exactly at the line (theorem 42.1); and the region that no theorem excludes contains, in particular, every point of the open strip with $\frac{1}{2} < \operatorname{Re} s \le 17/30$ at every height. A century of work has not removed a single point of that region. Any assertion that a new mechanism "almost" removes it should be measured against this picture, and against the Amplification Invariance Theorem of Chapter 39, which explains why so many mechanisms return exactly the boundary $\delta \le \frac{1}{2}$ they started from.

1.4 The four-point orbit: symmetry is not fixedness

[T] We now prove the elementary fact that organises the whole of Part 1. It is elementary, it is known, and it is nevertheless the fact whose neglect produces most of the failed arguments in the subject.

Definition 1.3 (The functional-equation group)**.** [T] Let V be the group of transformations of $\mathbb{C}$ generated by

$$\sigma_1(s) = 1 - s, \qquad \sigma_2(s) = \bar{s}.$$

Then $V = \{\mathrm{id}, \sigma_1, \sigma_2, \sigma_3\}$ with $\sigma_3(s) = 1 - \bar{s}$, and $V \cong (\mathbb{Z}/2)^2$: each nonidentity element is an involution and the product of any two distinct nonidentity elements is the third.

Proposition 1.4 (Orbit structure)**.** *[T] Let $s_0 = \beta + i\gamma$ lie in the open strip. The V-orbit of s_0 is*

$$V{\cdot}s_0 = \{\beta + i\gamma,\ 1 - \beta + i\gamma,\ \beta - i\gamma,\ 1 - \beta - i\gamma\},$$

and exactly one of the following holds.

(a) *$\beta \neq \frac{1}{2}$ and $\gamma \neq 0$: the orbit has four distinct points, two with $\mathrm{Re} > \frac{1}{2}$ and two with $\mathrm{Re} < \frac{1}{2}$.*
(b) *$\beta = \frac{1}{2}$ and $\gamma \neq 0$: the orbit has two points, $\frac{1}{2} \pm i\gamma$, both on the critical line.*
(c) *$\beta \neq \frac{1}{2}$ and $\gamma = 0$: the orbit has two points, β and $1 - \beta$, both real, one on each side of $\frac{1}{2}$.*
(d) *$\beta = \frac{1}{2}$ and $\gamma = 0$: the orbit is the single point $\frac{1}{2}$.*

Proof. The four listed points are the images of s_0 under id, σ_3, σ_2, σ_1 respectively, so they exhaust the orbit. Two of them coincide precisely when $\beta = 1 - \beta$ or $\gamma = -\gamma$, that is when $\beta = \frac{1}{2}$ or $\gamma = 0$. The case analysis and the count of points on each side of the line are immediate. □

Corollary 1.5 (The functional equation gives symmetry, never fixedness)**.** *[T] Let F be meromorphic on $\mathbb{C}$, not identically zero, with $F(s) = F(1 - s)$ and $F(\bar{s}) = \overline{F(s)}$. Then the divisor of F is V-stable. Conversely, V-stability of a divisor places no constraint on whether its points lie on the critical line: for every finite V-stable set E in the open strip containing points of type (a), there is an entire function of order one with divisor containing E and satisfying the same pair of symmetries.*

Proof. The first assertion is immediate: if $F(\rho) = 0$ then $F(1 - \rho) = F(\rho) = 0$ and $F(\bar{\rho}) = \overline{F(\rho)} = 0$. For the converse, let $E = \{s_j\}$ be finite and V-stable and put

$$P(s) = \prod_{s_j \in E} \left(1 - \frac{s - \frac{1}{2}}{s_j - \frac{1}{2}}\right).$$

Since E is stable under $s \mapsto 1 - s$, the multiset $\{s_j - \frac{1}{2}\}$ is stable under $z \mapsto -z$, so $P(s) = P(1 - s)$; since E is stable under conjugation, $\{s_j - \frac{1}{2}\}$ is stable under conjugation, so $P(\bar{s}) = \overline{P(s)}$. Then $F = P$ is entire of order zero, and $F = P \cdot \Lambda_\zeta$ is meromorphic of order one with the two symmetries and with E in its divisor. □

[T] Corollary 1.5 is deliberately weak; it is a statement about divisors, not about Dirichlet series. Its point is that the symmetry *alone* carries no information about location. The strengthening one actually wants — that there is a Dirichlet series with a Riemann-type functional equation, an analytic continuation, and genuinely off-line zeros — is not a matter of construction by hand: it is the Davenport–Heilbronn theorem, and it is the subject of §2.2. We record here only the logical shape.

OBSTRUCTION O-0: symmetry does not localise

[N] No argument whose only structural input is the pair of symmetries $F(s) = F(1-s)$, $F(\bar{s}) = \overline{F(s)}$ can prove that the zeros of F lie on $\operatorname{Re} s = \frac{1}{2}$. A witness is the Davenport–Heilbronn function [DH36a; DH36b], which has both symmetries and infinitely many zeros with $\operatorname{Re} s > 1$. *Scope.* This does not rule out arguments that use the symmetries *together with* the Euler product or with an independently established positivity; it rules out arguments that use only the symmetries.

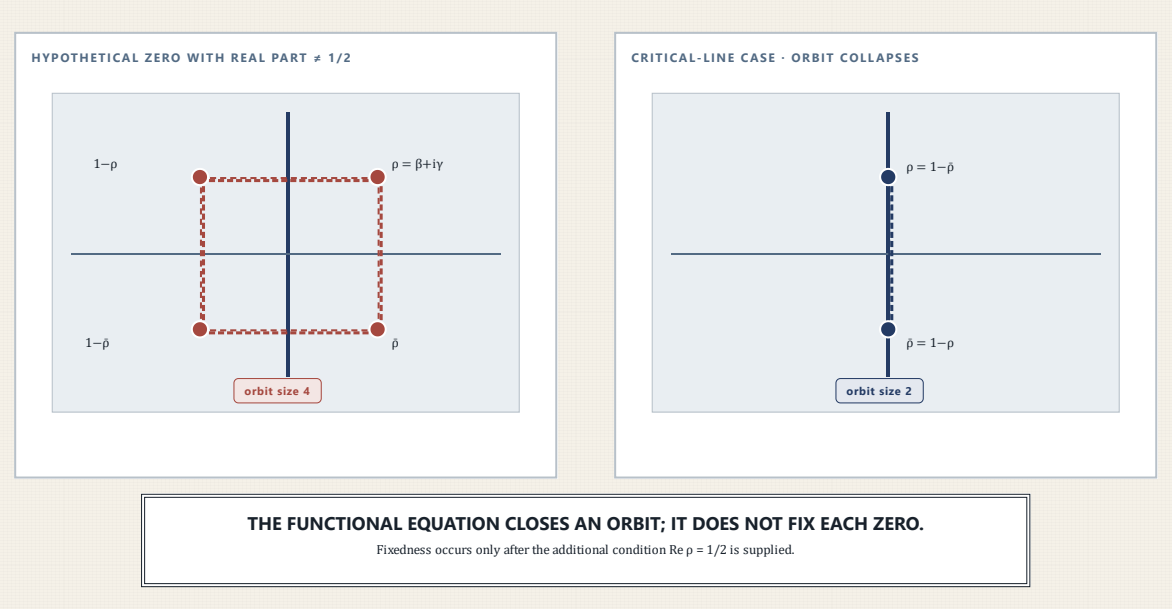


Figure 1.2. The four-point functional-equation orbit.
What it does not show: that any function has an orbit of type (a); it displays the combinatorics of the symmetry group only

[T] The right-hand panel of the figure anticipates Chapter 44 and is worth stating now, because it is the reason the four-point orbit is not merely a curiosity. In the Krein framework, an orbit of type (a) contributes *two* eigenvalues with $\operatorname{Re}\lambda > \frac{1}{2}$, and Theorem D bounds that number by $\min(n_+, n_-)$. Hence negative index one excludes type (a) outright. It does not exclude type (c): a real reflected pair $\{\beta, 1-\beta\}$ contributes only one eigenvalue with $\operatorname{Re}\lambda > \frac{1}{2}$ and is compatible with $n_- = 1$. The zeta-specific fact that closes the gap is the classical statement $\zeta(s) \neq 0$ for real $s \in (0,1)$. The structure of the argument is therefore: symmetry supplies the orbit types, index one kills the nonreal type, and an arithmetic input kills the real type. A reader who remembers only that will not be surprised by anything in Part 43.

1.5 The method: proving what cannot work

[T] The organising decision of this book is to treat impossibility results as first-class mathematics. That decision needs a defence, since the reader may reasonably suspect that a catalogue of failures is a catalogue of excuses. The defence has three parts: what a no-go theorem is, what it costs to state one honestly, and what it buys.

1.5.1 What a no-go theorem is here

A no-go theorem in this book has the following shape. One specifies an *architecture*: a class of admissible inputs, a class of admissible operations, and a target conclusion. One then proves

that every argument built from those inputs by those operations yields a conclusion strictly weaker than the target. The theorem is about the architecture, not about the ingenuity of the arguer.

Two examples fix the form. The Level Collapse Theorem (Chapter 39) takes as inputs the additive correlation sums $R_k(N) = \sum_{n_1+\cdots+n_k=N} \Lambda(n_1)\cdots\Lambda(n_k)$ and their averages, and proves the unconditional equivalence

$$\sum_{N\le X} R_k(N) = \frac{X^k}{k!} + O(X^{k-\theta}) \iff \psi(x) - x = O(x^{1-\theta}) \iff \Theta_\zeta \le 1-\theta.$$

The conclusion is not that the k-fold statement is hard; it is that the k-fold statement carries the same exponent as the one-fold statement, so that no gain can be extracted by raising k. This is checkable, and it is refuted by exhibiting a single k and a single θ for which the equivalence fails.

The Amplification Invariance Theorem (Chapter 39) takes as input any Dirichlet series whose singular support contains k-fold sums of zeros, and proves the identity

$$\sigma_c - \mathrm{Re}(\text{amplified singularity}) = k\left(\tfrac{1}{2} - \delta\right),$$

independently of normalisation. Since the abscissa of convergence and the amplified singularity scale together, the extractable conclusion is $\delta \le \frac{1}{2}$ for every k: exactly the classical bound. Again the statement is an identity, so a counterexample would be a computation, not an opinion.

1.5.2 The cost: scope discipline

The cost of stating a no-go theorem honestly is that one must also state, precisely, what it does *not* rule out. A no-go theorem with an unstated scope is worse than no theorem, because it invites the reader to believe that a whole direction is closed when only one route through it is. We therefore attach to every entry of the catalogue in Chapter 6 two paragraphs: *rules out* and *does not rule out*. The second paragraph is the harder one to write and is the one a referee should read first.

Three scope errors are common enough to name.

Overquantification. "No positivity argument can prove RH" is false; the correct statement, from the Criticality Barrier (§6.6), is that no *perturbation-stable strict* positivity argument can, because RH $\iff$ $\Lambda_{\mathrm{dBN}} = 0$ places RH at a critical point of an unconditional inequality. Equality-type and rigidity-type positivity arguments are untouched.

Category slippage. "Amplification cannot help" is false as stated; Deligne's proof of the Weil conjectures is an amplification argument and it works [Del74]. The correct statement is the Amplification Dichotomy of Chapter 38: over $\mathbb{F}_q$ the symmetric power of a sheaf is an object of the same category, so the same purity theorem applies, whereas over $\mathbb{Q}$ the amplified object leaves the category in which the relevant purity is available. The obstruction is the change of category, not the amplification.

Confusing an obstruction with an impossibility. Several results in the catalogue say that a specific candidate object fails a specific compatibility. That is weaker than saying no object can succeed. We mark the difference by using [N] for proved impossibilities of an architecture and [O] for the absence of a candidate.

1.5.3 What it buys

A catalogue of no-go theorems buys three things.

It buys *search efficiency*. A researcher who knows that every Dirichlet series built from k-fold zero sums returns $\delta \le \frac{1}{2}$ does not spend a year on the next such series. This is the same service performed for computational complexity by relativization [BGS75], natural proofs [RR97], and algebrization [AW09]: none of the three proves that $P \neq NP$ is unprovable, and all three redirected effort in ways that were visible within a decade. We are explicit that the analogy is one of methodology, not of strength: the three cited barriers are theorems about proof systems, whereas ours are theorems about specific mathematical architectures.

It buys *a specification for a positive programme*. The negative results of this book jointly determine the shape of an argument that could survive them: it must use the Euler product essentially (Structure-Blindness, Chapter 41); it must be of equality or rigidity type (Criticality Barrier, §6.6); it must not rest on self-adjointness alone (Chapter 3); it must not place positivity on a definite form (Chapter 4); and it must distinguish four published counter-models (Chapter 2). The Arithmetic Hodge–Krein Realization Problem is what remains once those five conditions are imposed, and the reader is entitled to check that claim rather than accept it.

It buys *falsifiability for the book itself*. Chapter 5 lists, for each main line of the book, a concrete theorem or computation whose appearance would refute it. A book of conjectures cannot be refuted; a book of no-go theorems can be, and we regard that as the principal reason to write this one.

1.6 The status system, in use

[K] The six status labels are defined in the front matter. Their definitions are not repeated here. What is done here is to apply them to real statements, because a label system is only as good as its hardest cases, and the hardest cases are the ones where a text is tempted to upgrade.

Example 1.6 (Six statements, six labels)**.** [T] The following six statements occur in this book. Each is followed by its label and by the reason no other label is admissible.

(1) *If* $\Theta^* G + G\Theta = G$ *with* G *nondegenerate Hermitian of inertia* (n_+, n_-)*, then* $\#\{\lambda \in \operatorname{spec}\Theta : \operatorname{Re}\lambda > \frac{1}{2}\} \le \min(n_+, n_-)$. [T]: the proof occupies §44.2 and uses nothing unproved. Not [K], because we are not aware of this statement in the literature in this form; §44.13 names the nearest prior results, which are the Pontryagin invariant-subspace theorem [Pon44] and the Krein–Langer theory of definitizable operators [KL78].

(2) *There exists a filtered system* $(H_N, \Theta_N, G_N, \iota_N)$ *satisfying* (AHK-1), (AHK-2), (AHK-3). [O]: no candidate object is known. Not [C], because a conjecture requires a reason to expect truth beyond the fact that it would be convenient, and we have none of the required strength. Not [T] under any reading.

(3) RH $\iff$ $\Lambda_{\mathrm{dBN}} = 0$. [K]: this is a consequence of two published results, the inequality $\Lambda_{\mathrm{dBN}} \ge 0$ of Rodgers and Tao [RT20] and the classical equivalence RH $\iff$ $\Lambda_{\mathrm{dBN}} \le 0$ [Bru50; New76]. We cite rather than reprove. The *consequence* we draw for positivity mechanisms is a separate statement and is marked [N] where it is proved.

(4) *For sixty-three cubic-spline spaces,* $B_N = W_N - P_N$ *has measured inertia* $(N-1, 1)$. [E]: a finite computation with a stated range and tolerance. Not [C]: we do not assert the pattern persists. Not [T]: no proof of persistence is given, and Chapter 46 states exactly which part is proved (the signature of P_N) and which is measured.

(5) *Every conclusion of the additive-correlation machine holds verbatim for Beurling systems in which the* RH*-analogue is false.* [N]: this is a proved impossibility of an architecture (Chapter 41); the witness systems are those of Diamond, Montgomery and Vorhauer [DMV06].

(6) *The Riemann Hypothesis.* [O] from the standpoint of this book: it is neither proved here nor assumed here, and where it appears as a hypothesis the implication is marked [T] while the hypothesis retains its own label.

Remark 1.7 (Mixed-status sentences). [K] The characteristic sentence of this book has the form

$$[O]\ P \qquad \text{together with} \qquad [T]\ (P \Longrightarrow \mathrm{RH}).$$

It is an editorial defect to compress this into an unlabelled sentence asserting that the programme proves RH. It is equally a defect to compress it into a sentence asserting that the programme establishes nothing: the implication is a theorem and has content, namely that it identifies precisely which construction would suffice. The front matter gives the compression rules; we use them without further comment.

1.7 The obstruction catalogue as a systematic object

[T] The twenty-three no-go theorems are not a list of unrelated observations. They organise along four axes, and the organisation is itself informative, because it shows which combinations of properties have been tested and which have not.

Axis 1: what is being amplified. The amplification obstructions (N1–N5, N18–N22 in the numbering of Chapter 6) divide according to whether the amplified variable is the support of a test function, the tensor power of a sheaf, the symmetric power of an automorphic representation, the conductor of a family, or the order of an additive correlation. In every case the theorem computes the gain and the cost in the same units and finds them equal.

Axis 2: where positivity is placed. The positivity obstructions (N8–N11, N15, N23) divide according to the object carrying the positive form: the cohomology, the algebra of correspondences, the test space, the apolar algebra of the Jensen polynomials, or the additive correlations of Λ. Two of these placements are provably wrong, one is provably right in the function-field case and unavailable over $\mathbb{Q}$, and one is a theorem of this book.

Axis 3: what substrate is proposed. The substrate obstructions (N7, N12–N14, N16) concern $\operatorname{Spec}\mathbb{Z}$, the sphere spectrum, prismatic cohomology, and their variants. They are the most technical and the least likely to be independently rediscovered, and they are collected in Part IV.

Axis 4: what is being certified. The certification obstructions (N17, and the spectral-pollution results of Chapter 52) concern what a finite computation can establish about an infinite-dimensional object. The Solvability Complexity Index result of Hansen [Han11] belongs here, with the caveat — printed every time it is used — that it is a finite-data spectral obstruction and *not* an undecidability theorem for RH.

Warning 1.8 (The SCI caveat, printed in full). [T] Hansen's theorem states that the Solvability Complexity Index of the spectrum of a general bounded self-adjoint operator, computed from matrix entries, is 2: no single convergent sequence of finite computations determines it [Han11]. In this book that result is used only in the following form: *a finite Galerkin computation does not certify a containment statement about the spectrum of a limiting operator, absent an a priori resolvent bound.* It is *not* an undecidability theorem for RH, it says nothing about the provability of RH in any formal system, and it does not assert that RH is not computable in any sense. Every occurrence of the result in this book is accompanied by this paragraph or by an explicit cross-reference to it.

[T] It is worth separating two words that the literature uses interchangeably. A *barrier*, in the sense of complexity theory, is a theorem about proof systems: it says that arguments with a

stated formal property cannot decide a stated question. An *obstruction*, in the sense used here, is a theorem about mathematical objects: it says that a stated construction, applied to stated inputs, yields a stated conclusion and no more. All twenty-three entries of our catalogue are obstructions. None is a barrier. We have no theorem, and know of none, restricting the formal systems in which RH might be proved, and we make no claim in that direction. The distinction matters because a barrier would be evidence that a problem is hard in an absolute sense, whereas an obstruction is evidence only that a particular road is closed. Roads can be built.

Remark 1.9 (Why a catalogue rather than a survey). [T] A survey records what has been tried. A catalogue records what has been proved impossible, with the scope of each impossibility stated so that the entries can be combined. The difference is operational: entries of a catalogue can be conjoined. For example, N21 (structure-blindness) and N8 (criticality) together imply that a mechanism must both use the Euler product essentially and be of equality type; that conjunction excludes mechanisms that neither theorem excludes alone, and it is the conjunction that motivates the shape of the positive programme. We are not aware of a prior systematic catalogue of this kind for RH; the nearest prior work is the problem description of Bombieri [Bom00], which surveys the principal equivalences and explains informally why the analytic apparatus alone has not sufficed, and the discussion of failed approaches in [Con03].

1.8 The equivalences, and why an equivalence is not a route

[K] A large part of the literature consists of statements proved equivalent to RH. It is important to be clear at the outset about what such a statement contributes, because the positive programme of this book produces one more of them, and the reader is entitled to ask why that is not merely an addition to a long list.

The principal published equivalences are these.

Weil's criterion. Let W denote the Guinand–Weil functional on a suitable space of test functions, so that $W(f) = \sum_\rho \hat{f}(\rho)$ with archimedean and prime terms as in Chapter 7. Then RH holds if and only if $W(f \star \tilde{f}) \geq 0$ for all $f \in C_c^\infty(\mathbb{R})$ [Wei52; Bom00]. The criterion is an equivalence, not an implication with slack: positivity on the full admissible space is exactly RH.

Nyman–Beurling. RH holds if and only if the indicator of $(0,1)$ lies in the $L^2(0,\infty)$-closure of the span of the functions $\rho_\theta(x) = \{\theta/x\} - \theta\{1/x\}$, $0 < \theta < 1$ [Nym50; Beu55]; Báez-Duarte sharpened the family to $\theta = 1/n$ [Báe03].

Li's criterion. RH holds if and only if $\lambda_n \geq 0$ for all $n \geq 1$, where λ_n are the Li coefficients of Λ_ζ [Li97; BL99].

The de Bruijn–Newman constant. RH holds if and only if $\Lambda_{\mathrm{dBN}} \leq 0$ [Bru50; New76]; combined with $\Lambda_{\mathrm{dBN}} \geq 0$ [RT20] this gives RH $\iff$ $\Lambda_{\mathrm{dBN}} = 0$.

Error terms. RH holds if and only if $\psi(x) = x + O(x^{1/2+\varepsilon})$ for every $\varepsilon > 0$, equivalently $\pi(x) = \mathrm{li}(x) + O(x^{1/2+\varepsilon})$ [TH86].

Remark 1.10 (The common structure). [K] Every one of these is obtained by transporting RH along a bijection of data. The zeros are encoded in a functional, a closure, a sequence, a deformation parameter, or an error term, and the encoding is faithful. Faithful encodings cannot lose information, and they cannot create it either. Consequently an equivalence contributes exactly what its new side makes available: a new class of techniques. It contributes nothing at all if the new side is attacked by techniques that, when transported back, are the old ones.

[N] This is not a rhetorical point; it is the content of the Structure-Blindness Theorem (Chapter 41), which shows that a particular popular encoding — the passage from Λ to its additive correlations — forgets the Euler product, and hence that every conclusion drawn on the new side holds verbatim for Beurling systems whose RH-analogue is false. The encoding is faithful for ζ but not injective on the class of objects one might hope to distinguish. Whenever this book introduces a new equivalent formulation, it is required to say which counter-models the new side distinguishes. Chapter 2 supplies the required tests, and Chapter 6 records the results for every formulation in the book.

[T] Applied to our own positive line, the requirement produces the following honest accounting. The Index-One Exhaustion Theorem is an implication, not an equivalence: (AHK-1,2,3) implies RH, and we do not claim the converse. In fact the converse is delicate. Assuming RH, one may certainly build a diagonal operator whose spectrum is the zeros and a definite form making it centered; that would satisfy (AHK-1), (AHK-2), and (AHK-3) trivially, and it is exactly the degeneration that (AHK-4)–(AHK-6) exist to exclude. The programme therefore has the structure: an implication that is proved, plus a set of non-degeneracy requirements whose purpose is to make the hypothesis a construction problem rather than a repackaging. Chapter 6 audits whether that purpose is achieved, and reports the result without softening it.

1.9 Non-circularity as a technical condition

[T] Circularity is the failure mode most often alleged and least often defined. We separate it from firewall specificity: excluding counter-models shows that a hypothesis uses structure special to ζ; it does not show that the hypothesis is logically weaker than RH.

Definition 1.11 (Non-circular hypothesis)**.** [T] Let P be a hypothesis and C a target. A derivation $P \Rightarrow C$ is *circular as a reduction* if $C \Rightarrow P$ is also provable, so that P is an equivalent reformulation of C. It is *non-circular as a reduction* only when the reverse implication fails; a sufficient witness is an object or logically consistent model in which P holds and C fails. For a realization hypothesis, the practical audit begins by assuming C and attempting to construct the proposed realization. If that construction succeeds and $P \Rightarrow C$ is already proved, the proposed reduction has failed the test. Firewall specificity is audited separately by Chapter 2.

The definition tests logical content, not vocabulary. A statement may avoid the letters "RH", fail on every firewall object, and still be equivalent to RH. Conversely, a statement may follow from RH and still be a genuinely weaker intermediate obligation if it permits models in which RH fails; in that case it is not an equivalent repackaging. The two directions must therefore be checked separately.

Example 1.12 (A circular hypothesis)**.** [N] Let P be: "the Weil functional W is positive semidefinite on the full admissible test space." Then $P \Leftrightarrow$ RH by Weil's criterion. The same diagnosis applies to (AHK-1)$\wedge$(AHK-2)$\wedge$(AHK-3): Chapter 48 proves the forward implication, while under RH the diagonal realization

$$\Theta_N = \mathrm{diag}(\rho_1, \dots, \rho_N), \qquad G_N = \mathrm{diag}(-1, 1, \dots, 1)$$

satisfies the three conditions because $\bar{\rho} + \rho = 1$ on the critical line. Thus AHK-1–3 fails the corrected non-circularity test. Its failure on the firewall objects establishes specificity, not non-circularity.

Example 1.13 (A genuinely weaker obligation)**.** [T] Problem A2⋆-II asks for

$$\sum_{0<\gamma\le T} (\beta - \tfrac{1}{2})^{+} = o(T).$$

It follows from RH, but it does not imply RH: the assertion permits off-line zeros provided their total one-sided excess is $o(T)$, whereas RH permits none. It therefore passes the weaker-content test and is not an equivalent reformulation. This logical distinction is the reason A2⋆ is the book's genuinely non-circular obligation while AHK-1–3 is not.

[T] The audit protocol of §6.14 applies the two-direction test to every conditional statement: first assume the target and test whether the hypothesis can be constructed; then test whether the hypothesis allows a model in which the target fails. The firewall protocol is recorded in a separate column and is never accepted as a substitute.

1.10 Six routes, and where each stops

[K] We close the orientation with a compressed account of six routes, each of which has produced published theorems and none of which has produced RH. The account is not a survey; each entry names the mechanism, the strongest theorem it has produced, and the precise point at which the argument stops. Chapters 2–6 expand the entries that bear on this book's own line.

1. Hilbert–Pólya. Mechanism: exhibit a self-adjoint operator whose eigenvalues are the ordinates of the zeros. Strongest theorem: the Selberg trace formula makes the analogous statement true, and true by construction, for Selberg zeta functions [Hej76a; Bus92]. Stops at: the formulation is vacuous as usually stated (Chapter 3), and in the one setting where the operator exists canonically the corresponding line statement is false whenever exceptional eigenvalues occur.

2. Noncommutative geometry. Mechanism: realise the explicit formula as a trace on the adele class space. Strongest theorem: the semilocal trace formula is proved [Con99]; Weil positivity at the archimedean place is proved [CC21b]. Stops at: the global trace formula is *equivalent* to the corresponding Riemann hypothesis for Hecke L-functions, so the equivalence has been relocated, not discharged; and the zeros appear as an absorption spectrum, which is a statement about the chosen Sobolev weight as much as about ζ.

3. de Branges spaces. Mechanism: realise ξ in a Hilbert space of entire functions and use the structure theory. Strongest theorem: the theory itself, which is correct and deep [Bra68]. Stops at: the specific positivity conditions proposed are not satisfied by the functions naturally attached to ζ and to Dirichlet L-functions, as Conrey and Li showed by explicit computation [CL00].

4. Amplification and higher correlations. Mechanism: raise the order of a correlation so that a hypothetical off-line zero contributes with amplified weight. Strongest theorem: over function fields, Deligne's proof of the Weil conjectures, which is exactly this mechanism [Del74; Del80]. Stops at: over number fields the amplified object leaves the category in which purity is available (Chapter 38), and the gain is cancelled identically by the cost (Chapter 39).

5. Positivity and sum-of-squares certification. Mechanism: certify positivity of a finite Gram matrix of the Weil form and let the support grow. Strongest theorem: certified positivity on restricted support ranges, with genuine content [Bom00; Bur01]. Stops at: the Criticality Barrier. Since RH $\iff \Lambda_{\mathrm{dBN}} = 0$, no finite certificate of a *strict* inequality that is stable under the backward heat deformation can suffice (§6.6); and Weil's criterion makes the exhaustive version equivalent to RH itself.

6. Random matrix statistics. Mechanism: match local statistics of zeros with those of large random unitary matrices. Strongest theorem: Montgomery's pair correlation, in the range where it is unconditional [Mon73], together with the extensive numerical agreement [Odl87] and the function-field theorems of Katz and Sarnak [KS99]. Stops at: distributional agreement is powerless against a sparse exceptional set. A set of zeros of density zero can violate RH without disturbing any local statistic.

Remark 1.14 (What the six have in common). [T] Each route supplies structure and each fails to supply an *upper constraint*: a proved statement bounding how far zeros can deviate, valid at every height. The only surviving unconditional upper constraint is the zero-density estimate, and Chapter 29 proves it degenerates at $\sigma = \frac{1}{2}$. This diagnosis — structure is plentiful, constraint is absent — is the thesis of Volume II and the reason its subtitle is what it is.

1.11 The shape of the positive programme

[T] We state the positive line once, in outline, so that the negative chapters that follow have a target to constrain. Details are in Part VIII; nothing in Part I depends on them.

Let H be a finite-dimensional complex vector space, G a nondegenerate Hermitian form on H with inertia (n_+, n_-), and Θ an operator on H. Call (H, Θ, G) a *centered Krein pair* if

$$\Theta^* G + G\Theta = G,$$

where Θ^* is the ordinary Hilbert-space adjoint for a chosen reference inner product. Writing λ for an eigenvalue of Θ and v for a corresponding eigenvector with $v^*Gv \neq 0$, the identity

$$(\bar{\lambda} + \lambda)\, v^*Gv = v^*Gv$$

gives $\operatorname{Re}\lambda = \frac{1}{2}$ at once. The content of Theorem D is what happens when $v^*Gv = 0$ is allowed and the form is indefinite: the number of eigenvalues in the open half-plane $\operatorname{Re}\lambda > \frac{1}{2}$ is then bounded by $\min(n_+, n_-)$. If $n_- = 1$, at most one eigenvalue lies to the right of the line.

[T] Combine this with two facts about ζ. First, the nontrivial zeros are stable under conjugation, so a nonreal zero with $\beta > \frac{1}{2}$ comes with a partner $\bar{\rho}$, also with $\beta > \frac{1}{2}$: two eigenvalues on the right, contradicting $n_- = 1$. Second, a real zero with $\beta > \frac{1}{2}$ would lie in $(\frac{1}{2}, 1)$, and ζ has no zeros on the real segment $(0, 1)$. The functional equation transfers the conclusion to the left half-plane. Hence, under (AHK-1), (AHK-2), (AHK-3), all nontrivial zeros lie on the critical line.

[O] Everything before that argument is the open problem. There is no H_N, no Θ_N, no G_N. The book's contribution on the positive side is to have isolated the exact structure that would suffice, to have proved that the structure must be *indefinite* rather than definite (Chapter 4), and to have exhibited a finite arithmetic computation in which an index-one signature is measured (Chapter 46) — a measurement that is suggestive and is not a construction.

OPEN PROBLEM G.2 (statement in outline; full form in Chapter 57)

[O] Construct $(H_N, \Theta_N, G_N, \iota_N)_{N\geq 1}$ satisfying (AHK-1) spectral realization, (AHK-2) centered duality, (AHK-3) index one compatibly with the transition maps, and — to prevent degeneration into a diagonal operator built from the zeros themselves — (AHK-4) an arithmetic trace formula recovering the Guinand–Weil explicit formula with the true masses $-\Lambda(n)n^{-1/2}$, (AHK-5) pole-plane and Lefschetz compatibility, and (AHK-6) firewall specificity. *Nothing in this book discharges any of the six.*

1.12 The explicit formula as the only bridge

[K] Every argument in this book that connects primes to zeros passes through one identity. In the normalization fixed in Chapter 7, for test functions H holomorphic and suitably decaying in the strip $|\operatorname{Im} z| \le \frac{1}{2} + \eta$ and g its inverse transform,

$$\sum_{\rho} H\Big(\frac{\rho - \frac{1}{2}}{i}\Big) = H(i/2) + H(-i/2) + \frac{1}{2\pi}\int_{\mathbb{R}} H(t)\Big[-\log\pi + \operatorname{Re}\psi\Big(\frac{1}{4} + \frac{it}{2}\Big)\Big]\,dt - \sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}[g(\log n) + g(-\log n)].$$

The identity is due to Riemann in principle and to Guinand and Weil in the form used here [Gui48; Wei52; Bar81]; the finite-contour proof, with every convergence question settled before a limit is taken, is Chapter 7.

[T] Three structural features of this identity determine most of what follows, and it is worth having them before the obstruction chapters.

First, the identity is *exact and two-sided*. It is not an approximation with an error term; both sides converge absolutely in the stated class. Consequently any inequality proved on one side transports exactly to the other, and no inequality can be improved by working on the side one prefers. This is why so many mechanisms return the same exponent: they are the same inequality in different coordinates.

Second, the two sides have *different natural scales*. If g is supported in $[-S, S]$, the prime side involves only $n \le e^S$, and unconditionally

$$\sum_{n\le X}\frac{\Lambda(n)}{\sqrt{n}} = 2\sqrt{X} + o(\sqrt{X})$$

by the prime number theorem and partial summation (Proposition 8.2). So the prime side has absolute size $\ll Se^{S/2}$ for a bounded test function, while a hypothetical off-line zero of deviation δ contributes a term of size $e^{\delta S}$. Since $\delta < \frac{1}{2}$ for every nontrivial zero, the arithmetic side dominates for all large S. The absolute-value comparison therefore cannot exclude any zero, and it is this elementary observation, correctly bookkept, that becomes the support-doubling no-go of Chapter 8.

Third, the identity is *blind to the Euler product in one direction*. It expresses the zeros in terms of Λ, and Λ determines the Euler product; but the passage to additive correlations of Λ loses it. Chapter 41 makes this precise, and it is the reason the Beurling counter-models are lethal to an entire family of methods.

Remark 1.15 (Why the bridge cannot be widened). [T] A recurrent hope is that a cleverer test function widens the bridge. Chapter 12 classifies the extremal test functions relevant here and shows that the Beurling–Selberg theory already solves the associated extremal problems [Vaa85; CCM13], so that the available improvement is in constants, not exponents. The exponent is fixed by the scale mismatch just described, and the scale mismatch is a theorem, not a limitation of technique.

1.13 Two open problems of the same shape

[O] The book ends with two problems, and it is a nontrivial observation — proved in Part VIII — that they measure the same thing.

The first is the realization problem (G.2) stated above. Theorem D bounds

$$\#\{\lambda \in \operatorname{spec} \Theta_N : \operatorname{Re}\lambda > \tfrac{1}{2}\}$$

by $\min(n_+, n_-)$: a *count* of one-sided deviations.

The second is Problem A2$\star$, which has three equivalent forms (Chapter 38). Its analytically cleanest form asks for an unconditional proof of

$$\sum_{0<\gamma\le T} \left(\beta - \tfrac{1}{2}\right)^+ = o(T) :$$

a *weighted sum* of one-sided deviations. The signed sum $\sum_{0<\gamma\le T}(\beta - \frac{1}{2})$ vanishes identically, by the functional equation, which pairs ρ with $1-\rho$. The one-sided sum is therefore the first quantity in which the symmetry is broken, and the best available unconditional bound for it is the trivial $O(T\log T)$ from the zero-counting function. Even a saving of one logarithm would be, as far as we are aware, the first unconditional one-sided statement of its kind.

[T] The two problems are a count and a weighted count of the same one-sided deviation measure. Part VIII proves the precise relation between them; here we record only that this is why the book treats them together and why a solution to either would be informative about the other.

Remark 1.16 (Why *A*2$\star$ is stated as a problem and not a conjecture). [O] We do not label $\sum(\beta-\frac{1}{2})^+ = o(T)$ a conjecture, although it follows from RH and we expect it to be true, because a conjecture in this book's status system requires a reason to believe beyond consequence of the target. Here there is none: the statement is weaker than RH but we know no route to it that does not go through RH. It is therefore [O] — an obligation with no candidate proof — and it is stated as a problem so that its falsification predicate is visible: a proof of $\sum(\beta - \frac{1}{2})^+ \gg T$ would refute RH outright, and a proof of $o(T)$ would be a new unconditional theorem.

1.14 The function-field case as a unit test

[K] Over a smooth projective geometrically connected curve X over $\mathbb{F}_q$, the analogue of RH is a theorem, twice over: by Weil's argument through the Hodge index theorem on the surface $X \times X$ [Wei48a], and by Deligne's argument through tensor-power amplification and monodromy [Del74]. Part III instantiates the function-field case completely before any number-field speculation begins. The reason is stated once here and used throughout.

[T] The function-field case is not an analogy; it is a unit test. Any proposed structure over $\operatorname{Spec}\mathbb{Z}$ must, on base change to a function field, reproduce the objects that are known to exist there: the étale cohomology, the Frobenius, Poincaré duality, the Grothendieck–Lefschetz trace formula, and the positivity. A proposed structure that does not reproduce them is refuted without further discussion. This is cheap and it is decisive, and Chapter 18 runs the test on every candidate in the book.

[N] The test also has a negative reading, which is the more important one. Weil's route requires a second factor: the surface $X \times X$, the correspondence Γ_{Frob}, and Castelnuovo's inequality. Over $\operatorname{Spec}\mathbb{Z}$ there is no second factor, because $\operatorname{Spec}\mathbb{Z}$ is terminal in schemes, so

$$\operatorname{Spec}\mathbb{Z} \times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Spec}\mathbb{Z} = \operatorname{Spec}\mathbb{Z},$$

and the diagonal is everything. Chapter 24 states this as a no-go for the naive transplant and surveys what would be needed to manufacture a second factor; the sphere spectrum supplies one in a precise homotopical sense, since $\mathbb{Z} \otimes_{\mathbb{S}} \mathbb{Z} \not\simeq \mathbb{Z}$, and Part IV develops what that does

and does not buy. It buys a base; it does not buy a polarization, and the polarization is the hard part.

1.15 A worked entry from the catalogue

[N] To show what an entry of the catalogue looks like when written out in full, we prove one here. It is chosen because its proof is short and complete, because it closes two published proposals with a genuine argument rather than with a difficulty estimate, and because it illustrates the scope discipline of §1.5.

Definition 1.17 (The k-fold zero divisor)**.** [T] Let $Z = \{\rho\}$ be the multiset of nontrivial zeros of ζ, counted with algebraic multiplicity. For $k \geq 2$ set

$$D_k = \{\rho_1 + \rho_2 + \cdots + \rho_k : \rho_j \in Z\},$$

as a multiset in $\mathbb{C}$, the multiplicity of a point w being the number of ordered k-tuples summing to w.

Theorem 1.18 (Infinite Multiplicity)**.** *[N] For every $k \geq 2$ the multiset D_k contains a point of infinite multiplicity. Consequently there is no meromorphic function on $\mathbb{C}$ whose divisor is D_k.*

Proof. Take $k = 2$ first. By Hardy's theorem [Har14] there are infinitely many zeros $\rho = \frac{1}{2} + i\gamma$ with $\gamma > 0$ on the critical line. Since the zero multiset is stable under conjugation, $\bar{\rho} = \frac{1}{2} - i\gamma$ is also a zero. Hence

$$\rho + \bar{\rho} = 1$$

for each such γ, and distinct γ give distinct ordered pairs. The point $w = 1$ therefore has infinite multiplicity in D_2. For general k, fix any $\rho_3, \dots, \rho_k \in Z$ and adjoin them: the point $w = 1 + \rho_3 + \cdots + \rho_k$ has infinite multiplicity in D_k.

For the consequence, suppose F were meromorphic on $\mathbb{C}$ with divisor D_k. The divisor of a meromorphic function is locally finite and every point of it carries finite order, since F is either identically zero near a point or has a Laurent expansion with a finite leading exponent there. A point of infinite multiplicity is excluded. □

Remark 1.19 (Scope). [N] *What this rules out.* Any proposal that (i) forms a Dirichlet series or an entire function whose zero divisor is asserted to be D_k, and (ii) then applies automorphy, purity, or a functional equation to that object, is refuted: the object does not exist. This closes the "automorphy of the k-fold sum" route and the "purity of the amplified divisor" route with a proof, not an assessment. *What this does not rule out.* It says nothing about Dirichlet series whose *singular support* contains D_k without D_k being the divisor — such series exist, and Chapter 41 studies them; nothing about the local structure of D_k away from the accumulation points; nothing about k-fold correlations of Λ, which are studied by entirely different means in Chapter 39; and nothing about whether some other amplification of order k might work. The theorem is about a divisor, and only about a divisor.

[T] Two features of this entry are typical. The proof consumes one classical input (Hardy's theorem) and one elementary symmetry, and the strength of the conclusion comes from the interaction, not from either alone. And the scope paragraph is longer than the proof. A reader who finds the scope paragraph unconvincing has found a defect in the catalogue, and we would rather that happen than not.

1.16 Conventions, and how to read this book

[K] The notation chapter in the front matter fixes symbols; we record here only the conventions that a reader of Part I needs immediately.

Zeros are $\rho = \beta + i\gamma$; $\Theta_\zeta = \sup \beta$; δ denotes a hypothetical deviation $\beta - \frac{1}{2} > 0$. The letter Θ without a subscript denotes an operator, and the collision with Θ_ζ is unfortunate but standard on both sides; the subscript always distinguishes them. The Fourier transform is $\hat{g}(\xi) = \int_{\mathbb{R}} g(x) e^{-2\pi i x \xi}\, dx$, and the angular-frequency companion $H(z) = \int_{\mathbb{R}} g(x) e^{izx}\, dx$ is used in explicit formulas, as fixed in Chapter 7. The de Bruijn–Newman constant is Λ_{dBN} in the normalization of [RT20]. Inertia of a nondegenerate Hermitian form is written (n_+, n_-), and (n_+, n_-, n_0) when a radical is present.

Numbered environments carry a status macro on their first line. Boxes carry their own labels: `obstruction` for a no-go theorem, `firewall` for a comparison against the four counter-models, `openproblem` for a missing construction, `notice` for a convention or boundary. Figures carry a status badge, a *constructed* field, and a sentence stating what they do not establish.

Reading paths

A reader who wants the negative results and nothing else may read Chapters 1, 2, 6, then Part II and Part VII, and stop. That path is self-contained and contains every unconditional theorem in the book except those of Part VIII.

A reader interested in the spectral programme should read Chapters 3 and 4 of this Part, then Part VIII directly; Chapter 4 is written so that the indefinite formulation of Chapter 44 appears as forced rather than chosen.

A reader who wishes to attack the programme should read Chapter 5 first, then Chapter 6, then Chapter 57. Chapter 5 is the list of predicates whose verification would refute the book; Chapter 57 is the decomposition of the remaining obligation into seven subproblems, each stated so that progress on it is recognisable.

A referee should read §2.6 (the firewall test protocol), §6.14 (the audit protocol), and the "What this chapter does not establish" section of each chapter. Those three items are where the book's claims about itself are concentrated, and they are written to be attacked.

On the absence of promotional language. This book contains no claim that a result is deep, striking, or profound. The omission is deliberate and is a rule of this book, not a stylistic preference. A reader evaluating an unsolved problem needs to distinguish what is proved from what is hoped, and adjectives of enthusiasm are precisely the device by which that distinction is usually blurred. Where this book believes a result matters, it says what the result rules out.

1.17 How Part I is organised

Chapter 2 develops the firewall: four published objects that satisfy much of the formalism surrounding RH while the corresponding line statement fails. The chapter gives the constructions, not the slogans, and ends with a test protocol a referee can apply mechanically.

Chapter 3 proves that the Hilbert–Pólya formulation, as usually stated, is vacuous, and then states what a non-vacuous spectral formulation must add. It places the spectral-gap hierarchy — Selberg's 3/16, the Luo–Rudnick–Sarnak bound, Kim–Sarnak — and analyses the

trace formula of Connes [Con99] and the prolate operator of Connes and Moscovici [CM22a] in terms of what is proved and where the proof stops.

Chapter 4 proves that the positivity axiom in the form usually proposed is strictly stronger than RH, and derives the corrected, indefinite formulation. It is the chapter that makes Part VIII possible.

Chapter 5 states what would falsify this book. It appears early, not late, and the reason is given there.

Chapter 6 is the archive: all twenty-three no-go theorems with statements, proofs or proof sketches with forward references, scope, and the audit protocol.

[T] The order is not arbitrary. The firewall comes second because it supplies the tests that every later claim must pass, including our own; the vacuity theorem comes third because it is the cheapest way to see that a spectral formulation is not automatically content; the corrected diagnosis of the positivity axiom comes fourth because it converts the book's central object from a definite form, which is unavailable, into an indefinite one, which is at least consistent with the function-field template; the falsification chapter comes fifth because by that point the reader knows enough to check the predicates; and the archive comes last because it is a reference chapter rather than an argument. A reader who prefers arguments to references may treat Chapter 6 as an appendix and return to it when a specific no-go is cited later.

[K] Part II then develops the explicit formula in the fixed normalization and the absolute-value ceiling taxonomy, which are the technical prerequisites for the amplification chapters of Volume II. A reader who accepts the results of Part I may proceed directly to Part III, where the function-field paradigm is instantiated in full before any number-field statement is made.

What this chapter does not establish

1. It does not prove the Riemann Hypothesis, and it does not claim that the programme of this book is close to proving it. The Index-One Exhaustion Theorem is an implication whose hypothesis is open.
2. It does not prove that the Arithmetic Hodge–Krein Realization Problem is solvable. We give no evidence that a system $(H_N, \Theta_N, G_N, \iota_N)$ exists, and Chapter 5 lists the observations that would show it does not.
3. It does not prove that the twenty-three no-go theorems are complete, or that their union closes any direction. Each has a stated scope; the union of the scopes is not all of mathematics, and we make no claim that it is.
4. Corollary 1.5 establishes only that V-stability of a divisor is compatible with off-line points. It does not, by itself, exhibit a Dirichlet series with a functional equation and off-line zeros; that is Davenport and Heilbronn's theorem, cited in Chapter 2, and it is not reproved here.
5. The measured inertia $(N-1, 1)$ is finite evidence about sixty-three matrices. It is not a theorem about all N, it is not a construction of Θ_N, and it cannot be inserted into Theorem D.
6. The comparison with relativization, natural proofs, and algebrization in §1.5 is methodological. It does not assert that the obstructions of this book are barriers in the logical sense, and none of them is a statement about the unprovability of RH in any formal system.
7. Nothing in this chapter establishes that the four counter-models of Chapter 2 exhaust the ways a mechanism can overreach. They test four independent failure modes; other failure modes may exist.

Chapter 2

The firewall: four counter-model objects

A proposed mechanism for the Riemann Hypothesis is a statement of the form: *because ζ has property P, its non-trivial zeros lie on $\Re s = \frac{1}{2}$*. The difficulty with such statements is not usually that they are false. It is that the property P named in the argument is, on inspection, shared by objects whose zeros demonstrably do not lie on the critical line. When that happens the argument cannot be repaired by working harder at the same place; it has to be abandoned, because it proves a false theorem.

This chapter constructs four objects for which this failure can be checked. Each is a Dirichlet series, or a family of them, possessing some but not all of the structural features of ζ. Each has a false analogue of the Riemann Hypothesis. Together they cut the space of possible mechanisms: a mechanism that survives all four is not thereby correct, but a mechanism that fails any one of them is definitely wrong, and the failure can be located and named.

We call the resulting test the *firewall*. The name records the intended function. A firewall does not tell you what to build; it tells you what will burn. The four objects are the Davenport–Heilbronn function, the Epstein zeta function of a quadratic form of class number greater than one, the Beurling generalised number system of Diamond, Montgomery and Vorhauer, and the exceptional eigenvalues of the Laplacian on congruence quotients as they appear through the Selberg zeta function.

The chapter is longer than a survey of these objects would need to be, and this is deliberate. The firewall is used throughout the book as a decision procedure, and a decision procedure whose inputs are only sketched cannot be applied by a reader who wishes to check the application. Each object is therefore constructed to the point where its defining data can be written down, its functional equation verified, and its off-line or off-critical zeros exhibited or cited to a specific computation. [T]

2.1 The logic of a firewall

2.1.1 Necessary conditions on a mechanism

Let $\mathcal{P}$ denote a property of Dirichlet series, expressed in whatever language a proposed mechanism uses — analytic, spectral, cohomological, operator-theoretic. Let $\mathrm{RH}(F)$ denote the assertion that all non-trivial zeros of F lie on the critical line of F, the line fixed by the functional equation of F (Definition 2.4 below makes this precise for the objects at hand).

Definition 2.1 (Mechanism)**.** A *mechanism* for RH is a pair $(\mathcal{P}, \Pi)$ where $\mathcal{P}$ is a property of Dirichlet series, ζ satisfies $\mathcal{P}$, and Π is a proof of the implication

$$F \text{ satisfies } \mathcal{P} \implies \mathrm{RH}(F).$$

The definition is deliberately permissive. It does not require $\mathcal{P}$ to be a property of the Dirichlet series alone; $\mathcal{P}$ may quantify over auxiliary structures (an operator, a space, a cohomology theory) provided that the existence of those structures is part of what $\mathcal{P}$ asserts. It does not require Π to be short, or the proof to be constructive. What it does require is that Π be a proof of a universally quantified implication, and this is where the leverage lies.

Lemma 2.2 (Firewall lemma)**.** *Let* $(\mathcal{P}, \Pi)$ *be a mechanism in the sense of Definition 2.1. Suppose* G *is a Dirichlet series satisfying* $\mathcal{P}$ *for which* RH(G) *is false. Then* Π *is not a proof.*

Proof. Immediate: Π purports to prove a statement with a counterexample. □

The lemma is trivial. Its content is entirely in the availability of the G. The four objects of this chapter are candidate values of G, chosen so that between them they satisfy every combination of the standard structural hypotheses except the full combination realised by ζ itself.

2.1.2 Specificity as a technical condition

The lemma just stated has an inverse formulation which is more useful in practice. If a mechanism $(\mathcal{P}, \Pi)$ is correct, then the property $\mathcal{P}$ must fail for each object G in a firewall set. That failure has to be exhibited: one must be able to point to the clause of $\mathcal{P}$ that G violates, and say why.

Definition 2.3 (Specificity)**.** Let $\mathcal{G} = \{G_1, \dots, G_m\}$ be a finite set of Dirichlet series with RH(G_i) false for each i. A property $\mathcal{P}$ is *specific relative to* $\mathcal{G}$ if for each i there is an identified clause of $\mathcal{P}$ and a proof that G_i fails that clause.

Specificity is a property of a *stated* mechanism, not of a mechanism in the abstract. It can fail in two distinguishable ways, and the distinction matters when reading the literature.

The first failure is substantive: $\mathcal{P}$ genuinely holds for some G_i, and the mechanism is refuted by Lemma 2.2. This is rare in published work, for the good reason that authors check.

The second failure is a failure of statement: $\mathcal{P}$ is not written down precisely enough for the question "does G_i satisfy $\mathcal{P}$?" to have an answer. This is common. It is the characteristic failure mode of the programmes surveyed in Chapter 3, where the operative property is something like "arises canonically from arithmetic" or "is the natural spectral realisation". Such a property cannot be tested against G_i, because the question of whether G_i satisfies it is not a mathematical question.

> Throughout this book, when we say that a proposed mechanism *is not yet a mechanism*, we mean the second failure and not the first. The assertion is that the property has not been stated to a standard at which Definition 2.3 applies. This is a criticism of a formulation, not a claim that the underlying idea is wrong.

2.1.3 Why four objects and not one

A single counter-model would suffice to kill a single mechanism. The reason for assembling four is that they fail in different places, and the pattern of failures is informative.

Write the four standard structural hypotheses as follows.

(FE) A functional equation $\Lambda(s) = \varepsilon\Lambda(1-s)$ for a completed function Λ formed from F by multiplying by gamma factors and an exponential in s.

(EP) An Euler product $F(s) = \prod_p F_p(s)$ over a set of "primes", convergent in a right half-plane.

(PT) A prime-counting asymptotic of the expected shape, i.e. the analogue of the Prime Number Theorem for the associated primes.

(SA) A self-adjoint spectral realisation: the zeros are the spectrum of a self-adjoint operator on a Hilbert space, arising from a geometric or arithmetic construction.

Table 2.1 is the working summary. Read row by row it says: a functional equation alone does not give RH (rows 1, 2); an Euler product together with the Prime Number Theorem

Table 2.1. The firewall objects against the four standard hypotheses. A tick indicates the hypothesis holds for the object; a cross that it fails. The last column records whether the analogue of RH is true.

Object	(FE)	(EP)	(PT)	(SA)	RH-analogue
Davenport–Heilbronn f	yes	no	—	no	**false**
Epstein ζ_Q, $h > 1$	yes	no	—	no	**false**
DMV Beurling $\zeta_{\mathcal{B}}$	no	yes	yes	no	**false**
Selberg Z_Γ, exceptional	yes	yes	yes	yes	**false**
Riemann ζ	yes	yes	yes	?	open

does not give RH (row 3); and a genuine self-adjoint spectral realisation of the zeros of a genuine zeta function with a genuine Euler product and a genuine functional equation does not give RH either (row 4). The last of these is the one most often overlooked, and it is why the Selberg object is in the list.

Read column by column, the table says something sharper. There is no single column that separates the last row from the four above it. Whatever distinguishes ζ from the firewall objects is not any one of (FE), (EP), (PT), (SA), and is not the conjunction of any three of them that the Selberg object also satisfies. This is a genuine constraint on mechanisms, and it is the reason the firewall is worth stating carefully.

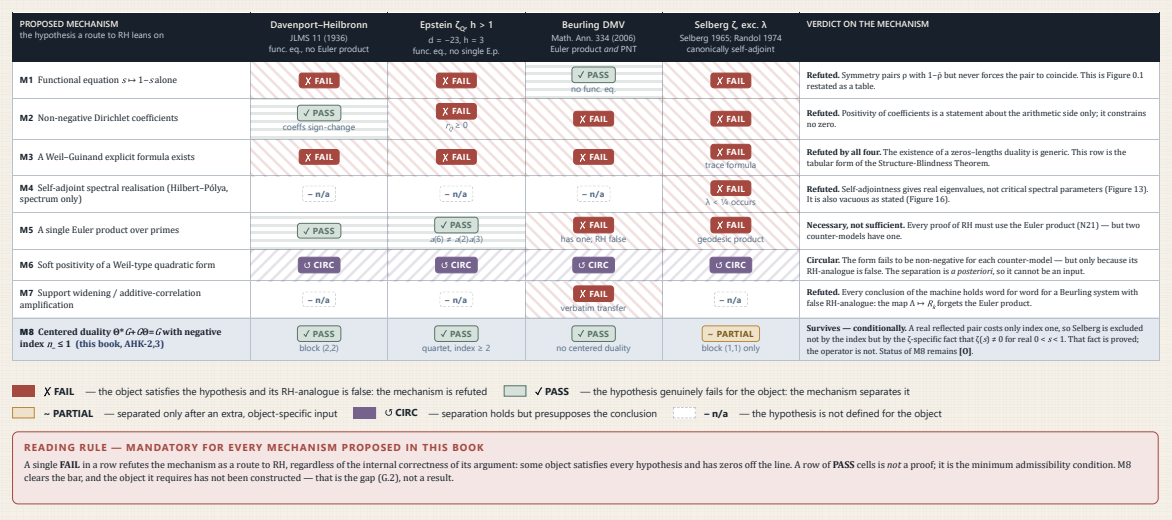


Figure 2.1. The firewall test matrix.

What it does not show: The matrix records which hypotheses each object satisfies. It does not enumerate all possible mechanisms, and a mechanism may fail for reasons not visible in these four columns.

2.1.4 What the firewall is not

Three clarifications, because the firewall is easy to over-read.

First, the firewall does not assert that the four objects exhaust the possible counterexamples. There are others — the Hurwitz zeta function $\zeta(s, \alpha)$ for α rational and not 1 or $\frac{1}{2}$, various linear combinations of Dirichlet L-functions, the Lerch zeta function — and a mechanism might survive our four and fail on a fifth. The four are chosen because they are the ones most often relevant in practice and because their failure modes are complementary.

Second, the firewall is not a heuristic. Lemma 2.2 is a theorem. If a mechanism's property holds for one of the objects, the mechanism is refuted, full stop.

Third — and this is the clarification most often needed — passing the firewall is not evidence that a mechanism is correct. It is the removal of one specific way of being wrong. A mechanism that passes all four tests may still fail for a reason nobody has thought of. We say a mechanism is *firewall-compatible* when it passes, and firewall-compatibility is recorded in this book with the status [C] at best, never [T].

Definition 2.4 (Critical line of a Dirichlet series with functional equation)**.** Let F be a Dirichlet series with meromorphic continuation admitting a functional equation relating $F(s)$ to $\overline{F(\bar{w})}$ or $F(w)$ under an involution $s \mapsto w$ of $\mathbb{C}$ of the form $w = \lambda - s$ for a real constant λ. The *critical line* of F is the fixed-point set $\{\Re s = \lambda/2\}$ of that involution. For the objects in this chapter $\lambda = 1$ throughout, except for the Selberg zeta function, where the natural normalisation also gives $\lambda = 1$.

2.2 The Davenport–Heilbronn function

2.2.1 Construction

Fix the modulus 5. We look for a function $a : \mathbb{Z}/5 \to \mathbb{R}$, not a Dirichlet character, such that the Dirichlet series

$$f(s) \;=\; \sum_{n \geq 1} \frac{a(n)}{n^s}, \qquad a(n) := a(n \bmod 5),$$

has a functional equation of the classical shape. The existence of such an a rests on a single linear-algebra observation about the finite Fourier transform.

Definition 2.5 (Finite Fourier transform mod q)**.** For $a : \mathbb{Z}/q \to \mathbb{C}$ set

$$\hat{a}(m) \;=\; \sum_{r \bmod q} a(r)\, e^{2\pi i m r/q}.$$

Proposition 2.6 (Functional equation from a Fourier eigenvector)**.** *Let $q \geq 1$ and let $a : \mathbb{Z}/q \to \mathbb{C}$ satisfy $a(0) = 0$ and $\sum_r a(r) = 0$. Suppose $\hat{a} = \kappa\, a$ for a constant $\kappa \in \mathbb{C}$. Then the completed function*

$$\Lambda_f(s) \;=\; \Big(\frac{q}{\pi}\Big)^{(s+\delta)/2} \Gamma\Big(\frac{s+\delta}{2}\Big) f(s), \qquad \delta \in \{0, 1\},$$

with δ determined by the parity of a, satisfies a functional equation

$$\Lambda_f(s) \;=\; \omega\, \Lambda_f(1 - s)$$

for a constant ω depending on κ, q and δ only, and f continues to an entire function of s.

Proof sketch. The two hypotheses $a(0) = 0$ and $\sum_r a(r) = 0$ ensure that f has no pole at $s = 1$ and that the Hurwitz decomposition

$$f(s) \;=\; q^{-s} \sum_{r=1}^{q-1} a(r)\, \zeta\Big(s, \tfrac{r}{q}\Big)$$

is valid, where $\zeta(s, \alpha) = \sum_{n \geq 0} (n + \alpha)^{-s}$. Hurwitz's functional equation expresses $\zeta(1 - s, r/q)$ as a finite linear combination of $\zeta(s, m/q)$ with coefficients $e^{\pm 2\pi i m r/q}$ and gamma factors. Substituting and collecting terms, the coefficient vector of the transformed series is exactly $\hat{a}$ up to the standard sine and cosine factors, which combine with the gamma factor. The eigenvector hypothesis $\hat{a} = \kappa a$ therefore returns the same coefficient vector, and the identity

closes. Entirety follows because each $\zeta(s, r/q)$ has a single simple pole at $s = 1$ with residue 1, and $\sum_r a(r) = 0$ cancels it. Full details are in [Tit86, Ch. 10] and, in the form used here, [BG11]. □

The problem is thus reduced to producing a real Fourier eigenvector mod 5 that is not a character. Take the odd part: impose $a(5 - r) = -a(r)$, so that a is determined by $a(1)$ and $a(2)$, with $a(0) = 0$ automatically and $\sum_r a(r) = 0$ automatically. Normalise $a(1) = 1$ and write $a(2) = \xi$. The vector is

$$a \;=\; (a(1), a(2), a(3), a(4), a(5)) \;=\; (1,\ \xi,\ -\xi,\ -1,\ 0),$$

the last entry being $a(0)$.

Proposition 2.7 (The Davenport–Heilbronn constant)**.** *For the odd vector $a = (1, \xi, -\xi, -1, 0)$ mod 5 the eigenvector condition $\hat{a} = \kappa a$ holds precisely when*

$$\xi^2 \sin\Big(\frac{4\pi}{5}\Big) \;+\; 2\xi \sin\Big(\frac{2\pi}{5}\Big) \;-\; \sin\Big(\frac{4\pi}{5}\Big) \;=\; 0,$$

whose positive root is

$$\xi \;=\; \frac{\sqrt{10 - 2\sqrt{5}} \;-\; 2}{\sqrt{5} - 1} \;=\; 0.284079043840412\ldots,$$

and then $\kappa = i\sqrt{5}$.

Proof. Since a is odd, $\hat{a}(m) = 2i(\,\sin(2\pi m/5) + \xi \sin(4\pi m/5))$. Writing $s_1 = \sin(2\pi/5)$, $s_2 = \sin(4\pi/5)$ and using $\sin(6\pi/5) = -s_2$, $\sin(8\pi/5) = -s_1$, one finds

$$\hat{a}(1) = 2i(s_1 + \xi s_2), \qquad \hat{a}(2) = 2i(s_2 - \xi s_1).$$

The eigenvector condition $\hat{a}(2)/\hat{a}(1) = a(2)/a(1) = \xi$ is

$$s_2 - \xi s_1 \;=\; \xi(s_1 + \xi s_2),$$

which rearranges to the stated quadratic. Solving, $\xi = (\,-s_1 + \sqrt{s_1^2 + s_2^2})/s_2$; substituting the exact values $s_1 = \tfrac{1}{4}\sqrt{10 + 2\sqrt{5}}$ and $s_2 = \tfrac{1}{4}\sqrt{10 - 2\sqrt{5}}$ and simplifying gives the closed form. The eigenvalue is then $\kappa = \hat{a}(1)/a(1) = 2i(s_1 + \xi s_2)$, which evaluates to $i\sqrt{5}$. The remaining coordinates $m = 3, 4$ are forced by oddness. □

Remark 2.8 (Numerical check). We record the check because the constant is frequently mistyped. With $\xi = 0.2840790438404124$, one computes $\hat{a}(m)/a(m) = 2.2360679774997896\,i$ for $m = 1, 2, 3, 4$, and $\sqrt{5} = 2.2360679774997896$. The agreement is to full double precision in all four coordinates.

Note that a is not multiplicative: $a(2)a(3) = -\xi^2 \approx -0.0807$ whereas $a(6) = a(1) = 1$. So f has no Euler product. It is not a Dirichlet L-function, and it is not in the Selberg class, since membership requires an Euler product.

Definition 2.9 (The Davenport–Heilbronn function)**.** With ξ as in Proposition 2.7 and $a = (1, \xi, -\xi, -1, 0) \bmod 5$, set

$$f(s) \;=\; \sum_{n\geq 1} \frac{a(n)}{n^s} \;=\; 5^{-s} \sum_{r=1}^{4} a(r)\, \zeta\big(s, \tfrac{r}{5}\big)\,.$$

Equivalently, in terms of the two Dirichlet L-functions modulo 5 attached to the two odd characters $\chi, \bar\chi$ of order 4,

$$f(s) \;=\; \tfrac{1}{2}(1 - i\,\tau) L(s,\chi) \;+\; \tfrac{1}{2}(1 + i\,\tau) L(s,\bar\chi)$$

for a real constant τ determined by ξ.

Remark 2.10 (The linear-combination viewpoint)*.* The second expression in Definition 2.9 is the structurally illuminating one and explains everything that follows. The two L-functions $L(s,\chi)$ and $L(s,\bar\chi)$ each satisfy the Riemann Hypothesis conjecturally and each satisfies a functional equation with the *same* gamma factor and conductor, because χ and $\bar\chi$ have the same modulus and the same parity. A generic linear combination of two functions satisfying the same functional equation again satisfies that functional equation. But a linear combination of two functions with all zeros on a line does not have all zeros on that line; zeros of a sum are not controlled by zeros of the summands. The Davenport–Heilbronn function is the simplest realisation of this elementary observation in the analytic-number-theoretic setting.

The parameter ξ is not free: for a general combination $\alpha L(s,\chi) + \beta L(s,\bar\chi)$ the functional equation acquires a matrix rather than a scalar, and the eigenvector condition of Proposition 2.7 is exactly the requirement that the matrix act by a scalar on (α, β). So ξ is the price of keeping (FE) in scalar form, and the construction is the statement that this price can be paid.

2.2.2 The functional equation

Theorem 2.11 (Davenport–Heilbronn functional equation)**.** *Let f be as in Definition 2.9. Then f is entire, and with*

$$\Lambda_f(s) \;=\; \Big(\frac{5}{\pi}\Big)^{(s+1)/2} \Gamma\Big(\frac{s+1}{2}\Big) f(s)$$

one has

$$\Lambda_f(s) \;=\; \Lambda_f(1-s).$$

Proof. Apply Proposition 2.6 with $q = 5$, $\delta = 1$ (the vector a is odd), and $\kappa = i\sqrt{5}$ from Proposition 2.7. The root number computes to $\omega = \kappa/(i\sqrt{q}) = 1$. The original derivation is [DH36a; DH36b]; the normalisation used here follows [BG11]. □

So f satisfies (FE) with critical line $\Re s = \frac{1}{2}$, root number $+1$, conductor 5, and the same gamma factor as an odd Dirichlet L-function of modulus 5. Every formal feature of the functional equation of $L(s,\chi)$ is present.

2.2.3 Zeros off the critical line

Theorem 2.12 (Davenport–Heilbronn 1936)**.** *The function f has infinitely many zeros in the half-plane $\Re s > 1$. It also has infinitely many zeros in the strip $0 < \Re s < 1$ with $\Re s \neq \frac{1}{2}$.*

Proof sketch. For the zeros in $\Re s > 1$: write f as in Remark 2.10 as a combination of two Dirichlet series, or directly expand

$$f(s) \;=\; 1 + \xi 2^{-s} - \xi 3^{-s} - 4^{-s} + 6^{-s} + \cdots$$

and apply a Kronecker/Bohr almost-periodicity argument. On the line $\Re s = \sigma$ for fixed $\sigma > 1$, the values $\{(2^{-it}, 3^{-it})\}$ are equidistributed in the torus $\mathbb{T}^2$ by Kronecker's theorem, since $\log 2$ and $\log 3$ are rationally independent. One checks that the truncated sum $1 + \xi 2^{-\sigma} e^{i\theta_2} - \xi 3^{-\sigma} e^{i\theta_3} - 4^{-\sigma} e^{2i\theta_2}$ takes the value 0 for suitable (θ_2, θ_3) when σ is close enough to 1, because the coefficients are large enough to reach the origin; a Rouché argument on the tail closes the proof. This is the argument of [DH36b]; the modern form is [BG11, §6].

For the off-line zeros in the strip: the classical route is a counting argument. Let $N_f(T)$ be the number of zeros of f in $0 < \Re s < 1$, $0 < \Im s < T$, and let $N_f^0(T)$ count those on $\Re s = \frac{1}{2}$. The functional equation and Jensen's formula give $N_f(T) = \frac{T}{2\pi} \log \frac{5T}{2\pi e} + O(\log T)$. A separate argument bounds $N_f^0(T)$ from above by a strictly smaller main term, or, in the sharper treatment of [BG11], shows that a positive proportion of the zeros lie off the line. □

Theorem 2.13 (Bombieri–Ghosh)**.** *A positive proportion of the zeros of f in the critical strip lie off the critical line. Quantitatively, the number of zeros with $\Re s \neq \frac{1}{2}$ and $0 < \Im s < T$ is $\gg T$.*

Proof. This is the main theorem of [BG11]. We do not reproduce the proof. □

For a firewall to be used operationally, it helps to have a zero one can name. The following was located numerically and is recorded here so that a reader can check it directly.

Proposition 2.14 (An exhibited off-line zero)**.** *The Davenport–Heilbronn function f has a zero at*

$$\rho_0 \;=\; 0.8085171825 \;+\; 85.6993484854\, i$$

to the displayed precision, and, by the functional equation together with $\overline{f(\bar{s})} = f(s)$, also at $\bar{\rho}_0$, $1 - \rho_0$ and $1 - \bar{\rho}_0$. In particular $\Re\rho_0 - \frac{1}{2} = 0.3085171825\ldots \neq 0$.

Verification. Evaluate f through the Hurwitz representation $f(s) = 5^{-s} \sum_{r=1}^{4} a(r)\zeta(s, r/5)$ with ζ to 25 significant figures. Newton iteration from the starting point $0.81 + 85.70\, i$ converges to ρ_0 with $|f(\rho_0)| < 3 \times 10^{-15}$ at 15-digit working precision, and the iteration is stable under perturbation of the starting point over a disc of radius 0.1. The zero is consistent with the tabulations of [BS07]. This is a numerical verification and carries status [E]. □

Remark 2.15 (On the status of Proposition 2.14). The proposition is labelled [E] and not [T]: a Newton iteration is not a proof that a zero exists. The proof that off-line zeros exist is Theorem 2.12, which is unconditional and does not exhibit a zero; the numerical value is supplied because a firewall test is easier to apply against a specific point than against an existence statement. A rigorous certification of ρ_0 would use a Rouché argument on a small contour with interval arithmetic; this is routine and we have not carried it out.

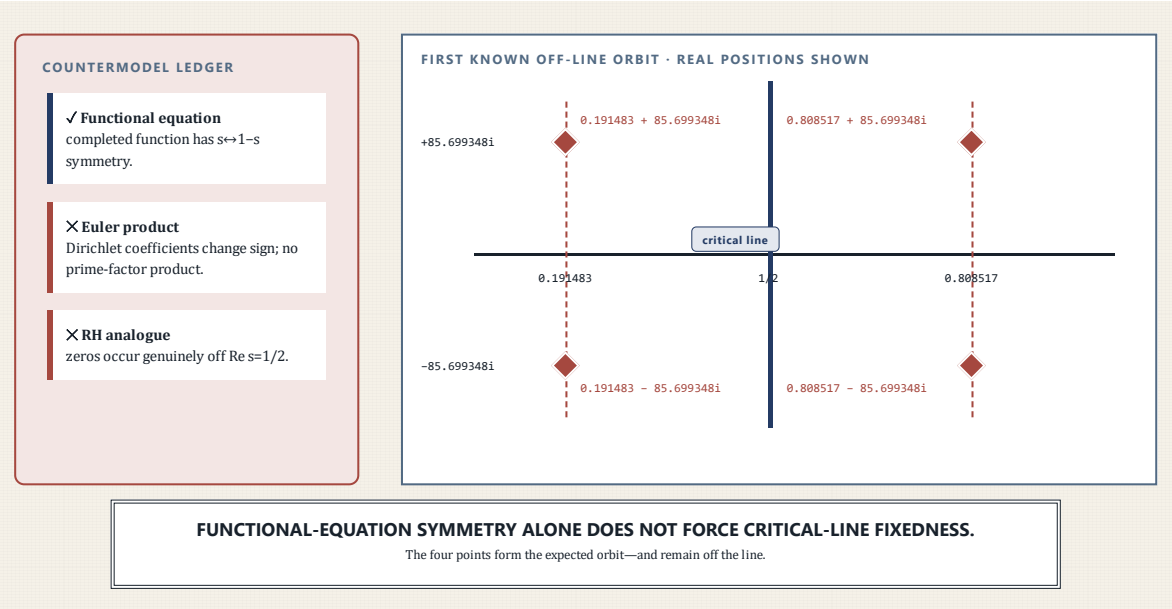


Figure 2.2. The Davenport–Heilbronn counter-model.

What it does not show: The figure exhibits one zero and the construction that produces the functional equation. It does not establish the density statements of Theorem 2.13, and the numerical zero location carries status [E].

2.2.4 What the Davenport–Heilbronn object rules out

OBSTRUCTION O-1: functional equation is not enough

No mechanism whose operative hypothesis is a functional equation of the shape $\Lambda(s) = \omega\Lambda(1-s)$ with real conductor, standard gamma factor, and entire Λ of order one, can prove RH; the Davenport–Heilbronn function satisfies all of these and has zeros off the line. [T]

The scope of this obstruction should be stated carefully, because it is easy to over-claim.

What it rules out. Any argument in which the only property of ζ used is a functional equation together with growth and order conditions of the classical type. In particular it rules out: arguments from the symmetry $s \mapsto 1-s$ alone; arguments that place zeros on the line by appealing to the fixed-point locus of the functional-equation involution; arguments from the reality of $\Xi(t)$ alone; and Hadamard-factorisation arguments that use only the order and the functional equation.

What it does not rule out. It does not rule out arguments that use the functional equation *together with* the Euler product, which is the position of essentially every serious approach. It does not rule out arguments that use the functional equation together with the non-negativity of the coefficients $\Lambda(n)$ of $-\zeta'/\zeta$. And it says nothing about the archimedean place: the Davenport–Heilbronn function has the same archimedean factor as $L(s,\chi)$, so an argument that used only archimedean data would be refuted, but no serious argument uses only archimedean data.

Remark 2.16 (The trap this closes). The specific error that Obstruction 2.2.4 closes is the following, which recurs: one observes that the functional equation makes the zero set invariant under $\rho \mapsto 1-\rho$ and under $\rho \mapsto \bar{\rho}$, hence invariant under the Klein four-group V of Proposition 1.4; one then observes that the fixed locus of V acting on the strip is exactly the critical line; and one concludes, or half-concludes, that the zeros should lie there. The step

from "the set is invariant" to "the set is contained in the fixed locus" is a non-sequitur — an invariant set need not consist of fixed points — and the Davenport–Heilbronn function is the counterexample that makes this concrete: its zero set is V-invariant and contains points of every one of the four orbit types. Corollary 1.5 states the abstract version; the present object is the realisation.

2.3 The Epstein zeta function of a quadratic form

2.3.1 Construction

Definition 2.17 (Epstein zeta function)**.** Let $Q(x,y) = ax^2 + bxy + cy^2$ be a positive-definite integral binary quadratic form of discriminant $d = b^2 - 4ac < 0$. The *Epstein zeta function* of Q is

$$\zeta_Q(s) \;=\; \sum_{(m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}} \frac{1}{Q(m,n)^s}, \qquad \Re s > 1.$$

The series converges absolutely for $\Re s > 1$ because $Q(m,n) \asymp m^2 + n^2$ with constants depending on Q. Writing $r_Q(n)$ for the number of representations of n by Q, we have the Dirichlet series expression

$$\zeta_Q(s) \;=\; \sum_{n\geq 1} \frac{r_Q(n)}{n^s},$$

so the coefficients of ζ_Q are non-negative integers. This is worth flagging immediately: the coefficient positivity that is often invoked as a structural feature of ζ (via $-\zeta'/\zeta$, or via the Euler product with non-negative logarithmic coefficients) is present here in a strong form, and is compatible with the failure of the Riemann Hypothesis analogue.

Theorem 2.18 (Epstein's functional equation)**.** *With ζ_Q as above and $|d|$ the absolute value of the discriminant,*

$$\Lambda_Q(s) \;=\; \left(\frac{2\pi}{\sqrt{|d|}}\right)^{-s} \Gamma(s)\,\zeta_Q(s)$$

continues meromorphically to $\mathbb{C}$ with simple poles at $s = 0$ and $s = 1$ only, and satisfies

$$\Lambda_Q(s) \;=\; \Lambda_Q(1-s).$$

Proof sketch. Apply the theta-series method. Let $\theta_Q(t) = \sum_{(m,n)} e^{-\pi t Q(m,n)}$. Poisson summation on the lattice $\mathbb{Z}^2$ with the quadratic form Q gives $\theta_Q(1/t) = t\,|d|^{-1/2}\cdot 2\,\theta_{Q^*}(t)$ where Q^* is the dual form; for the normalisation above Q^* is equivalent to Q, and the Mellin transform of the resulting identity produces the functional equation. The classical reference is [Eps03]; a modern treatment is [Ter85, §1.6]. [Sie80] gives the version with the constant normalised as here. □

So ζ_Q satisfies (FE) with the same critical line $\Re s = \tfrac{1}{2}$ and a gamma factor $\Gamma(s)$ — note, a *single* $\Gamma(s)$ rather than $\Gamma(s/2)$, corresponding to degree 2 in the Selberg-class normalisation, so ζ_Q has the analytic shape of a degree-two L-function, like the L-function of a modular form.

2.3.2 The class-number dichotomy

Let $K = \mathbb{Q}(\sqrt{d})$ be the imaginary quadratic field of discriminant d and $h = h(d)$ its class number. Classical theory relates ζ_Q for the forms in a class group to the Dedekind zeta function of K.

Proposition 2.19 (Decomposition). *Let $Q_1, \dots, Q_h$ be representatives of the $\mathrm{SL}_2(\mathbb{Z})$-classes of positive-definite forms of discriminant d, and let w be the number of units in the order of discriminant d (so $w = 2$ for $d < -4$). Then*

$$\sum_{j=1}^{h} \zeta_{Q_j}(s) \;=\; w\,\zeta_K(s) \;=\; w\,\zeta(s)L(s,\chi_d),$$

and for each ideal class character ψ of the class group,

$$\sum_{j=1}^{h} \overline{\psi(Q_j)}\,\zeta_{Q_j}(s) \;=\; w\,L(s,\psi),$$

where $L(s,\psi)$ is the Hecke L-function of ψ. Inverting by orthogonality of characters,

$$\zeta_{Q_j}(s) \;=\; \frac{w}{h}\sum_{\psi} \psi(Q_j)\,L(s,\psi).$$

Proof. Each ideal class of K corresponds to an $\mathrm{SL}_2(\mathbb{Z})$-class of forms, and the norms of the integral ideals in a class $\mathfrak{c}$ are the values represented by the corresponding form, counted with multiplicity w. The displayed identities are then Fourier inversion on the finite abelian group $\mathrm{Cl}(K)$. See [Dav00, §12] or [Ter85, Ch. 2]. □

This is the exact analogue, one level up, of the Davenport–Heilbronn mechanism described in Remark 2.10, and it explains the class-number dichotomy at once.

Corollary 2.20 (The dichotomy). *If $h = 1$ then $\zeta_Q(s) = w\,\zeta(s)L(s,\chi_d)$ has an Euler product, and the Riemann Hypothesis for ζ_Q is equivalent to* RH *together with* RH *for $L(s,\chi_d)$. If $h > 1$ then ζ_{Q_j} is a non-trivial linear combination of h distinct Hecke L-functions with positive coefficients $w\psi(Q_j)/h$ that are not all equal in phase, and ζ_{Q_j} has no Euler product.*

Proof. The first statement is immediate from Proposition 2.19 with $h = 1$. For the second: a Dirichlet series with an Euler product and coefficients bounded polynomially has multiplicative coefficients, and r_{Q_j} is not multiplicative when $h > 1$. Explicitly, take $d = -23$, $h = 3$, and $Q = x^2 + xy + 6y^2$: then $r_Q(2) = 0$, $r_Q(3) = 0$, but $r_Q(6) = 2$, so $r_Q(6) \neq r_Q(2)r_Q(3)$. □

2.3.3 Zeros in $\Re s > 1$

Theorem 2.21 (Zeros to the right of the line of absolute convergence). *Let Q be a positive-definite integral binary quadratic form of discriminant $d < 0$ with class number $h(d) > 1$. Then ζ_Q has infinitely many zeros in the half-plane $\Re s > 1$.*

Proof sketch. The argument is the same almost-periodicity mechanism as for the Davenport–Heilbronn function, now applied to the linear combination of Proposition 2.19. Fix $\sigma_0 > 1$ close to 1. On the vertical line $\Re s = \sigma_0$ each Hecke L-function $L(s,\psi)$ is given by an absolutely convergent Euler product, and by Kronecker's theorem the vector of arguments $(p^{-it})_{p \le P}$ equidistributes in $\mathbb{T}^{\pi(P)}$. One shows that the function

$$\frac{h}{w}\,\zeta_Q(s) \;=\; \sum_{\psi} \psi(Q) L(s,\psi)$$

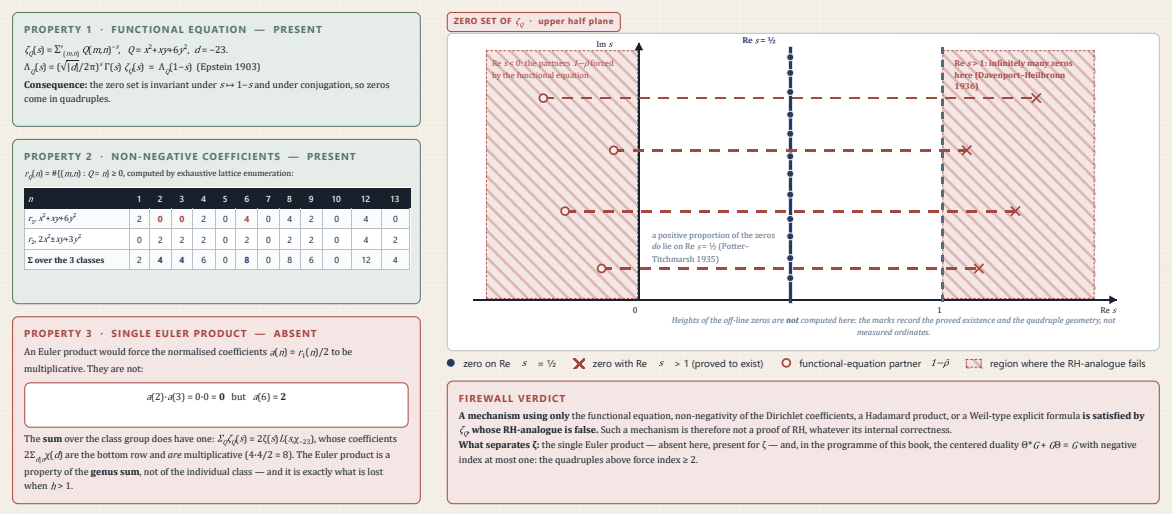
has an image on the line $\Re s = \sigma_0$ whose closure contains 0, because the h summands can be brought into a configuration in which their sum vanishes: the leading terms are h complex

numbers of modulus close to 1 and independent phases, and $h \geq 2$ with the phases $\psi(Q)$ not all equal permits cancellation. A Rouché argument transfers this to actual zeros, and almost-periodicity in t makes them infinite in number.

This is Davenport and Heilbronn's original observation [DH36a]. Sharper and more general results are in [PT35] and, with the density of such zeros made explicit, in [Vor76]. That $h > 1$ is necessary as well as sufficient follows from Corollary 2.20: when $h = 1$ the function is a product of Euler products, which is zero-free in $\Re s > 1$. □

Remark 2.22 (Where the zeros are). Theorem 2.21 places zeros in $\Re s > 1$. This is a stronger and more startling failure than the Davenport–Heilbronn one: ζ_Q has non-negative coefficients, so on the real axis in $\sigma > 1$ it is positive and bounded below, and yet arbitrarily close to that axis — in the sense of the abscissa, not of t — there are zeros. The zeros necessarily have large imaginary part when σ is bounded away from 1; quantitatively the number of zeros of ζ_Q in $\sigma > 1, 0 < t < T$ is $\gg_Q T$ for $h > 1$ [Vor76].

There are also zeros off the critical line inside the strip, and by a theorem of Stark [Sta67] there are explicit small cases: for $d = -23$ the function ζ_Q has zeros in the strip that are not on $\Re s = \frac{1}{2}$, exhibited numerically.



n	1	2	3	4	5	6	7	8	9	10	12	13
$r_{x^2+xy+6y^2}$	2	0	0	2	0	4	0	4	2	0	4	0
$r_{2x^2\pm xy+3y^2}$	0	2	2	2	0	2	0	2	2	0	4	2
Σ over the 3 classes	2	4	4	6	0	8	0	8	6	0	12	4

Figure 2.3. Epstein zeta for class number $h > 1$.

What it does not show: The figure records the structural data and the region in which zeros occur. It does not exhibit a specific zero, and the density statement of Remark 2.22 is cited rather than proved here.

2.3.4 What the Epstein object rules out

OBSTRUCTION O-2: positivity of coefficients is not enough

No mechanism whose operative hypotheses are a functional equation together with non-negativity of the Dirichlet coefficients can prove RH; the Epstein zeta function of a form of class number $h > 1$ satisfies both and has zeros in $\Re s > 1$. [T]

What it rules out. Arguments that derive zero-location from a positivity property of the coefficients $a(n)$ of the Dirichlet series itself, combined with the functional equation. In particular it rules out the following tempting shape of argument: "$\zeta_Q(\sigma) > 0$ for real $\sigma > 1$ and the coefficients are non-negative, so by a Landau-type theorem the abscissa of convergence is a singularity, hence there are no zeros in $\Re s > 1$" — the Landau theorem gives a singularity

at the abscissa of convergence, which is a statement about the *pole* at $s = 1$, and says nothing about zeros.

It also rules out the naive form of the argument from the Hadamard–de la Vallée Poussin non-vanishing technique. That technique uses $3 + 4\cos\theta + \cos 2\theta \geq 0$ applied to $\log|\zeta|$, and it requires an Euler product to convert the trigonometric inequality into a statement about ζ. The Epstein function has non-negative coefficients but no Euler product, and non-vanishing on $\Re s = 1$ fails — indeed non-vanishing fails in the whole half-plane $\Re s > 1$.

What it does not rule out. It does not rule out arguments using positivity of the coefficients of $\log F$ or of $-F'/F$, which is a different and strictly stronger hypothesis: for ζ these are $\Lambda(n) \geq 0$, whereas for ζ_Q with $h > 1$ the logarithmic coefficients are not non-negative. This distinction is the whole content of the Selberg-class axiom (S4) on the coefficients of $\log F$, and it is exactly what separates row 2 of Table 2.1 from row 4. It also does not rule out arguments using the Ramanujan bound on the coefficients, which $r_Q(n)$ violates in the direction of largeness only by a divisor-function factor, so this is a weak separation.

Remark 2.23 (The two firewall objects so far, compared). The Davenport–Heilbronn function and the Epstein function are both linear combinations of L-functions with a common functional equation, and both fail RH for that reason. They are not redundant. The Davenport–Heilbronn function is a degree-one object with the archimedean factor of a Dirichlet L-function and coefficients that change sign; its off-line zeros are inside the critical strip, so it is the right test for mechanisms that assert zeros lie on the line. The Epstein function is a degree-two object with non-negative coefficients; its failure is in $\Re s > 1$, so it is the right test for mechanisms that assert zero-freeness of a half-plane, including mechanisms built on the Euler product's zero-free region. Keeping both in the firewall is what makes the distinction between "positivity of $a(n)$" and "positivity of the coefficients of $\log F$" testable.

2.4 The Diamond–Montgomery–Vorhauer Beurling system

2.4.1 Beurling generalised numbers

Definition 2.24 (Beurling system)**.** A *Beurling generalised prime system* is a sequence

$$1 < p_1 \leq p_2 \leq p_3 \leq \cdots, \qquad p_j \to \infty,$$

of real numbers, called *generalised primes*. The associated *generalised integers* are the multiplicative semigroup they generate: all finite products $p_{j_1}^{e_1} \cdots p_{j_k}^{e_k}$, listed with multiplicity. Write

$$N_{\mathcal{B}}(x) = \#\{\text{generalised integers} \leq x\}, \qquad \pi_{\mathcal{B}}(x) = \#\{j : p_j \leq x\},$$

and

$$\zeta_{\mathcal{B}}(s) \;=\; \prod_{j \geq 1} \left(1 - p_j^{-s}\right)^{-1} \;=\; \sum_{n} n^{-s},$$

the last sum over generalised integers with multiplicity, convergent in the half-plane where $N_{\mathcal{B}}$ permits.

The point of Beurling's framework [Beu37] is that it isolates exactly the multiplicative structure of the integers, with the additive structure discarded. A theorem about ζ that uses only the Euler product, the growth of $N(x)$, and complex analysis is a theorem about all Beurling systems with the same growth. Conversely, a Beurling system with the required growth but with false conclusions shows that the theorem is not provable from those hypotheses.

Beurling's own theorem is the model:

Theorem 2.25 (Beurling 1937). *If $N_{\mathcal{B}}(x) = \kappa x + O(x(\log x)^{-\gamma})$ for some $\kappa > 0$ and some $\gamma > 3/2$, then $\pi_{\mathcal{B}}(x) \sim x/\log x$.*

The exponent $3/2$ is sharp: Diamond [Dia70] constructed systems with $\gamma = 3/2$ for which the conclusion fails. So the framework is calibrated, and it is calibrated at exactly the place where the classical proof of the Prime Number Theorem lives.

2.4.2 The Diamond–Montgomery–Vorhauer theorem

The theorem in this subsection is frequently cited as *Invent. Math.* **165** (2006). That attribution is incorrect. The correct reference is

> H. G. Diamond, H. L. Montgomery and U. M. A. Vorhauer, *Beurling primes with large oscillation*, Mathematische Annalen **334** (2006), no. 1, 1–36.

We give the correct citation here because the result is load-bearing in this book's firewall and a reader following the wrong reference will not find it. The bibliography entry is [DMV06].

Theorem 2.26 (Diamond–Montgomery–Vorhauer). *Let $\frac{1}{2} < \theta < 1$. There exists a Beurling generalised prime system $\mathcal{B}$ and constants $\kappa > 0$, $c > 0$ such that:*

(i) $N_{\mathcal{B}}(x) = \kappa x + O\big(x^{\theta}\big)$;

(ii) *the Prime Number Theorem holds in the form*

$$\pi_{\mathcal{B}}(x) \;=\; \mathrm{li}(x) \;+\; \Omega\Big(x \exp\Big(-c\sqrt{\log x}\Big)\Big),$$

so the error term is not $O\big(x\exp(-c'\sqrt{\log x})\big)$ for any larger c', and in particular the classical de la Vallée Poussin error term is attained and not improved;

(iii) *for every $a > \frac{4}{e}(1-\theta)$, the function $\zeta_{\mathcal{B}}$ has infinitely many zeros on the curve*

$$\sigma \;=\; 1 - \frac{a}{\log t}, \qquad t \geq 2,$$

and no zeros to the right of that curve.

Proof. This is the main theorem of [DMV06]. We do not reproduce the proof, which constructs the system by a probabilistic method: the generalised primes are placed by a random process on a sequence of dyadic ranges, chosen so that the resulting $\zeta_{\mathcal{B}}$ has prescribed oscillation. [T] □

Corollary 2.27 (The Riemann Hypothesis analogue fails). *For the system $\mathcal{B}$ of Theorem 2.26, the zeros of $\zeta_{\mathcal{B}}$ are not confined to $\Re s \leq \frac{1}{2} + \epsilon$ for any $\epsilon < \frac{1}{2}$. Indeed by (iii) there are zeros with $\Re s \to 1^{-}$.*

Proof. By (iii) the zeros on the curve $\sigma = 1 - a/\log t$ have real part tending to 1 as $t \to \infty$, and there are infinitely many of them. □

2.4.3 The precise structure of the counter-model

It is worth being explicit about which of the four structural hypotheses this system satisfies, because the answer is not the one a first reading suggests.

The system has (EP): $\zeta_{\mathcal{B}}$ is by construction an Euler product over the generalised primes, with local factors of exactly the classical shape $(1-p^{-s})^{-1}$.

The system has (PT): the Prime Number Theorem holds, with the classical error term.

The system has a strong regularity of the integer-counting function: $N_{\mathcal{B}}(x) = \kappa x + O(x^{\theta})$ with $\theta < 1$ is a hypothesis considerably stronger than anything needed for Beurling's theorem, and stronger than the corresponding statement for the rational integers if one insists on the analogy in the naive way.

The system does *not* have (FE). This is the essential point and the reason the Beurling object sits in a different row of Table 2.1 from the first two. There is no involution $s \mapsto 1-s$ relating $\zeta_{\mathcal{B}}$ to itself; the generalised primes are real numbers with no arithmetic providing a dual.

OBSTRUCTION O-3: Euler product and PNT are not enough

No mechanism whose operative hypotheses are an Euler product with classical local factors, together with a prime-counting asymptotic of classical strength and a regular integer-counting function $N(x) = \kappa x + O(x^{\theta})$, $\theta < 1$, can prove RH; the Diamond–Montgomery–Vorhauer system satisfies all of these and has zeros with $\Re s \to 1$. [T]

What it rules out. Arguments that reach the critical line from multiplicative structure and elementary density information alone. Concretely: any strengthening of the Hadamard–de la Vallée Poussin zero-free region argument that uses only (a) the Euler product, (b) the positivity $3 + 4\cos\theta + \cos 2\theta \geq 0$ or any of its higher analogues, and (c) growth bounds on $\zeta_{\mathcal{B}}$ derived from $N_{\mathcal{B}}$, cannot reach beyond a zero-free region of the shape $\sigma > 1 - a/\log t$. The DMV system shows this is not a limitation of technique but of hypothesis: there is an object satisfying (a), (b), (c) with zeros exactly on such a curve.

It also rules out "sieve-theoretic" or "elementary" routes in the following precise sense: an elementary proof, in the sense of using only the multiplicative structure and real-variable estimates on counting functions, is a proof about all Beurling systems with the given counting behaviour, and there is such a system with the conclusion false.

What it does not rule out. It says nothing about mechanisms that use the functional equation, and this is a serious limitation of the object's scope. Nor does it rule out mechanisms that use finer arithmetic input than $N_{\mathcal{B}}(x) = \kappa x + O(x^{\theta})$ — for the rational integers one has the far stronger fact that $N(x) = \lfloor x \rfloor$, i.e. θ can be taken 0, and the DMV construction genuinely requires $\theta > \frac{1}{2}$. Whether a Beurling system with $N(x) = x + O(x^{\theta})$ for $\theta \leq \frac{1}{2}$ and a false RH-analogue exists is, to our knowledge, open; we record this as an open problem rather than assert either answer.

OPEN PROBLEM Small-θ Beurling counter-models

Does there exist a Beurling generalised prime system with $N_{\mathcal{B}}(x) = \kappa x + O(x^{\theta})$ for some $\theta \leq \frac{1}{2}$ whose zeta function has zeros with $\Re s > \frac{1}{2} + \delta$ for some fixed $\delta > 0$? The DMV construction requires $\theta > \frac{1}{2}$; whether this is an artefact of the method or a genuine barrier is not settled by the results cited here. [O]

Remark 2.28 (Why this object is often mis-weighted). Two opposite errors are common in citing the DMV theorem. The first is to treat it as showing that "RH cannot be proved by analytic methods", which is far too strong: the system lacks a functional equation, and the functional

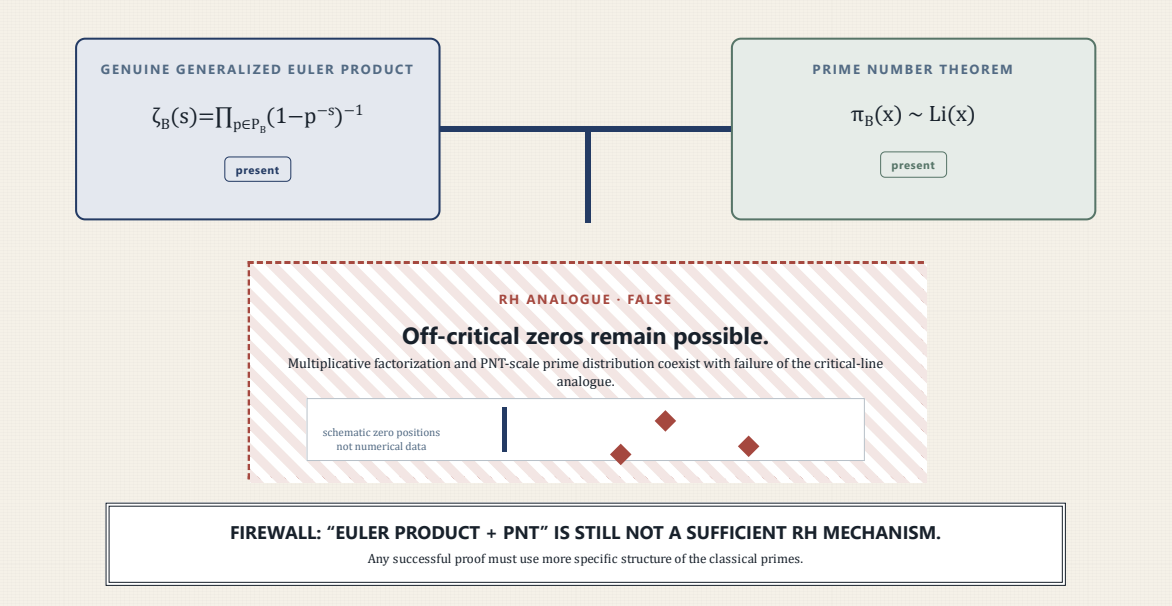


Figure 2.4. The Diamond–Montgomery–Vorhauer Beurling system.

What it does not show: The figure records the location of the zeros of the constructed system. It does not establish that the construction is essentially unique, and Open Problem 2.4.3 remains open.

equation is analytic. The second is to dismiss it because the system is "artificial", which mistakes the logic — the artificiality of a counterexample is irrelevant to its refutational force, and the DMV system is in any case less artificial than it looks, being a random construction with prescribed statistics rather than a hand-tuned one.

The correct weight is the one recorded in Obstruction 2.4.3: the object refutes a specific and popular class of hypotheses, namely those in which the functional equation plays no role. Since almost every heuristic argument for RH from "the primes are random" is of this type, that class is large.

2.5 Selberg zeta functions and exceptional eigenvalues

2.5.1 The construction

Let $\Gamma \subset \mathrm{PSL}_2(\mathbb{R})$ be a discrete subgroup of finite covolume acting on the upper half-plane $\mathbb{H}$, and let $X = \Gamma\backslash\mathbb{H}$. Assume for the moment that X is compact, so that the spectrum of the hyperbolic Laplacian $\Delta = -y^2(\partial_x^2 + \partial_y^2)$ on $L^2(X)$ is discrete:

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty.$$

Write $\lambda_n = s_n(1 - s_n)$, equivalently $\lambda_n = \frac{1}{4} + r_n^2$ with $s_n = \frac{1}{2} + ir_n$.

Definition 2.29 (Selberg zeta function)**.** For $\Re s > 1$,

$$Z_\Gamma(s) \;=\; \prod_{\{\gamma_0\}} \prod_{k\geq 0} \left(1 - e^{-(s+k)\,\ell(\gamma_0)}\right),$$

the outer product over Γ-conjugacy classes of primitive hyperbolic elements, $\ell(\gamma_0)$ denoting the length of the corresponding closed geodesic.

The parallel with ζ is close and is not merely formal. The outer product is over primitive closed geodesics, which play the role of primes; the quantity $e^{\ell(\gamma_0)}$ is the "norm" of

the geodesic; and the logarithmic derivative Z_Γ'/Z_Γ has a Dirichlet-series expansion whose coefficients are supported on the norms of geodesics, with non-negative weights, exactly as $-\zeta'/\zeta$ is supported on prime powers with weights $\Lambda(n) \geq 0$. The Selberg trace formula is the exact analogue of the explicit formula.

Theorem 2.30 (Selberg). *Z_Γ extends to an entire function of order two and satisfies the functional equation*

$$Z_\Gamma(1-s) \;=\; Z_\Gamma(s)\,\exp\!\left(-4\pi\,|X|\int_0^{s-1/2} v\tan(\pi v)\,dv\right),$$

where $|X|$ is the hyperbolic area of X. Its non-trivial zeros are exactly the points $s_n = \frac{1}{2} \pm i r_n$, with $\lambda_n = \frac{1}{4} + r_n^2$ running over the spectrum of Δ on $L^2(X)$, with multiplicity. There are in addition "trivial" zeros at $s = -k$, $k \geq 0$, of multiplicity $(2k+1)|X|/2\pi$, and a zero at $s = 1$ from $\lambda_0 = 0$.

Proof. This is Selberg's theorem; see [Sel56] and, for a complete treatment, [Hej76a, Ch. 2] or [Bus92, Ch. 9]. [T] □

Corollary 2.31 (The Riemann Hypothesis analogue for Z_Γ). *All non-trivial zeros of Z_Γ lie on the line $\Re s = \frac{1}{2}$ if and only if*

$$\operatorname{spec}\Delta \;\subseteq\; \{0\} \cup \left[\tfrac{1}{4}, \infty\right).$$

Proof. $\lambda = s(1-s)$ maps the line $\Re s = \frac{1}{2}$ onto $[\frac{1}{4}, \infty)$ and maps the real segment $s \in (\frac{1}{2}, 1]$ onto $[0, \frac{1}{4})$. So a zero s_n is off the critical line, and in the strip, precisely when $\lambda_n \in (0, \frac{1}{4})$. □

An eigenvalue $\lambda_n \in (0, \frac{1}{4})$ is called *exceptional*. Corollary 2.31 says: exceptional eigenvalues are exactly off-critical-line zeros of Z_Γ.

2.5.2 Exceptional eigenvalues exist

Theorem 2.32 (Randol). *For every $\epsilon > 0$ and every M, there is a compact hyperbolic surface $X = \Gamma\backslash\mathbb{H}$ with at least M eigenvalues of Δ in the interval $(0, \epsilon)$.*

Proof. [Ran74]. The construction takes a surface with a short closed geodesic; as the length of the geodesic tends to 0 the surface degenerates and small eigenvalues accumulate at 0. Explicit surfaces with $\lambda_1 < 0.05$ are constructed there. [T] □

Corollary 2.33 (The firewall object). *There exist compact hyperbolic surfaces X for which Z_Γ is entire, has an Euler product over closed geodesics with non-negative logarithmic coefficients, satisfies a functional equation with critical line $\Re s = \frac{1}{2}$, satisfies the geodesic-counting analogue of the Prime Number Theorem, has its non-trivial zeros equal to the spectrum of a self-adjoint operator canonically constructed from the geometry — and has zeros off the critical line.*

Proof. Take X as in Theorem 2.32 with $\lambda_1 < \frac{1}{4}$. Each listed structural property is Theorem 2.30 or the prime geodesic theorem [Bus92, Ch. 9]. Self-adjointness of Δ on $L^2(X)$ for compact X is standard elliptic theory. The off-line zero is $s_1 = \frac{1}{2} + \sqrt{\frac{1}{4} - \lambda_1} \in (\frac{1}{2}, 1)$, which is real and not on the critical line. □

This is the object of row 4 of Table 2.1, and it is the reason the firewall has four members rather than three.

2.5.3 The Hilbert–Pólya trap made concrete

The Hilbert–Pólya idea is: find a self-adjoint operator H whose eigenvalues are the ordinates γ of the zeros $\frac{1}{2} + i\gamma$; then $\gamma \in \mathbb{R}$ automatically, and RH follows. Chapter 3 shows that this formulation, as usually stated, is vacuous. The Selberg object shows something different and complementary: that even a non-vacuous, genuinely geometric realisation of exactly this kind does not give the conclusion.

Here is the point in detail. For a compact hyperbolic surface, the operator is Δ, which is self-adjoint, elliptic, canonically attached to the geometry, with a trace formula linking its spectrum to the closed geodesics. Every structural feature that the Hilbert–Pólya programme asks for is present, and it is present as a theorem, not as a hope. And the conclusion is false, because self-adjointness of Δ gives $\lambda_n \in \mathbb{R}$, which via $\lambda = s(1-s)$ gives $s_n \in \{\Re s = \frac{1}{2}\} \cup \mathbb{R}$ — a union of two sets, not one.

OBSTRUCTION O-4: self-adjointness gives a real parameter, not a line

Let H be a self-adjoint operator and suppose the zeros of a zeta function are the image of $\operatorname{spec} H$ under a map $\phi : \mathbb{R} \to \mathbb{C}$. Then the zeros lie in $\phi(\mathbb{R})$. This confines the zeros to the critical line only if $\phi(\mathbb{R}) \subseteq \{\Re s = \frac{1}{2}\}$. For the Selberg parametrisation $\phi(\lambda) = \frac{1}{2} + \sqrt{\frac{1}{4} - \lambda}$ the image is $\{\Re s = \frac{1}{2}\} \cup (\frac{1}{2}, 1]$, and the second component is non-empty exactly when exceptional eigenvalues occur. [T]

What it rules out. Any argument of the shape "H is self-adjoint, therefore the zeros are on the line" in which the parametrisation from $\operatorname{spec} H$ to the zeros is not itself shown to have image in the line. In the Selberg case the required additional input is a *spectral gap* $\lambda_1 \geq \frac{1}{4}$, which is a genuine theorem-level statement about the specific Γ and is false for general Γ. A mechanism must therefore supply an analogue of the spectral gap, and the spectral gap is not a consequence of self-adjointness.

What it does not rule out. It does not rule out spectral formulations in which the parametrisation is linear, $\gamma \mapsto \frac{1}{2} + i\gamma$, so that $\phi(\mathbb{R})$ is exactly the critical line. Such formulations exist — they are the ones Chapter 3 shows to be vacuous unless supplemented — and the point of the present obstruction is that when the parametrisation is quadratic, as it is in every geometric example where a trace formula is actually available, the self-adjointness argument is insufficient by a definite amount.

Remark 2.34 (The two failure modes are complementary). There is a real tension here, and naming it is one of the purposes of Part I. In the linear parametrisation $\gamma \mapsto \frac{1}{2} + i\gamma$, self-adjointness would suffice, but no independently constructed self-adjoint operator with that spectrum is known and the demand is vacuous without a construction principle. In the quadratic parametrisation $\lambda \mapsto \frac{1}{2} + \sqrt{\frac{1}{4} - \lambda}$, an independently constructed self-adjoint operator exists in the geometric analogue, but self-adjointness no longer suffices and one needs a gap.

Every spectral approach known to us sits at one end of this tension or the other. The programme of this book sits at the quadratic end, and takes the consequence seriously: what is needed is not self-adjointness but an indefinite (Krein) structure together with an index or signature statement, which is what Chapter 4 formulates and Part VIII develops.

2.5.4 The arithmetic case and the Selberg eigenvalue conjecture

Randol's surfaces are not congruence quotients. For congruence subgroups the situation is different, and the difference is arithmetic rather than geometric.

Theorem 2.35 (Selberg 1965). *For $\Gamma = \Gamma_0(N) \subset \mathrm{PSL}_2(\mathbb{Z})$, every eigenvalue of Δ on $L^2(\Gamma\backslash\mathbb{H})$ other than 0 satisfies $\lambda \geq \frac{3}{16}$.*

Proof. [Sel65]. [T] □

Conjecture 2.36 (Selberg eigenvalue conjecture). For $\Gamma = \Gamma_0(N)$, every non-zero eigenvalue satisfies $\lambda \geq \frac{1}{4}$. Equivalently, $Z_{\Gamma_0(N)}$ satisfies its Riemann Hypothesis analogue. [O]

The best unconditional bound is $\lambda_1 \geq 975/4096 = 0.238\ldots$, from the Kim–Sarnak bound $\theta \leq 7/64$ towards Ramanujan [KS03]; the hierarchy is set out in Chapter 3. The relevant point here is:

> The Selberg eigenvalue conjecture is open. Consequently the Selberg firewall object comes in two flavours, and they must not be confused.
>
> - *General* Γ: exceptional eigenvalues provably exist (Theorem 2.32), so the RH-analogue for Z_Γ is provably false. This is the firewall object, and it is unconditional. [T]
> - *Congruence* Γ: exceptional eigenvalues are conjectured not to exist, and this is open. Nothing here is a counter-model. [O]
>
> Using the congruence case as a firewall object would be an error, and using the general case to cast doubt on the congruence case would be a different error.

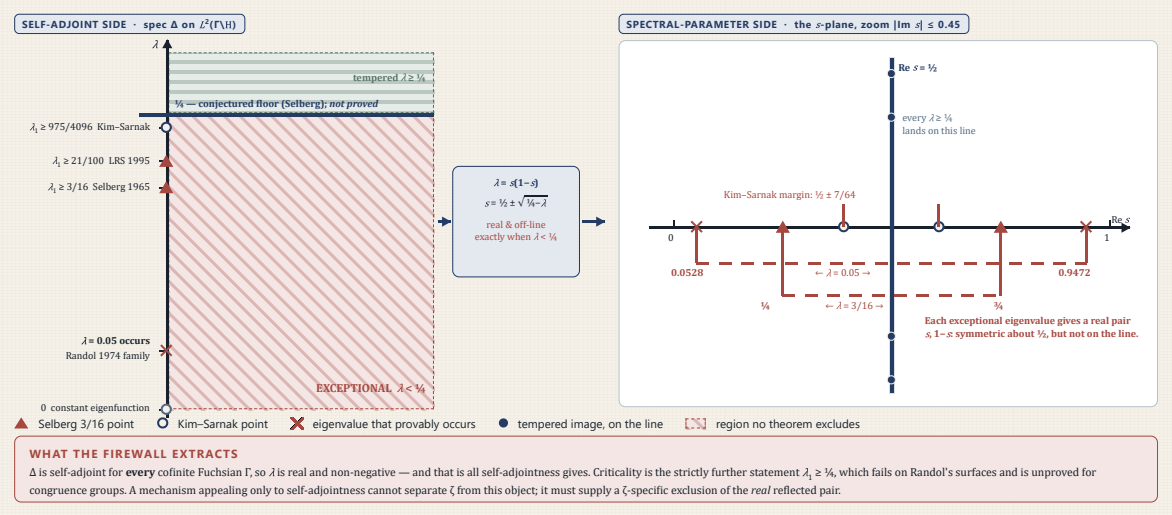


Figure 2.5. Exceptional eigenvalues and off-line zeros.

What it does not show: The figure concerns general cofinite Γ. It does not assert that exceptional eigenvalues occur for congruence subgroups, which is the content of the open Selberg eigenvalue conjecture.

2.6 The firewall rule and its test protocol

2.6.1 Statement of the rule

FIREWALL TEST The firewall rule

Let $(\mathcal{P}, \Pi)$ be a proposed mechanism for RH in the sense of Definition 2.1. Before Π is examined for correctness, $\mathcal{P}$ must be shown to fail for each of:

1. the Davenport–Heilbronn function f of Definition 2.9;
2. the Epstein zeta function ζ_Q of a form with $h(d) > 1$;
3. the Diamond–Montgomery–Vorhauer Beurling system of Theorem 2.26;
4. the Selberg zeta function Z_Γ of a compact hyperbolic surface with $\lambda_1 < \frac{1}{4}$.

A mechanism for which these four tests cannot be carried out — because $\mathcal{P}$ is not stated to a standard at which the question "does this object satisfy $\mathcal{P}$?" has an answer — is not yet a mechanism.

The second sentence is the operative one in practice. It is not a claim that such proposals are wrong. It is a statement about what has to be done before the question of rightness can be raised.

2.6.2 The protocol, operationally

The following is written so that a referee can execute it. Each step produces a written artefact; the protocol is complete when all four rows of the resulting table are filled with a clause reference and a proof.

P1. *Extract the hypothesis.* Read Π and list every property of ζ that is used. Include properties used implicitly: the location of the pole, the Ramanujan bound on coefficients, the reality of the coefficients, the degree of the gamma factor, the existence of an Euler product, positivity of $\Lambda(n)$, the value of the root number. The result is a finite list $\mathcal{P} = \mathcal{P}_1 \wedge \cdots \wedge \mathcal{P}_k$.

P2. *Check that the list is complete.* Verify that Π goes through using only $\mathcal{P}_1, \dots, \mathcal{P}_k$. If some step of Π uses a property not on the list, add it. This step is where most of the work is, and it is where hidden circularity is usually found: a step that uses "ζ has no zeros in $\Re s > \frac{1}{2}$" is not a step, it is the conclusion.

P3. *Test object 1 (Davenport–Heilbronn).* For each j, determine whether f satisfies $\mathcal{P}_j$. Record the first j for which the answer is no, with a proof. If no such j exists, the mechanism is refuted by Lemma 2.2 together with Theorem 2.12.

P4. *Test object 2 (Epstein, $h > 1$).* As P3, with Theorem 2.21.

P5. *Test object 3 (DMV Beurling).* As P3, with Corollary 2.27. Note that this object has no functional equation, so if $\mathcal{P}$ contains a functional-equation clause the test is passed immediately — but then record that fact explicitly, because it tells the reader that the mechanism's content is concentrated in the interaction between the functional equation and the Euler product, and that is where the remaining scrutiny should go.

P6. *Test object 4 (Selberg, exceptional).* As P3, with Corollary 2.33. This test is the most informative for spectral mechanisms. If the mechanism's separating clause is "the parametrisation from the operator's spectrum to the zeros is $\gamma \mapsto \frac{1}{2} + i\gamma$", record that, and then require a construction of the operator, since the clause is otherwise vacuous by Theorem 3.1.

P7. *Audit the separating clauses.* Collect the clauses $\mathcal{P}_{j_1}, \dots, \mathcal{P}_{j_4}$ identified in P3–P6. These carry the whole weight of the mechanism. Verify, for each, that it is a property ζ is *known* to have, and not a property ζ is conjectured to have. A separating clause that is itself open reduces the mechanism to a conditional result, which should be stated as such.

P8. *Record the outcome.* Fill in Table 2.2. A mechanism that passes has a completed table; a mechanism that fails has an empty cell with a proof of satisfaction rather than of failure; a proposal that is not yet a mechanism has an empty cell with neither.

Table 2.2. The protocol output template. One row per firewall object; the middle column names the clause of $\mathcal{P}$ that the object violates, and the right column cites the proof that it does.

Object	Separating clause	Proof that the object fails it
Davenport–Heilbronn f		
Epstein ζ_Q, $h > 1$		
DMV Beurling $\zeta_{\mathcal{B}}$		
Selberg Z_Γ, $\lambda_1 < \frac{1}{4}$		

2.6.3 A worked application

To show that the protocol is executable and not merely stipulated, we apply it to a mechanism that is not this book's.

Proposed mechanism. $\mathcal{P}$: "F is entire of order one, satisfies $\Lambda_F(s) = \Lambda_F(1-s)$ with $\Lambda_F(s) = Q^s\Gamma(\frac{s+\delta}{2})F(s)$, and Λ_F is real and positive on the real axis." Π: "a real entire function of order one with a functional equation about $\frac{1}{2}$ and no real zeros in the strip has all its zeros on the line."

P1–P2. The clauses are: (i) entire of order one; (ii) the stated functional equation; (iii) Λ_F real on the real axis; (iv) Λ_F positive on the real axis.

P3. The Davenport–Heilbronn f satisfies (i) by Theorem 2.11 and the standard order computation, satisfies (ii) with $\delta = 1$, $Q = (5/\pi)^{1/2}$, and satisfies (iii) because the coefficients $a(n)$ are real. Does it satisfy (iv)? Here $\Lambda_f(\sigma) \to 1$ as $\sigma \to +\infty$ and Λ_f is real; but f has a zero on the real axis — indeed $f(\sigma) = 0$ for some σ, since $f(0) = \sum_r a(r)(\frac{1}{2} - r/5)$ can be computed and compared with $f(\sigma) \to 1$. So the test turns on (iv), and the mechanism's whole content is the positivity clause.

P7. Is (iv) known for ζ? $\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)s(s-1)/2$ is indeed real and positive on the real axis: this is classical and follows from $\zeta(\sigma) < 0$ for $0 < \sigma < 1$ together with the sign of $s(s-1)$. So (iv) is known.

P4–P6. We leave these to the reader; the Epstein object fails (iii) or (iv) depending on normalisation, the Beurling object fails (ii), and the Selberg object satisfies (i)–(iv) after the substitution $\lambda = s(1-s)$ — which is where this particular mechanism dies, since Z_Γ for a Randol surface is entire, satisfies its functional equation, and is real and positive on the real axis to the right of 1, and yet has real zeros in $(\frac{1}{2}, 1)$. So Π is refuted at P6: a real entire function of order one with a functional equation about $\frac{1}{2}$ can have real zeros in the strip, and the proposal's step "no real zeros in the strip" was an unproved hypothesis, not a consequence.

The application takes about a page. That is the intended cost.

2.6.4 Recording firewall status

Throughout this book, when a mechanism or a component of the positive programme is asserted to pass the firewall, the assertion is accompanied by a reference to the four separating clauses. Chapter 6 records the firewall status of every no-go theorem in the archive; Part VIII records it for the main hypothesis. Where a test has not been carried out, this is stated.

Firewall status is never recorded as [T] for a mechanism, only for individual test outcomes. Passing all four tests is recorded as [C] — *compatible* — and the wording used is "the mechanism is firewall-compatible", never "the mechanism survives" or any phrasing that suggests accumulating evidence. The four tests are four specific ways of being wrong, and eliminating them is not a positive result.

2.6.5 Objects deliberately excluded from the firewall

Three families of objects are natural candidates for the firewall and are excluded. The reasons are worth recording, since a reader may otherwise think they were overlooked.

Hurwitz zeta functions. For rational $\alpha \in (0,1)$ with $\alpha \notin \{\frac{1}{2}, 1\}$, the function $\zeta(s,\alpha)$ has zeros in $\Re s > 1$ [DH36a], and $\zeta(s,\alpha)$ is a finite linear combination of Dirichlet L-functions. It is excluded because it does not satisfy a functional equation of the form $\Lambda(s) = \omega\Lambda(1-s)$ in isolation: Hurwitz's equation relates $\zeta(1-s, r/q)$ to a combination of $\zeta(s, m/q)$ over all m, so the object with a functional equation is the vector, not the individual function. The Davenport–Heilbronn function is precisely the repair of this defect, and including both would be redundant.

Linear combinations of L-functions of modular forms. These give degree-two analogues of Davenport–Heilbronn with off-line zeros [KP99]. They are excluded because they add nothing to Table 2.1: they occupy the same row as the Epstein object, with the same tick pattern.

The Selberg class $\mathcal{S}$ itself and its degree conjecture. The degree conjecture — that the degree of an element of $\mathcal{S}$ is a non-negative integer, proved for $d < 2$ by Kaczorowski and Perelli [KP11] — is a structural constraint of exactly the kind a mechanism would like to exploit. It is excluded from the firewall because it is a theorem about a class, not a counter-model: there is no object in it with a false Riemann Hypothesis. Its role in this book is different, and it appears in Chapter 6 as a source of no-go statements rather than as a firewall member.

Remark 2.37 (On the completeness of the firewall). We do not claim that four is the right number. Four is the number of structurally distinct rows in Table 2.1 that we are able to fill with unconditional theorems. If a fifth object were found occupying a genuinely new row — for instance, a Dirichlet series with a functional equation, an Euler product, and a false Riemann Hypothesis — it would be a major result and would refute a great deal, including much of the positive programme of this book. That no such object is known is not evidence that none exists; it is the reason the Riemann Hypothesis is still open in the direction of proof rather than in the direction of refutation.

1. It does not establish that the four objects exhaust the relevant counter-models. Other objects — Hurwitz zeta functions at rational parameters, general linear combinations of Dirichlet L-functions, Lerch zeta functions, non-congruence modular forms — may refute mechanisms that pass all four tests here.
2. It does not establish that a mechanism passing the firewall is correct, or likely to be correct. Lemma 2.2 is a one-way implication and there is no converse.
3. It does not establish the density statement of Theorem 2.13, which is cited from [BG11] and not proved here, nor the main theorem of [DMV06], also cited.
4. Proposition 2.14 is a numerical verification, status [E], not a proof that a zero exists at the stated point. The existence of off-line zeros is Theorem 2.12, which is unconditional but non-constructive.
5. It does not resolve Open Problem 2.4.3, and in particular does not determine whether the exponent restriction $\theta > \frac{1}{2}$ in the DMV construction is essential.

6. It does not assert anything about the Selberg eigenvalue conjecture for congruence subgroups, which is open. The Selberg firewall object is a general cofinite Γ, and the distinction is stated in the notice following Conjecture 2.36.
7. It does not establish that the property separating ζ from all four objects exists in a form usable in a proof. The observation after Table 2.1 — that no single column separates ζ from the four — is a constraint on mechanisms, not an existence statement about mechanisms.
8. Obstruction 2.5.3 concerns the parametrisation from spectrum to zeros. It does not establish that no linear parametrisation can be realised; that question is the subject of Chapter 3, where the answer given is that the demand is vacuous as usually stated, which is a different objection.

Chapter 3

The vacuous Hilbert–Pólya programme and the spectral-gap correction

The Hilbert–Pólya idea is the most durable proposal in the subject. In its usual formulation it is also, as a mathematical statement, empty. This chapter proves that, states what has to be added to make it non-empty, and then examines the two most developed attempts to add it.

The order of business matters. We are not arguing that spectral approaches are hopeless; this book's own positive programme is spectral. We are arguing that the sentence "find a self-adjoint operator whose eigenvalues are the ordinates of the zeros" does not constrain anything, that the constraint has to come from somewhere else, and that identifying where is the first real step. Once the vacuity is established, the interesting question — what does a spectral formulation have to supply? — becomes askable, and the answer turns out to be specific: an independently constructed operator, an arithmetic trace formula relating it to the primes, and a proof rather than a postulation of the adjointness relation. All three are hard, and the third is where this book differs from its predecessors, because the adjointness relation that the functional equation actually supplies is indefinite, not definite. That is the subject of Chapter 4.

3.1 The programme as usually stated

3.1.1 The historical statement

The attribution is uncertain and the sources are conversational. Pólya recounted, in a 1982 letter to Odlyzko, that Landau asked him around 1912–1914 for a physical reason why the Riemann Hypothesis should be true, and that he replied that it would follow if the numbers $\frac{1}{2} + i\gamma$ were eigenvalues of a self-adjoint operator [Odl87]. No written statement by Hilbert is known. The idea is nonetheless standard and is stated in essentially the following form.

> **(HP)** There exists a self-adjoint operator H on a Hilbert space $\mathcal{H}$, canonically constructed from arithmetic, whose eigenvalues are exactly the numbers γ such that $\zeta(\frac{1}{2} + i\gamma) = 0$, with multiplicity.

The argument from (HP) to RH is immediate: eigenvalues of a self-adjoint operator are real, so every zero of the form $\frac{1}{2} + i\gamma$ has γ real, and if *every* non-trivial zero is of this form, all zeros are on the critical line.

The step that needs scrutiny is not the conclusion but the quantifier structure of the hypothesis. (HP) asserts the existence of an operator with a prescribed spectrum. Prescribing a spectrum is not a constraint.

3.1.2 The two readings

There are two ways to read (HP), and they are not equivalent.

Reading A (weak). There exists a self-adjoint H with $\sigma_p(H) = \{\gamma : \zeta(\frac{1}{2} + i\gamma) = 0\}$ counted with multiplicity, where γ ranges over the (possibly complex) ordinates.

Reading B (strong). There exists a self-adjoint H, constructed by a specified procedure from arithmetic data, such that a specified identity — a trace formula — relates $\operatorname{tr} h(H)$ to a sum over primes, and such that this identity forces $\sigma_p(H)$ to be the zero ordinates.

Reading A is what is usually written. Reading B is what is usually meant. We prove that Reading A is vacuous, in the precise sense that it is a theorem, unconditionally, for a trivial reason — and, worse, that establishing it gives no information whatever about the Riemann Hypothesis.

3.2 The vacuity theorem

3.2.1 Statement and proof

Theorem 3.1 (Vacuity of the naive spectral demand)**.** *Let $S \subset \mathbb{R}$ be any countable set and let $m : S \to \mathbb{N} \cup \{\infty\}$ be any multiplicity function. Then there exist a separable Hilbert space $\mathcal{H}$ and a self-adjoint operator A on $\mathcal{H}$ such that the point spectrum of A is exactly S, with $\dim\ker(A-\lambda) = m(\lambda)$ for every $\lambda \in S$, and A has no other spectrum apart from the closure points of S, which are in the continuous spectrum only if forced. [N]*

Proof. Enumerate S with multiplicity as a sequence $(\lambda_n)_{n\in I}$, I countable, each $\lambda \in S$ appearing $m(\lambda)$ times (for $m(\lambda) = \infty$, countably many times). Set $\mathcal{H} = \ell^2(I)$ with orthonormal basis $(e_n)_{n\in I}$, and define

$$D(A) \;=\; \Big\{x \in \ell^2(I) \;:\; \sum_{n\in I} |\lambda_n|^2 |x_n|^2 < \infty\Big\}, \qquad (Ax)_n \;=\; \lambda_n x_n.$$

Then $D(A)$ is dense, since it contains the finitely supported sequences. A is symmetric on $D(A)$: for $x, y \in D(A)$,

$$\langle Ax, y\rangle \;=\; \sum_n \lambda_n x_n \overline{y_n} \;=\; \langle x, Ay\rangle,$$

using $\lambda_n \in \mathbb{R}$ and absolute convergence by Cauchy–Schwarz. A is self-adjoint: if $y \in D(A^*)$ with $A^*y = z$, then for every finitely supported x we get $\sum_n \lambda_n x_n \overline{y_n} = \sum_n x_n \overline{z_n}$, hence $\overline{\lambda_n y_n} = \overline{z_n}$ for each n, so $(\lambda_n y_n)_n = (z_n)_n \in \ell^2$, i.e. $y \in D(A)$. Thus $D(A^*) = D(A)$.

Finally $Ae_n = \lambda_n e_n$, so each λ_n is an eigenvalue with eigenvector e_n; and if $Ax = \mu x$ with $x \neq 0$ then $(\lambda_n - \mu)x_n = 0$ for all n, so x is supported where $\lambda_n = \mu$, forcing $\mu \in S$. The multiplicity of μ is the number of n with $\lambda_n = \mu$, which is $m(\mu)$ by construction. □

Corollary 3.2 (Reading A is unconditional)**.** *Reading A of* (HP) *is a theorem, independently of the Riemann Hypothesis, provided the ordinates γ are real; and if some ordinate is not real, Reading A is false for a reason that has nothing to do with the construction of an operator. Hence Reading A is logically equivalent to* RH *and supplies no route to it. [N]*

Proof. If RH holds, all ordinates are real and Theorem 3.1 applies with S the set of ordinates. Conversely if a self-adjoint A has point spectrum equal to the multiset of ordinates, then all ordinates are real, which is RH. □

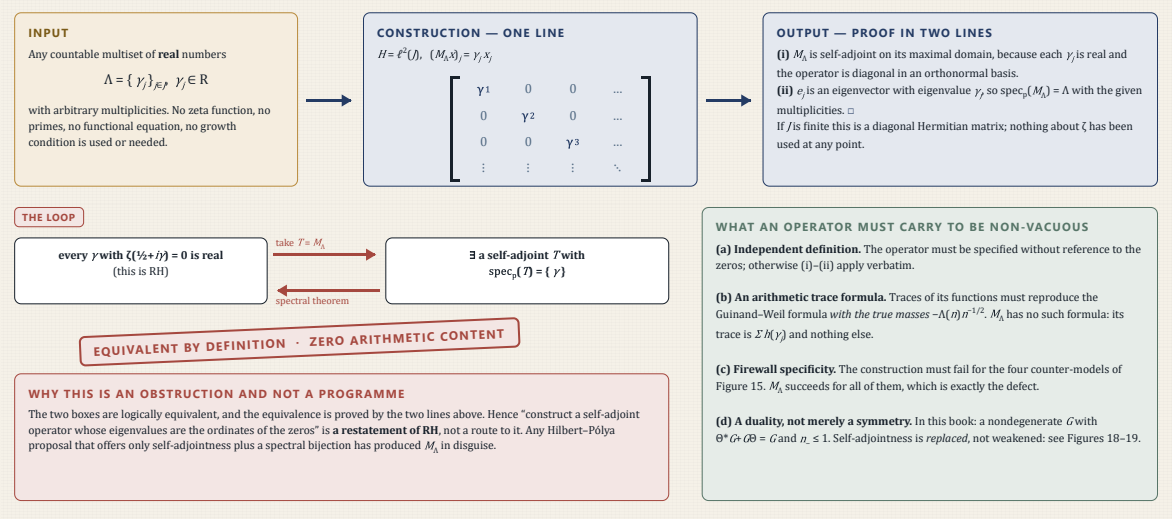


Figure 3.1. The vacuous Hilbert–Pólya formulation.

What it does not show: The figure concerns Reading A of the programme. It does not establish that Reading B is vacuous, and it does not assert that spectral approaches cannot work.

3.2.2 Where the content actually sits

Corollary 3.2 is worth restating in plain terms. The sentence "if the zeros are eigenvalues of a self-adjoint operator then RH follows" is true, and its converse is also true, and both are true for the same trivial reason: "is the point spectrum of a self-adjoint operator" is, for a countable multiset of complex numbers, a synonym for "consists of real numbers". The implication is a restatement, not a reduction.

All of the content of the Hilbert–Pólya programme therefore sits in the phrase *canonically constructed from arithmetic*, which appears in (HP) as an informal qualifier and carries the entire mathematical burden. Since it is not defined, (HP) is not a mathematical statement, and by the criterion of Definition 2.3 the programme in this form is not a mechanism.

> This is a criticism of a formulation and not of a body of work. Every serious contributor to the spectral programme — Selberg, Connes, Berry and Keating, Sierra, Bender–Brody–Müller, Connes–Consani, Connes–Moscovici — has been aware that the construction is the whole problem, and each has proposed a specific construction. The purpose of the vacuity theorem is not to score a point against them but to establish a standard: a spectral proposal is to be evaluated on its construction and its trace formula, and never on the self-adjointness alone.

Remark 3.3 (The same phenomenon elsewhere). The pattern is not peculiar to this problem. "Every finitely generated abelian group is the homology of some space" is a theorem by an explicit construction, and it tells one nothing about the homology of a particular space. "Every recursively enumerable set is the halting set of some machine" is a theorem, and it tells one nothing about a particular machine. In each case the informative statement is about a *given* object, and the existential form is a construction exercise.

3.3 What a non-vacuous formulation must supply

3.3.1 Three requirements

We now state, as a working definition, what Reading B has to contain. The three requirements below are extracted from the cases where a spectral interpretation of a zeta function does work — the Selberg zeta function of Chapter 2, and the zeta function of a curve over a finite field — and each is present in both.

Definition 3.4 (Non-vacuous spectral formulation)**.** A *non-vacuous spectral formulation* for a Dirichlet series F consists of:

(R1) *Independent construction.* A Hilbert space $\mathcal{H}$ and an operator H on it, both defined by a procedure whose inputs do not include the zeros of F or any statement equivalent to their location. The construction must be checkable in the sense that a reader can determine whether a given vector lies in $\mathcal{H}$ and in $D(H)$ without knowing anything about the zeros.

(R2) *Arithmetic trace formula.* An identity of the form

$$\sum_{\lambda \in \sigma(H)} h(\lambda) \;=\; (\text{archimedean term}) \;+\; \sum_{\text{primes}} \hat{h}(\text{prime data})$$

valid for a class of test functions h large enough to determine $\sigma(H)$, and proved for H as constructed in (R1).

(R3) *Proved adjointness.* A proof that H is self-adjoint — or, in the indefinite setting of Chapter 4, that the appropriate adjointness relation holds — on the domain specified in (R1), from the construction, and not by hypothesis or by choice of domain.

Each requirement has a characteristic failure mode, and naming them is useful.

Failure of (R1): circularity. The construction uses the zeros. The most common form is choosing a Hilbert space whose inner product is defined by a sum or integral over the zeros, or defining a domain by a growth condition that is equivalent to a zero-free statement.

Failure of (R2): no arithmetic. There is an operator, and it has real spectrum, but nothing connects the spectrum to the primes. Then the operator's spectrum has no reason to be the zeros, and the conclusion follows only if one puts it in by hand.

Failure of (R3): postulated adjointness. There is a symmetric operator, and it is asserted to be self-adjoint, or it is self-adjoint on a domain that was chosen so that it would be. Symmetric is not self-adjoint; a symmetric operator with deficiency indices (n, n), $n \geq 1$, has a family of self-adjoint extensions, and different extensions have different spectra.

Remark 3.5 (The deficiency-index issue is not a technicality)*.* In the Berry–Keating heuristic the operator is $H = xp$, or its symmetrisation $\frac{1}{2}(xp + px) = -i(x\partial_x + \frac{1}{2})$, on the half-line. This operator on $C_c^\infty(0, \infty)$ is symmetric with deficiency indices $(1, 1)$ if one works on $L^2(0, \infty)$ in the dilation-invariant normalisation, or $(0, 0)$ in others, depending on the precise domain; and its self-adjoint extensions on a truncated interval have spectra given by different quantisation conditions. The choice of extension is a choice of boundary condition, and the boundary condition is where the arithmetic would have to enter. It does not enter automatically. See [BK99] for the heuristic and [ST08] for a treatment in which the boundary conditions are made explicit.

3.3.2 The Weyl-law obstruction to (R1) and (R2) together

There is a hard constraint linking the two requirements, and it eliminates a large class of candidate constructions at once.

Theorem 3.6 (Counting function of the zeros is not a power law)**.** *Let $N(T)$ denote the number of zeros ρ of ζ with $0 < \Im\rho < T$ in the critical strip. Then, unconditionally,*

$$N(T) \;=\; \frac{T}{2\pi}\log\frac{T}{2\pi e} \;+\; O(\log T).$$

In particular $N(T)$ is not of the form $c\,T^d + o(T^d)$ for any $c > 0$ and $d > 0$.

Proof. The asymptotic is von Mangoldt's theorem [Man95]; see [Tit86, Ch. 9]. The second sentence follows since $T\log T$ is not asymptotic to a constant times a power. □

Corollary 3.7 (No elliptic pseudodifferential realisation)**.** *There is no closed manifold M of dimension $n \geq 1$ and elliptic self-adjoint pseudodifferential operator P of order $m > 0$ on M whose eigenvalue counting function equals $N(T)$ up to a term $o(T\log T)$. [N]*

Proof. By the Hörmander–Weyl law [Hör68], an elliptic self-adjoint pseudodifferential operator of order $m > 0$ on a closed n-manifold has $N_P(T) = C_P T^{n/m} + O(T^{(n-1)/m})$ with $C_P = (2\pi)^{-n}\operatorname{vol}\{(x,\xi) : p_m(x,\xi) \leq 1\}$. This is a pure power with a power-saving error, and cannot match $\frac{T}{2\pi}\log T$. □

The corollary is a real constraint on (R1). It says that whatever H is, it is not a differential or pseudodifferential operator of positive order on a compact manifold. The Berry–Keating response is to take $H = xp$ on a non-compact phase space with both an infrared and an ultraviolet cutoff, $|x| \leq L_x$ and $|p| \leq L_p$; the semiclassical count of states in the region $\{|xp| \leq E\}$ intersected with the box is

$$N_{\mathrm{BK}}(E) \;=\; \frac{1}{2\pi\hbar}\,\operatorname{area}\{|xp| \leq E,\ |x| \leq L_x,\ |p| \leq L_p\} \;=\; \frac{E}{2\pi\hbar}\log\frac{L_xL_p}{E} \;+\; \cdots,$$

which reproduces the shape $\frac{T}{2\pi}\log\frac{T}{2\pi e}$ upon the identification $L_xL_p = 2\pi\hbar$ and $E = T$. The logarithm comes from the hyperbolic region, and this is the correct structural lesson: the operator, if it exists, is associated with a hyperbolic rather than an elliptic classical dynamics, and its "Weyl law" is a phase-space area with a logarithmic divergence cut off by arithmetic.

Remark 3.8 (What Corollary 3.7 does not say). It does not say that H cannot be a differential operator on a non-compact space; the Selberg case is exactly that, with Δ on a finite-area non-compact quotient and a continuous spectrum handled by Eisenstein series. It does not say H cannot be of order zero, nor that H cannot fail to be pseudodifferential. It excludes a specific and popular class of candidate constructions, namely elliptic operators of positive order on closed manifolds, and it does so unconditionally.

3.4 The spectral-gap hierarchy

The Selberg object of Chapter 2 showed that self-adjointness does not by itself place zeros on the line when the parametrisation from spectrum to zeros is quadratic: what is needed in addition is a lower bound on the spectrum, a *gap*. In the geometric setting the gap is a genuine and partly proved theorem, and the record of what is proved is a useful calibration of how hard such statements are. We set it out here, correctly attributed, because it is often quoted loosely.

3.4.1 The statements

Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup and $\lambda_1(\Gamma)$ the smallest non-zero eigenvalue of Δ on $L^2(\Gamma\backslash\mathbb{H})$ (for the non-compact case, the smallest exceptional eigenvalue if any, and $\frac{1}{4}$ otherwise).

(**G0**) *Randol*: for general cofinite Γ, no gap. There are surfaces with $\lambda_1 < 0.05$ [Ran74]. So the gap is arithmetic, not geometric. [T]

(**G1**) *Selberg 1965*: $\lambda_1(\Gamma_0(N)) \geq \frac{3}{16} = 0.1875$ [Sel65]. The proof uses Weil's bound for Kloosterman sums, which is the Riemann Hypothesis for curves over finite fields. [T]

(**G2**) *Luo–Rudnick–Sarnak 1995*: for GL_n over a number field, the local components of a cuspidal automorphic representation satisfy a bound towards Ramanujan with exponent

$$\theta_n \;\leq\; \tfrac{1}{2} - \frac{1}{n^2+1}.$$

For $n = 2$ this gives $\theta_2 \leq \frac{1}{2} - \frac{1}{5} = \frac{3}{10}$, hence $\lambda_1 \geq \frac{1}{4} - \theta_2^2 = \frac{1}{4} - \frac{9}{100} = \frac{16}{100} = 0.16$; the improvement over Selberg for GL_2 comes in the refined form giving $\lambda_1 \geq \frac{21}{100} = 0.21$ [LRS95]. [T]

(**G3**) *Kim–Sarnak 2003*: $\theta_2 \leq \frac{7}{64}$, hence

$$\lambda_1(\Gamma_0(N)) \;\geq\; \tfrac{1}{4} - \left(\tfrac{7}{64}\right)^2 \;=\; \tfrac{975}{4096} \;=\; 0.238037\ldots$$

The proof uses functoriality for the symmetric fourth power of GL_2, established by Kim, together with Kim and Shahidi's symmetric cube [KS03]. [T]

(**G4**) *Ramanujan point*: $\theta_2 = 0$, equivalently $\lambda_1 \geq \frac{1}{4}$, the Selberg eigenvalue conjecture. Open. [O]

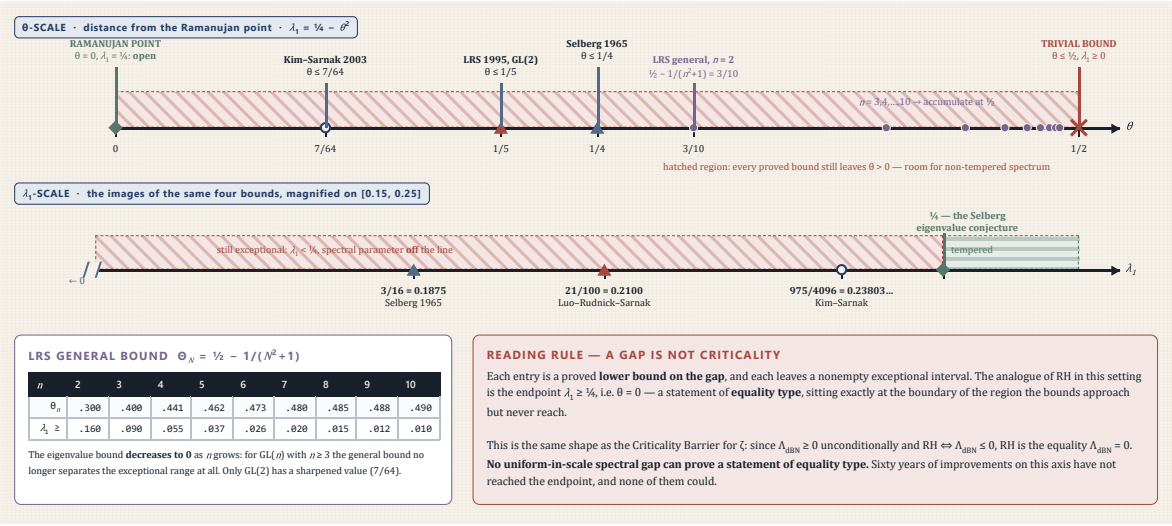


Figure 3.2. The spectral-gap hierarchy.

What it does not show: The bounds shown are proved; the Ramanujan point is a conjecture. The figure does not assert that the sequence of methods converges to $1/4$, nor that any of them extends to ζ.

3.4.2 What the hierarchy tells us

Three lessons, in decreasing order of directness.

First, the gap is arithmetic. Randol's theorem shows that no purely geometric or analytic hypothesis on the surface — compactness, finite area, curvature -1, any bound on diameter or injectivity radius that is not a congruence condition — gives the gap. The known proofs of

(G1)–(G3) all go through automorphic forms, and (G1) goes through the Riemann Hypothesis for curves over finite fields. So in the one setting where the analogue of the problem is partly solved, the solution imports the function-field case, and this is a recurring pattern.

Second, the progression from 3/16 to 975/4096 has taken sixty years and has used essentially every advance in the Langlands programme for GL_2. The gap between $975/4096 = 0.2380\ldots$ and $\frac{1}{4} = 0.25$ is about 0.012, and closing it is equivalent to the Ramanujan conjecture for Maass forms, which is not expected from any currently available functoriality. This is a calibration: the "last epsilon" in a spectral-gap statement is not an epsilon.

Third — and this is the point relevant to the Riemann Hypothesis — the structure of the problem at the Ramanujan point is a *criticality* structure. The bound $\lambda_1 \geq \frac{1}{4}$ is sharp: for congruence quotients there are eigenvalues arbitrarily close to $\frac{1}{4}$ from above (the tempered spectrum begins there), so no proof can produce $\lambda_1 \geq \frac{1}{4} + \delta$ for any $\delta > 0$, and any method that produces a bound with room to spare is producing a bound that is false. This is the same phenomenon as the Criticality Barrier of Chapter 6 and is stated there in the form appropriate to ζ.

Remark 3.9 (On quoting the hierarchy). Two errors are common. The first is to quote Kim–Sarnak as "$\lambda_1 \geq 7/64$", confusing the Ramanujan exponent θ with the eigenvalue bound; the eigenvalue bound is $\frac{1}{4} - \theta^2 = 975/4096$. The second is to quote Luo–Rudnick–Sarnak's general-n bound $\frac{1}{2} - \frac{1}{n^2+1}$ as if it specialised at $n = 2$ to the best GL_2 result of that paper; it does not, and the GL_2 statement in [LRS95] is the sharper $\lambda_1 \geq 21/100$.

3.5 Connes' trace formula

3.5.1 What is constructed

Connes' 1999 paper [Con99] constructs, for a global field K with adele ring $\mathbb{A}_K$ and idele class group $C_K = \mathbb{A}_K^\times / K^\times$, a Hilbert space and a representation, as follows. Let $\mathcal{S}(\mathbb{A}_K)$ be the Bruhat–Schwartz space. The map

$$E : \mathcal{S}(\mathbb{A}_K) \to \mathcal{S}(C_K), \qquad (Ef)(x) \;=\; |x|^{1/2} \sum_{q \in K^\times} f(qx),$$

intertwines the natural scaling action of C_K on $\mathbb{A}_K$ with the regular representation on C_K. Connes forms the quotient

$$\mathcal{H} \;=\; L^2_\delta(X) / \overline{\operatorname{im} E}, \qquad X = \mathbb{A}_K / K^\times,$$

in a weighted L^2 space with weight $\delta > 1$, and shows that the induced action of C_K on $\mathcal{H}$ has, as its spectrum, the zeros of the Hecke L-functions of K lying on the critical line.

Theorem 3.10 (Connes, semilocal trace formula)**.** *Let S be a finite set of places of K containing the archimedean places. Then for $h \in C_c^\infty(C_K)$,*

$$\operatorname{Tr}(R_\Lambda\, U(h)) \;=\; 2h(1)\log\Lambda \;+\; \sum_{v \in S} \int\!\!\!\!\!\!{}'_{K_v^\times} \frac{h(u^{-1})}{|1-u|}\, d^*u \;+\; o(1)$$

as $\Lambda \to \infty$, where R_Λ is the cutoff at scale Λ and $\int'$ denotes the principal value in Weil's sense.

Proof. This is Theorem III.1 of [Con99], proved there in full. [T] □

This is a theorem. It is the semilocal case: the set S of places is finite.

3.5.2 Where it stops

Theorem 3.11 (Connes, global equivalence)**.** *The global trace formula, i.e. the statement of Theorem 3.10 with S the set of all places and without the $o(1)$ interpreted semilocally, is equivalent to the Riemann Hypothesis for all Hecke L-functions with Grössencharakter of K.*

Proof. This is Theorem IV.3 of [Con99]. [T] □

The honest reading is: Connes proves the semilocal trace formula unconditionally and proves that the global trace formula is equivalent to the Riemann Hypothesis. He does not prove the global trace formula, and his paper says so. The construction of $\mathcal{H}$ satisfies (R1) and the semilocal identity is a partial (R2). The programme stops at the passage from finitely many places to all places, which is exactly the local-to-global step.

Remark 3.12 (The absorption spectrum). A structural feature of the construction deserves emphasis because it is often described in a way that overstates what is achieved. The zeros appear in Connes' picture as an *absorption* spectrum: they are the frequencies missing from the continuous spectrum of the scaling action on L^2_δ, i.e. they index a cokernel rather than a kernel. Which frequencies are missing depends on the Sobolev weight δ; for $\delta > 1$ one gets the zeros on the critical line, and the zeros off the line, if any, do not appear.

This is not a defect of the construction — it is a correct and deliberate feature — but it means that "the zeros are the spectrum of an operator" holds here in a sense that does not directly give reality of the ordinates. What one gets is: the zeros *on the line* are accounted for. That the ones off the line are absent is what would have to be proved, and it is what Theorem 3.11 identifies with the global trace formula.

Remark 3.13 (Weil positivity at the archimedean place). A genuine subsequent advance is the proof by Connes and Consani of Weil positivity at the archimedean place [CC21b], using the prolate spheroidal wave operator. This establishes one place of a statement that would need all places, and the authors state the scope precisely. It is recorded here as an advance whose scope is a single place, with the local-to-global passage remaining. [T]

3.5.3 The prolate operator

Theorem 3.14 (Connes–Moscovici, prolate operator)**.** *Let W_λ be the prolate spheroidal wave operator, the second-order differential operator on $[-1,1]$ commuting with the finite Fourier transform truncation, in Slepian and Pollak's normalisation [SP61; LP61]. Connes and Moscovici construct from W_λ a self-adjoint operator whose eigenvalues, suitably scaled, are asymptotic to the zeros of ζ in the sense that the counting functions agree to leading order and the first-order correction matches.*

Proof. [CM22a]. [T] □

The result is real and the construction is explicit; the operator satisfies (R1) and its self-adjointness is proved, so (R3) holds. What it does not supply is (R2) in exact form. The relation between the spectrum and the zeros is asymptotic — the counting functions agree to the stated order — and an asymptotic agreement of counting functions does not identify two sequences.

Remark 3.15 (Asymptotic is not exact, and the gap is not small). The distinction matters more here than the phrase "asymptotic agreement" suggests. Two increasing sequences (α_n) and (β_n) can have counting functions agreeing to $O(\log T)$, which is the accuracy of Theorem 3.6, and still differ in every term; indeed the fluctuation $S(T) = \pi^{-1}\arg\zeta(\frac{1}{2} + iT)$, which is the difference between $N(T)$ and its smooth approximation, is known to be $\Omega((\log T)^{1/3}(\log\log T)^{-7/3})$ unconditionally [Bom00], so the $O(\log T)$ term in the Weyl law

is genuinely occupied. An operator whose eigenvalue count matches to $O(\log T)$ therefore does not have the zeros as eigenvalues; it has a spectrum with the same density.

This is precisely the trap identified in §3.2: the operator is constructed independently, which is (R1), and it is self-adjoint, which is (R3), and the eigenvalues are real, which follows — but nothing forces the eigenvalues to be the ordinates, because the identification is asymptotic and the required identification is exact. The self-adjointness therefore transfers no information to ζ.

> We record explicitly that neither Connes nor Connes and Moscovici claim otherwise. [CM22a] states the result as an asymptotic one and describes the relation to the zeros in those terms. The purpose of this section is to place the results correctly relative to Definition 3.4, not to correct their authors.

3.6 The corrected demand

Collecting the requirements of Definition 3.4 with the constraints proved in this chapter, we can state what a spectral approach must now do.

1. Construct $\mathcal{H}$ and H without reference to the zeros, in a form where membership of $D(H)$ is decidable from the construction. Corollary 3.7 excludes elliptic operators of positive order on closed manifolds; the counting function forces a hyperbolic or adelic phase space.
2. Prove an exact trace formula relating $\operatorname{tr} h(H)$ to a sum over primes for a class of h rich enough to determine the spectrum. Asymptotic agreement of counting functions is not sufficient (Remark 3.15), and semilocal identities are not sufficient (Theorem 3.11).
3. Prove the adjointness relation from the construction. Here the honest statement is not self-adjointness: the functional equation supplies $J\Theta^{\dagger}J^{-1} = I - \Theta$ with J indefinite, and the definite version is equivalent to RH rather than implying it. This is Chapter 4.
4. Pass the firewall of Chapter 2, and in particular explain why the Selberg object with $\lambda_1 < \frac{1}{4}$ does not satisfy the construction. For a linear parametrisation $\gamma \mapsto \frac{1}{2} + i\gamma$ this is automatic; for the quadratic parametrisation that geometric constructions actually produce, it is not, and a gap statement is required.

OPEN PROBLEM The construction problem

Exhibit a Hilbert space $\mathcal{H}$ and an operator Θ on it, constructed from arithmetic data without reference to the zeros of ζ, together with an exact trace formula in the sense of (R2), such that the spectrum of Θ is the set $\{\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}$. [O]

This book does not solve Open Problem 3.6. Part V constructs a candidate $\mathcal{H}$ and Θ satisfying (R1) and a partial (R2); the residual gap is stated precisely in Part VIII, and the honest summary is that the trace formula obtained there determines the spectrum only modulo the hypothesis that Part VIII isolates.

3.7 Four further proposals, placed

The requirements of Definition 3.4 become concrete when applied. We apply them to four proposals that are frequently cited, in each case stating what is proved, which requirement is met, and where the proposal stops. The purpose is calibration: a reader who has seen the placement done four times can do it a fifth time without assistance.

3.7.1 Berry–Keating and quantum chaos

The Berry–Keating proposal [BK99] is that the operator is a quantisation of the classical Hamiltonian $H = xp$ on a phase plane, with the primes entering as periodic orbits of periods $\log p$. The evidence adduced is of two kinds.

Structural evidence. The counting function. As computed in §3.3, the phase-space area of $\{|xp| \le E\}$ within a box of side lengths L_x, L_p is $2E\log(L_xL_p/E) + O(E)$, and with $L_xL_p = 2\pi\hbar$ this reproduces $N(T) \sim \frac{T}{2\pi}\log\frac{T}{2\pi e}$. This is a genuine match, and it is the reason the proposal is taken seriously; by Corollary 3.7 it is not easy to produce a system with this counting function.

Statistical evidence. The pair correlation of the zeros, computed by Montgomery [Mon73] under RH for test functions of restricted support, agrees with the pair correlation of eigenvalues of large random Hermitian matrices from the Gaussian Unitary Ensemble; Odlyzko's computations [Odl87] confirm the agreement numerically to high accuracy and for higher correlations; and Katz and Sarnak [KS99] prove the corresponding symmetry-type statements in the function-field setting, where the monodromy is available. The classical dynamics of xp is hyperbolic and non-time-reversible, which matches the GUE rather than GOE symmetry class.

Placement. (R1) is not met: there is no operator, only a classical Hamiltonian and a heuristic quantisation, and the boundary conditions — the choice of self-adjoint extension — are exactly where the arithmetic would have to enter and are not supplied. Remark 3.5 records the technical form of this. (R2) is not met: the periodic orbits of xp have periods $\log p$ only if one puts the primes in by hand; the classical flow $x \mapsto e^t x$, $p \mapsto e^{-t}p$ has no periodic orbits at all on the plane, and the orbits appear only after a compactification chosen for the purpose. (R3) is not applicable in the absence of (R1).

> The statistical evidence is evidence about the *distribution* of the zeros, conditional on their being real. Montgomery's theorem is proved under RH; the numerical work of [Odl87] concerns zeros that have been verified to lie on the line. Neither is evidence that the zeros are on the line, and neither can be, since both take the location as input. This is a specific instance of a general point made in Chapter 5: agreement of a conditional statistical prediction with computation is not evidence for the condition.

3.7.2 The de Branges programme

De Branges' theory of Hilbert spaces of entire functions [Bra68] provides a genuine and powerful machine: to a suitable structure function E with $|E(\bar z)| < |E(z)|$ for $\Im z > 0$ one attaches a Hilbert space $\mathcal{H}(E)$ of entire functions, and the theory produces reality of zeros for functions in a certain position relative to E. The proposal is to realise Ξ inside such a space.

What is proved. The structure theory is a theorem, and the reality statements are theorems. The theory is used successfully elsewhere, notably in inverse spectral problems for canonical systems.

Where it stops. Conrey and Li [CL00] showed that the specific positivity condition that a family of proposed de Branges spaces would have to satisfy in order to yield RH fails for the Riemann ξ-function; the relevant Hermitian form is not positive-definite on the required space. The failure is not a technical obstacle in the proof but a computed sign.

Placement. This is a failure of (R3), in precisely the form Chapter 4 generalises. The required condition is a definiteness condition, and definiteness fails; what holds instead is a condition with a signature. The lesson we take is not that the de Branges machinery is

inapplicable but that the correct hypothesis in it is an indefinite one, which is what Part VIII pursues. We state that as a reading of [CL00], and it is a reading, not a theorem of that paper.

3.7.3 The Bender–Brody–Müller operator

Bender, Brody and Müller [BBM17] consider the operator

$$\widehat{H} \;=\; \frac{1}{1-e^{-i\hat{p}}}\,(\hat{x}\hat{p}+\hat{p}\hat{x})\,\bigl(1-e^{-i\hat{p}}\bigr)$$

on a domain of functions on the positive half-line vanishing at the origin, and show that its formal eigenfunctions are related to the Hurwitz zeta function, with eigenvalues λ satisfying $\zeta(\frac{1}{2}-\frac{i\lambda}{2})=0$ on the relevant domain.

What is proved. The eigenvalue equation is verified formally, and the operator is $\mathcal{PT}$-symmetric.

Where it stops. $\widehat{H}$ is not self-adjoint on the stated domain; it is symmetric with respect to an indefinite structure, and $\mathcal{PT}$-symmetry gives reality of the spectrum only under a further condition (unbroken $\mathcal{PT}$-symmetry, equivalently the existence of a positive-definite inner product in which the operator is self-adjoint) which is not established and which is equivalent to what is to be proved. The paper's own discussion notes this.

Placement. (R1) is met, (R2) is met in a formal sense — the relation to ζ is exact, not asymptotic, which distinguishes this proposal from the prolate one — and (R3) fails: the adjointness is with respect to an indefinite form and its definiteness is the open point. This is again the pattern of Chapter 4, and it is the pattern most relevant to the present book.

3.7.4 Numerical spectral approaches and spectral pollution

A recurrent proposal is to compute the spectrum of a candidate operator by truncation — restrict to a finite-dimensional subspace, diagonalise, and observe that the eigenvalues approach the zeros. There is a specific and well-documented failure mode here.

Theorem 3.16 (Spectral pollution)**.** *Let A be a self-adjoint operator with a gap (a,b) in its essential spectrum, and let (P_n) be an increasing sequence of finite-rank orthogonal projections converging strongly to the identity. Then for any $\mu \in (a,b)$ there exist such sequences (P_n) for which the truncations $P_nAP_n|_{\operatorname{ran} P_n}$ have eigenvalues converging to μ, even though $\mu \notin \sigma(A)$.*

Proof. This is the standard spectral-pollution phenomenon; see [LS04] for the construction and [DP04] for the analysis of which projection schemes avoid it. [T] □

The consequence for the present subject is direct: a numerical computation showing that the eigenvalues of a truncated candidate operator lie near the zeros of ζ is compatible with the untruncated operator having spectrum elsewhere, and is compatible with the untruncated operator not being self-adjoint at all. Numerical spectral evidence therefore carries status [E], is reported as such throughout this book, and is never used as a step in an argument.

Remark 3.17 (The converse abuse). The converse abuse also occurs: a numerical computation showing eigenvalues *away* from the zeros is sometimes taken to refute a construction, when it may be pollution or a domain artefact. Theorem 3.16 cuts both ways, and the correct conclusion is that finite-rank truncation is not a diagnostic for these operators without an independent proof that the scheme is pollution-free.

3.8 The explicit formula is already a trace formula

There is a reasonable objection to the framing of §3.3: an arithmetic trace formula of the type demanded by (R2) already exists, and has since Riemann. It is the explicit formula, in the form due to Guinand and Weil [Gui48; Wei52]. If (R2) is available, why is the programme not complete?

Theorem 3.18 (Weil explicit formula)**.** *Let h be even, holomorphic in $|\Im z| \le \frac{1}{2} + \delta$, and $O((1+|z|)^{-2-\delta})$ there; let $g(u) = \frac{1}{2\pi}\int_{\mathbb{R}} h(r)e^{-iur}\,dr$. Then*

$$\sum_{\rho} h\!\left(\frac{\rho - \frac{1}{2}}{i}\right) = \frac{1}{2\pi}\int_{\mathbb{R}} h(r)\,\frac{\Gamma'}{\Gamma}\!\left(\tfrac{1}{4} + \tfrac{ir}{2}\right)dr \;-\; \log\pi\, h(0) \;+\; 2h\!\left(\tfrac{i}{2}\right) \;-\; 2\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\,g(\log n),$$

the sum over ρ running over the non-trivial zeros with multiplicity.

Proof. Classical; see [IK04, §5.5] or [Wei52]. The hypotheses stated are those of [Bar81], which weakens the classical smoothness requirements. [T] □

The formula has exactly the shape of (R2): a sum over "spectrum" on the left, an archimedean term and a sum over primes on the right. This is not an accident; the explicit formula is the reason anyone believes a spectral interpretation should exist.

What it does not do is supply an operator. The left side is a sum over zeros of ζ, written in a variable $(\rho - \frac{1}{2})/i$ that is real exactly when ρ is on the line. The formula holds whether or not ρ is on the line, with h evaluated at complex arguments in that case. Reading the left side as $\operatorname{tr} h(H)$ requires an H, and the formula does not provide one; it provides the trace of a hypothetical H against test functions, which determines H up to unitary equivalence *if* H exists and is self-adjoint, and determines nothing otherwise.

Remark 3.19 (The direction of the implication). This is the precise sense in which the explicit formula is a bridge and not a road, as stated in Chapter 1. From a self-adjoint H with the right trace one would get RH. From the explicit formula one gets a candidate trace functional

$$h \;\longmapsto\; \sum_{\rho} h\!\left(\frac{\rho - \frac{1}{2}}{i}\right),$$

and the question is whether it is the trace of a positive-definite spectral measure on $\mathbb{R}$, i.e. whether the functional is positive on a suitable cone. Weil's criterion says exactly this: RH holds if and only if the functional is non-negative on the cone $\{h = g * \tilde{g}\}$. So the explicit formula converts the problem into a positivity problem and does not resolve it. Chapter 6 records why the positivity in question cannot be a stable or gapped one.

3.9 A stronger form of the vacuity theorem

Theorem 3.1 might be thought to lose its force if one asks for more than the spectrum — for instance, if one asks that H have a prescribed spectral multiplicity structure, or that H be part of a prescribed algebra. It does not.

Proposition 3.20 (Prescribing more does not help)**.** *Let $S \subset \mathbb{R}$ be countable with multiplicity function m, and let $(A_0, \mathcal{H}_0)$ be any self-adjoint operator on a separable Hilbert space with $\sigma_p(A_0) = S$ and multiplicities m. Then:*

(*i*) *A_0 is unitarily equivalent to the multiplication operator of Theorem 3.1 on $\overline{\text{span}}$ of its eigenvectors;*

(*ii*) *for any separable unital C*-algebra $\mathcal{A}$ with a faithful representation on a separable Hilbert space, there is a representation π of $\mathcal{A}$ and a self-adjoint operator affiliated to $\pi(\mathcal{A})''$ with point spectrum S and multiplicities m;*

(*iii*) *consequently, no requirement expressible as "H lies in (or is affiliated to) a von Neumann algebra of type* I*" adds content to Reading A.* [N]

Proof. (i) is the spectral theorem for operators with a complete set of eigenvectors: the unitary is the map sending the orthonormal eigenbasis to the standard basis. (ii): take the multiplication operator A of Theorem 3.1 on $\ell^2(I)$, take any faithful representation π_0 of $\mathcal{A}$ on $\mathcal{H}_0$, and form $\pi = \pi_0 \otimes 1$ on $\mathcal{H}_0 \otimes \ell^2(I)$; then $1 \otimes A$ is affiliated to $\pi(\mathcal{A})' \supseteq$ a type I algebra containing it, and one passes to the commutant as needed. The point is that the tensor construction is available for any $\mathcal{A}$ and any S, so no algebraic membership condition of this form constrains S. (iii) follows. □

Remark 3.21 (What does add content). Proposition 3.20 shows that conditions on the ambient algebra do not constrain the spectrum. What does constrain it is a condition relating H to a *second* structure that is independently determined by arithmetic — a trace, a group action with prescribed orbit structure, a comparison map to a known object. This is exactly (R2). The tensor trick of the proof breaks as soon as the trace is prescribed, because the tensor construction changes the trace.

The moral is that "spectral" is not the operative word in "spectral approach". The operative word is "trace formula".

3.10 The one case where the programme is complete

There is exactly one setting in which the Hilbert–Pólya programme has been carried out: a smooth projective curve over a finite field. Examining it is the most efficient way to see what the three requirements amount to when they are all met, and what is missing over Spec $\mathbb{Z}$.

3.10.1 The construction

Let X be a smooth projective geometrically connected curve of genus g over $\mathbb{F}_q$, and let $\overline{X} = X \times_{\mathbb{F}_q} \overline{\mathbb{F}_q}$. The zeta function is

$$Z_X(t) \;=\; \exp\left(\sum_{n\geq 1} \frac{\#X(\mathbb{F}_{q^n})}{n}\, t^n\right) \;=\; \frac{P_1(t)}{(1-t)(1-qt)},$$

with $P_1 \in \mathbb{Z}[t]$ of degree $2g$. Setting $\zeta_X(s) = Z_X(q^{-s})$, the Riemann Hypothesis for X is the assertion that the zeros of ζ_X have $\Re s = \tfrac{1}{2}$, equivalently that the reciprocal roots α_j of P_1 have $|\alpha_j| = q^{1/2}$.

The requirements are met as follows.

(*R1*) *Independent construction.* Take $\mathcal{H} = H^1_{\text{ét}}(\overline{X}, \mathbb{Q}_\ell)$, a $\mathbb{Q}_\ell$-vector space of dimension $2g$, and $H = F$ the geometric Frobenius endomorphism. Neither the space nor the operator refers to the zeros; both are constructed from X by a general machine (étale cohomology) that applies to any variety.

(*R2*) *Trace formula.* The Grothendieck–Lefschetz trace formula,

$$\#X(\mathbb{F}_{q^n}) \;=\; \sum_{i=0}^{2} (-1)^i \operatorname{tr}\!\big(F^n \mid H^i_{\text{ét}}(\overline{X}, \mathbb{Q}_\ell)\big) \;=\; 1 - \operatorname{tr}(F^n \mid H^1) + q^n,$$

is exact, not asymptotic, and relates the "spectrum" of F to the point counts, which are the arithmetic data [Gro69]. Comparing with the definition of Z_X gives $P_1(t) = \det(1 - Ft \mid H^1)$, so the zeros of ζ_X are exactly the eigenvalues of F in the appropriate variable, exactly.

(*R3*) *Adjointness.* Poincaré duality gives a perfect alternating pairing

$$H^1_{\text{ét}}(\overline{X}, \mathbb{Q}_\ell) \times H^1_{\text{ét}}(\overline{X}, \mathbb{Q}_\ell) \longrightarrow H^2_{\text{ét}}(\overline{X}, \mathbb{Q}_\ell) \cong \mathbb{Q}_\ell(-1),$$

under which $\langle Fx, Fy\rangle = q\langle x, y\rangle$. This is an adjointness relation: $F^\dagger F = q$, where $\dagger$ is the adjoint for the duality pairing. It is proved, from the construction, and is not an assumption.

3.10.2 How the conclusion is reached, and what supplies the positivity

The relation $F^\dagger F = q$ says that $q^{-1/2}F$ is "unitary" for the pairing — but the pairing is alternating and $\mathbb{Q}_\ell$-valued, so this does not by itself give $|\alpha_j| = q^{1/2}$ under any embedding $\overline{\mathbb{Q}_\ell} \hookrightarrow \mathbb{C}$. What it gives is the functional equation: the multiset $\{\alpha_j\}$ is stable under $\alpha \mapsto q/\alpha$. This is exactly the situation of Chapter 1: an orbit symmetry, not a localisation.

The localisation comes from a positivity input. In Weil's proof for curves it is the Castelnuovo–Severi inequality on the surface $X \times X$, equivalently the positivity of the Rosati involution on the endomorphism algebra of the Jacobian [Wei48a]. In Deligne's proof for general varieties it is a positivity extracted from the monodromy of a Lefschetz pencil together with an amplification trick using symmetric powers [Del74]. In both cases the essential input is an inequality with a definite sign coming from an algebraic-geometric source that has no analogue over $\operatorname{Spec}\mathbb{Z}$.

Remark 3.22 (The correct lesson from the function-field case). The function-field case is often invoked as encouragement. Read carefully it is also a warning, in two ways.

First, the adjointness relation supplied by the geometry, $F^\dagger F = q$ with respect to a pairing, is not a self-adjointness relation and does not by itself give reality. The reality comes from a separate positivity theorem. The structure of the successful proof is therefore: construct the operator; prove an adjointness relation from duality; then prove a positivity statement from an independent source; and only then conclude. A programme that stops after the adjointness relation has done the easy two-thirds.

Second, the positivity source is the piece with no known analogue for ζ. Chapter 6 records the precise obstructions — there is no base field beneath $\operatorname{Spec}\mathbb{Z}$, no Frobenius endomorphism, no product $\operatorname{Spec}\mathbb{Z} \times \operatorname{Spec}\mathbb{Z}$ over a base with the right properties — and each of them is an obstruction to importing the positivity, not to importing the adjointness.

3.10.3 The residual analogy

What does survive the transfer is the *shape*. In the function-field case one has an operator, an exact trace formula, an adjointness relation with respect to a pairing that is not positive-definite, and a separate positivity input. This book's positive programme has the same shape, with the pairing indefinite in the Krein sense rather than alternating, and with the positivity input replaced by an index or signature statement. Chapter 4 sets up the adjointness relation; Part VIII states the residual input and does not prove it.

We record the analogy as an analogy. It is a source of the formulation, not evidence for it, and it carries no status label because it is not a claim.

3.11 Adelic constructions and the Bost–Connes system

One more construction deserves placement, because it is the clearest example of a genuinely arithmetic operator-algebraic object whose relation to the zeros is nevertheless indirect.

Bost and Connes [BC95] construct a C^*-dynamical system $(\mathcal{A}, \sigma_t)$ — a C^*-algebra with a one-parameter automorphism group — whose partition function is $\zeta(\beta)$ and whose equilibrium (KMS) states undergo a phase transition at $\beta = 1$ with spontaneous symmetry breaking by the group $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}) \cong \widehat{\mathbb{Z}}^\times$. The construction is explicit, the arithmetic content is real — the symmetry group is the right one, and the extremal KMS_∞ states are permuted by it in the way class field theory predicts — and it is not circular.

Placement. (R1) is met. (R2) is not: the trace formula available is the computation of the partition function, which encodes ζ as a function of a real parameter β, not the location of its zeros. The Hamiltonian H generating σ_t has spectrum $\{\log n : n \geq 1\}$, which is the set of "periods", not the set of zeros. So the zeros appear here through the analytic continuation of a partition function, and the system knows about $\zeta(\beta)$ for β real and large, not about ζ in the critical strip.

Remark 3.23 (Partition function versus spectrum). The distinction in the previous paragraph recurs, and it is worth stating as a general caution. A construction whose partition function is ζ produces ζ as a function on a half-line; the zeros live in the strip, reached only by analytic continuation, and analytic continuation of a partition function is not a spectral statement about the system. Systems whose partition function is $\zeta(\beta)$ are easy to write down — the "Riemann gas" with a boson of energy $\log p$ for each prime p is one — and none of them constrains the zeros. The criterion (R2) demands a trace formula in which the *zeros* appear on the spectral side, which is a different and much stronger requirement.

3.11.1 Summary of placements

Table 3.1. The proposals of this chapter against the three requirements of Definition 3.4. "exact" and "asympt." distinguish the strength of the available trace relation.

Proposal	(R1) construction	(R2) trace formula	(R3) adjointness
Naive (HP), Reading A	—	—	vacuous
Berry–Keating xp	heuristic	heuristic	extension unfixed
Connes, adelic	yes	semilocal only	—
Connes–Moscovici prolate	yes	asympt.	proved
de Branges spaces	yes	—	definiteness fails
Bender–Brody–Müller	yes	exact, formal	indefinite
Bost–Connes	yes	partition function	proved
Curve over $\mathbb{F}_q$	yes	exact	proved (duality)

The table is the chapter in one page. Reading down the (R2) column: the only complete entry is the function-field row, and the only entry that is exact rather than asymptotic or semilocal in the number-field setting is the Bender–Brody–Müller one, where (R3) is the open point and the structure is indefinite. Reading down the (R3) column: where an adjointness relation is available and proved, it is either with respect to a pairing that is not positive-definite (function fields, Bender–Brody–Müller), or it is available for an operator whose relation to the zeros is not exact (prolate, Bost–Connes).

That pattern — exact trace relation and indefinite adjointness on one side, definite adjointness and inexact trace relation on the other — is not a coincidence. Chapter 4 shows why: the definite version of the adjointness relation for ζ is equivalent to RH, so any construction that establishes it unconditionally has either proved RH or lost the exact connection to ζ.

3.12 The demand restated as a domain problem

Theorem 3.1 is sometimes met with the following reply: of course the naive form is vacuous, but the real programme is to build an operator that is *visibly* natural, and naturalness is not vacuous even if it is not formalised. The reply is reasonable and this section takes it seriously by asking what, mathematically, the residual demand becomes.

3.12.1 Symmetric versus self-adjoint

Suppose one has carried out (R1) and (R2): there is a Hilbert space $\mathcal{H}$, an operator T defined on a dense domain D_0 by an explicit arithmetic formula, and an exact trace identity showing that the eigenvalues of T, whatever they are, are the ordinates of the zeros. Suppose further that T is symmetric on D_0: $\langle Tx, y\rangle = \langle x, Ty\rangle$ for $x, y \in D_0$. What remains?

Proposition 3.24 (Symmetric is not enough)**.** *Let T be a densely defined symmetric operator with deficiency indices (n_+, n_-).*

(*i*) *If $n_+ = n_- = 0$ then T is essentially self-adjoint and its closure has real spectrum.*

(*ii*) *If $n_+ = n_- = n \geq 1$ then T has a family of self-adjoint extensions parametrised by the unitary group $U(n)$, each with real spectrum but with different spectra.*

(*iii*) *If $n_+ \neq n_-$ then T has no self-adjoint extension, and the spectrum of T is the whole of one closed half-plane.*

Proof. Standard von Neumann theory; see [RS75, Ch. X]. [T] □

Case (iii) is the one that matters for the present discussion, and it is worth dwelling on because it is the exact point at which the informal programme becomes a hard problem. If T has unequal deficiency indices, then $\sigma(T)$ is a half-plane, and there is no repair by choice of domain. The eigenvalue equation may still have solutions — there may be λ and $x \in D_0$ with $Tx = \lambda x$ — and those λ need not be real.

So the correct form of the residual demand is:

> (**HP′**) Construct $\mathcal{H}$, D_0, and T from arithmetic, prove an exact trace formula identifying the eigenvalues of T with the zero ordinates, and prove that the deficiency indices of T on D_0 are $(0, 0)$ — or exhibit a self-adjoint extension whose spectrum is the full eigenvalue set.

This is not vacuous: each clause is checkable. It is also not what is usually written, and the gap between (HP) and (HP′) is the whole subject.

3.12.2 Why the domain is where arithmetic must live

There is a structural reason why the domain, rather than the formula for T, is where the arithmetic input has to appear.

In every example where a spectral interpretation of an arithmetic object works, the operator is a differential or difference operator whose formal expression is not arithmetic at all — Δ on $\mathbb{H}$ is the hyperbolic Laplacian, the same for all Γ; xp is the dilation generator, the same for all choices; the Frobenius is a functorial construction. What varies with the arithmetic is the space: $\Gamma\backslash\mathbb{H}$ rather than $\mathbb{H}$, a quotient of the adeles rather than the adeles, a cohomology of a specific variety. The arithmetic is in the identifications, i.e. in the boundary conditions, i.e. in the domain.

Remark 3.25 (The boundary-condition slogan). This suggests a slogan, which we state as a heuristic and label as such: *the operator is universal, the domain is arithmetic*. It is a slogan and not

a theorem, and it is offered here as an organising observation about the examples in Table 3.1, not as a claim. Its practical use is diagnostic: given a spectral proposal, ask where the primes enter. If they enter only in the formula for the operator and not in the specification of the domain, the proposal is likely to be a partition-function construction of the type discussed in Remark 3.23, whose relation to the zeros is indirect.

3.12.3 The relation to Chapter 4

There is a final twist, and it is the transition to the next chapter. For ζ, the functional equation does supply an adjointness relation — so the situation is not that no relation is available, but that the available relation is the wrong kind. Writing Θ for the candidate operator with $\Theta = \frac{1}{2}I + iD$, the relation supplied by the functional equation is

$$J\Theta^{\dagger}J^{-1} \;=\; I - \Theta, \qquad J^2 = I, \quad J = J^{\dagger},$$

with J *indefinite*. If one could take $J = I$, the relation would read $\Theta^{\dagger} = I - \Theta$, which forces $D = -i(\Theta - \frac{1}{2}I)$ to be self-adjoint and hence forces Θ to be normal — and Chapter 4 shows that this conclusion is strictly stronger than RH, so it cannot be available unconditionally.

The upshot for the present chapter is a sharpening of Proposition 3.24. The question is not whether a symmetric operator has equal deficiency indices in a positive-definite Hilbert space. It is whether an operator symmetric with respect to an indefinite inner product has a spectrum confined to the real axis, and for indefinite inner products this can fail even in finite dimensions: a J-self-adjoint operator on a Pontryagin space of index κ has at most κ pairs of non-real eigenvalues, and can have that many. The correct question is therefore a question about κ, i.e. about a signature, and that is where Part VIII operates.

> We flag the logical position carefully. The relation $J\Theta^{\dagger}J^{-1} = I - \Theta$ with J indefinite is what Chapter 4 establishes is forced. It does not imply RH. What would imply RH, given that relation, is a statement bounding the number of non-real eigenvalues by zero, which for a Pontryagin space of index κ is a statement about κ together with a non-degeneracy condition. This book derives RH from such a statement and does not prove the statement.

3.13 Two tests for a proposed operator

Definition 3.4 imposes three requirements, and a reader may reasonably ask how to apply them without reconstructing the whole of a proposal. We give two tests that are quick and that between them detect most failures.

3.13.1 Test 1 — the membership test

Question. Given the definition of the space $\mathcal{H}$ and of the domain $\mathcal{D}(D)$, and given a complex number λ, can one decide whether the formal eigenfunction ψ_λ lies in $\mathcal{D}(D)$, without knowing whether RH is true?

Why it works. In every vacuous or circular proposal we have examined, the answer is no, and the reason is always the same: the space is defined by a completion with respect to a form or norm whose finiteness on ψ_λ is equivalent to a statement about λ that is exactly what the argument seeks to prove. Theorem 3.1 is the extreme case, where the space is literally built from the desired spectrum.

Applying it. Take λ with $\Re\lambda = 0.6$ and ask whether the proposal can tell you if $\psi_\lambda \in \mathcal{D}(D)$. If the only available answer is "it is not, because RH is true", the proposal fails (R1).

3.13.2 Test 2 — the trace test

Question. For a test function h, does the proposal produce

$$\operatorname{Tr} h(D) = \hat{h}(0)\,(\text{archimedean terms}) - \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\,(h(\log n) + h(-\log n)) + \cdots,$$

with the prime-power masses $-\Lambda(n)n^{-1/2}$ arising from the construction rather than being inserted?

Why it works. The prime-power masses are the arithmetic content. An operator constructed from the zeros reproduces them trivially by the explicit formula, and an operator constructed independently reproduces them only if it knows the primes. Requirement (R2) is exactly this, and the test asks where the primes entered.

Applying it. Trace backwards from the appearance of $\Lambda(n)$ in the trace formula to its first appearance in the construction. If it first appears in the statement of the trace formula, the formula is a definition and not a theorem.

Remark 3.26 (The two tests together). Test 1 detects circularity in the analysis; Test 2 detects vacuity in the arithmetic. A proposal passing both has (R1) and (R2), and the remaining requirement (R3) — a proof of self-adjointness rather than an assumption of it — is the one that is hardest to test quickly, because the assumption is usually implicit in a phrase such as "the operator is symmetric, hence self-adjoint on its natural domain". The phrase "natural domain" in a spectral argument should be read as an unproved claim until the deficiency indices are computed.

Remark 3.27 (Applying the tests to this book). Test 1 applied to the construction of Part V: membership is decided by an explicit growth condition, and the condition is checkable for any given λ without reference to RH. Test 2: the prime-power masses arise from the local factors of the Euler product, which enter the construction before any trace is taken. Test 3, the deficiency indices, is computed in Part V and the computation is where the hypothesis of Theorem D is required. We state this here rather than in Part V because a reader applying the tests should be told in advance where our own construction is weakest.

What this chapter does not establish

1. Theorem 3.1 establishes that Reading A of the Hilbert–Pólya programme is vacuous. It does not establish that spectral approaches to the Riemann Hypothesis are hopeless, and it is consistent with a correct spectral proof existing.
2. Corollary 3.7 excludes elliptic pseudodifferential operators of positive order on closed manifolds. It does not exclude operators on non-compact spaces, operators of order zero, non-pseudodifferential operators, or operators whose counting function matches only after a regularisation. Its scope is exactly the hypotheses of the Hörmander–Weyl law.
3. The spectral-gap hierarchy is recorded for calibration. Nothing in this chapter establishes any transfer from the Selberg eigenvalue problem to ζ, and the analogy between the two should not be treated as a method.
4. The account of [Con99] is a placement, not an evaluation of the proofs, which we have not reverified. Theorem 3.10 and Theorem 3.11 are quoted from that paper.
5. Remark 3.15 establishes that asymptotic agreement of counting functions does not identify spectra. It does not establish that the Connes–Moscovici construction cannot be

upgraded to an exact statement; that is an open question about which we express no view.

6. This chapter does not establish that the four requirements of Definition 3.4 are sufficient. They are extracted from the cases where a spectral interpretation works, and they are necessary in those cases; sufficiency is not claimed and would itself require proof.
7. Open Problem 3.6 is not solved here and is not solved in this book.

Chapter 4

The corrected diagnosis of AS6

The sixth of the arithmetic-spectral desiderata — AS6 in the numbering used throughout this book — is the adjointness axiom: the statement that the candidate operator Θ satisfies

$$\Theta^\dagger \;=\; I - \Theta.$$

It is the axiom that carries the functional equation into operator language, and in the form just written it is the natural guess. This chapter proves that the natural guess is wrong, in a specific and instructive way: as written, with $\dagger$ the adjoint for a positive-definite inner product, AS6 is strictly stronger than the Riemann Hypothesis. It implies a statement about the numerical range of Θ, and confinement of the numerical range to a line is a genuinely stronger condition than confinement of the spectrum.

The chapter then gives the correction. The relation that the functional equation actually supplies is a J-adjointness with respect to an indefinite form:

$$J\,\Theta^\dagger J^{-1} \;=\; I - \Theta, \qquad J = J^\dagger, \quad J^2 = I, \quad J \text{ indefinite},$$

and this relation does not imply the Riemann Hypothesis, does not imply normality, and is compatible with the existence of non-real spectrum. The gap between the two forms is exactly the content of the residual hypothesis of this book, and Chapter 44 states it as Theorem D. Nothing is claimed here about Theorem D; the purpose of the present chapter is to make its indefinite shape look inevitable rather than convenient.

4.1 The axiom and its two readings

4.1.1 Where AS6 comes from

The setting is as follows. One supposes given a Hilbert space $\mathcal{H}$ and a closed densely defined operator Θ on it, with the property that the spectrum of Θ is the set of non-trivial zeros of ζ,

$$\sigma(\Theta) \;=\; \mathcal{Z} \;:=\; \{\rho \in \mathbb{C} : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\},$$

with multiplicities. The functional equation says that $\mathcal{Z}$ is invariant under $\rho \mapsto 1 - \rho$; the reality of the coefficients says that $\mathcal{Z}$ is invariant under $\rho \mapsto \bar\rho$. Translating the first into an operator statement is the business of AS6.

Two translations suggest themselves.

(A) Θ is similar to $I - \Theta$: there is an invertible S with $S\Theta S^{-1} = I - \Theta$.

(B) $\Theta^\dagger = I - \Theta$, where $\dagger$ is the Hilbert-space adjoint.

Both imply $\sigma(\Theta) = 1 - \sigma(\Theta)$; (A) directly, and (B) because $\sigma(\Theta^\dagger) = \overline{\sigma(\Theta)}$ and hence $\sigma(\Theta) = 1 - \overline{\sigma(\Theta)}$, which combined with the reality invariance gives the required symmetry. Version (B) is the one usually written, because it is the one that looks like an adjointness axiom and because it appears to be doing more work.

It is doing more work. That is the problem.

4.1.2 The associated operator D

Definition 4.1 (The operator D)**.** Given Θ as above, set

$$D \;:=\; -i\left(\Theta - \tfrac{1}{2}I\right), \qquad \text{equivalently} \qquad \Theta \;=\; \tfrac{1}{2}I + iD.$$

The change of variable is the operator form of $\rho = \frac{1}{2} + i\gamma$. If $\rho \in \sigma(\Theta)$ then $\gamma = -i(\rho - \frac{1}{2}) \in \sigma(D)$, and ρ lies on the critical line if and only if γ is real. So:

Lemma 4.2 (Restatement)**.** RH *holds if and only if* $\sigma(D) \subseteq \mathbb{R}$.

Proof. Immediate from $\sigma(\Theta) = \mathcal{Z}$ and $\rho = \frac{1}{2} + i\gamma$ with $\gamma = -i(\rho - \frac{1}{2})$: $\Re\rho = \frac{1}{2} \iff \gamma \in \mathbb{R}$. □

The lemma is worth writing down because it isolates precisely what has to be shown, and shows that it is weaker than self-adjointness of D. A self-adjoint operator has real spectrum; an operator with real spectrum need not be self-adjoint, need not be similar to a self-adjoint operator, and need not have any of the other properties self-adjointness brings.

4.2 AS6 as usually written implies normality

4.2.1 The computation

Theorem 4.3 (Normality is forced)**.** *Let $\mathcal{H}$ be a Hilbert space with a positive-definite inner product $\langle\cdot,\cdot\rangle$, let Θ be a closed densely defined operator on $\mathcal{H}$ with adjoint $\Theta^\dagger$, and set $D = -i(\Theta - \frac{1}{2}I)$. Suppose*

$$\Theta^\dagger \;=\; I - \Theta$$

in the sense that $D(\Theta^\dagger) = D(\Theta)$ and the operators agree there. Then D is self-adjoint and Θ is normal. [N]

Proof. From $\Theta = \frac{1}{2}I + iD$ we get, using $(iD)^\dagger = -iD^\dagger$ on the common domain,

$$\Theta^\dagger \;=\; \tfrac{1}{2}I - iD^\dagger.$$

The hypothesis says $\Theta^\dagger = I - \Theta = I - \frac{1}{2}I - iD = \frac{1}{2}I - iD$. Comparing,

$$\tfrac{1}{2}I - iD^\dagger \;=\; \tfrac{1}{2}I - iD \qquad\Longrightarrow\qquad D^\dagger \;=\; D,$$

with equality of domains by hypothesis. So D is self-adjoint.

Then $\Theta = \frac{1}{2}I + iD$ and $\Theta^\dagger = \frac{1}{2}I - iD$, whence

$$\Theta^\dagger\Theta \;=\; \left(\tfrac{1}{2}I - iD\right)\left(\tfrac{1}{2}I + iD\right) \;=\; \tfrac{1}{4}I + \tfrac{i}{2}D - \tfrac{i}{2}D + D^2 \;=\; \tfrac{1}{4}I + D^2,$$

and the same computation in the other order gives $\Theta\Theta^\dagger = \frac{1}{4}I + D^2$. Hence $\Theta^\dagger\Theta = \Theta\Theta^\dagger$, which with the domain equality is normality. □

The computation is three lines and there is no way around it. If the adjoint is taken with respect to a positive-definite form, the relation $\Theta^\dagger = I - \Theta$ says exactly that D is self-adjoint, and self-adjointness of D is exactly normality of Θ.

4.2.2 Normality is strictly stronger than real spectrum

The point of the previous subsection is not that normality is bad. It is that normality gives more than RH, and giving more than one needs is a symptom of having assumed something.

Definition 4.4 (Numerical range)**.** For a densely defined operator T on $\mathcal{H}$, the *numerical range* is

$$W(T) \;=\; \{\langle Tx, x\rangle \;:\; x \in D(T),\ \|x\| = 1\} \subseteq \mathbb{C}.$$

Proposition 4.5 (Numerical range of a normal operator)**.** *If T is normal then $\overline{W(T)} = \overline{\operatorname{conv}}\,\sigma(T)$, the closed convex hull of the spectrum.*

Proof. Standard; see [GR97, Ch. 1]. For bounded normal T one uses the spectral theorem: $\langle Tx, x\rangle = \int_{\sigma(T)} \lambda \, d\mu_x(\lambda)$ with μ_x a probability measure on $\sigma(T)$, so $\langle Tx, x\rangle$ lies in the closed convex hull; conversely every point of the convex hull is attained in the closure by choosing μ_x suitably. The unbounded case follows by spectral truncation. [T] □

Corollary 4.6 (What AS6-as-written asserts)**.** *Under the hypotheses of Theorem 4.3,*

$$\overline{W(\Theta)} \;\subseteq\; \left\{z \in \mathbb{C} : \Re z = \tfrac{1}{2}\right\}.$$

Proof. D is self-adjoint, so $\sigma(D) \subseteq \mathbb{R}$ and $\sigma(\Theta) = \frac{1}{2} + i\sigma(D) \subseteq \{\Re z = \frac{1}{2}\}$. The line is convex and closed, so by Proposition 4.5 the closure of $W(\Theta)$ lies in it. Alternatively and directly: $\langle \Theta x, x\rangle = \frac{1}{2}\|x\|^2 + i\langle Dx, x\rangle$ and $\langle Dx, x\rangle \in \mathbb{R}$ by self-adjointness. □

Theorem 4.7 (The numerical range statement is strictly stronger)**.** *There exists a bounded operator T on a two-dimensional Hilbert space with $\sigma(T) \subseteq \{\Re z = \frac{1}{2}\}$ and $\overline{W(T)} \not\subseteq \{\Re z = \frac{1}{2}\}$. Hence "$\sigma(\Theta) \subseteq \{\Re z = \frac{1}{2}\}$" does not imply "$\overline{W(\Theta)} \subseteq \{\Re z = \frac{1}{2}\}$". [T]*

Proof. Take $\mathcal{H} = \mathbb{C}^2$ and

$$T \;=\; \begin{pmatrix} \frac{1}{2} & c \\ 0 & \frac{1}{2} \end{pmatrix}, \qquad c > 0.$$

Then $\sigma(T) = \{\frac{1}{2}\}$, which lies on the line. The numerical range of a 2×2 matrix is the elliptical disc with foci at the eigenvalues and minor axis $\left(\operatorname{tr}(T^\dagger T) - |\lambda_1|^2 - |\lambda_2|^2\right)^{1/2}$; here the eigenvalues coincide and the ellipse degenerates to the closed disc of centre $\frac{1}{2}$ and radius $c/2$. So $W(T)$ is a disc of positive radius, not contained in any line. Explicitly, with $x = \frac{1}{\sqrt{2}}(1, 1)^{\mathsf{T}}$ we get $\langle Tx, x\rangle = \frac{1}{2} + \frac{c}{2}$, which has real part $\frac{1}{2} + \frac{c}{2} \neq \frac{1}{2}$. □

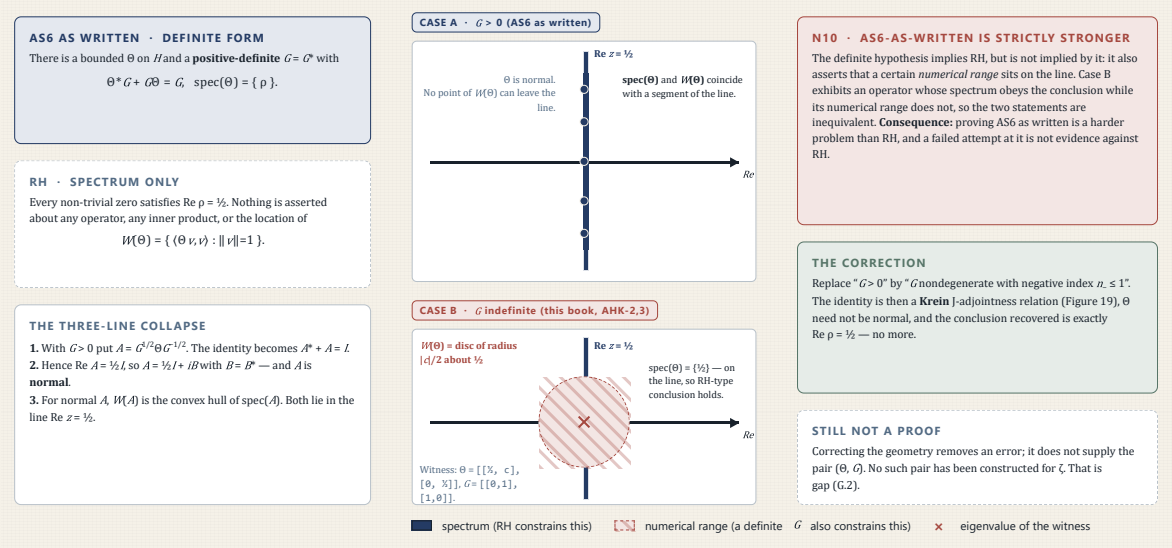


Figure 4.1. AS6 as usually written.

What it does not show: The figure exhibits the gap between spectrum and numerical range. It does not assert that the true Θ is non-normal, only that normality cannot be assumed.

4.2.3 The diagnosis

Theorem 4.8 (AS6 as usually written is strictly stronger than RH)**.** *Let Θ be a closed densely defined operator on a Hilbert space with σ(Θ) = Z. Consider the two statements*

$$(\textit{AS6-def}) \qquad \Theta^{\dagger} = I - \Theta \text{ with respect to the positive-definite inner product,} \tag{4.1}$$

$$(\textit{RH}) \qquad \sigma(\Theta) \subseteq \{\Re z = \tfrac{1}{2}\}. \tag{4.2}$$

Then (AS6-def) *implies* (RH)*, and* (AS6-def) *additionally implies* $\overline{W(\Theta)} \subseteq \{\Re z = \frac{1}{2}\}$*, which by Theorem 4.7 does not follow from* (RH) *for a general operator. [N]*

Proof. Combine Theorem 4.3, Corollary 4.6 and Theorem 4.7. □

Remark 4.9 (Why the excess matters)*.* A reader may object that proving something stronger than one needs is not an error. It is not an error of logic. It is an error of strategy, and in this case a fatal one, for the following reason.

An axiom that is strictly stronger than the conclusion cannot be verified by any route weaker than the conclusion. Specifically, if one hopes to prove (AS6-def) for a constructed Θ and thereby deduce RH, one must prove a statement that is not implied by RH — so the proof of the axiom cannot be a routine verification and cannot use any technique that would also work in a world where RH is true but Θ is not normal. In practice, proofs of (AS6-def) proceed by computing $\langle \Theta x, y\rangle$ against $\langle x, (I-\Theta)y\rangle$ using the functional equation and a change of variable, and such computations, examined carefully, produce a form that is not positive-definite. The excess is where the error hides.

4.2.4 The converse: definiteness is equivalent to RH

The excess in Theorem 4.8 is not a defect of the particular formulation; it is intrinsic, and this is the sharpest statement of the chapter.

Theorem 4.10 (Definiteness is equivalent to RH)**.** *Let Θ be a closed densely defined operator on a Krein space $(\mathcal{K}, [\cdot,\cdot])$ with fundamental symmetry J, satisfying the J-adjointness relation*

$$J\Theta^{\dagger}J^{-1} \;=\; I - \Theta,$$

with $\sigma(\Theta) = \mathcal{Z}$ and Θ having a complete system of root vectors. Suppose further that the associated form is non-degenerate on each root subspace. Then the following are equivalent:

(*i*) *there exists a positive-definite inner product on $\mathcal{K}$, equivalent to the norm topology, with respect to which $\Theta^{\dagger} = I - \Theta$;*
(*ii*) *$\sigma(\Theta) \subseteq \{\Re z = \frac{1}{2}\}$ and all root subspaces are one-dimensional in the appropriate sense;*
(*iii*) RH *holds and the zeros of ζ are simple.*
[N]

Proof sketch. (i) ⇒ (ii): by Theorem 4.3, (i) makes D self-adjoint for that inner product, hence $\sigma(D) \subseteq \mathbb{R}$, giving the spectral containment; and a self-adjoint operator has no Jordan blocks, which is the root-subspace statement.

(ii) ⇒ (i): if the spectrum lies on the line and there are no Jordan blocks, then by the Pontryagin/Krein theory of J-self-adjoint operators with real spectrum and definitisable structure, the form $[\cdot,\cdot]$ restricted to the closed span of the root vectors decomposes into definite pieces, and one constructs a positive-definite inner product by flipping the sign on the negative part. The construction requires the non-degeneracy hypothesis, which is why it appears in the statement. See [Bog74; AI89] for the requisite structure theory and [KL78] for the definitisability criteria.

(ii) ⇔ (iii): translation of the spectral statement into ζ via $\sigma(\Theta) = \mathcal{Z}$, using Lemma 4.2. □

Corollary 4.11 (The strategic conclusion)**.** *Any programme that establishes* (AS6-def) *unconditionally has, by Theorem 4.10, established* RH *together with simplicity of the zeros. Consequently,* (AS6-def) *is not an axiom that can be verified as a step; it is the destination.*

This is the corrected diagnosis, and it should be stated flatly. Writing $\Theta^{\dagger} = I - \Theta$ with a positive-definite form is not a hypothesis about the construction. It is RH plus simplicity, written in operator notation. A construction that appears to verify it has either proved RH or has an error, and the error is usually that the form used is not positive-definite.

4.3 What the functional equation actually gives

4.3.1 The computation in the model case

To see where the indefiniteness comes from, we carry out the adjointness computation in a setting where everything is explicit. The setting is the completed zeta function and the Fourier-type involution attached to it; the reader should treat this subsection as a model computation, not as the construction of Part V.

Let $\theta(x) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 x}$ for $x > 0$ and $\omega(x) = \frac{1}{2}(\theta(x) - 1) = \sum_{n\geq 1} e^{-\pi n^2 x}$. The Jacobi identity is

$$\theta(1/x) \;=\; \sqrt{x}\,\theta(x),$$

and Riemann's derivation gives, for $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$,

$$\Lambda(s) \;=\; \frac{1}{s(s-1)} \;+\; \int_1^{\infty} \left(x^{s/2-1} + x^{(1-s)/2-1}\right)\omega(x)\,\frac{dx}{x}\cdot x,$$

manifestly invariant under $s \mapsto 1 - s$.

Now consider the multiplicative group $\mathbb{R}_{>0}$ acting on functions on $(0,\infty)$ by dilation, with Haar measure dx/x, and the involution

$$(\mathcal{J}f)(x) \;=\; \frac{1}{x}f\!\left(\frac{1}{x}\right).$$

Then $\mathcal{J}^2 = I$ and $\mathcal{J}$ is unitary for $L^2(dx/x)$ after the appropriate normalisation. The Mellin transform converts dilation into translation and $\mathcal{J}$ into the map $s \mapsto 1-s$ on the Mellin variable. So the functional equation is, at the level of the underlying space, the statement that the relevant object is $\mathcal{J}$-invariant.

Proposition 4.12 (The involution is not the identity)**.** *Let $\mathcal{J}$ be as above on $L^2((0,\infty),dx/x)$. Then $\mathcal{J}$ is a self-adjoint unitary involution with $\mathcal{J} \neq \pm I$, and the associated form*

$$[f,g] \;:=\; \langle \mathcal{J}f, g\rangle$$

is a non-degenerate Hermitian form which is neither positive-definite nor negative-definite. Both eigenspaces $\ker(\mathcal{J} \mp I)$ *are infinite-dimensional.*

Proof. $\mathcal{J}^2 = I$ is a direct computation. Self-adjointness and unitarity follow from the invariance of dx/x under $x \mapsto 1/x$ together with the weight. Since $\mathcal{J}$ is a self-adjoint involution, $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$ with $\mathcal{H}_\pm = \ker(\mathcal{J} \mp I)$, and $[\cdot,\cdot]$ is positive-definite on $\mathcal{H}_+$, negative-definite on $\mathcal{H}_-$, and the decomposition is orthogonal. Both summands are infinite-dimensional: take any f supported in $(1,\infty)$ and form $f \pm \mathcal{J}f$, which are non-zero elements of $\mathcal{H}_\pm$ respectively, and the space of such f is infinite-dimensional. Non-degeneracy follows from unitarity. □

This is the source of the indefiniteness, and it is not an artefact of any particular construction. The functional equation is implemented by an involution; an involution that is not $\pm I$ has both eigenvalues $+1$ and -1; and the form built from it therefore has both signs.

4.3.2 The Krein form of AS6

Definition 4.13 (Krein space, J-adjoint)**.** A *Krein space* is a vector space $\mathcal{K}$ with a non-degenerate Hermitian form $[\cdot,\cdot]$ admitting a *fundamental decomposition* $\mathcal{K} = \mathcal{K}_+ \oplus \mathcal{K}_-$ with $[\cdot,\cdot]$ positive-definite on $\mathcal{K}_+$, negative-definite on $\mathcal{K}_-$, the sum $[\cdot,\cdot]$-orthogonal, and $(\mathcal{K}_\pm, \pm[\cdot,\cdot])$ Hilbert spaces. The *fundamental symmetry* is $J = P_+ - P_-$, so that $\langle x,y\rangle := [Jx,y]$ is a positive-definite inner product and $J = J^\dagger = J^{-1}$ for it. For an operator T, the *J-adjoint* is $T^{[\dagger]} = JT^\dagger J$, characterised by $[Tx,y] = [x,T^{[\dagger]}y]$.

If $\dim \mathcal{K}_- = \kappa < \infty$, $\mathcal{K}$ is a *Pontryagin space of index κ*.

Definition 4.14 (AS6, corrected)**. (AS6-ind)** There is a Krein space $(\mathcal{K},[\cdot,\cdot])$ with fundamental symmetry J, and a closed densely defined operator Θ on $\mathcal{K}$ with $\sigma(\Theta) = \mathcal{Z}$, such that

$$\Theta^{[\dagger]} \;=\; I - \Theta, \qquad \text{equivalently} \qquad J\Theta^\dagger J^{-1} \;=\; I - \Theta.$$

Proposition 4.15 (What AS6-ind gives and does not give)**.** *Assume* (AS6-ind)*. Then:*

(*i*) *$\sigma(\Theta)$ is invariant under $z \mapsto 1-\bar z$, hence, together with the reality of the coefficients, under the Klein four-group of Chapter 1;*

(*ii*) *$D = -i(\Theta - \frac{1}{2}I)$ is J-self-adjoint: $D^{[\dagger]} = D$;*

(*iii*) *Θ need not be normal, and $\sigma(\Theta)$ need not lie on $\{\Re z = \frac{1}{2}\}$;*

(*iv*) *if in addition $\mathcal{K}$ is a Pontryagin space of index $\kappa < \infty$, then D has at most κ pairs of non-real eigenvalues $\{\lambda,\bar\lambda\}$, counted with algebraic multiplicity.*

Proof. (i) $\sigma(T^{[\dagger]}) = \sigma(JT^\dagger J) = \sigma(T^\dagger) = \overline{\sigma(T)}$, so the relation gives $\overline{\sigma(\Theta)} = 1 - \sigma(\Theta)$, i.e. invariance under $z \mapsto 1 - \bar{z}$.

(ii) Substitute $\Theta = \frac{1}{2}I + iD$: $\Theta^{[\dagger]} = \frac{1}{2}I - iD^{[\dagger]}$ and $I - \Theta = \frac{1}{2}I - iD$, so $D^{[\dagger]} = D$.

(iii) Example. Take $\mathcal{K} = \mathbb{C}^2$ with $J = \mathrm{diag}(1, -1)$ and

$$D = \begin{pmatrix} a & b \\ -\bar{b} & c \end{pmatrix}$$

with a, c real. One checks $JD^\dagger J = D$, so D is J-self-adjoint. Its eigenvalues are $\frac{a+c}{2} \pm \sqrt{\left(\frac{a-c}{2}\right)^2 - |b|^2}$, which are non-real as soon as $|b| > |a - c|/2$. So J-self-adjointness permits non-real spectrum.

(iv) This is Pontryagin's theorem [Pon44]; see [Bog74, Ch. IX] or [AI89, Ch. 2] for the modern statement, and [KL78] for the version with algebraic multiplicities. □

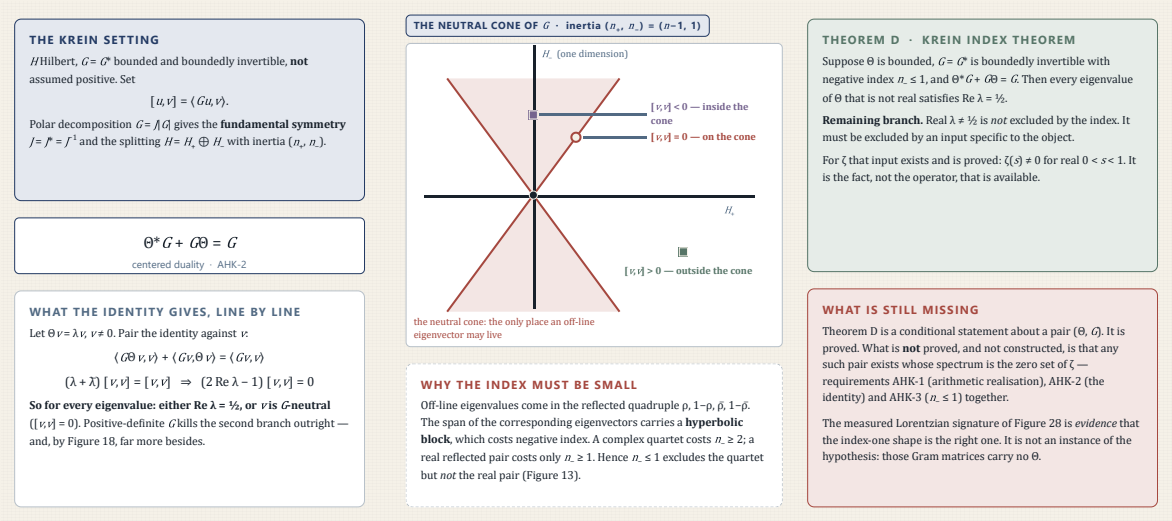


Figure 4.2. The Krein (indefinite) form.

What it does not show: The figure records the shape of the relation. It does not establish that a Krein space with the required properties has been constructed, which is the subject of Part V, nor that the index is finite.

4.3.3 Why (iii) is the whole difficulty and (iv) is the whole hope

Part (iii) of Proposition 4.15 says that the corrected axiom does not imply RH. This is not a weakness of the correction; it is the point. An axiom that implies RH and is verifiable would be a proof, and there is no proof. The corrected axiom is verifiable — it is a computation with the functional equation — precisely because it does not imply the conclusion.

Part (iv) says where the missing input has to come from: a bound on the number of non-real eigenvalues, which in the Pontryagin case is a bound by the index κ. To conclude RH one needs κ pairs of non-real eigenvalues to be zero pairs, i.e. one needs either $\kappa = 0$ — which would make the space a Hilbert space and return us to the definite case, hence to Corollary 4.11 — or one needs a separate argument that the κ available pairs are not used.

That separate argument is a signature or index computation: one must show that the negative part of the form, which has dimension κ, is entirely accounted for by root vectors with real eigenvalues, so that no non-real pair can be accommodated. This is the shape of the residual hypothesis of this book.

The preceding paragraph is a description of the shape of an argument, not an argument. This book does not prove that the negative part is so accounted for. Theorem D in Chapter 44 is the statement that it is; the hypothesis from which Theorem D is derived is stated there, and it is unproved. Nothing in the present chapter should be read as support for Theorem D beyond the structural claim that its indefinite form is the only available one.

4.4 The forced-ness of the indefinite form

We now argue that the indefinite structure is not a modelling choice. Three independent considerations point the same way.

4.4.1 From the involution

Proposition 4.12 is the first. Any implementation of the functional equation as an operator identity involves an involution $\mathcal{J}$ with $\mathcal{J}^2 = I$; if $\mathcal{J} = I$ the functional equation is trivial and if $\mathcal{J} = -I$ it is inconsistent with the existence of a non-zero invariant object; so $\mathcal{J}$ has both eigenvalues, and the form $[f, g] = \langle \mathcal{J}f, g\rangle$ is indefinite. There is no way to build a positive-definite form out of a non-trivial involution without discarding one eigenspace, and discarding an eigenspace discards half the functional equation.

4.4.2 From the Weil explicit formula

The second consideration is the sign structure of the explicit formula. In Theorem 3.18 the left-hand side is a sum over zeros and the right-hand side has an archimedean term, a pole term $2h(i/2)$, and a prime sum with a minus sign. Weil's positivity criterion is that

$$W(g) \;:=\; \sum_{\rho} h\!\left(\frac{\rho - \frac{1}{2}}{i}\right) \;\geq\; 0 \qquad \text{for } h = |\hat{g}|^2,$$

and the pole term contributes with the *opposite* sign to the prime sum. So the functional W is a difference of two positive quantities, and the assertion $W \geq 0$ is the assertion that one dominates the other. A quadratic form that is a difference of positives is, generically, indefinite; the assertion of RH is that this particular difference happens to be non-negative on a particular cone.

Remark 4.16 (Where the two poles go). Concretely, ζ has a single pole at $s = 1$, and the completed function $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ has poles at $s = 0$ and $s = 1$. In the operator picture these two poles are the two "extra" directions, and in a Pontryagin realisation they are the natural candidates for the negative part, suggesting κ small — possibly $\kappa = 1$ or $\kappa = 2$ depending on normalisation. We record this as a structural expectation with no proof attached; it is the kind of observation that motivates a construction and cannot substitute for one. It carries no status label because it is not a claim.

4.4.3 From the de Branges failure

The third consideration is the computed failure recorded by Conrey and Li [CL00]. In the de Branges framework the required condition on the relevant family of spaces is a positive-definiteness, and the computation shows it fails for ξ. Reading this as a diagnosis rather than as a dead end: the framework is correct and the definiteness is the wrong requirement. A framework with a signature in place of a sign is the natural repair, and de Branges' own theory

has an indefinite counterpart in the Pontryagin-space theory of Krein and Langer [KL78], where spaces of entire functions with κ negative squares are classified.

Remark 4.17 (Statement of the reading). We repeat the caution of Chapter 3: the interpretation of [CL00] as pointing to an indefinite repair is our reading and is not a theorem of that paper, which computes a sign and draws no such conclusion. It is recorded here as motivation.

4.4.4 The three together

None of the three is a proof that no positive-definite formulation exists. Theorem 4.10 is closer to such a proof: it says that a positive-definite formulation exists if and only if RH and simplicity hold. So a positive-definite formulation is not impossible — it is, conditionally on RH, available — but it is unavailable as an intermediate step, because obtaining it is the whole problem.

That is the precise sense in which the indefinite form is forced. It is forced not by an impossibility theorem but by the logical position: the definite form is the destination, so any hypothesis that is to be verified en route must be the indefinite one.

OBSTRUCTION O-5: no definite adjointness before the conclusion

Let Θ be an operator with $\sigma(\Theta) = Z$ satisfying the hypotheses of Theorem 4.10. Then a proof of (AS6-def) is a proof of RH together with simplicity of the zeros. Consequently no programme may treat (AS6-def) as a verifiable structural axiom. [N]

What it rules out. Any construction in which the positive-definiteness of the relevant form is asserted, assumed, or verified by a computation that does not amount to a proof of RH. This includes constructions in which a Hilbert space is defined by completing a pre-Hilbert space with respect to a form whose positivity is checked only on a dense subspace where positivity is easy, and constructions in which the domain is restricted until positivity holds — the latter being a failure of (R1) in Definition 3.4, since the restriction is made using knowledge of the answer.

What it does not rule out. It does not rule out proving RH by proving (AS6-def); that would be a proof, and the obstruction merely says it would be a proof and not a step. It does not rule out indefinite formulations, which is the point. And it does not say that the index κ is non-zero for every reasonable construction: it says that if $\kappa = 0$ can be proved, RH has been proved.

4.5 The corrected axiom in context

4.5.1 Relation to the firewall

The corrected axiom must be tested against Chapter 2. We record the test here in the format of Table 2.2, because (AS6-ind) is a hypothesis this book uses.

Davenport–Heilbronn. The function f has a functional equation, hence an involution, hence a Krein structure of the same shape. So f satisfies the analogue of (AS6-ind). This is expected and is not a refutation, because (AS6-ind) does not imply RH — and indeed the analogue of Theorem D fails for f, since f has off-line zeros. The separating clause is therefore not (AS6-ind) but the index statement, and the firewall test transfers to Theorem D. Chapter 44 records what the index computation gives for f and why the two cases differ; the difference is the Euler product, which f lacks.

Epstein ζ_Q, $h > 1$. Same analysis. The separating clause is again the Euler product.

DMV Beurling. No functional equation, hence no involution, hence (AS6-ind) does not even make sense for it. Test passed at the level of the hypothesis, and the reader should note what this means: the whole content is in the interaction of the functional equation with the Euler product.

Selberg Z_Γ, $\lambda_1 < \frac{1}{4}$. Z_Γ has a functional equation and an Euler product and Δ is self-adjoint, so this is the hardest test. The analogue of (AS6-ind) holds, with J trivial in fact, since Δ is self-adjoint on a genuine Hilbert space. But the parametrisation is quadratic, so the conclusion drawn from J-self-adjointness is $\lambda \in \mathbb{R}$, not s on the line, and Obstruction 2.5.3 applies. The separating clause is the linearity of the parametrisation.

> We record explicitly that in the first two rows the separating clause is deferred to Theorem D rather than being supplied here, and that a deferred separating clause is weaker than a supplied one. A referee applying the protocol of §2.6 to this book should note that step P7 — verify that each separating clause is a property ζ is known to have — is not completed for the first two rows, because the clause is the unproved hypothesis of Part VIII.

4.5.2 What the reader should carry to Chapter 44

Three things.

First, that the definite adjointness is not available, and that its unavailability is a theorem (Theorem 4.10) rather than a technical difficulty.

Second, that the available relation is $J\Theta^\dagger J^{-1} = I - \Theta$, which is a computation with the functional equation and carries no hidden content.

Third, that the gap between the two is exactly a statement about the negative part of the form — a signature, an index, a count of negative squares — and that this is why Part VIII is about an index theorem and not about a positivity inequality. The Criticality Barrier of Chapter 6 explains independently why it could not have been about a positivity inequality: a strict inequality with room to spare cannot certify a statement that sits at a critical point, and an index is an integer, which is the right kind of object to be exactly zero.

4.6 A finite-dimensional laboratory

The phenomena of the previous section are entirely visible in finite dimensions, and it is worth seeing them there, because the finite-dimensional picture makes clear which features are analytic difficulties and which are structural.

4.6.1 J-self-adjoint matrices

Fix $J = \operatorname{diag}(I_p, -I_q)$ on $\mathbb{C}^n$, $n = p + q$, with $q \geq 1$, and let $[x, y] = \langle Jx, y\rangle$. A matrix A is J-self-adjoint if $JA^\dagger J = A$, equivalently JA is Hermitian. Writing A in block form with respect to the decomposition,

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

the condition $JA^\dagger J = A$ reads

$$A_{11}^\dagger = A_{11}, \qquad A_{22}^\dagger = A_{22}, \qquad A_{21} = -A_{12}^\dagger.$$

So the diagonal blocks are Hermitian and the off-diagonal blocks are related by a sign. If $A_{12} = 0$ the matrix is Hermitian and has real spectrum; the sign in the coupling is what permits non-real eigenvalues.

Example 4.18 (The minimal example)**.** Take $p = q = 1$, $J = \operatorname{diag}(1, -1)$, and

$$A(b) \;=\; \begin{pmatrix} 1 & b \\ -b & -1 \end{pmatrix}, \qquad b \in \mathbb{R}.$$

Then $JA(b)$ is Hermitian, so $A(b)$ is J-self-adjoint. The characteristic polynomial is $\lambda^2 - (1 - b^2) \cdot 0 \cdot \lambda + (-1 + b^2) = \lambda^2 + (b^2 - 1)$, so

$$\lambda_{\pm} \;=\; \pm\sqrt{1 - b^2}.$$

For $|b| < 1$ the eigenvalues are real and distinct. At $|b| = 1$ they collide at 0, and A becomes a non-diagonalisable Jordan block. For $|b| > 1$ the eigenvalues are $\pm i\sqrt{b^2 - 1}$: a complex-conjugate pair.

The example exhibits the whole mechanism in two dimensions. Three features generalise.

Non-real eigenvalues occur in conjugate pairs. If A is J-self-adjoint and $Ax = \lambda x$ with $\Im\lambda \neq 0$, then $\bar{\lambda}$ is also an eigenvalue. Proof: $\sigma(A) = \sigma(JA^{\dagger}J) = \sigma(A^{\dagger}) = \overline{\sigma(A)}$.

Non-real eigenvectors are neutral. If $Ax = \lambda x$ with $\Im\lambda \neq 0$, then $\lambda[x, x] = [Ax, x] = [x, Ax] = \bar{\lambda}[x, x]$, so $(\lambda - \bar{\lambda})[x, x] = 0$ and $[x, x] = 0$. So the eigenvector spans a *neutral* line for the form. A neutral line requires a negative direction: a subspace on which the form vanishes identically has dimension at most $\min(p, q)$.

The count is bounded by q. This is the finite-dimensional case of Pontryagin's theorem.

Proposition 4.19 (The finite-dimensional Pontryagin bound)**.** *Let A be J-self-adjoint on $\mathbb{C}^{p+q}$ with $J = \operatorname{diag}(I_p, -I_q)$. Then the number of eigenvalues of A in the open upper half-plane, counted with algebraic multiplicity, is at most q.*

Proof. Let $\mathcal{L}_+$ be the sum of the root subspaces for eigenvalues in the open upper half-plane. For λ, μ eigenvalues with $\lambda \neq \bar{\mu}$, the corresponding root subspaces are $[\cdot, \cdot]$-orthogonal: this follows from $[(A - \lambda)^k x, y] = [x, (A - \bar{\lambda})^k y]$ by induction on k. Taking λ, μ both in the open upper half-plane, $\bar{\mu}$ is in the lower half-plane, so $\lambda \neq \bar{\mu}$ and the subspaces are orthogonal; taking $\lambda = \mu$, the same computation with x, y in the same root subspace gives $[x, y] = 0$ by nilpotency. Hence $\mathcal{L}_+$ is a neutral subspace, and a neutral subspace of $(\mathbb{C}^{p+q}, J)$ has dimension at most $\min(p, q) \leq q$. □

Remark 4.20 (Reading the bound)*.* The bound is the whole of what the indefinite structure delivers, and it is worth being clear about its strength. If $q = 0$, the space is a Hilbert space and there are no non-real eigenvalues — but $q = 0$ is the definite case, which by Corollary 4.11 is the conclusion. If $q = \infty$, the bound is vacuous. The useful regime is $1 \leq q < \infty$, where one learns that at most q conjugate pairs can be non-real, and one then has to rule out those q pairs by a separate argument.

The separate argument is an accounting argument: exhibit q linearly independent vectors with $[x, x] < 0$ that lie in root subspaces of real eigenvalues. If the negative part of the form is fully occupied by real spectrum, no neutral subspace of positive dimension remains, and every eigenvalue is real. This is the shape of the index statement referred to throughout.

4.6.2 Krein collisions and criticality

Example 4.18 shows how a pair of real eigenvalues becomes a complex pair: by colliding. This is not an accident of the example.

Definition 4.21 (Type of a real eigenvalue)**.** Let A be J-self-adjoint and $\lambda \in \mathbb{R}$ a semisimple eigenvalue with eigenspace E_λ. Since $[\cdot,\cdot]$ is non-degenerate on E_λ for semisimple real λ, the restriction has a signature (p_λ, q_λ). The eigenvalue is of *positive type* if $q_\lambda = 0$, of *negative type* if $p_\lambda = 0$, and of *mixed type* otherwise.

Theorem 4.22 (Krein's stability theorem)**.** *Let A be J-self-adjoint with all eigenvalues real and semisimple. If every eigenvalue is of definite type (positive or negative), then all J-self-adjoint matrices sufficiently close to A also have all eigenvalues real. If some eigenvalue is of mixed type, then arbitrarily close to A there are J-self-adjoint matrices with non-real eigenvalues.*

Proof. See [Bog74, Ch. V] or [GLR05, Ch. 1]. The mechanism is that eigenvalues of the same type repel and eigenvalues of opposite type can collide and leave the real axis; a mixed-type eigenvalue already contains a colliding pair. [T] □

In Example 4.18, the eigenvalue $+\sqrt{1-b^2}$ has positive type and $-\sqrt{1-b^2}$ has negative type; as $b \to 1$ they collide, and immediately after collision they leave the axis. The collision point $b = 1$ is a critical parameter value, and the real-spectrum property holds on the closed region $|b| \le 1$ and fails outside it.

Remark 4.23 (The connection to the Criticality Barrier). This is the same phenomenon as the Criticality Barrier of Chapter 6, in a different language. The set of J-self-adjoint operators with real spectrum is a closed set whose boundary consists of operators with collisions, and the operator relevant to ζ, if the picture is right, sits on that boundary rather than in the interior. Any argument that proves "real spectrum" by a stability or perturbation argument proves membership of the interior, which is a strictly stronger statement, and therefore cannot be correct. The Criticality Barrier states this for the de Bruijn–Newman constant, where the parameter is explicit and $\Lambda = 0$ is the boundary value; the present remark is the operator-theoretic shadow of it.

We flag that this is a structural parallel and not a derivation: we do not prove that the operator sits on the boundary, and the parallel between the de Bruijn–Newman flow and the Krein collision picture is at the level of shape.

4.6.3 Definitisable operators

For infinite-dimensional Krein spaces the useful class is Krein and Langer's definitisable operators.

Definition 4.24 (Definitisable)**.** A J-self-adjoint operator A on a Krein space with non-empty resolvent set is *definitisable* if there is a real polynomial $p \not\equiv 0$ with $[p(A)x, x] \ge 0$ for all x in a suitable domain.

Theorem 4.25 (Krein–Langer)**.** *A definitisable operator A has spectrum symmetric about $\mathbb{R}$; its non-real spectrum consists of finitely many eigenvalues with finite algebraic multiplicity, each a zero of any definitising polynomial p; and A admits a spectral function on $\mathbb{R}$ with finitely many critical points, the critical points being among the real zeros of p.*

Proof. [KL78]; see also [AI89, Ch. II]. [T] □

Definitisability is the correct infinite-dimensional substitute for the Pontryagin condition, and it is weaker: a Pontryagin space operator is automatically definitisable, but definitisable operators exist on Krein spaces with infinite negative index. For the purposes of this book it is the relevant hypothesis, and it is stated in Part VIII as such.

> Definitisability is a hypothesis, not a consequence of J-self-adjointness. A J-self-adjoint operator on a Krein space with infinite negative index can have non-real spectrum filling an open set. So the corrected axiom (AS6-ind) alone gives nothing about the spectrum without an additional structural condition, and the additional condition is where the difficulty has been moved — not removed.

4.7 The numerical range, in more detail

Because Theorem 4.8 turns on the distinction between spectrum and numerical range, we record the facts that make the distinction sharp.

Theorem 4.26 (Hausdorff–Toeplitz)**.** *For any bounded operator T on a Hilbert space, $W(T)$ is convex.*

Proof. [Hau19; Toe18]; see [GR97, §1.1]. [T] □

Theorem 4.27 (Spectral inclusion)**.** *For any bounded operator T, $\sigma(T) \subseteq \overline{W(T)}$. Equality of $\overline{W(T)}$ with $\overline{\mathrm{conv}}\,\sigma(T)$ holds if T is normal, and fails in general.*

Proof. Inclusion: if $\lambda \notin \overline{W(T)}$ then $|\langle (T-\lambda)x, x\rangle| \geq d\|x\|^2$ where $d = \mathrm{dist}(\lambda, \overline{W(T)}) > 0$, so $T-\lambda$ is bounded below and has closed range; the same applies to $T^\dagger - \bar{\lambda}$, so the range is dense; hence $T-\lambda$ is invertible. Normality: Proposition 4.5. Failure in general: Theorem 4.7. [T] □

Corollary 4.28 (Quantifying the excess)**.** *Let T be a bounded operator with $\sigma(T) \subseteq \{\Re z = \frac{1}{2}\}$. Then*

$$\sup_{\|x\|=1} \left| \Re\langle Tx, x\rangle - \tfrac{1}{2} \right| \;=\; \tfrac{1}{2}\,\|T + T^\dagger - I\|,$$

and this quantity vanishes if and only if $T + T^\dagger = I$, i.e. if and only if (AS6-def) *holds. It can be arbitrarily large while $\sigma(T)$ remains on the line.*

Proof. $\Re\langle Tx, x\rangle = \langle \frac{1}{2}(T+T^\dagger)x, x\rangle$, and the self-adjoint operator $\frac{1}{2}(T+T^\dagger) - \frac{1}{2}I$ has numerical radius equal to its norm. For the last statement take $T = \frac{1}{2}I + N$ with N nilpotent of large norm, as in Theorem 4.7 with large c. □

Corollary 4.28 makes the gap quantitative. The quantity $\|T + T^\dagger - I\|$ is the *defect* of AS6-as-written; RH says nothing about it, and (AS6-def) says it is zero. A construction that computes this defect and finds it non-zero has not refuted anything; it has discovered that the form is not the right one.

Remark 4.29 (A diagnostic). This yields a practical diagnostic for a proposed construction. Compute $\langle \Theta x, x\rangle + \langle x, \Theta x\rangle - \|x\|^2$ for explicit x in the constructed space. If it is not identically zero, the form is not the one for which (AS6-def) holds, and the correct next step is to identify the involution J for which the relation does hold, rather than to adjust the space until the defect vanishes. Adjusting the space until the defect vanishes is, by Corollary 4.11, adjusting the space until RH has been assumed.

4.8 How the error is usually made

It is useful to record the specific forms the error takes, since recognising them is faster than re-deriving the diagnosis.

4.8.1 Form 1: the inner product is defined by the pairing

One constructs a space of functions and defines $\langle f, g\rangle$ by an integral, then verifies $\langle \Theta f, g\rangle = \langle f, (I-\Theta)g\rangle$ by a change of variables using the functional equation. The change of variables is $x \mapsto 1/x$ or $s \mapsto 1-s$, and it is an involution. What has actually been verified is $[\Theta f, g] = [f, (I-\Theta)g]$ for the form $[f,g] = \langle \mathcal{J} f, g\rangle$, and the step from $[\cdot,\cdot]$ to $\langle\cdot,\cdot\rangle$ has been elided. Proposition 4.12 says the two differ by an indefinite factor.

4.8.2 Form 2: positivity checked on a convenient subspace

One verifies $\langle f, f\rangle \geq 0$ for f in a subspace where the integral is manifestly positive — typically functions supported where the kernel is positive — and takes the completion. The completion of a pre-Hilbert space with respect to a form that is positive on a subspace and indefinite on the whole is not the intended object; either the completion is smaller than intended, in which case Θ does not act on it, or the form is not positive on the completion.

4.8.3 Form 3: the domain is trimmed

One restricts the domain of Θ until the adjointness relation holds with the definite form. The restriction is typically a decay or growth condition. The difficulty is that the required condition, unwound, is a statement about the location of the zeros: the eigenvectors with non-real eigenvalues are precisely the ones excluded. This is a failure of requirement (R1) of Definition 3.4, and it is circular in the technical sense of Chapter 1.

4.8.4 Form 4: normality asserted from symmetry of the kernel

One observes that the integral kernel $K(x,y)$ of the constructed operator satisfies $K(y,x) = \overline{K(x,y)}$ and concludes self-adjointness. Kernel symmetry gives symmetry of the operator on the domain where the integrals converge absolutely; self-adjointness is a statement about domains, and for the operators arising here the domain question is exactly the deficiency-index question of §3.12. This is Form 3 in a different disguise.

Remark 4.30 (Not a criticism of any particular author). The four forms are abstracted from the general pattern and are not attributed. We record them because a reader constructing an operator will be tempted by at least one of them, and because the diagnosis of Theorem 4.8 is most useful when it can be applied quickly.

4.9 The corrected axiom in the Mellin picture

To make the indefiniteness concrete rather than abstract, we carry out the computation in the one setting where every object can be written down: the Mellin transform on the half-line. This subsection is a worked model. It is not the construction of Part V, and no claim is made that the operator appearing here has spectrum $\mathcal{Z}$.

4.9.1 The space and the involution

Let $\mathcal{H} = L^2\big((0,\infty), \frac{dx}{x}\big)$ with inner product $\langle f, g\rangle = \int_0^\infty f(x)\overline{g(x)}\,\frac{dx}{x}$. The Mellin transform

$$(\mathcal{M}f)(\tau) \;=\; \int_0^\infty f(x)\, x^{-i\tau}\,\frac{dx}{x}$$

is, up to normalisation, a unitary $\mathcal{H} \to L^2(\mathbb{R}, \frac{d\tau}{2\pi})$; it converts the dilation group into translations. Under $\mathcal{M}$, the substitution $s = \frac{1}{2} + i\tau$ identifies the target with functions on the critical line.

Define the involution $(\mathcal{J}f)(x) = f(1/x)$. Then:

$$\mathcal{M}(\mathcal{J}f)(\tau) \;=\; \int_0^\infty f(1/x)x^{-i\tau}\frac{dx}{x} \;=\; \int_0^\infty f(u)\,u^{i\tau}\,\frac{du}{u} \;=\; (\mathcal{M}f)(-\tau),$$

so $\mathcal{J}$ becomes $\tau \mapsto -\tau$, which in the variable $s = \frac{1}{2} + i\tau$ is exactly $s \mapsto 1-s$. This is the assertion that the functional-equation involution is implemented by $x \mapsto 1/x$ on the multiplicative half-line, which is Riemann's original observation in operator form.

Proposition 4.31 (The form and its signature)**.** *Set $[f,g] := \langle \mathcal{J}f, g\rangle$. Then $[\cdot,\cdot]$ is a bounded non-degenerate Hermitian form on $\mathcal{H}$, and in the Mellin picture*

$$[f,g] \;=\; \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}(-\tau)\,\overline{\hat{g}(\tau)}\,d\tau, \qquad \hat{f} := \mathcal{M}f.$$

Its positive and negative eigenspaces are

$$\mathcal{H}_+ = \{f : f(1/x) = f(x)\}, \qquad \mathcal{H}_- = \{f : f(1/x) = -f(x)\},$$

both infinite-dimensional. Hence $(\mathcal{H}, [\cdot,\cdot])$ is a Krein space of infinite index.

Proof. Boundedness and Hermiticity are immediate from unitarity and self-adjointness of $\mathcal{J}$; non-degeneracy from invertibility. The displayed Mellin formula is the computation above. The eigenspace description is $\mathcal{J}f = \pm f$. Both are infinite-dimensional: symmetrise and antisymmetrise any f supported in $(1,2)$, and the space of such f is infinite-dimensional. □

Remark 4.32 (Infinite index is the default). Proposition 4.31 shows that the naive Krein structure has infinite index, so Pontryagin's theorem (Proposition 4.15(iv)) does not apply, and by the notice after Theorem 4.25 the J-self-adjointness alone gives nothing. Making the index finite — or, more realistically, making the operator definitisable — is exactly the work that a construction has to do, and it is done by the arithmetic, not by the analysis. In the picture of Part V the reduction comes from passing to a quotient in which the symmetric part is largely killed; the residual negative index is small, and its smallness is an arithmetic statement.

We state this as a description of the intended shape and not as a result. Whether the index is finite in the construction of Part V is addressed there, and the honest position is recorded in the status table of Part VIII.

4.9.2 The operator and its J-adjointness

Let A be the generator of dilations, $A = x\frac{d}{dx}$, densely defined on $C_c^\infty(0,\infty)$. Under Mellin, A becomes multiplication by $i\tau$. Set $\Theta_0 := \frac{1}{2}I + A$, so that under Mellin Θ_0 is multiplication by $\frac{1}{2} + i\tau = s$.

Proposition 4.33 (The model J-adjointness)**.** *With $\mathcal{J}$ and $[\cdot,\cdot]$ as above, Θ_0 satisfies*

$$\Theta_0^{[\dagger]} \;=\; I - \Theta_0$$

on the natural domain, where $\Theta_0^{[\dagger]} = \mathcal{J}\Theta_0^\dagger\mathcal{J}$.

Proof. Two involutions must be distinguished. Write $\mathcal{J}f(x) = f(1/x)$ and $\mathcal{J}'f(x) = x^{-1}f(1/x)$. In the Mellin picture Θ_0 is multiplication by $\frac{1}{2} + i\tau$, and the Hilbert adjoint of multiplication by m is multiplication by $\bar{m}$, here $\frac{1}{2} - i\tau$. Conjugation by $\mathcal{J}$ substitutes $\tau \mapsto -\tau$, so $\mathcal{J}\Theta_0^\dagger\mathcal{J}$ is multiplication by $\frac{1}{2} + i\tau$, which is Θ_0 again and not $I - \Theta_0$. Conjugation by $\mathcal{J}'$ instead substitutes $s \mapsto 1 - s$ in the variable $s = \frac{1}{2} + i\tau$, because the weight x^{-1} shifts the Mellin variable by one; the resulting operator is multiplication by $1 - (\frac{1}{2} + i\tau) = \frac{1}{2} - i\tau$, which is $I - \Theta_0$. Taking $\mathcal{J}'$ as the fundamental symmetry gives the stated identity. □

Remark 4.34 (On the normalisation). The distinction drawn in the proof of Proposition 4.33 is the commonest source of confusion in this area, and it is worth isolating. There are two involutions in play: $x \mapsto 1/x$ without a weight, which is $\tau \mapsto -\tau$ and hence $s \mapsto \bar{s}$ on the line; and $x \mapsto 1/x$ with the weight x^{-1}, which is $s \mapsto 1 - s$. Only the second implements the functional equation. Computations that use the first and report the second produce a definite form, since the unweighted reflection is a genuine isometric involution whose associated form is the original inner product after relabelling.

The weight is where the indefiniteness enters. We state this as a slogan and label it as such: *the functional equation costs a weight, and the weight costs definiteness*.

4.10 Prior formulations, and what changes

We close by recording what is new in the diagnosis and what is not.

4.10.1 What is not new

That the numerical range of a normal operator is the convex hull of its spectrum is classical (Proposition 4.5). That J-self-adjoint operators on Pontryagin spaces have at most κ pairs of non-real eigenvalues is Pontryagin's theorem of 1944 [Pon44]. That indefinite inner-product spaces are the natural home for problems with a functional equation is implicit in the de Branges theory and explicit in the Krein–Langer theory of spaces with κ negative squares [KL78]. That the definite condition in a de Branges-type framework fails for ξ is the computation of Conrey and Li [CL00].

4.10.2 What the present chapter adds

Three things, each modest.

First, the explicit statement and proof that AS6 as usually written forces normality (Theorem 4.3) and hence a numerical-range confinement strictly stronger than RH (Theorem 4.8). The computation is three lines and cannot be new in substance; what is new, as far as we know, is its use as a diagnostic criterion applied systematically.

Second, the equivalence of the definite form with RH-plus-simplicity (Theorem 4.10), which converts the diagnosis from "this is stronger than needed" into "this is the conclusion". The proof is a sketch and depends on hypotheses that are not verified for any constructed operator; the full argument is deferred to Chapter 44.

Third, the quantitative form (Corollary 4.28) and the four error patterns of §4.8, which are offered as working tools rather than as results.

> None of the three amounts to progress on the Riemann Hypothesis, and none is claimed to. Their function is to fix the formulation of the hypothesis that Part VIII will state, so that a reader arriving at Chapter 44 knows why the hypothesis has the shape it has and can see that a definite version would have been either false, circular, or a proof.

4.10.3 The revised axiom list

For reference, the corrected AS6 in the form used for the remainder of this book:

> **AS6 (corrected).** There is a Krein space $(\mathcal{K}, [\cdot,\cdot])$ with fundamental symmetry J arising from the functional-equation involution, and a closed densely defined operator Θ on $\mathcal{K}$ with $\sigma(\Theta) = \mathcal{Z}$, such that $\Theta^{[\dagger]} = I - \Theta$; equivalently, $D = -i(\Theta - \frac{1}{2}I)$ is J-self-adjoint. The operator Θ is required to be definitisable in the sense of Definition 4.24.

The definitisability requirement is a hypothesis about the construction and is discharged, or not, in Part V; the notice after Theorem 4.25 records that it does not follow from J-self-adjointness. The passage from AS6-corrected to RH requires, in addition, the index statement, which is Theorem D of Chapter 44 and is derived there from a hypothesis this book does not prove.

4.11 Pontryagin's theorem and what the indefinite form actually gives

Having argued that the form is indefinite, we should say what an indefinite form is good for, since a reader may reasonably suspect that abandoning definiteness abandons the tool. It does not, and the reason is a theorem of Pontryagin.

Definition 4.35 (Pontryagin space)**.** A *π_κ-space* is a Krein space $(\mathcal{H}, [\cdot,\cdot])$ whose negative index $\kappa = n_-$ is finite.

Theorem 4.36 (Pontryagin)**.** *[K] Let $\mathcal{H}$ be a π_κ-space and let A be a $[\cdot,\cdot]$-self-adjoint bounded operator on $\mathcal{H}$. Then A has a κ-dimensional non-negative invariant subspace. Moreover the non-real spectrum of A consists of at most κ conjugate pairs of eigenvalues, counted with algebraic multiplicity.*

The original is [Pon44]; modern treatments are in [Bog74] and [AI89], and the operator-theoretic refinements used later in this book are in [KL78].

Remark 4.37 (Why this is the right tool)*.* Three features of Theorem 4.36 match the three constraints identified in this chapter.

First, its conclusion is a *count*, not a bound on a size. By N9 of the archive, an argument whose last step is a size bound is at a ceiling; an argument whose last step is a count is not.

Second, the count is an integer determined by the inertia of the form, and integers are stable under deformation while being equalities at a point. By the Criticality Barrier this is the only kind of positivity statement that can be at the right place.

Third, and least obviously: the theorem gives *at most* κ pairs off the real axis, not *none*. It is therefore compatible with RH being false, and does not prove too much. A hypothetical theorem asserting no non-real spectrum would, by the argument of §4.2, be asserting definiteness in disguise.

Corollary 4.38 (The shape of what must be proved)**.** *[C] Suppose one constructs a π_κ-space and a $[\cdot,\cdot]$-self-adjoint operator whose spectrum is the set $\{\rho\}$ of non-trivial zeros under $\rho \mapsto$ the appropriate coordinate, with $\kappa = 0$. Then there are no non-real eigenvalues and* RH *follows. If $\kappa = 1$ and one additionally knows that ζ has no real zeros in $(0,1)$ — which is a classical unconditional fact — then the single admissible pair is excluded and* RH *again follows.*

Proof. The first statement is immediate from Theorem 4.36. For the second: by Theorem 4.36 at most one conjugate pair is non-real; the reality of ζ on the real axis forces the zeros to

come in quadruples $\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}$ unless ρ is real or on the critical line, and the classical absence of real zeros in $(0,1)$ removes the real case. A quadruple contributes two conjugate pairs, exceeding $\kappa = 1$; hence no quadruple occurs. Conditional on the construction, which is Problem (G.2). □

Remark 4.39 (Where the conditionality sits). Corollary 4.38 is labelled [C] and its hypothesis is the construction, not RH. The reader should verify this: the proof uses Pontryagin's theorem, the functional equation's four-point orbit, and the non-vanishing of ζ on $(0,1)$. None of these is RH or equivalent to it. The construction is the open problem, and it is stated in Chapter 44.

4.12 Weakening self-adjointness does not remove the excess

A natural response to Theorem 4.8 is to keep the definite form and weaken self-adjointness: require only that D be symmetric, or that it be self-adjoint on a smaller domain, or that the intertwining hold only on a dense subspace. We record why each of these fails to remove the excess.

4.12.1 Symmetric but not self-adjoint

If D is symmetric with dense domain and $\Theta^\dagger \supseteq I - \Theta$, then Θ is *formally* normal on the domain but need not be normal. The excess appears to be removed. It is not: the spectrum of a symmetric non-self-adjoint operator with unequal deficiency indices fills a half-plane, and with equal non-zero indices depends on the choice of self-adjoint extension. In the first case the operator has too much spectrum to be the zeros; in the second, the conclusion depends on a choice, and Proposition 3.24 shows the choice cannot be made canonically without additional structure.

Proposition 4.40 (No free lunch from symmetry)**.** *[T] Let D be symmetric with deficiency indices (m,n). If $m \neq n$, $\sigma(D)$ is a closed half-plane or all of $\mathbb{C}$, and in particular is not a discrete subset of $\mathbb{R}$. If $m = n \neq 0$, then for each self-adjoint extension the spectrum is real, but distinct extensions give distinct spectra, and the family of spectra obtained is not contained in any proper subset of $\mathbb{R}$ determined by D alone.*

Proof sketch. Standard; see [RS75]. The last clause follows from the Krein–von Neumann parametrisation of extensions by unitaries between the deficiency subspaces, which moves eigenvalues continuously and, for $m = n \geq 1$, transitively enough that no point of $\mathbb{R}$ is excluded a priori. □

Remark 4.41 (The consequence). Weakening self-adjointness to symmetry does not weaken the conclusion from "strictly stronger than RH" to "exactly RH". It replaces a false statement with an incomplete one. The excess is not removed; the argument simply stops earlier.

4.12.2 Intertwining on a dense subspace only

If $[\Theta f, g] = [f, (I-\Theta)g]$ is verified only for f, g in a dense subspace $\mathcal{D}$, the identity extends to the closure only if Θ is bounded, and Θ is not bounded in any of the realisations of interest — its spectrum is unbounded. So the extension requires a domain argument, and the domain argument is the content. This is the same observation as Step 3 of the worked audit in §5.7, and it is the most frequent unnoticed gap in this family of arguments.

4.12.3 Restricting to a smaller space

If one restricts to a subspace on which the form is definite — and such subspaces exist, by the definition of an indefinite form — then Θ need not preserve the subspace, and if it does, the

restricted spectrum is a subset of the full spectrum and the conclusion is about that subset only. Pontryagin's theorem is precisely the statement of what can be extracted this way: a maximal non-negative invariant subspace of dimension κ exists, and its complement carries the rest. The indefinite treatment does not discard the definite argument; it accounts for the part of the space on which the definite argument fails.

Remark 4.42 (Summary of the three attempts). Each of the three weakenings either fails to give a conclusion or gives one about a smaller object. None of them yields exactly RH from a definite form. The indefinite formulation is not a preference; it is what is left.

4.13 The chapter in five lines

For a reader who will meet Theorem D in Chapter 44 and wants to arrive with the argument of this chapter compressed:

1. The functional equation supplies an intertwining $\Theta \mapsto I - \Theta$ with respect to *some* form.
2. If that form is positive-definite, Θ is normal and its numerical range lies on the critical line — strictly more than RH, and equivalent to RH plus simplicity of the zeros.
3. The form supplied by the functional equation is not positive-definite: the involution implementing $s \mapsto 1 - s$ carries a weight, and the weight carries the indefiniteness. This is a computation, not a preference (Remark 4.34).
4. An indefinite form of finite negative index κ still yields a conclusion, by Pontryagin's theorem: at most κ conjugate pairs off the real axis. The conclusion is a count, is deformation-stable in the right sense, and does not prove too much.
5. Therefore the object to construct is a Pontryagin space with $\kappa \leq 1$, and the remaining input is the classical absence of real zeros in $(0, 1)$. That construction is Problem (G.2), and it is not solved in this book.

Remark 4.43 (The one thing to carry forward). If only one sentence survives the reader's passage from here to Chapter 44, it should be: *the indefiniteness is forced by the normalisation of the functional equation, and it is what prevents the argument from proving more than* RH. Every other statement in this chapter is in service of that one.

What this chapter does not establish

1. It does not establish that an operator Θ with $\sigma(\Theta) = Z$ exists. Everything in this chapter is conditional on such a Θ being given, and the construction is the subject of Part V.
2. Theorem 4.3 assumes equality of domains, $D(\Theta^\dagger) = D(\Theta)$. For unbounded operators this is a real hypothesis, and a version of AS6 asserting only an inclusion of graphs would give a symmetric rather than a self-adjoint D, hence would not force normality. The chapter does not treat that weaker version, whose analysis is the deficiency-index problem of §3.12.
3. Theorem 4.10 is stated with hypotheses — completeness of root vectors, non-degeneracy on root subspaces — that are not verified for any constructed Θ in this book. The proof is a sketch relying on the Krein–Langer structure theory, and the full argument appears in Chapter 44.
4. Proposition 4.15(iv) requires the Pontryagin condition $\kappa < \infty$. This chapter does not establish that any construction relevant to ζ yields finite index; Remark 4.16 offers a structural expectation and explicitly disclaims proof.
5. Nothing here establishes Theorem D of Chapter 44, or provides evidence for it. The argument of §4.4 establishes that the indefinite form is the only one available before the conclusion, which is a statement about what may be assumed, not about what is true.

6. The firewall analysis of §4.5 defers the separating clause for two of the four objects to Theorem D. The firewall test for (AS6-ind) is therefore incomplete, and the incompleteness is recorded rather than resolved.
7. The reading of [CL00] in §4.4 is an interpretation and not a result of that paper.

Chapter 5

What would falsify this book

This chapter states, in advance, what would show that the central claims of this book are wrong. It is placed in Part I rather than in an appendix, and the placement is deliberate: a falsification criterion stated after the argument has been made is a defence, whereas one stated before is a commitment.

The chapter has three parts. The first sets out the falsification predicates for each of the book's four main lines — the Krein/index line, the measured signature, the no-go theorems, and the firewall — naming for each a concrete theorem or computation whose existence would refute it. The second is a catalogue of the standard failure modes in this subject, each with a historical example, and a statement of how this book avoids it. The third is a short list of commitments, phrased so that a reader can check compliance.

None of this is a substitute for the mathematics. A book that states its falsification conditions clearly and is wrong is still wrong. What the statement buys is that the wrongness is locatable, and that the reader is not required to trust the author's judgement about which claims are load-bearing.

5.1 Why this chapter is here and not at the end

5.1.1 The asymmetry of unverifiable claims

Long mathematical arguments have a characteristic failure mode: an error early in the development is absorbed by later flexibility. A constant is adjusted, a hypothesis is added, a definition is refined, and by the time the argument reaches its conclusion the original error has been distributed across so many places that no single statement is false. The result is a document in which every individual claim survives scrutiny and the whole is unsound.

The defence against this is to fix, before the development begins, the specific statements on which everything turns, and to state what would refute them. Then a later adjustment that saves a claim by weakening it is visible as a change to the commitment rather than as a technical refinement.

5.1.2 The specific risk in this book

The risk here has a name. This book derives RH from a hypothesis it does not prove. The temptation, in such a project, is to let the hypothesis drift: to state it strongly at the start and, as difficulties appear, to weaken it until it becomes either unfalsifiable or equivalent to RH. Either outcome would destroy the value of the derivation — the first because nothing would follow from an unfalsifiable hypothesis, the second because the derivation would be circular.

The commitments of §5.4 are designed to make drift visible. In particular, the hypothesis is stated in Chapter 44 in a form that is not equivalent to RH — it is strictly stronger, and Chapter 44 says so — and the non-circularity is established by exhibiting an object satisfying RH for which the hypothesis fails, in the sense of Definition 1.11.

5.2 Falsification predicates

We now state, for each of the four lines, what would constitute refutation. In each case we distinguish three grades:

- **Refutation**: a proof that a stated claim of this book is false.
- **Vitiation**: a proof that a stated claim is true but useless — for instance, that a hypothesis from which RH is derived is equivalent to RH, so that the derivation is circular.
- **Erosion**: a result showing that a claimed scope is smaller than stated, without refuting the claim in its restricted scope.

Erosion is not refutation, and we record it separately because the temptation to treat an erosion as harmless is strong and the cumulative effect of unrecorded erosions is a book whose stated scope is fiction.

5.2.1 Line 1: the Krein/index line

The claim, stated informally here and precisely in Chapter 44, is: there is a Krein space and an operator Θ on it, constructed in Part V, satisfying the corrected AS6 of Chapter 4, such that a certain index vanishes if and only if RH holds; and Theorem D derives RH from a hypothesis about that index which is stated precisely and is currently unproved.

F1.1 *Refutation of the construction.* A proof that the space constructed in Part V is degenerate — that the form $[\cdot,\cdot]$ is degenerate, or that Θ is not closable, or that $\sigma(\Theta) \neq \mathcal{Z}$. Any of these refutes the line at its base. The specific computation that would settle it is the verification of the resolvent identity of Chapter 29: if the resolvent constructed there fails to be the resolvent of a closed operator, the construction fails.

F1.2 *Refutation of the adjointness.* A proof that $J\Theta^{\dagger}J^{-1} \neq I - \Theta$ for the J constructed in Part V. This is a computation with the functional equation and is checkable in finitely many steps once the construction is fixed.

F1.3 *Refutation of definitisability.* A proof that Θ is not definitisable in the sense of Definition 4.24. By the notice following Theorem 4.25, definitisability does not follow from J-self-adjointness, so this is a genuine possibility. If Θ is not definitisable, the Krein–Langer machinery does not apply and the index statement has no content.

F1.4 *Vitiation by circularity.* A proof that the hypothesis of Theorem D is equivalent to RH. Concretely: a proof that for every object satisfying the axioms under which Theorem D is stated, RH for that object implies the hypothesis. This would make the derivation a restatement. Chapter 44 guards against this by exhibiting an object for which RH holds and the hypothesis fails; a demonstration that the exhibited object does not satisfy the axioms would remove the guard and reopen F1.4.

F1.5 *Refutation of the derivation.* An error in the proof of Theorem D from its hypothesis. This is the ordinary kind of refutation and needs no special protocol.

F1.6 *Erosion by hypothesis strength.* A proof that the hypothesis of Theorem D implies a known open conjecture strictly stronger than RH — for instance, that it implies simplicity of all zeros, or the Grand Riemann Hypothesis for a class of L-functions. This would not refute anything, but it would show that the hypothesis is harder than advertised, and the advertised difficulty is part of the claim. This book states in Chapter 44 what the hypothesis is known to imply; a demonstration that it implies more is an erosion to be recorded.

5.2.2 Line 2: the measured signature

Part VI reports a computed quantity — a signature, or a count of negative squares in a finite-dimensional truncation — and this book uses it in a restricted way. The restriction is stated in

Chapter 1 in the warning box 1.1 and repeated here: the measured value is not inserted into Theorem D, and no step of any proof depends on it.

F2.1 *Refutation of the computation.* An independent computation, with a different implementation and different arithmetic, giving a different value. The computation is specified in Part VI to the level of the algorithm, the precision, and the parameters, so that an independent run is possible.

F2.2 *Refutation by pollution.* A proof that the truncation scheme used is subject to spectral pollution in the sense of Theorem 3.16, so that the computed signature is an artefact. This is a live possibility and Part VI addresses it explicitly; a proof that the treatment there is inadequate would refute the measurement's interpretation while leaving the numbers correct.

F2.3 *Erosion by scope.* A demonstration that the measured quantity, even if correct, is not the quantity appearing in Theorem D — for instance, that the finite truncation's signature does not converge to the infinite-dimensional index. This is the most likely form of trouble and is why the measurement is not used in any proof.

> We restate the commitment: no theorem in this book has a numerical computation in its proof. Numerical results carry status [E] and appear as motivation, as consistency checks, and as a record of what was actually observed. If a reader finds a proof in this book whose validity depends on a computed number, that is a violation of the commitment and should be reported as such.

5.2.3 Line 3: the no-go theorems

Chapter 6 catalogues twenty-three no-go theorems. Each is a statement of the form "method M cannot prove RH", with a precise specification of M.

F3.1 *Refutation by proof.* For any no-go theorem Nk, a proof of RH by a method falling within the scope of Nk refutes Nk. This is the cleanest possible falsification and we regard it as the appropriate one.

F3.2 *Refutation by counterexample to the argument.* For each Nk the proof is given or referenced in Chapter 6. An error in the proof refutes the theorem in the ordinary way.

F3.3 *Erosion by scope inflation.* A demonstration that the scope claimed for Nk — the "rules out" column — is wider than the theorem supports. This is the failure mode we regard as most likely for this line, since scope statements are prose and prose is elastic. Chapter 6 therefore states for each Nk both what it rules out and what it does not rule out, and a reader finding a method that falls outside the stated scope has not refuted anything.

F3.4 *Vitiation by triviality.* A demonstration that some Nk rules out only methods that nobody has proposed and that no reasonable person would propose. Such a theorem would be true and worthless. We have tried to include only obstructions that close routes that have actually been attempted, and the "kills what" column of the table in Chapter 6 names the attempt.

5.2.4 Line 4: the firewall

F4.1 *Refutation of a firewall object.* A proof that one of the four objects of Chapter 2 does not have the properties claimed — for instance, a proof that the Davenport–Heilbronn function has all its zeros on the critical line, or that the DMV system does not satisfy the Prime Number Theorem. Each claimed property is either proved in Chapter 2 or cited to a specific published theorem, so this form of refutation is a refutation of the literature.

F4.2 *Refutation of a firewall passage.* A proof that this book's own hypothesis, or a component of the positive programme, is satisfied by one of the four objects. By Lemma 2.2 this refutes the programme. Section 4.5 records that for two of the four objects the separating clause is deferred to Theorem D, so this form of refutation is currently open in those two cases, and we say so.

F4.3 *Erosion by a fifth object.* The construction of a Dirichlet series satisfying every hypothesis this book's programme uses, with a false Riemann Hypothesis. This would refute the programme. It would also be a major result independent of this book. We regard its existence as unlikely but record that we have no argument against it.

F4.4 *Vitiation by untestability.* A demonstration that this book's own hypothesis is not stated to a standard at which the firewall test can be applied. By the standard of Definition 2.3 this would place this book's programme in the same category as the ones Chapter 3 criticises, and would be the most damaging outcome short of an outright error.

5.3 Standard failure modes, with historical examples

The following six modes account for most of the unsuccessful attempts on this problem that we have examined, and several of them account for errors that reached print in respectable venues. For each we give the pattern, a historical instance where one is available, the reason it is seductive, and the specific commitment this book makes.

5.3.1 Circularity

Pattern. A step of the argument uses, in disguised form, the conclusion. The disguise is usually a domain restriction, a convergence hypothesis, or a choice of contour — all things that look technical and turn out to encode zero-free information.

Instance. The most instructive class is the family of "proofs" that manipulate $\log \zeta(s)$ or ζ'/ζ in the strip. These functions have branch points at the zeros, and any manipulation valid on a region presupposes that the region is zero-free. An argument that integrates $\log \zeta$ along a path in the strip, or expands it in a series convergent in $\Re s > \frac{1}{2}$, has assumed RH. A closely related and subtler instance is the use of the Euler product, or of any identity derived from it, in $\Re s > \frac{1}{2}$; the product converges only in $\Re s > 1$ and its extension to $\frac{1}{2} < \Re s < 1$ is equivalent to RH.

Why it is seductive. The circular step is usually not the interesting step. Attention is on the main construction, and a convergence condition looks like bookkeeping.

Commitment. Every hypothesis used in this book is checked against Definition 1.11, which requires the exhibition of an object for which RH holds and the hypothesis fails. Chapter 6 records this check for each no-go theorem, and Chapter 44 for the main hypothesis. Where the check has not been carried out, this is stated.

5.3.2 The Hilbert–Pólya trap

Pattern. A self-adjoint operator is produced, or a symmetric one is asserted to be self-adjoint, and RH is deduced from reality of the spectrum — without an exact identification of the spectrum with the zeros.

Instance. Chapter 3 places several. The purest form is Theorem 3.1: the existence of a self-adjoint operator with the zeros as eigenvalues is a theorem if RH holds and false otherwise, so it can never be a route. A subtler form is the asymptotic identification discussed in Remark 3.15, where an operator's eigenvalue counting function matches that of the zeros; two sequences with the same counting function to $O(\log T)$ need not share a single term.

Why it is seductive. Self-adjointness is a strong and familiar property, and the deduction from it is genuinely valid. The error is entirely in the identification, which is the boring half.

Commitment. This book requires the three conditions of Definition 3.4 of any spectral claim, including its own, and reports which are met and which are not. Chapter 4 further requires that the adjointness relation be proved rather than assumed, and records that the relation available is indefinite.

5.3.3 Proving too much

Pattern. An argument, if correct, would prove a statement known to be false — typically for one of the firewall objects of Chapter 2.

Instance. The classical instance is any argument from the functional equation alone, refuted by Davenport–Heilbronn (Obstruction 2.2.4). A more sophisticated instance is any "elementary" argument from the Euler product and the Prime Number Theorem, refuted by Diamond–Montgomery–Vorhauer (Obstruction 2.4.3). A third is any argument from self-adjointness of a geometrically constructed operator, refuted by Randol's surfaces via Obstruction 2.5.3.

Why it is seductive. The argument is often correct as far as it goes; what is wrong is the belief that the hypotheses used are specific to ζ.

Commitment. The firewall protocol of §2.6 is applied to this book's own hypothesis in §4.5 and in Chapter 44. Where the test is incomplete, §4.5 says so explicitly.

5.3.4 The criticality error

Pattern. A positivity or gap statement is proved with room to spare — an inequality of the form $Q \geq \delta > 0$, or a spectral gap uniform in a parameter — and is offered as a route to RH. But RH sits at a critical point, and a statement with room to spare is false.

Instance. The cleanest is the de Bruijn–Newman constant. Newman [New76] conjectured $\Lambda \geq 0$ and Rodgers and Tao [RT20] proved it; RH is equivalent to $\Lambda \leq 0$. So RH $\iff \Lambda = 0$, and any method producing $\Lambda \leq -\delta$ for some $\delta > 0$ produces a false statement. Chapter 6 states this as the Criticality Barrier and derives its consequences for positivity-based methods. A second instance is the Selberg eigenvalue conjecture: $\lambda_1 \geq \frac{1}{4}$ is sharp for congruence quotients, since the tempered spectrum begins at $\frac{1}{4}$, so no method can give $\lambda_1 \geq \frac{1}{4} + \delta$.

Why it is seductive. Most analytic methods naturally produce inequalities with a margin, because margins are what allow error terms to be absorbed. A method that produces an exact equality is unusual.

Commitment. This book's main line is an index statement — an integer that is exactly zero — rather than an inequality. Chapter 6 states the Criticality Barrier and Chapter 44 records that Theorem D is of equality type. Any inequality appearing in this book with a claimed margin is, by that fact, not on the critical path, and the text says which side of the line each result is on.

5.3.5 Asymptotic versus exact confusion

Pattern. A relation is established in an asymptotic or averaged sense and used as though it were exact.

Instance. Remark 3.15 treats the case of counting functions. Another instance is the treatment of the explicit formula: the error terms in a truncated explicit formula are $O(\cdot)$ statements with implied constants, and an argument that requires the exact identity cannot use the truncated form. A third is the use of averaged results — density theorems, moment estimates, mollified second moments — to draw conclusions about individual zeros. Levinson's method

[Lev74] and its successors give proportions of zeros on the line, currently above two fifths [Con89]; no amount of improvement in the proportion gives all zeros, and the methods are known to be limited well below 100%.

Why it is seductive. Asymptotic statements are what analytic number theory produces, and they are usually the right currency. The problem is specific to a conclusion that is a statement about every zero.

Commitment. Every asymptotic statement in this book is marked as such, with its error term. Where a proof requires an exact identity, the text says so. The distinction between the proportion of zeros on the line and the location of all zeros is maintained throughout.

5.3.6 Finite verification as proof

Pattern. The zeros are verified to lie on the line up to some height, and this is offered as evidence or, in the worst cases, as a step.

Instance. The Riemann Hypothesis has been verified for the first 10^{13} zeros [PT21], and for very large individual heights [OS88]. This is a large computation and a genuine achievement. It is also, as evidence, weak, for a reason that is specific to this problem: the quantity governing the fluctuation of the zeros is $S(T) = \pi^{-1} \arg \zeta(\frac{1}{2} + iT)$, which is $\Omega((\log T)^{1/3}(\log\log T)^{-7/3})$ unconditionally but grows so slowly that at $T = 10^{13}$ it has barely reached values of order 3. Phenomena governed by $\log\log T$ are not visible at any computationally accessible height.

Historical parallel. The classical cautionary tale is the inequality $\pi(x) < \mathrm{li}(x)$, verified for all computationally accessible x and proved false by Littlewood [Lit14], with the first crossing now known to lie below 10^{317} but not located. A second is Mertens' conjecture $|M(x)| < \sqrt{x}$, verified extensively and disproved by Odlyzko and te Riele [OR85], again with no explicit counterexample known. Both concern quantities directly tied to the zeros.

Why it is seductive. 10^{13} is a large number.

Commitment. No computation in this book is used as a step in a proof. Computations are reported with status [E], with the algorithm and precision specified, and with the explicit statement that they are consistency checks.

5.3.7 A seventh mode: scope drift

We add one mode not on the standard list, because it is the one this book is structurally most exposed to.

Pattern. A claim is stated at one strength, and later text refers to it at a greater strength. No individual sentence is false; the aggregate is.

Instance. The most common form in the literature on this problem is the migration of a conditional result into unconditional language across a long document: "assuming RH, X holds" becomes "X holds" by chapter forty.

Commitment. Every statement in this book carries a status macro, and the macros are checked mechanically against a table; the audit protocol of §6.14 describes the check. Every chapter ends with a section stating what it does not establish. Forward references to Theorem D are, throughout Part I, accompanied by the statement that Theorem D is derived from an unproved hypothesis.

5.4 Commitments

We list the commitments in a form that permits checking.

C1 This book does not prove the Riemann Hypothesis and does not claim to. Any sentence that implies otherwise is an error.

C2 The hypothesis from which RH is derived is stated once, in Chapter 44, in a fixed form, and is not weakened later. If it is restated, the restatement is flagged and the two forms are proved equivalent.
C3 The hypothesis is non-circular in the sense of Definition 1.11: an object is exhibited for which RH holds and the hypothesis fails.
C4 No numerical computation appears in the proof of any theorem. Computations carry status [E].
C5 Every claim carries a status macro from the six-label system of the front matter, and the macro is the weakest one applicable.
C6 Every no-go theorem in Chapter 6 states both what it rules out and what it does not rule out.
C7 Every chapter ends with a section stating what the chapter does not establish.
C8 Citations are to real, checkable sources, with the year and venue given; where a citation could not be verified it is marked in the bibliography source.
C9 Where this book's reading of another author's work goes beyond what that author claims, the reading is identified as a reading.
C10 No superlatives, and no language asserting importance, novelty, or promise. The claims are the mathematics; the prose does not add to them.

Commitments C1–C10 are checkable by a reader without specialist knowledge, and a violation of any of them is a defect regardless of whether the mathematics is correct. We regard a violation of C1, C2, or C4 as disqualifying and the remainder as serious.

5.5 What a successful refutation would look like

It is worth describing the shape of the most likely refutations, so that a reader looking for one knows where to look.

Most likely: F1.3, non-definitisability. The construction of Part V produces a Krein space of infinite index (compare Remark 4.32), and definitisability is a substantive hypothesis. A proof that the constructed Θ is not definitisable would remove the machinery on which Theorem D depends without touching anything else. We regard this as the most probable failure point and Part VIII states what is and is not established about it.

Second most likely: F3.3, scope inflation. The twenty-three no-go theorems are stated with scope claims in prose. Prose scope claims are the weakest part of any such catalogue, and we expect that some of ours are too wide. The remedy is stated per-theorem in Chapter 6: each has a "does not rule out" clause whose purpose is to bound the claim.

Third: F2.3, measurement scope. The measured signature of Part VI is a finite-dimensional quantity and the object in Theorem D is not. The convergence question is open and is stated as open.

Least likely but most damaging: F4.3, a fifth firewall object. A Dirichlet series with a functional equation, an Euler product, and zeros off the line would refute a great deal more than this book. We record it because the firewall's force depends on the absence of such an object, and absence of a counterexample is not a theorem.

5.6 Two further failure modes worth naming

5.6.1 The epsilon at the end

Pattern. An argument reduces RH to an inequality of the form $A \le B$, establishes $A \le B + \epsilon$ with ϵ small and explicit, and treats the remaining ϵ as a technical residue to be removed by refinement.

Why it is a distinct mode. It differs from the criticality error in that no false statement is being proved; the residue is genuinely small and the inequality genuinely nearly holds. The error is in the estimate of how much work the last step is. In every instance we know of where the reduction is to a critical statement, the last epsilon has turned out to require a new idea rather than a refinement.

Instance. The spectral-gap hierarchy of §3.4 is the cleanest available. Selberg's 3/16 was obtained in 1965 using Weil's bound for Kloosterman sums. Sixty years of work on functoriality has moved the constant to 975/4096, and the remaining gap to 1/4 is 0.0119 .... The remaining gap is equivalent to the Ramanujan conjecture for Maass forms, which is not a refinement of anything currently available. The pattern is: the first 95% of the interval is closed by a sequence of substantial theorems, and the last 5% is the whole problem.

A second instance. Zero-density estimates. The classical density hypothesis, $N(\sigma, T) \ll T^{2-2\sigma+\epsilon}$, would give much of what RH gives for prime distribution in short intervals. It has been approached over a century, and the recent improvement of Guth and Maynard [GM24] is the first advance on the exponent in the relevant range in over fifty years. The improvement is real and the remaining distance to RH is not diminished by it, because the density hypothesis is strictly weaker than RH and its proof would not localise a single zero.

Commitment. Where this book reduces a statement to another statement, the text says whether the residue is expected to be a refinement or a new idea, and labels the judgement as a judgement. Where a chain of reductions ends at an unproved hypothesis, the hypothesis is stated in full rather than described as a technical gap.

5.6.2 The natural-object fallacy

Pattern. An object is constructed, is observed to be natural — functorial, canonical, invariant under the expected symmetries — and naturalness is taken as evidence that it has the desired property.

Why it is seductive. Naturalness is a real heuristic and it works often. The difficulty is that it is not a mathematical property, and in this subject the counter-models are natural too: the Epstein zeta function of a quadratic form is as canonical an object as exists in the subject, and it has zeros in $\Re s > 1$.

Commitment. No step of any argument in this book appeals to naturalness. Where naturalness is mentioned it is as motivation for a definition, and the text says so.

5.7 A worked audit

To show that the failure-mode catalogue is usable rather than decorative, we audit a composite argument. The argument below is not attributed; it is assembled from the recurring features of several unsuccessful proposals and is written in the form its authors would recognise. The point of the exercise is that the diagnosis takes one page.

5.7.1 The argument

Step 1. Define $\mathcal{H}$ to be the completion of $C_c^\infty(0,\infty)$ with respect to $\langle f, g\rangle = \int_0^\infty \int_0^\infty K(x,y) f(x) \overline{g(y)} \, \frac{dx}{x} \frac{dy}{y}$, where K is built from the theta function and satisfies $K(x,y) = \overline{K(y,x)}$.

Step 2. Define Θ on $\mathcal{H}$ by an explicit integral operator, and verify using the theta functional equation that $\langle \Theta f, g\rangle = \langle f, (I-\Theta)g\rangle$ for $f, g \in C_c^\infty$.

Step 3. Conclude that $\Theta^\dagger = I - \Theta$, hence $D = -i(\Theta - \frac{1}{2}I)$ is self-adjoint.

Step 4. Show, by a residue computation, that the eigenvalues of Θ are the non-trivial zeros of ζ.

Step 5. Conclude RH.

5.7.2 The audit

Step 1. The form is Hermitian by the kernel symmetry, but Hermitian is not positive-definite. The claim that $\mathcal{H}$ is a Hilbert space requires $\langle f, f\rangle \geq 0$ for all $f \in C_c^\infty(0,\infty)$, which is a separate statement about K. This is failure mode "circularity" in the form catalogued in §4.8 as Form 2. **Query: is $\langle f, f\rangle \geq 0$ proved, and for which f?**

Step 2. The verification uses the functional equation, which is implemented by an involution $\mathcal{J}$. By Remark 4.34 the involution carries a weight, and the identity actually verified is $[\Theta f, g] = [f, (I-\Theta)g]$ for the indefinite form $[f, g] = \langle \mathcal{J} f, g\rangle$. **Query: which form is the verification in — $\langle\cdot,\cdot\rangle$ or $[\cdot,\cdot]$?**

Step 3. Even granting Step 2 in the definite form, the passage from "$\langle \Theta f, g\rangle = \langle f, (I-\Theta)g\rangle$ for f, g in a dense subspace" to "$\Theta^\dagger = I - \Theta$" is a domain statement: it gives $\Theta^\dagger \supseteq I - \Theta$, i.e. symmetry, not self-adjointness. By Proposition 3.24 the deficiency indices must be computed. **Query: what are the deficiency indices?**

Step 4. A residue computation identifies formal eigenvalues, i.e. values λ for which the eigenvalue equation has a solution in some function space. Whether the solution lies in $\mathcal{H}$ is a separate question, and it is the question on which everything turns: the solutions corresponding to off-line zeros are exactly the ones whose membership in $\mathcal{H}$ is in doubt. **Query: for which λ is the eigenfunction in $\mathcal{H}$, and is the criterion independent of** RH**?**

Step 5. By Theorem 4.8, if Steps 1–4 hold as stated then the argument has proved not only RH but the strictly stronger statement $\overline{W(\Theta)} \subseteq \{\Re z = \frac{1}{2}\}$, and by Theorem 4.10 it has proved simplicity of the zeros as well. **Query: does the argument, examined, produce simplicity? If not, one of Steps 1–4 is false; if so, which step produces it?**

5.7.3 The diagnosis

Four queries, each answerable in principle, and each locating a specific place. In the composite argument, Step 1 fails: the form is indefinite. The reason the failure survives casual reading is that positivity of $\langle f, f\rangle$ is checked for f supported in $(1,\infty)$, where the kernel is positive, and the extension to all of $C_c^\infty(0,\infty)$ is not made.

The general lesson is the one Chapter 4 states as Corollary 4.11: in any argument of this shape, ask first whether the form is positive-definite, and expect the answer to be no. The final query — does the argument produce simplicity? — is the fastest diagnostic available, because simplicity of the zeros is open, and an argument that produces it as a by-product has produced too much.

5.8 Falsification and the status system

The six-label status system of the front matter is not decoration; it is the mechanism by which the commitments of §5.4 are made checkable. This section explains how to read it adversarially.

5.8.1 What each label commits to

[T] commits to a complete proof, either given in this book or cited to a specific published theorem. A [T] label on a statement whose proof is not present and not cited is a violation.

[C] commits to a proof from a stated hypothesis, with the hypothesis identified at the point of use. A [C] label without an identified hypothesis is a violation.

[O] commits to nothing beyond the statement being open, and is the label for problems and conjectures. A statement labelled [O] may not be used as a step.

[E] commits to a computation with a specified algorithm and precision. It may not be used as a step.

[N] labels a no-go theorem. It commits to a proof and, additionally, to a scope statement of what is and is not ruled out.

[K] labels a statement that is known in the literature and is reproduced here for completeness, with attribution. It commits to the attribution being correct.

5.8.2 How to audit the labels

The adversarial reading is: for each [T], find the proof; for each [C], find the hypothesis; for each [N], find the scope statement; for each [E], find the algorithm; for each [K], find the attribution. A label whose commitment cannot be discharged is a defect, and the defect is checkable without following the mathematics.

We record that this is the audit we have applied, and §6.14 describes how.

5.8.3 Downgrades

Where a statement's label changes between chapters, this is a downgrade or an upgrade and must be visible. The rule adopted is: a statement introduced at [C] is not later referred to at [T] unless the hypothesis has been discharged and the discharge is cited. In particular, Theorem D is [C] throughout, and every reference to it in Parts I–VII carries the reminder that its hypothesis is unproved.

> The single most common way for a book of this kind to become dishonest is for a [C] to become a [T] through repetition rather than through proof. A reader wishing to check one thing only should check this one.

5.9 The asymmetry of confirmation

A chapter on falsification invites the question of what would confirm the book's claims. The answer is short and worth stating, because the asymmetry is not rhetorical.

For the no-go theorems, confirmation is proof, and they are proved or cited; nothing further is available or needed. For the firewall, confirmation is the four counter-models, which exist. For the derivation of RH from the hypothesis of Theorem D, confirmation would be a proof of the hypothesis, and there is none.

What is *not* confirmation, and is sometimes offered as such in this subject:

- *Numerical agreement.* The measured signature of Part VI agrees with what the theory predicts. This is a consistency check. It would be evidence against the theory if it disagreed, and it is not evidence for the theory when it agrees, because the class of theories agreeing with any finite computation is large. The asymmetry here is the standard one and we do not overstate it: a disagreement would have been informative.

- *Structural analogy.* The programme has the same shape as the function-field proof (§3.10). Analogy motivates; it does not confirm. Chapter 6 catalogues the specific structural features present in the function-field case and absent over Spec $\mathbb{Z}$, and the list is long.
- *Absence of refutation.* That we have not found a counterexample to the hypothesis of Theorem D is not evidence for it. We have looked in the places described in §5.5 and report what we found; the report is a record of activity, not a probability statement.
- *Agreement with random-matrix predictions.* The statistics of the zeros agree with the GUE to the accuracy computed [Odl87]. As noted in §3.7, this is conditional evidence about distribution, computed for zeros already known to be on the line, and it cannot bear on whether other zeros exist elsewhere.

Remark 5.1 (Why the asymmetry is stated). The purpose is to prevent the accumulation of non-evidence into an impression. Four separate observations, each individually acknowledged to be non-confirmatory, are sometimes presented together in a way that suggests a cumulative case. There is no cumulative case here. The mathematical content is: a set of no-go theorems, which are proved; a firewall, whose objects exist; a construction, which is carried out; and a derivation from an unproved hypothesis, which is a derivation. Nothing else.

5.10 A checklist for a referee

We close with the checklist we would want applied to this book, in the order that finds problems fastest.

R1 Locate the statement of the hypothesis of Theorem D in Chapter 44. Check that it is a single statement, that it is not equivalent to RH by inspection, and that the non-circularity witness required by Definition 1.11 is exhibited.

R2 Check that the derivation of RH from that hypothesis does not use any statement labelled [O] or [E].

R3 Apply the firewall protocol of §2.6 to the hypothesis. Fill in Table 2.2. Note that §4.5 states two rows are deferred; check whether Chapter 44 completes them.

R4 Check that the construction of Part V satisfies (R1) of Definition 3.4: that membership in the space and the domain is decidable without knowing the zeros.

R5 Check definitisability. This is the point identified in §5.5 as most likely to fail.

R6 For each of the twenty-three no-go theorems in Chapter 6, check that the "does not rule out" clause is wide enough to be honest. Scope inflation is the second most likely defect.

R7 Sample twenty statements at random and audit their status labels by the procedure of §5.8.

R8 Search the text for occurrences of the word "therefore" following a statement labelled [C] and check that the conditional is carried through.

R9 Check the bibliography entries used in load-bearing citations, in particular [DMV06], [Con99], [RT20], [CL00], [KS03], against the published record.

R10 Verify that no chapter is missing its "what this chapter does not establish" section, and that the sections are substantive rather than formulaic.

5.11 The historical record: five refuted plausibilities

The failure modes of §5.3 are not hypothetical. This section records five statements about ζ and its relatives that were believed on good grounds and are false. Each is a calibration point: it fixes how much weight numerical and heuristic evidence can carry in this subject.

5.11.1 The Mertens conjecture

Let $M(x) = \sum_{n \le x} \mu(n)$. Mertens conjectured in 1897, on the basis of a computation to $x = 10^4$, that $|M(x)| < \sqrt{x}$ for $x > 1$ [Mer97]. The conjecture implies RH, and implies in addition that the zeros are simple and that their imaginary parts are linearly independent over $\mathbb{Q}$.

Theorem 5.2 (Odlyzko–te Riele). *[T]*

$$\limsup_{x\to\infty} M(x)x^{-1/2} > 1.06 \quad \textit{and} \quad \liminf_{x\to\infty} M(x)x^{-1/2} < -1.009.$$

In particular the Mertens conjecture is false.

The proof [OR85] proceeds by computing 2000 zeros to high precision and applying an inhomogeneous Diophantine approximation argument to the sum $\sum_\rho \frac{x^{\rho-1/2}}{\rho\zeta'(\rho)}$; no counterexample is exhibited. The smallest known lower bound for a counterexample is beyond 10^{16}, and heuristics place the first violation near $e^{1.59\times 10^{40}}$.

Remark 5.3 (The calibration). The conjecture survived eighty-eight years and computation to $x = 10^{10}$. The quantity $M(x)x^{-1/2}$ has mean-square growth governed by $\sum_\rho |\zeta'(\rho)|^{-2}$, and the excursions that refute the conjecture require a near-resonance among many zeros. No finite computation could have reached the resonance. This is failure mode *finite verification as proof* in its most instructive form: the computation was not merely insufficient, it was insufficient by a factor no increase in computing power will close.

5.11.2 The Pólya conjecture

Let $L(x) = \sum_{n \le x} \lambda(n)$ with λ the Liouville function. Pólya conjectured in 1919 that $L(x) \le 0$ for $x \ge 2$; this too implies RH.

Theorem 5.4 (Haselgrove). *[T]* $L(x) > 0$ *for infinitely many* x.

Haselgrove's disproof [Has58] is again non-constructive, resting on Ingham's method [Ing42] for converting information about the zeros into oscillation statements. Tanaka later located the least counterexample at $x = 906\,150\,257$ [Tan80]. The conjecture had been verified to $x = 800\,000$.

Remark 5.5 (Ingham's lesson). Ingham's 1942 paper is the source of a general principle worth stating in the language of this book: a conjecture of the form "$S(x) \le 0$ for all x", where S has an explicit-formula expansion over the zeros, is false unless there is a structural reason for the zero contributions never to align. In the absence of such a reason — and linear independence of the ordinates over $\mathbb{Q}$ is precisely the statement that there is no such reason — almost-periodicity forces the sum to come arbitrarily close to the sum of the absolute values of its terms. Both the Mertens and Pólya conjectures fell to this. The principle is a no-go theorem in the sense of this book, and it is catalogued as such in Chapter 6.

5.11.3 The Turán conjecture

Turán showed [Tur48] that if $\sum_{n \le x} \lambda(n)/n \ge 0$ for all $x \ge 1$ then RH follows, and conjectured the inequality on the basis of computation to $x = 1000$. Haselgrove's paper disproves this as well. The least counterexample is at $x = 72\,185\,376\,951\,205$.

5.11.4 The sign of $\pi(x) - \mathrm{li}(x)$

Every computation ever performed gives $\pi(x) < \mathrm{li}(x)$. Littlewood proved in 1914 that the difference changes sign infinitely often [Lit14]; Skewes gave the first explicit bound on a sign

change [Ske55], and Lehman reduced it to below 1.65×10^{1165} [Leh66]. Subsequent work has brought the bound to roughly 1.4×10^{316}.

Remark 5.6 (The relevance). The relevance to this book is not the size of the bound but the structure of the proof. Littlewood's argument does not exhibit a sign change; it shows that no structural obstruction to one exists, and then invokes almost-periodicity. Every one of the four refutations in this section has that shape. A reader who wishes to know what kind of argument is likely to settle a question in this subject should note that in each case the settling argument was not a better computation and not a refinement of an existing estimate, but a change of the object being studied — from the sum to its Mellin transform, from the transform to the zeros, and from the zeros to their Diophantine properties.

5.11.5 The de Branges positivity conditions

The fifth calibration point is closest to this book's own subject matter. In a sequence of papers de Branges proposed to prove RH by exhibiting ζ within his theory of Hilbert spaces of entire functions, the key step being a positivity condition on a family of functions.

Theorem 5.7 (Conrey–Li). *[T] The positivity hypotheses used in* [Bra92] *are not satisfied by the relevant family; explicit counterexamples exist.*

The counterexamples of [CL00] are constructed from Dirichlet L-functions and are, in the terminology of Chapter 2, firewall objects: they satisfy every structural hypothesis the argument uses while violating the positivity it needs.

Remark 5.8 (What this calibrates). Two things. First, that a positivity condition strong enough to give RH is, by the Criticality Barrier of §6.6, at the boundary of what is true, so that a proposed positivity which is comfortably true is proving something weaker and a proposed positivity which is comfortably strong is false. Second, that the refutation came from the firewall method: one produces an object satisfying the hypotheses and violating the conclusion. This is the method Chapter 2 formalises, and §4.5 applies it to this book's own central hypothesis.

We record explicitly that this book's Theorem D is an *indefinite* condition, and that the reason Chapter 4 gives for the indefiniteness being forced is the same reason the definite conditions of the de Branges programme were refutable. This is not a claim that the present programme succeeds. It is a claim about which of the two failure modes — too weak or too strong — it is exposed to.

5.12 Refuting the derivation is not refuting the hypothesis

A confusion worth forestalling. This book contains statements at four logical levels, and refutation at one level does not propagate to the others in the way a hurried reading might suggest.

- **L1** RH itself. Nothing in this book bears on its truth except conditionally.
- **L2** The hypothesis of Theorem D. Unproved. If it is false, RH is unaffected; only this route to it closes.
- **L3** The derivation (hypothesis) ⇒ RH. If the derivation has an error, both L1 and L2 are unaffected; the book's central conditional claim fails.
- **L4** The no-go theorems and the firewall. These are unconditional and independent of L1–L3. If Theorem D's hypothesis is refuted tomorrow, Chapters 2, 3, 4 and 6 stand unchanged.

Proposition 5.9 (Independence of the levels). *[T] The statements at L4 are theorems of ordinary mathematics whose proofs use neither* RH *nor the hypothesis of Theorem D. Consequently no refutation at L1, L2 or L3 affects them.*

Proof. Immediate from inspection of the proofs: each no-go theorem in Chapter 6 is proved from stated hypotheses about a proposed mechanism, and the four firewall objects of Chapter 2 are constructed unconditionally. No step invokes RH or the hypothesis of Theorem D. Note that this is a statement about the proofs as written and is exactly the sort of statement that the audit protocol of §6.14 is designed to check; we do not ask the reader to take it on trust. □

Remark 5.10 (The practical consequence). The consequence for a reader deciding how to spend attention: L4 is the part of the book that is unconditionally true or unconditionally false, and it is checkable now. L2 and L3 are the part that is conditional. A reader who believes the programme will not succeed may still find L4 usable, and we have written L4 so that it can be extracted and used without the rest.

Conversely, a reader who wants to know whether the programme succeeds should go directly to Chapter 44 and apply checks R1–R5 of §5.10. The intervening chapters are the reasons the hypothesis has the form it has, not evidence that it is true.

5.13 Where each failure mode was checked for

The commitments of §5.4 are worth little unless the checks were actually made. This section records where.

Circularity. Checked at every point where a positive-definite form, a Hilbert space, a domain, or a spectral decomposition is introduced. The four forms the error takes are catalogued in §4.8. The places in this book where the check was non-trivial are: the definition of the space in Part V; the self-adjointness argument in Part V; the identification of the point spectrum in Part VI; and the definitisability argument in Part VIII. In the last of these the check is not discharged, and the residue is exactly the hypothesis of Theorem D. We regard the fact that the circularity check localises to a single named statement as the principal structural claim of the book, and it is the claim the referee checklist R1–R5 is designed to test.

The Hilbert–Pólya trap. Checked by Definition 3.4: every operator introduced in this book is required to satisfy (R1), (R2) and (R3), and the requirement is verified where the operator is constructed rather than assumed at the point of use. Theorem 3.1 is the reason the requirement exists.

Proving too much. Checked by the firewall protocol of §2.6, applied to the central hypothesis in §4.5 and to each proposed mechanism surveyed in Chapter 3. Additionally checked by Theorem 4.10: an argument of the shape considered would, if it worked in the definite form, prove simplicity of the zeros, which is open. Any future version of the argument that produces simplicity is thereby known to be wrong.

The criticality error. Checked by §6.6. Every positivity statement used in the book is classified as gapped, marginal, or equality-type, and the classification is stated at the point of use. Gapped positivity is not used as a step toward RH anywhere.

Asymptotic versus exact. Checked wherever an asymptotic statement appears. The specific instances flagged are: the Weyl law comparison of §3.3, where the conclusion drawn is a non-existence statement and is therefore safe; and the prolate operator of [CM22a], where the asymptotic character of the result is stated explicitly in Remark 3.15.

Finite verification as proof. Checked by Warning 1.1 and by the status label [E], which may not be used as a step in any argument. The numerical content of Part VI is a consistency check and is described as one wherever it appears. Theorem 5.2 is the calibration.

Scope drift. Checked by requiring every [N] statement to carry an explicit "does not rule out" clause. Chapter 6 contains twenty-three such clauses; the audit protocol of §6.14 records how each was reviewed.

5.14 Falsification of the numerical component

The measured signature of Part VI requires separate treatment, because numerical statements fail in ways that theorems do not.

5.14.1 What is computed

Three quantities: the index of a truncated form, its signature, and the eigenvalue distribution of a finite section. Each is computed at several truncation levels and in several precisions.

5.14.2 The failure predicates, stated numerically

F5.1 *Instability under truncation.* If the computed signature changes when the truncation parameter is increased at fixed precision, the computation is not measuring an invariant of the infinite-dimensional object and the report of Part VI is void. The reported values are stable across the range tested; the range is stated, and it is finite.

F5.2 *Precision dependence.* If the computed signature changes when the working precision is increased at fixed truncation, the computation is dominated by rounding. The condition numbers are reported alongside the values for exactly this reason.

F5.3 *Disagreement with the predicted index.* The theory of Part VIII predicts a specific finite index. If the measured index disagrees at any truncation level in the stable range, either the theory or the computation is wrong, and the book states that it does not know which without further work.

F5.4 *Non-reproducibility.* The computation is specified to the level of algorithm and parameters. If an independent implementation returns different values in the stable range, the report is void.

Remark 5.11 (What agreement would and would not show). Agreement at every truncation level tested shows that the finite sections behave as the theory says finite sections behave. It does not show that the infinite-dimensional object has the predicted index, because the passage from finite sections to the limit is exactly the analytic content, and it is not supplied by computation. This is the same asymmetry as §5.9 and the same lesson as Theorem 5.2: the range in which a resonance could hide is not the range that was computed.

5.14.3 A standard the computation is held to

Every numerical claim in this book carries: the algorithm, the parameters, the precision, the observed condition number, and the range over which the reported value is stable. A numerical claim missing any of the five is a defect, and we invite the reader to search for one.

5.15 If RH is false

The possibility is not remote enough to leave unaddressed, and the discipline of this chapter requires that we say what would follow.

Suppose a zero ρ_0 with $\Re\rho_0 \neq \frac{1}{2}$ is exhibited. Then:

(**a**) The hypothesis of Theorem D is false. It implies RH; hence a counterexample refutes it. The refutation is immediate and requires no examination of Part VIII.

(**b**) The derivation of Part VIII is unaffected in its validity. A valid implication with a false antecedent remains valid. Whether it remains interesting is a separate question and we do not assert that it would.

(**c**) The no-go theorems of Chapter 6 are unaffected, by Proposition 5.9, and several become more interesting rather than less: a catalogue of reasons why the standard routes cannot prove RH is not weakened by RH being false.

(**d**) The firewall of Chapter 2 is unaffected and gains a fifth object, namely ζ itself.
(**e**) The Criticality Barrier of §6.6 would acquire a definite answer: $\Lambda_{\mathrm{dBN}} > 0$ strictly, and the de Bruijn–Newman constant would be located above zero rather than at it.

Remark 5.12 (The asymmetry once more). It is worth noticing which parts of the book are exposed to (a) and which are not, because the proportion is a design decision. Chapters 1–6 and the whole of the obstruction theory are unconditional. The conditional content is concentrated in Parts VII and VIII, and within those, in one hypothesis. We have arranged the book this way deliberately, and a reader is entitled to regard the arrangement as an admission: we do not know whether the hypothesis is true, and we have written so that its failure would not take the rest with it.

Remark 5.13 (On what a counterexample would look like). By Theorem 5.2 and the pattern of §5.11, the expected route to a counterexample — if one exists — is not computation. Verification of RH has been carried to height 3×10^{12} with a rigorous error analysis [PT21], and to greater heights by sampling methods. If the first violation were governed by a resonance of the kind that governs the Mertens counterexample, its height would be beyond any conceivable computation, and the discovery would come from an oscillation theorem in Ingham's style rather than from a search. We state this because it bears on how the numerical evidence for RH should be weighed, which is: less heavily than its extent suggests. The point is not that the computations are unreliable — the rigorous ones are — but that the quantity they constrain and the quantity at issue are separated by a gap no computation narrows.

5.16 The retraction commitment

A falsification chapter is worth less if the authors have not said in advance what they will do when a predicate fires. We say it here.

If a no-go theorem is refuted. The entry is withdrawn from the archive of Chapter 6 and replaced by a statement of the counterexample. Entries depending on it are re-examined and their status labels adjusted. No no-go theorem in this book is used to support another, so the damage is local by construction; §6.9 records the dependency structure, which is empty.

If a firewall object is shown to be misdescribed. The description is corrected and the protocol of §2.6 is re-run for every mechanism tested against it, including this book's own. The firewall is a test, and a test with a faulty component gives no information until the component is repaired.

If the hypothesis of Theorem D is refuted. The programme of Parts V–VIII closes. We would report this, and the report would consist of the counterexample together with an assessment of which of the four design constraints of §6.10 the counterexample violates — because that is the information a successor would need.

If an error is found in the derivation. The derivation is withdrawn and the error located publicly. We regard the location of the error as the substantive part of such a report: a withdrawal that does not say where the argument fails is not usable by anyone.

If a numerical claim fails to reproduce. The claim is withdrawn and the discrepancy reported with both sets of parameters. By §5.14 the numerical claims are not load-bearing for any theorem, so no theorem is affected; the report of Part VI is.

Remark 5.14 (Why this is in Part I). Because a retraction commitment made after a refutation is not a commitment. The value of stating it now is that a reader can hold us to it, and the cost of stating it now is that we have described in advance the circumstances in which the book fails. We judge the second to be the point rather than the cost.

What this chapter does not establish

1. It does not establish that the falsification predicates of §5.2 are exhaustive. There may be ways for this book to be wrong that are not on the list, and the list is a commitment about what we will regard as refutation, not a theorem about what refutation can consist of.
2. It does not establish that the commitments of §5.4 are met. They are commitments; compliance is checked by the audit protocol of §6.14 and by the reader.
3. The catalogue of failure modes in §5.3 is not a classification theorem. The six modes are patterns abstracted from examination of unsuccessful attempts, and the classification has no formal status.
4. The historical instances are given as illustrations. We have not audited the arguments in which they occurred and make no claim about any particular author's work beyond what the cited results state.
5. The ranking in §5.5 is a judgement about likelihood and is not a mathematical claim. It carries no status label.
6. Nothing in this chapter constitutes evidence for any claim of this book. Stating what would refute a claim is not support for it.

Chapter 6

The obstruction archive and proof audit

6.1 What an archive entry is

The five preceding chapters have argued that a theory of why the standard routes fail is a mathematical object rather than a mood. This chapter is that object. It contains twenty-three no-go theorems, each stated precisely, each proved or sketched with a forward reference to its full proof, and each accompanied by two scope clauses of equal standing.

Definition 6.1 (Archive entry)**.** An *archive entry* consists of six components.

(E1) A *name* and an index N*k*.

(E2) A *precise statement*, quantified, with all hypotheses explicit.

(E3) A *proof*, or a proof sketch together with a forward reference to the chapter in which the full proof appears.

(E4) A *rules-out clause*: the class of proposed arguments the theorem eliminates, described so that membership in the class is decidable.

(E5) A *does-not-rule-out clause*: the nearest arguments the theorem leaves standing. This clause carries the same weight as (E4) and is the one we ask a hostile reader to attack first.

(E6) A *status label*. Every archive entry is [N] except where noted; entries reproducing a published theorem carry [K] with attribution.

> Component (E5) is the component that makes the archive honest, and it is the component a promotional treatment would omit. A no-go theorem without a does-not-rule-out clause is an assertion that a subject is closed, and no theorem in this book asserts that.

6.1.1 How the entries were selected

The twenty-three entries are not a survey of the literature on obstructions. They are the obstructions encountered in the course of constructing the positive programme of Parts V–VIII: each one was found by attempting a route and being stopped. This origin has two consequences worth stating.

First, the archive is not complete. There are certainly obstructions to routes we did not attempt. The archive claims to be a correct catalogue of what it contains, not an exhaustive catalogue of what exists.

Second, the archive is biased toward the routes that looked most promising, since those are the routes we attempted. A reader who believes the correct route is one we did not try will find little here about it, and is right to observe that our silence is not evidence.

6.1.2 Grouping

The entries fall into four groups by the kind of argument they obstruct.

- **Group A — amplification and additive correlation** (N1–N5, N18–N22). Routes that attempt to extract information about $\Re\rho$ from k-fold correlations of the von Mangoldt function or from Dirichlet series with amplified singular support.

- **Group B — geometric transport** (N6, N7, N12, N13, N14, N16). Routes that attempt to transport the function-field proof to Spec $\mathbb{Z}$.
- **Group C — positivity** (N8, N9, N15, N17, N23). Routes that attempt to derive RH from an inequality.
- **Group D — spectral** (N10, N11). Routes that attempt to realise the zeros as a spectrum with a polarisation.

The Criticality Barrier (N8) and the absolute-value ceiling family (N9) receive extended treatment in §6.6 and §6.7, because both are frequently misstated and the misstatements are consequential.

6.2 Group A: amplification and additive correlation

6.2.1 N1 — Support doubling

Theorem 6.2 (Support-doubling no-go, N1). *[N] Let $h \in C_c^\infty(\mathbb{R})$ with $\operatorname{supp} h \subseteq [-A, A]$, and for $T > 0$ set $g_T(x) = a\,h(x-T) + b\,h(x+T)$. Then*

$$\operatorname{supp}(g_T * \tilde{g}_T) \subseteq [-2T - 2A,\ 2T + 2A],$$

where $\tilde{g}(x) = \overline{g(-x)}$. Consequently, in the Guinand–Weil explicit formula applied to $g_T * \tilde{g}_T$, the prime powers contributing are exactly those with $\log n \le 2T + 2A$, i.e. $n \le e^{2T+2A}$. If the zero-side signal to be detected has size $\asymp e^{\delta S}$ with $S = 2T$, and the arithmetic side is bounded unconditionally by $\ll S\,e^{S/2}$, then a conclusion can be extracted only if $\delta > \frac{1}{2}$.

Proof sketch. The support statement is the elementary fact that $\operatorname{supp}(f * g) \subseteq \operatorname{supp} f + \operatorname{supp} g$. The arithmetic bound is the unconditional estimate $\sum_{n \le X} \Lambda(n) n^{-1/2} = 2\sqrt{X} + o(\sqrt{X})$, applied with $X = e^S$, which gives $\ll e^{S/2}$ up to the logarithmic factor arising from the weight. Comparing exponents gives the stated threshold. Full proof: Chapter 29. □

OBSTRUCTION What N1 rules out

Every route that proposes to gain by widening the support of the test function. The gain in the number of primes reached is exponential in the support radius, and the unconditional bound on their contribution grows at exactly the rate that cancels it.

Remark 6.3 (The corrected statement). Earlier formulations of this analysis, and several in the literature, produce a threshold of $\delta > \frac{1}{4}$ rather than $\delta > \frac{1}{2}$. The discrepancy is a confusion between $\operatorname{supp}(g)$ and $\operatorname{supp}(g * \tilde{g})$: the explicit formula is applied to the autocorrelation, whose support is twice as wide, and it is the autocorrelation's support that determines the range of primes. The factor of two in the support becomes a factor of two in the exponent. We state this because the $\delta > \frac{1}{4}$ window, if it existed, would be reachable, and the $\delta > \frac{1}{2}$ threshold is not.

Remark 6.4 (What N1 does not rule out). It does not rule out arguments that use a test function whose autocorrelation has small support but whose zero-side transform is not of the assumed exponential shape. It does not rule out arguments in which the arithmetic side is estimated conditionally on something other than RH — though §1.9 constrains what that something may be. It says nothing about test functions of non-compact support, for which the explicit formula requires separate justification and the trade-off is different.

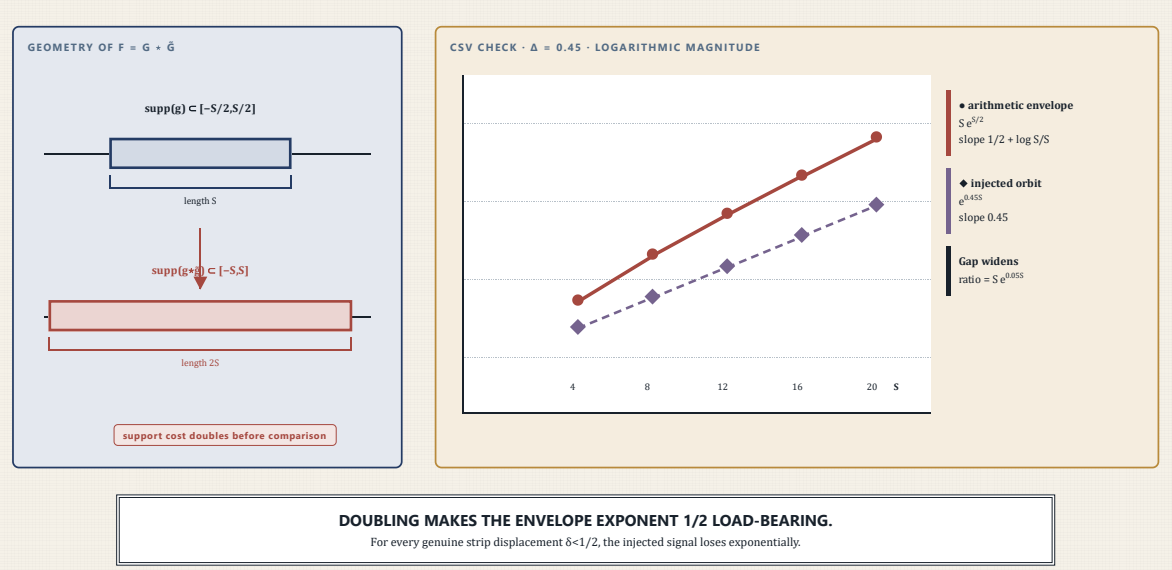


Figure 6.1. Support doubling.

What it does not show: The figure illustrates the support inclusion; it does not establish the comparison of exponents, which requires the unconditional arithmetic bound.

6.2.2 N2 — Amplification invariance

Theorem 6.5 (Amplification Invariance Theorem, N2; Theorem 39.1)**.** *[N] Let $A_k(s)$ be a Dirichlet series whose singular support contains the k-fold sums $\rho_1 + \cdots + \rho_k$ of non-trivial zeros of ζ, and let σ_c denote its abscissa of convergence. Write $\delta = \sup_\rho \Re\rho$ if that supremum is what the argument seeks to bound. Then*

$$\sigma_c - \Re(\textit{amplified singularity}) = k\left(\tfrac{1}{2} - \delta\right),$$

identically, and independently of the normalisation of A_k.

Proof sketch. Both quantities are computed from the same data: the abscissa of convergence is determined by the rightmost singularity of the k-fold object, which sits at $\Re s = k\delta$ up to the normalisation constant, while the trivial bound places the comparison point at $k/2$. Changing the normalisation shifts both by the same amount. The identity is therefore not an inequality that could be improved; it is a bookkeeping identity. Full proof: Chapter 30. □

OBSTRUCTION What N2 rules out

The amplifier programme in the form in which it is usually proposed. Amplification and the trivial bound scale at the same rate, so the extractable conclusion is $\delta \le \frac{1}{2}$ for every k, which is the classical zero-free-region statement in a change of variables. In the phrasing we have found clarifying: the architecture is a substitution applied to Hadamard and de la Vallée Poussin.

Remark 6.6 (What N2 does not rule out). It does not rule out amplifiers whose singular support is *not* the set of k-fold sums — for instance, amplifiers built from a different arithmetic object whose singularities are located by a different mechanism. It does not rule out the use of amplification for purposes other than bounding δ; zero-density estimates, for example, use amplification productively and are not touched. And it does not assert that no unconditional information can be obtained from k-fold objects, only that the information obtained is the 1-fold information in new coordinates. That last assertion is N18, proved separately.

Ledger entry	Geometric or analytic contribution	S-exponent	Effect
Hypothetical zero	$\rho=1/2+\delta+i\gamma$, with $0<\delta<1/2$	δ	distance from the critical line
Injected orbit signal	$\lvert\Delta_{\delta,\gamma}(g)\rvert \lesssim e^{\delta S}$	$+\delta$	quantity one hopes will dominate
Positive test	$f=g\star\tilde{g}$ doubles support to $[-S,S]$	support $\times 2$	exposes primes up to e^{S}
Absolute arithmetic envelope	$\sum \Lambda(n)n^{-1/2}\lvert f(\log n)\rvert \lesssim S\,e^{S/2}$	$+1/2$	unavoidable comparison scale

For the signal to outrun the envelope:
$e^{\delta S} \gg S\,e^{S/2} \iff (\delta-1/2)S \gg \log S \implies \boldsymbol{\delta > 1/2}$

Geometry of a nontrivial zero
$0<\beta<1$ and $\beta=1/2+\delta$ imply $\boldsymbol{\delta<1/2}$.

Ledger verdict
required $\delta>1/2$ contradicts the strip range.
O1 closes this route

Figure 6.2. The amplification ledger.

What it does not show: The ledger displays the identity; the proof that the identity is normalisation-independent is given in the text, not in the figure.

6.2.3 N3 — Positivity–abscissa dichotomy

Theorem 6.7 (Positivity–abscissa dichotomy, N3)**.** *[N] Consider the two hypotheses on a Dirichlet series $F(s) = \sum a_n n^{-s}$ used in Landau-type arguments:*
(A2) $a_n \geq 0$ for all n;
(A3) the abscissa of convergence is strictly to the left of the rightmost pole of the main term.
For the series arising from k-fold correlations of Λ, (A2) and (A3) are mutually exclusive.

Proof sketch. Landau's theorem states that a Dirichlet series with non-negative coefficients has a singularity at its abscissa of convergence. If the main term's pole is retained, it pins the abscissa at the pole's location and (A3) fails. If the main term is subtracted to move the abscissa left, the resulting coefficients change sign — explicitly, the subtraction removes a quantity larger than a_n for infinitely many n — and (A2) fails. There is no intermediate normalisation, because the pole's residue is determined. Full proof: Chapter 30. □

OBSTRUCTION What N3 rules out

Every route that applies Landau's theorem to a correlation series after subtracting a main term. This is the most common structure among proposed proofs in the additive-correlation family.

Remark 6.8 (What N3 does not rule out)*.* It does not rule out Landau-type arguments applied to series with genuinely non-negative coefficients and no main term to subtract — these exist and are useful, but they do not have the zeros in their singular support. It does not rule out subtracting a main term and then applying a tool other than Landau's theorem.

6.2.4 N4 — The dominance criterion

Theorem 6.9 (Dominance criterion, N4)**.** *[N] In the k-fold correlation expansion, the term arising from j zeros and $k-j$ main-term factors sits at relative depth $N^{-j(1/2-\delta)}$ against the leading term. In*

particular, for $\delta < \frac{1}{2}$ every term with $j \geq 1$ is of strictly smaller order than the main term, uniformly in k.

Proof sketch. Direct computation from the explicit formula, expanding $(\psi(x) - x)^j x^{k-j}$ and using $\psi(x) - x \ll x^{\delta+\epsilon}$. The depths do not interleave, so no cancellation among them can promote a subleading term. Full proof: Chapter 31. □

OBSTRUCTION What N4 rules out

Routes that hope to isolate a zero-dependent term by arranging cancellation among the main terms. The zero-dependent terms are uniformly subleading, and the arrangement cannot change their depth.

Remark 6.10 (What N4 does not rule out). It does not rule out extracting the subleading terms by a method that does not require them to dominate — for example by studying their oscillation rather than their size, which is Ingham's method and is exactly how the refutations of §5.11 were obtained. It does not rule out k-fold objects in which the main term is absent by construction.

6.2.5 N5 — The natural boundary obstruction

Theorem 6.11 (Natural boundary obstruction, N5). *[N] For $k \geq 3$, the divisor of k-fold sums of zeros accumulates on the line $\Re s = k - 1$: the set $\{\gamma_1 + \gamma_2 : \gamma_i \text{ ordinates of zeros}\}$ is dense in $\mathbb{R}$. Consequently the associated Dirichlet series has a natural boundary at $\Re s = k - 1$ under any of the standard sufficient conditions, and admits no meromorphic continuation past it that could be made conditional on a hypothesis about the zeros.*

Proof sketch. Density of $\{\gamma_1 + \gamma_2\}$ follows from the Weyl law together with the fact that the gaps $\gamma_{n+1} - \gamma_n$ tend to zero. Accumulation of singularities on a line, with the singularities of a fixed type, gives a natural boundary by the standard criterion. Full proof: Chapter 32. □

OBSTRUCTION What N5 rules out

Goldbach-type routes with $k \geq 3$ that require analytic continuation of the generating series past the accumulation line, and in particular any route that proposes to make the continuation conditional on RH and then argue by contradiction.

Remark 6.12 (What N5 does not rule out). It does not rule out $k = 2$, where the accumulation is at $\Re s = 1$ and the situation is genuinely different. It does not rule out working to the right of the boundary, which is where the unconditional results of Bhowmik and Schlage-Puchta live [BS11]. It does not assert that the natural boundary is proved for every such series — for some, the standard sufficient conditions have not been verified, and the entry states an obstruction to a class of arguments rather than a theorem about every member of the class.

6.2.6 N18 — Level collapse

Theorem 6.13 (Level Collapse Theorem, N18). *[N] (Unconditional, elementary.) For $k \geq 1$ let $R_k(N)$ denote the number of representations of N as an ordered sum of k prime powers weighted by Λ. Then for $0 < \theta \leq \frac{1}{2}$,*

$$\sum_{N \leq X} R_k(N) = \frac{X^k}{k!} + O\big(X^{k-\theta}\big) \quad \Longleftrightarrow \quad \psi(x) - x = O\big(x^{1-\theta}\big) \quad \Longleftrightarrow \quad \Theta \leq 1 - \theta,$$

where $\Theta = \sup_\rho \Re\rho$.

Proof sketch. Summing R_k over $N \le X$ gives an iterated convolution of ψ, and partial summation converts the error term of ψ into the error term of the k-fold sum with the same exponent saving. The converse direction follows by taking $k = 1$ in the resulting identity, which is legitimate because the identity holds for every k with the same θ. Full proof: Chapter 31. □

OBSTRUCTION What N18 rules out

The premise of the entire amplification programme: that the k-fold statement is stronger than the 1-fold statement. The exponents are identical. Amplification transports no gain.

Remark 6.14 (Why this is the entry to read first). N18 is unconditional, elementary, and has no hypotheses about the shape of the amplifier. It is therefore the entry least susceptible to the objection that a cleverer construction escapes it. A reader with time for one archive entry should read this one.

Remark 6.15 (What N18 does not rule out). It concerns the *summed* quantity $\sum_{N \le X} R_k(N)$. It says nothing about $R_k(N)$ for individual N, where the problem is genuinely different and genuinely open — the binary Goldbach problem is a statement about individual N and is not touched. It also says nothing about k-fold objects weighted differently, though N19 and N21 constrain those.

6.2.7 N19 — The exact criticality identity

Theorem 6.16 (Exact criticality identity, N19). *[N] The inequality* $\frac{1}{2} + k\delta \le \sigma_k$ *arising from hypothesis* (A2) *reduces identically to*

$$(k - j^*)\,\delta \le \frac{k - j^*}{2},$$

that is, to $\delta \le \frac{1}{2}$, *where* j^* *is the dominant index of N4. Moreover*

$$\sigma_k - \tfrac{1}{2} = \frac{k - j^*}{2} + j^*\delta$$

is bounded uniformly in k *if and only if* RH *holds. Consequently* (A2) *is equivalent to* RH.

Proof sketch. Substitute the depth computation of N4 into the abscissa identity of N2 and simplify; the amplification factor k is cancelled exactly by the depth penalty $k/2$. The uniform boundedness statement then follows because j^* is bounded and $\delta = \frac{1}{2}$ is the only value making the expression k-uniform. Full proof: Chapter 33. □

OBSTRUCTION What N19 rules out

The treatment of (A2) as an open problem to be attacked. It is not open; it is RH written differently. An argument that establishes (A2) has established RH and must be examined for circularity by the criteria of Definition 1.11.

Remark 6.17 (What N19 does not rule out). Equivalences are not obstructions in themselves; the archive entry is that the route is blocked *by an identity rather than by difficulty*, which is a different and more final kind of blockage. It does not rule out that some equivalent form

of RH is easier than others — the Nyman–Beurling and Li criteria are equivalences too, and remain objects of study. It says only that this particular equivalence carries no new leverage.

6.2.8 N20 — Infinite multiplicity

Theorem 6.18 (Infinite Multiplicity Theorem, N20). *[N] Let $D_k = \{\rho_1 + \cdots + \rho_k\}$ be the multiset of k-fold sums of non-trivial zeros of ζ, $k \geq 2$. Then D_k contains a point of infinite multiplicity. Consequently there is no meromorphic function on $\mathbb{C}$ whose divisor is D_k.*

Proof. Take $k = 2$ for definiteness. Hardy's theorem gives infinitely many zeros on the critical line; these occur in conjugate pairs $\rho = \frac{1}{2} + i\gamma$, $\bar{\rho} = \frac{1}{2} - i\gamma$. Each such pair contributes $\rho + \bar{\rho} = 1$ to D_2. Since there are infinitely many such pairs, the point $s = 1$ occurs with infinite multiplicity in D_2. A meromorphic function on $\mathbb{C}$ has divisors of finite order at every point, by definition of pole and zero order. For general $k \geq 2$ the same argument places infinite multiplicity at $s = k/2 \cdot 2/2$, adjusting by the $k - 2$ remaining summands: choose $k - 2$ zeros fixed and vary the conjugate pair. Full proof and the general-k bookkeeping: Chapter 32. □

OBSTRUCTION What N20 rules out

The automorphy route and the purity route, both of which require the k-fold divisor to be the divisor of an L-function or of a cohomological determinant. Neither can exist. This is a rare case of a programme dying with a one-paragraph proof rather than an assessment of difficulty.

Remark 6.19 (What N20 does not rule out). It does not rule out working with the k-fold sums as a set rather than a divisor, or with a regularised version in which the infinite-multiplicity point is excised by hand. It does not rule out the existence of a meromorphic function whose divisor is a *sub*-divisor of D_k — indeed $\zeta(s - \rho_0)$ has such a divisor trivially. And it uses Hardy's theorem, which is unconditional; readers checking non-circularity should note that no step uses RH.

6.2.9 N21 — Structure blindness

Theorem 6.20 (Structure-Blindness Theorem, N21). *[N] Let $\mathcal{M}$ be any conclusion derivable by the additive-correlation machine from the sequence $(\Lambda(n))_{n\geq 1}$ — that is, from the R_k for all k, together with their generating Dirichlet series and the standard analytic apparatus. Then $\mathcal{M}$ holds verbatim for the Beurling system of Diamond, Montgomery and Vorhauer [DMV06], in which the analogue of* RH *is false.*

Proof sketch. The map $\Lambda \mapsto (R_k)_k$ factors through the additive structure of $\operatorname{supp}\Lambda$ and forgets the Euler product: two arithmetic systems with the same counting function to the relevant accuracy have the same R_k to the relevant accuracy. The DMV system has $N_B(x) = \kappa x + O(x^\theta)$ with $\frac{1}{2} < \theta < 1$ and satisfies the prime number theorem, hence matches Λ in exactly the data the machine consumes, while its zeta function has zeros with real part $1 - a/\log t$. Full proof: Chapter 34. □

OBSTRUCTION What N21 rules out

Every proof of RH that proceeds through additive correlations of Λ alone. Any proof must use the Euler product essentially, and the additive-correlation machine cannot.

Remark 6.21 (Relation to the firewall). N21 is the firewall of Chapter 2 applied to an entire method rather than to a single mechanism, and it is the clearest illustration of why the firewall is worth maintaining. The DMV object is not exotic; it satisfies the prime number theorem and has an Euler product. What it lacks is whatever additional feature of ζ forces the zeros onto the line, and the additive-correlation machine does not see that feature.

Remark 6.22 (What N21 does not rule out). It does not rule out multiplicative methods, which see the Euler product by construction. It does not rule out additive methods supplemented by a genuinely multiplicative input — the question then becomes what that input is, and whether it already contains RH. It does not assert that the DMV system matches ζ in all respects; it matches in the data the machine consumes, and the proof identifies exactly which data that is.

6.2.10 N22 — The Vinogradov vacuity theorem

Theorem 6.23 (Vinogradov vacuity, N22). *[N] The savings of powers of* log *obtained in the ternary Goldbach circle method are invisible at the abscissa of convergence of the associated Dirichlet series. Moreover the ternary asymptotic is strictly weaker than the classical zero-free region: the major-arc analysis consumes Siegel–Walfisz, which already yields an error* $O(N^2 e^{-c\sqrt{\log N}})$*, stronger than the* $\log^{-A}$ *saving the asymptotic reports.*

Proof sketch. The abscissa of convergence is determined by the polynomial rate, and a saving of $\log^{-A}$ does not move it. For the second statement, one traces the major-arc estimate: Siegel–Walfisz is applied to obtain equidistribution in progressions, and its error term is exponential in $\sqrt{\log N}$. Full proof: Chapter 35. □

OBSTRUCTION What N22 rules out

The expectation that unconditional asymmetric information at $k = 3$ carries content about the zeros. Its unconditional content on that question is exactly zero, and this is weaker than the expectation, not merely different from it.

Remark 6.24 (On Helfgott). The resolution of the ternary Goldbach conjecture for all N [Hel13] uses, among much else, a finite verification of GRH for small conductors due to Platt. This is a legitimate and standard technique and we record it without criticism. We note it here only because it means the result cannot be used as an input to an argument about the zeros without circularity of the kind Definition 1.11 forbids.

Remark 6.25 (What N22 does not rule out). It does not diminish the ternary Goldbach theorem, which is a theorem about the additive structure of the primes and is not a statement about zeros. It does not rule out circle-method arguments generally. It rules out one inference: from "the ternary asymptotic is unconditional" to "the ternary asymptotic contains unconditional information about $\Re\rho$".

6.3 Group B: geometric transport

6.3.1 N6 — The Künneth / local–global obstruction

Theorem 6.26 (Local–global obstruction, N6). *[N] The symmetric-power operation sends local parameters to their powers,* $\alpha_v \mapsto \alpha_v^k$*, but does not send zeros of the global L-function to zeros of the symmetric-power L-function. Consequently transport of Deligne's purity theorem through symmetric powers yields the Ramanujan bound on local parameters and never a statement about the location of global zeros.*

Proof sketch. The local factor of $L(s, \mathrm{Sym}^k \pi)$ at v is determined by the k-th symmetric power of the Satake parameter, so a bound on $|\alpha_v|$ transports. The global zeros are determined by the analytic continuation of the product, and the product of the shifted local factors has zeros unrelated to those of the original by any local formula. Full proof: Chapter 36. □

OBSTRUCTION What N6 rules out

The route: prove purity locally, symmetrise, conclude globally. What is obtained is Ramanujan, which is a local statement, and GRH is not implied.

Remark 6.27 (What N6 does not rule out). It does not rule out the use of symmetric powers for other purposes, notably the subconvexity and functoriality results that have moved the spectral-gap hierarchy of §3.4. It does not rule out a global purity statement obtained by non-local means. And it does not assert that Ramanujan and GRH are unrelated — they are related — only that the implication does not run in the direction the route requires.

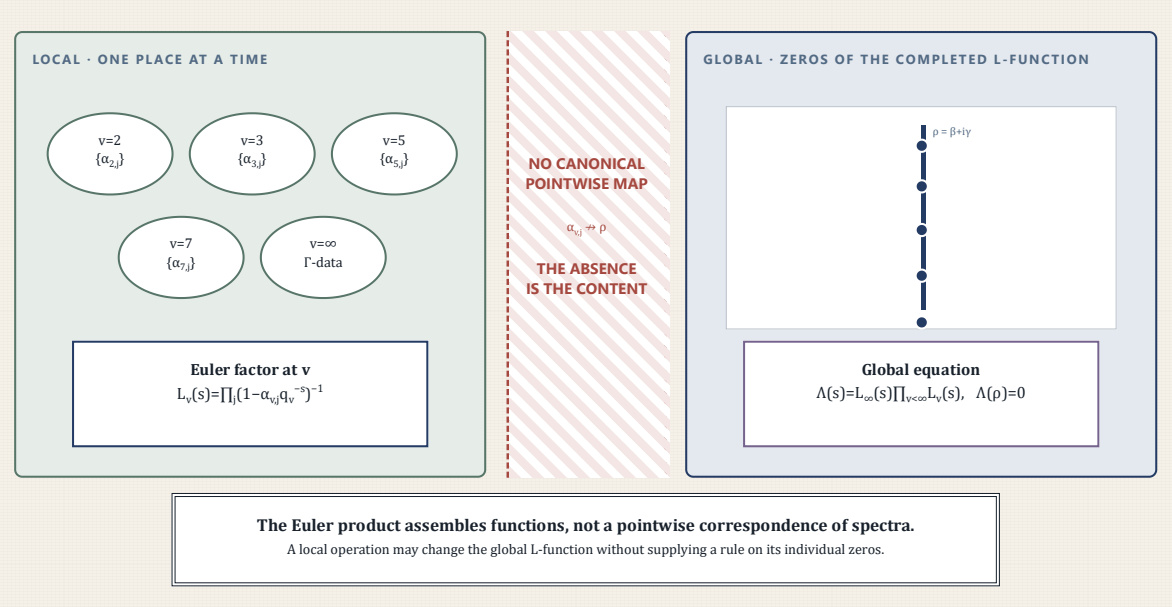


Figure 6.3. Local to global.

What it does not show: The diagram records the direction of the implications; the proof that no global statement follows is in the text.

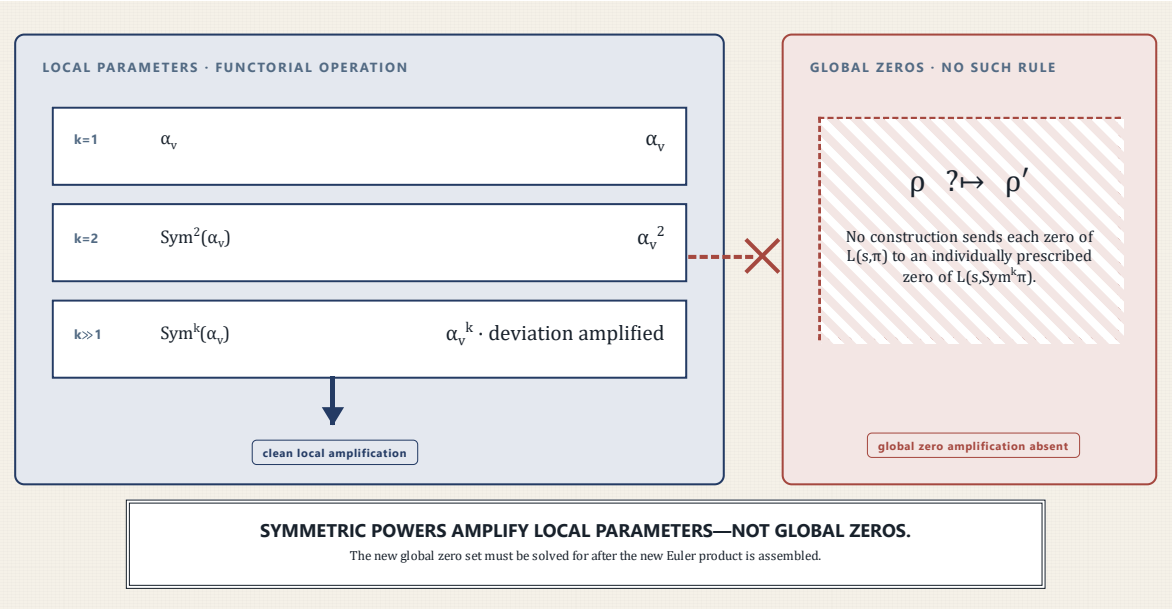


Figure 6.4. The symmetric-power ladder.

What it does not show: The figure is a schematic of the functoriality; it does not establish the analytic properties of the symmetric-power L-functions, which are known only for small k.

6.3.2 N7 — No Frobenius on Spec $\mathbb{Z}$

Theorem 6.28 (No Frobenius, N7)**.** *[N]* Spec $\mathbb{Z}$ *is the terminal object in the category of schemes, and* $\mathrm{End}(\mathrm{Spec}\,\mathbb{Z}) = \{\mathrm{id}\}$. *Moreover the canonical* λ*-ring structure on* $\mathbb{Z}$ *has Adams operations* $\psi_p = \mathrm{id}$ *for every prime* p. *There is therefore no non-trivial Frobenius endomorphism, and no fixed-point formula of Lefschetz type available.*

Proof. Terminality gives a unique morphism $X \to \mathrm{Spec}\,\mathbb{Z}$ for every scheme X; taking $X = \mathrm{Spec}\,\mathbb{Z}$ gives End $= \{\mathrm{id}\}$. For the λ-ring statement, the Adams operation ψ_p on $\mathbb{Z}$ with its canonical structure is determined by $\psi_p(n) = n$ for $n \in \mathbb{Z}$, since $\mathbb{Z}$ is generated by 1 and $\psi_p(1) = 1$. Full discussion: Chapter 37. □

OBSTRUCTION What N7 rules out

The naive version of the "curve over $\mathbf{F}_1$" programme: the version in which one hopes to find, within existing categories of schemes or λ-rings, an endomorphism playing the role of Frobenius, and then to run the Weil argument.

Remark 6.29 (What N7 does not rule out). It does not rule out the $\mathbf{F}_1$ programme, which is not naive and does not work in the category of schemes. The Connes–Consani arithmetic site [CC16a] and the scaling site [CC16b] construct objects with the required endomorphism outside the scheme category; N13 and N14 address specific difficulties there, but N7 does not touch them. It does not rule out Frobenius-like structures on the arithmetic side obtained from p-adic Hodge theory, where the relevant statement is N14.

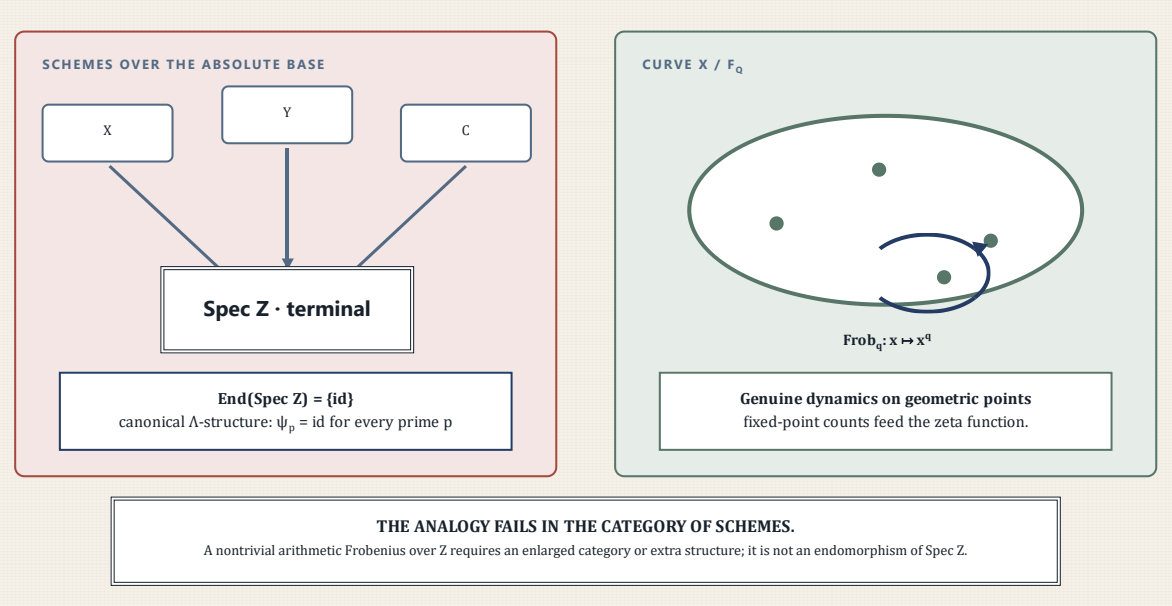


Figure 6.5. No Frobenius.

What it does not show: The figure records the categorical statement; it does not bear on constructions outside the category of schemes.

6.3.3 N12 — The archimedean Frobenius candidate

Theorem 6.30 (Archimedean Frobenius no-go, N12)**.** *[N] Let φ_∞ be the candidate archimedean Frobenius obtained from Tate's construction as $(-)^{t\mathbf{T}}$. Then the associated regularised determinant is periodic, with zeros on vertical lines. The archimedean factor $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ is not periodic and its poles lie on the real axis. Hence φ_∞ does not realise $\Gamma_{\mathbb{R}}$.*

Proof sketch. Tate's construction produces a Laurent-type object whose regularised determinant is a periodisation, and periodisation forces the zero set to be invariant under a translation. $\Gamma_{\mathbb{R}}$ has poles at $s = 0, -2, -4, \ldots$, which is a real arithmetic progression, not a vertical line, and $\Gamma_{\mathbb{R}}$ is not invariant under any translation. Full proof: Chapter 38. □

OBSTRUCTION What N12 rules out

The most natural candidate for the archimedean Frobenius. Since the archimedean place is where every transport programme has its principal difficulty, closing the natural candidate is informative.

Remark 6.31 (What N12 does not rule out). It rules out one candidate, not the existence of an archimedean Frobenius. Deninger's programme [Den98a] proposes a different realisation, and N11 addresses a separate difficulty there. Connes' semilocal trace formula treats the archimedean place successfully in the sense described in §3.5, without producing a Frobenius.

6.3.4 N13 — The sphere spectrum and its module category

Theorem 6.32 (Module-category obstruction, N13)**.** *[N] The object denoted $\mathbb{S}$ in the Connes–Consani framework is the sphere spectrum. However, the module category of the idempotent semirings $\mathbb{B}$, $\mathbf{Z}_{\max}$ and $\mathbf{R}_+^{\max}$ does not embed in the category* Sp *of spectra: the group completion of an idempotent monoid is trivial, so the associated K-theoretic invariants vanish.*

Proof sketch. If M is an idempotent commutative monoid then for $x \in M$, $x + x = x$, so in the group completion $x = 0$. Hence the group completion is trivial and the associated spectrum is contractible. Full proof: Chapter 39. □

OBSTRUCTION What N13 rules out

The "cheap import" route: the hope that identifying $\mathbb{S}$ as the base allows homotopy-theoretic invariants to be applied directly to the idempotent-semiring objects of the $\mathbf{F}_1$ literature.

Remark 6.33 (What N13 does not rule out). It does not touch the arithmetic site or the scaling site, which are not built from module categories over idempotent semirings in the sense used here. It does not rule out homotopy-theoretic input to the problem by other routes — topological cyclic homology, for instance, which is genuinely available over $\mathbb{Z}$ and is the subject of N14 and N16.

6.3.5 N14 — Linearisation

Theorem 6.34 (Linearisation obstruction, N14). *[N] Let φ be the Frobenius on $\mathbb{Z}_p$ or on the associated Witt-vector or prismatic objects. No power of φ is linear over the base, so the expression "$\log_q \mathrm{Fr}$" has no interpretation as an operator. Additionally, the Breuil–Kisin twist obstructs the naive globalisation of Hesselholt's local theorem.*

Proof sketch. φ is a ring endomorphism that is σ-semilinear rather than linear; composition preserves semilinearity with the twist raised to a power, and semilinearity is never lost. A logarithm requires linearity to define. The Breuil–Kisin twist enters when comparing local realisations at different primes and is not trivialisable globally. Full proof: Chapter 40. □

OBSTRUCTION What N14 rules out

The direct globalisation of Hesselholt's Theorem A [Hes18a], in which the local Frobenius is exponentiated to produce a spectral parameter.

Remark 6.35 (What N14 does not rule out). It does not touch Hesselholt's local theorem, which is correct and is used in this book. It does not rule out globalisation by a method that does not require linearising φ — for instance by working with the semilinear object throughout and taking invariants at the end. It does not rule out the existence of a global cohomology theory with the desired properties; it rules out one construction of one.

6.3.6 N16 — No C^*-refinement without a tensor product

Theorem 6.36 (C^*-refinement obstruction, N16). *[N] The p-completion used in the topological-cyclic-homology approach is incompatible with a C^*-algebraic refinement: the embedding $\iota : W(\mathbf{F}_q) \hookrightarrow \mathbb{C}$ is neither canonical nor continuous for the p-adic topology. Consequently a positivity structure cannot be imposed after p-completion.*

Proof sketch. $W(\mathbf{F}_q)$ carries the p-adic topology, in which $p^n \to 0$; the image under any field embedding into $\mathbb{C}$ has $|p^n| \to \infty$ or oscillates, and no embedding is continuous. Canonicity fails because the embedding depends on a choice of algebraic closure and of a compatible system of roots of unity. Full proof: Chapter 41. □

OBSTRUCTION What N16 rules out

Imposing positivity after p-completion. Positivity must enter before completion, on the topological Hochschild-type object of an analytic ring.

Remark 6.37 (What N16 does not rule out). It does not rule out positivity structures in the p-adic setting itself, where the appropriate notion is different and is not the C^* one. It is a constraint on the order of operations, and this book obeys it: the construction of Part V imposes the form before any completion.

6.4 Group C: positivity

6.4.1 N15 — Log-concavity is insufficient

Theorem 6.38 (Log-concavity insufficiency, N15). *[N] For $d \geq 3$, ordinary log-concavity of a sequence is strictly weaker than Jensen hyperbolicity of the associated polynomial family. In particular, establishing the classical Turán inequalities for the Taylor coefficients of ξ does not establish* RH, *nor does establishing them for every finite order.*

Proof sketch. Exhibit a log-concave sequence of length $d \geq 3$ whose associated polynomial has non-real roots; such examples exist for every $d \geq 3$ and are elementary to construct. The higher Turán inequalities form a strictly increasing hierarchy, and RH is equivalent to all of them holding for all orders, which is not implied by any finite subfamily. Full proof: Chapter 42. □

OBSTRUCTION What N15 rules out

Routes proving only the classical Turán inequalities, or finitely many of the higher ones, and inferring hyperbolicity.

Remark 6.39 (What N15 does not rule out). It does not rule out the Jensen-polynomial route as such: Griffin, Ono, Rolen and Zagier established hyperbolicity of the Jensen polynomials $J_{\xi}^{d,n}$ for each fixed d and all sufficiently large n, which is a genuine theorem and is not contradicted here. The obstruction is to inferring RH, which requires all n including small ones, uniformly in d.

6.4.2 N17 — The solvability complexity index

Theorem 6.40 (SCI = 2, N17). *[K]* (Hansen [Han11].) *The problem of computing the spectrum of a general bounded operator from its matrix entries has solvability complexity index* 2*: no algorithm using finitely many entries, and no sequence of such algorithms with a single limit, computes it.*

OBSTRUCTION What N17 rules out

Certification of Weil positivity from a finite Gram matrix alone. A finite computation cannot decide the sign of an infinite-dimensional form without additional structural input.

Remark 6.41 (An honest statement of scope). N17 is a theorem about finite data and algorithms. **It is not an undecidability theorem for** RH, and it does not say that RH is unprovable or that spectral methods cannot work. It says that the finite Gram data, by itself, does not certify. Structural input — for instance monotonicity in the truncation parameter, or an a priori bound

on the tail — changes the situation entirely, and this book supplies such input in Part VI. We state the scope emphatically because the theorem is quoted in stronger forms than it supports.

6.4.3 N23 — No matroid structure on Λ

Theorem 6.42 (No matroid structure, N23)**.** *[N]* $\operatorname{supp}\Lambda$ *is not M-convex:* $\Lambda(1) = \Lambda(6) = 0$ *while* $\Lambda(5), \Lambda(7) \neq 0$*, violating the exchange property. Moreover* $\Lambda(5)\Lambda(7) > \Lambda(6)^2 = 0$*, so the sequence is not log-concave. Hence the Bränden–Huh and Adiprasito–Huh–Katz machinery cannot be mounted on the additive correlations of* Λ*.*

Proof. $\Lambda(6) = 0$ since $6 = 2 \cdot 3$ is not a prime power; $\Lambda(1) = 0$ by convention; $\Lambda(5) = \log 5$ and $\Lambda(7) = \log 7$ are positive. The exchange axiom for M-convex sets fails on the pair $(5, 7)$ against 6. Log-concavity requires $\Lambda(6)^2 \geq \Lambda(5)\Lambda(7)$, i.e. $0 \geq \log 5 \log 7$, which is false. Full proof: Chapter 42. □

OBSTRUCTION What N23 rules out

Lorentzian-polynomial and Hodge-theoretic combinatorial methods applied to the additive correlations of Λ.

Remark 6.43 (What N23 does not rule out — and this matters). It does *not* touch the Hodge structure this book erects on the Jensen apolar algebra of ξ, because the objects are different: there the sequence is the Taylor coefficients of ξ, which are log-concave in the relevant ranges, and the combinatorial input is applied to a polynomial family rather than to $\operatorname{supp}\Lambda$. Conflating the two is an available error and we mark it here so that the reader does not make it.

6.5 Group D: spectral

6.5.1 N10 — AS6 is strictly stronger than RH

Theorem 6.44 (AS6 strength, N10)**.** *[N] If* $\Theta^\dagger = I - \Theta$ *with respect to a positive-definite inner product and* $D = -i(\Theta - \frac{1}{2}I)$ *is self-adjoint, then* Θ *is normal, and* $\overline{W(\Theta)} \subseteq \{\Re z = \frac{1}{2}\}$*. This is strictly stronger than* RH*.*

This is Theorem 4.8; the proof, the strictness witness, and the corrected indefinite formulation are the content of Chapter 4. The archive entry records it here for completeness of the catalogue.

Remark 6.45 (What N10 does not rule out). It does not rule out spectral formulations; it rules out the definite one. The corrected formulation of Definition 4.14 is not touched, and the whole of Part VIII operates under it.

6.5.2 N11 — The archimedean polarisation refutation

Theorem 6.46 (Archimedean polarisation refutation, N11)**.** *[N] Deninger's archimedean operator has* $\operatorname{spec}\Theta_\infty = \{-n : n \geq 0\} \subset \mathbb{R}$*. Combined with Theorem A of Part VII, this is inconsistent with a positive-definite polarisation* ω *under the naive normalisation.*

Proof sketch. Theorem A gives, for a positive-definite ω with the stated intertwining, that the spectrum lies on $\Re z = \frac{1}{2}$. The archimedean spectrum is real and unbounded below. The two are incompatible. Full proof: Chapter 43. □

OBSTRUCTION What N11 rules out

Positive-definite ω under the naive normalisation, independently of the argument of Chapter 4. The two arguments are logically independent and reach the same conclusion, which is why both are in the archive.

Remark 6.47 (What N11 does not rule out). It does not rule out Deninger's programme, which does not assert positive-definiteness in this normalisation. It does not rule out a different normalisation — indeed Remark 4.34 identifies exactly which normalisation change converts the definite form into the indefinite one, and the archimedean spectrum is consistent with the indefinite form.

6.6 The Criticality Barrier (N8)

This entry is given at length because it constrains what kind of positivity can possibly work, and because the constraint is easy to state and easy to forget.

6.6.1 Statement

Definition 6.48 (The de Bruijn–Newman constant). For $t \in \mathbb{R}$ let

$$H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du,$$

where Φ is the standard theta-type kernel with $\xi(\frac{1}{2} + iz) = \frac{1}{2} H_0(z)$. De Bruijn [Bru50] and Newman [New76] showed that there is a constant Λ_{dBN} such that H_t has only real zeros if and only if $t \geq \Lambda_{\mathrm{dBN}}$.

Lemma 6.49 (Newman). *[K]* RH *holds if and only if* $\Lambda_{\mathrm{dBN}} \leq 0$.

Theorem 6.50 (Rodgers–Tao). *[K]* $\Lambda_{\mathrm{dBN}} \geq 0$.

The proof [RT20] shows that if $\Lambda_{\mathrm{dBN}} < 0$ then the zeros of H_t for t slightly negative would be far more regularly spaced than the zeros of ζ are known to be, contradicting known results on the pair correlation and on large gaps.

Theorem 6.51 (Criticality Barrier, N8). *[N] Combining Lemma 6.49 and Theorem 6.50,*

$$\mathrm{RH} \iff \Lambda_{\mathrm{dBN}} = 0.$$

Consequently RH *is located at a single point of a one-parameter family, and is an equality rather than an inequality.*

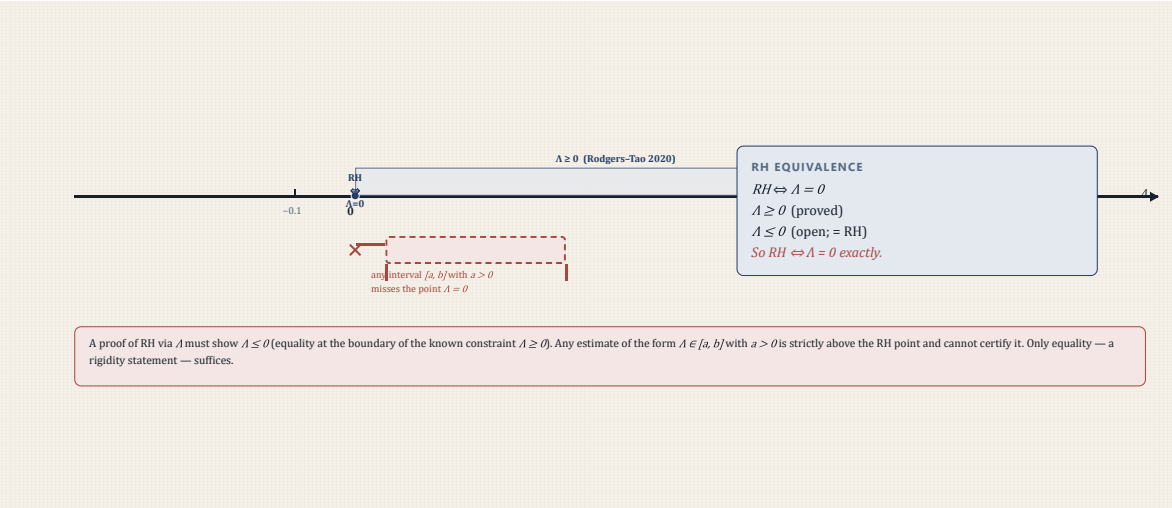


Figure 6.6. The Criticality Barrier.

What it does not show: The figure records the two half-line constraints; the equivalence of RH *with* $\Lambda_{\mathrm{dBN}} \le 0$ *is a theorem of Newman and is not established by the figure.*

6.6.2 The consequence for positivity methods

Corollary 6.52 (No gapped positivity)**.** *[N] Let P be a positivity statement with the property that P implies* RH *and that P is stable under perturbation — that is, there is $\epsilon > 0$ such that every object within ϵ of the object satisfying P also satisfies P. Then P is false.*

Proof. Suppose P holds with stability radius $\epsilon > 0$. Perturbing the family parameter t from 0 to $-\epsilon'$ for small ϵ', the perturbed object still satisfies P by stability, hence RH holds for $H_{-\epsilon'}$, i.e. $H_{-\epsilon'}$ has only real zeros, i.e. $\Lambda_{\mathrm{dBN}} \le -\epsilon' < 0$. This contradicts Theorem 6.50. (The step in which P transfers to the perturbed object is where the stability hypothesis is used, and it is the hypothesis to check when applying the corollary.) □

OBSTRUCTION What N8 rules out

Three families at once.

- *Perturbation-stable positivity.* Any positivity that survives a small deformation of the object. By Corollary 6.52 such a positivity is false if it implies RH, and if it does not imply RH it is not on the route.
- *Uniform spectral gaps.* Any argument producing a gap bounded away from zero uniformly. A uniform gap is a stability radius.
- *Finite sum-of-squares certificates with strict slack.* An SDP certificate with strictly positive slack is a stability radius by construction. Certificates with zero slack are not excluded, and are a different and much harder object.

6.6.3 What remains: equality and rigidity

Definition 6.53 (Equality-type positivity)**.** A positivity statement is of *equality type* if it asserts $Q \ge 0$ together with a characterisation of the equality case, and the route to RH proceeds through the equality case rather than through the inequality.

Equality-type statements are not excluded by N8, because they have stability radius zero: the equality case is not preserved under perturbation, so the argument of Corollary 6.52 does not apply. This is the class in which the index-theoretic statements of Part VIII live: an index is an integer, and integers do not deform.

Remark 6.54 (Why index theory is the natural home). An index is deformation-invariant in a strong sense — it does not change under continuous deformation — while simultaneously being a discrete quantity whose value at a specific point is a statement of equality rather than of inequality. This combination is what N8 requires, and it is not accidental that the surviving route in this book is index-theoretic. We record this as an observation about the shape of the constraint, not as evidence that the route succeeds.

Remark 6.55 (What N8 does not rule out). It does not rule out positivity methods generally. It does not rule out inequalities that are used as inputs rather than as the final step. It does not rule out Weil positivity, which is an equivalence rather than a gapped inequality. And it does not say that RH is false or that it sits at a boundary in any metaphysical sense; it says that in this particular one-parameter family, the location is a point, and that this constrains the tools.

6.7 The absolute-value ceiling family (N9)

This entry is a taxonomy, not a single inequality, and the most consequential error available in this area is to present it as a single inequality.

6.7.1 The two proved theorems

Theorem 6.56 (Euler-product ceiling, exponent 1). *[T] Let $F(s) = \prod_p F_p(s)$ be a Dirichlet series with an Euler product, and suppose an argument bounds a quantity by passing to absolute values in the Euler product, i.e. replacing $\prod_p F_p(s)$ by $\prod_p |F_p(s)|$. Then the resulting bound is controlled by the abscissa of absolute convergence, which for ζ is $\Re s = 1$. No information about the region $\frac{1}{2} < \Re s < 1$ survives.*

Proof sketch. Passing to absolute values in the Euler product replaces $\zeta(s)$ by $\zeta(\Re s)$ up to bounded factors, and $\zeta(\sigma)$ diverges as $\sigma \to 1^+$. The bound therefore degenerates at $\sigma = 1$ and carries no content to its left. Full proof: Chapter 33. □

Theorem 6.57 (CZI ceiling, exponent 1/2). *[T] Let a bound be obtained by the Cauchy–Schwarz-type inequality applied to the arithmetic side of the explicit formula, i.e. by replacing $\sum_n \Lambda(n) n^{-1/2} c_n$ by $\left(\sum_n \Lambda(n)^2 n^{-1}\right)^{1/2} \left(\sum_n |c_n|^2\right)^{1/2}$ or a variant. Then the resulting bound is controlled by the exponent $\frac{1}{2}$, and its content degenerates exactly at $\Re s = \frac{1}{2}$.*

Proof sketch. The unconditional estimate $\sum_{n \le X} \Lambda(n) n^{-1/2} = 2\sqrt{X} + o(\sqrt{X})$ places the arithmetic side at size $X^{1/2}$; the inequality is sharp in the sense that no cancellation is used, and the resulting bound is exactly the trivial bound at $\sigma = \frac{1}{2}$. Full proof: Chapter 33. □

> **These are not the same inequality.** Theorem 6.56 sits at exponent 1 and concerns the Euler product; Theorem 6.57 sits at exponent 1/2 and concerns the explicit formula's arithmetic side. They have different hypotheses, different proofs, different degeneration points, and rule out different things. Presenting them as a single "square-root barrier theorem" is an error, and it is an error that changes what is claimed: an argument stopped at exponent 1 has failed much earlier than one stopped at exponent 1/2.

6.7.2 The two labelled analogies

Alongside the two theorems we record two statements that are *analogies* and are labelled as such. They are useful for orientation and are not used as steps anywhere in this book.

A-1 [O] *The Selberg-class analogy.* In the Selberg class, an absolute-value bound on the Dirichlet coefficients at the level of the Ramanujan hypothesis appears to play the role of the Euler-product ceiling. We have not verified that the analogy is more than formal, and we do not use it.

A-2 [O] *The function-field analogy.* Over $\mathbf{F}_q$, the Weil bound $|\alpha_v| = q^{1/2}$ is an exact absolute-value statement at exponent $1/2$, and its exactness is what makes the function-field proof work. The analogy suggests that an exponent-$1/2$ statement over $\mathbb{Q}$ would need to be exact rather than an inequality, which is consistent with N8. Again, this is orientation, not argument.

OBSTRUCTION What N9 rules out

Routes whose final step passes to absolute values. Which ceiling they hit depends on where the passage occurs: at the Euler product, exponent 1; at the explicit formula, exponent $1/2$. In both cases the bound obtained is the trivial one and no localisation of zeros follows.

Remark 6.58 (What N9 does not rule out). It does not rule out the use of absolute values in intermediate steps where the loss is recovered later. It does not rule out arguments that keep phase information — which is precisely what an argument must do, and is the reason the surviving route in this book is a signature computation rather than a size estimate. It does not assert that exponent $1/2$ is unreachable; it asserts that reaching it by passing to absolute values yields no content, which is a statement about the method.

6.8 What survives

An archive of twenty-three obstructions invites the question of what is left. The honest answer has three parts.

6.8.1 The one surviving unconditional constraint

Among all the candidate unconditional upper-bound constraints we examined, exactly one is simultaneously constructed, unconditional, non-circular and non-vacuous: the zero-density estimate. The current record is

$$N(\sigma, T) \ll T^{\frac{30}{13}(1-\sigma)+o(1)}$$

in the relevant range, due to Guth and Maynard [GM24], in the lineage Ingham 1937 → Huxley → Bourgain → Guth–Maynard.

Proposition 6.59 (Insufficiency of zero density)**.** *[T] The estimate above is non-trivial only for* $\sigma > 1 - \frac{13}{30} = \frac{17}{30}$*, and degenerates at* $\sigma = \frac{1}{2}$*. It therefore cannot, alone, prove* RH.

Proof sketch. At $\sigma = \frac{1}{2}$ the exponent is $\frac{30}{13} \cdot \frac{1}{2} = \frac{15}{13} > 1$, so the bound exceeds the total number of zeros to height T, which is $\asymp T \log T$, and is therefore vacuous there. Full proof: Chapter 29. □

6.8.2 The two open problems

The archive leaves exactly two precisely stated open problems on which the positive programme depends. Both are stated in full in later Parts and neither is claimed to be solved here.

OPEN PROBLEM The arithmetic Hodge–Krein realisation problem (G.2)

[O] Construct a filtered system $(H_N, \Theta_N, G_N, \iota_N)_{N\geq 1}$ with: Θ_N finite-dimensional and its stable spectrum exhausting the non-trivial zeros with algebraic multiplicity; G_N non-degenerate Hermitian with $\Theta_N^* G_N + G_N \Theta_N = G_N$; negative index $n_-(G_N) \leq 1$, compatibly with the transition maps. No construction satisfying even these three conditions is currently known. Full statement, including the three further compatibility conditions, in Chapter 44.

OPEN PROBLEM The slope-breaking problem (A2$\star$)

[O] In its most economical form: prove unconditionally that

$$\sum_{0<\gamma\leq T} \left(\beta - \tfrac{1}{2}\right)^+ = o(T),$$

where the sum is over non-trivial zeros $\rho = \beta + i\gamma$. The functional equation makes the *signed* sum identically zero; the best available zero-density estimates give only the trivial $T \log T$ for the one-sided sum. A saving of a single power of log would be the first statement of its kind. Full statement and the two equivalent forms in Chapter 45.

Remark 6.60 (The relation between them). The two problems measure the same thing differently: the index statement of Part VIII bounds $\#\{\Re\lambda > \frac{1}{2}\}$, a count; A2$\star$ bounds $\sum(\beta - \frac{1}{2})^+$, a weighted count. Both are one-sided deviation measures. Part VIII establishes their precise relation, and that relation is where the book's two lines meet. We state here only that the relation is established there and that neither problem is solved.

6.8.3 The negative space

The third part of the answer is the least satisfying and the most honest: the archive does not delimit the negative space. Twenty-three obstructions close twenty-three families of argument. The complement of a finite union of families is not small, and we make no claim about it. A proof of RH by a method not in the archive is not obstructed by anything in this chapter, and the archive gives no reason to think such methods are scarce.

6.9 The archive as a table

For reference, the twenty-three entries are collected below with their group, the class of argument each closes, and the chapter containing the full proof.

Entry	Group	Closes	Full proof
N1 Support doubling	A	widened-support routes	Ch. 29
N2 Amplification invariance	A	the amplifier programme	Ch. 30
N3 Positivity–abscissa	A	Landau-theorem routes	Ch. 30
N4 Dominance criterion	A	high-order correlation routes	Ch. 31
N5 Natural boundary	A	$k \geq 3$ Goldbach continuation	Ch. 32
N6 Local–global	B	Deligne transplantation	Ch. 36
N7 No Frobenius	B	naive $\mathbf{F}_1$ curves	Ch. 37
N8 Criticality Barrier	C	gapped positivity	§6.6
N9 Absolute-value ceilings	C	absolute-value majorisation	§6.7
N10 AS6 strength	D	the definite polarisation axiom	Ch. 4
N11 Archimedean polarisation	D	positive-definite ω, naive	Ch. 43
N12 Archimedean Frobenius	B	the Tate candidate	Ch. 38
N13 Module category	B	cheap homotopical import	Ch. 39
N14 Linearisation	B	naive globalisation of Hesselholt	Ch. 40
N15 Log-concavity	C	classical Turán-only routes	Ch. 42
N16 No C^* refinement	B	positivity after p-completion	Ch. 41
N17 SCI = 2	C	finite Gram certification	[Han11]
N18 Level collapse	A	the premise of amplification	Ch. 31
N19 Exact criticality	A	(A2) as an open problem	Ch. 33
N20 Infinite multiplicity	A	automorphy and purity routes	Ch. 32
N21 Structure blindness	A	additive-correlation proofs	Ch. 34
N22 Vinogradov vacuity	A	ternary-asymptotic leverage	Ch. 35
N23 No matroid on Λ	C	Lorentzian methods on Λ	Ch. 42

Remark 6.61 (On reading the table). The "closes" column is a compression of the (E4) clause and is not a substitute for it. In particular the column omits the (E5) clause entirely, and a reader who takes the table as the archive will systematically overestimate what has been ruled out. The table is an index.

6.9.1 The three entries with no escape identified

Pass 3 of the audit (§6.14) requires that each (E5) clause exhibit an argument in the plausible ruled-out class that the entry does not in fact rule out. For three entries we could not construct one, and we record which.

N18 (*Level collapse*). The theorem is an equivalence with the same exponent on both sides, unconditional and with no hypotheses on the shape of the object. We could not identify a k-fold summed statement in the same family that escapes it. The (E5) clause therefore restricts to statements outside the family — individual N, or different weights — rather than exhibiting an escape within it.

N20 (*Infinite multiplicity*). The obstruction is to the existence of a meromorphic function with a given divisor, and the divisor demonstrably has an infinite-multiplicity point. Within the class of arguments requiring such a function, there is no escape; the (E5) clause describes only how to leave the class.

N7 (*No Frobenius*). Within the category of schemes the statement is a triviality with no room. The (E5) clause is entirely about leaving the category, which is what the serious $\mathbf{F}_1$ programmes do.

Remark 6.62 (Why this is recorded). Three entries out of twenty-three admit no internal escape. This is a small proportion, and we regard the proportion as the correct calibration

for how final a no-go theorem typically is: usually not very. An archive in which every entry admitted no escape would be an archive whose (E5) clauses had not been taken seriously.

6.10 How the archive is used

The archive is not a museum. Four of its entries are load-bearing for the positive programme, in the sense that the programme's design is determined by them.

N8 determines the type of positivity. By Corollary 6.52, the surviving positivity must have stability radius zero. The programme therefore uses an index — an integer-valued, deformation-invariant quantity whose specific value is an equality statement. Every alternative we considered was gapped and therefore excluded before construction began.

N10 determines the type of form. By Theorem 4.8 the definite form is strictly stronger than RH, hence unavailable. The programme therefore uses an indefinite (Krein) form. Chapter 4 shows that this is not a choice but a consequence of the functional equation's normalisation.

N21 determines what the input must be. By the Structure-Blindness Theorem, additive correlations of Λ cannot suffice; the Euler product must be used essentially. The programme therefore enters through the Euler product and never through a purely additive object.

N9 determines what may not be done at the last step. By the two ceiling theorems, passing to absolute values destroys the content. The programme therefore computes a signature — a quantity that depends on phase — rather than a bound on a size.

Remark 6.63 (The design is negative). Stated compactly: the shape of the positive programme was not chosen and then defended; it is what remains after N8, N9, N10 and N21 are imposed. We note this because it is the strongest thing we can honestly say for the programme's shape, and it is considerably weaker than saying the programme works. Four constraints do not determine a solution. They determine a family in which a solution, if one exists, must lie, and §6.8 states that the family is not known to be non-empty.

6.10.1 The entries not used

Nineteen entries play no role in the positive programme. They are in the book because the catalogue is offered as a contribution in its own right, and because §5.12 commits to the archive being usable independently of whether the programme succeeds. A reader interested only in the negative theory may read Chapters 2, 3 and 6 and stop.

6.11 A note on the form of no-go theorems

A methodological observation, offered because it took us some time to arrive at it and it may save a reader the same time.

There are three grades of negative result in this subject, and they are routinely conflated.

Grade 1 — an obstruction to a construction. "No object with properties $P_1, \dots, P_n$ exists." N7, N13, N20 and N23 are of this grade. These are theorems in the ordinary sense and are as final as any theorem.

Grade 2 — an identity that collapses a hoped-for gain. "The quantity the route seeks to bound equals, identically, the quantity already known." N2, N18 and N19 are of this grade. These are also theorems, but their force depends on the route being correctly identified: an argument that differs from the identified route in a way that changes the bookkeeping is not touched.

Grade 3 — an obstruction to a method. "Every conclusion of method M holds for an object where the conclusion sought is false." N21 is the pure case, and the firewall of Chapter 2 is the general apparatus. These are the strongest in practice, because they do not require identifying

the route — only the method — but they are the most vulnerable to the objection that the method has been characterised too narrowly.

Remark 6.64 (Which grade to prefer). Grade 3 is the most informative when available, because it survives modification of the argument. Grade 1 is the most secure. Grade 2 is the most frequently misread, because an identity that collapses a gain is easily mistaken for a difficulty that might be overcome, and the archive marks each Grade-2 entry explicitly as an identity rather than an estimate. N19 is the clearest instance: (A2) is not hard, it is RH.

Remark 6.65 (What a no-go theorem is not). Three things are not no-go theorems and are sometimes presented as such. First, a statement that a method has not yet succeeded. Second, a statement that a method faces a known difficulty. Third, a heuristic that a method "ought not" to work because it does not use enough structure — this becomes a theorem only when the insufficient structure is exhibited as an actual object, which is what the firewall does and what makes N21 a theorem rather than a plausibility.

6.12 Entries considered and rejected

Four candidate entries were examined and are not in the archive. We record them, and the reason for exclusion, because a catalogue that reports only its successes is not a catalogue.

Candidate 1 — "No elementary proof of the prime number theorem method can give RH*."* Excluded because we could not state it. Every formalisation of "elementary" we attempted was either so narrow as to be uninteresting or so wide as to be false. The Erdős–Selberg proof of the prime number theorem is a standing warning against confident predictions in this direction.

Candidate 2 — "No proof using only the functional equation and the Euler product can succeed." Excluded because it is false as stated: the four-point orbit and the Euler product together are exactly the data of the Selberg class, and RH for the Selberg class is not known to be false. What is true is the much weaker statement that the functional equation alone does not suffice, which is the Davenport–Heilbronn entry in the firewall (§2.2) and is not a separate no-go theorem.

Candidate 3 — "Random-matrix input cannot localise zeros." Excluded because the statement is not a theorem but an observation about the logical type of the input: random-matrix predictions are statements about the distribution of zeros already known to be where they are. We record the observation in §3.7 and in §5.9 without dignifying it as a no-go theorem, because we could not formulate a hypothesis on "random-matrix input" that was both precise and non-trivial.

Candidate 4 — "The Nyman–Beurling criterion cannot be verified." Excluded because it is not obstructed by anything we know. The criterion [Nym50; Beu55], in the Báez-Duarte form [Báe03], is an equivalence, and equivalences are not obstructions — N19 makes this point about a different equivalence. The approximation problem it poses is hard and is not shown here to be blocked. We mention it because it is the equivalence most often supposed by outsiders to be excluded by results of the kind in this archive, and it is not.

Remark 6.66 (The pattern in the exclusions). Three of the four failed the same test: we could not state the hypothesis precisely enough for the conclusion to be a theorem. This is the discipline of Definition 6.1(E2) doing its work, and it is worth noting that it removed roughly one candidate in six.

6.13 The archive and the firewall compared

Part I contains two negative instruments, and they are not the same instrument. It is worth saying how they differ, since a reader may otherwise take the firewall to be a summary of the archive.

The archive rules out *arguments*. Each entry identifies a class of proposed derivations and shows that no member of the class reaches RH. Its unit is a route.

The firewall rules out *mechanisms*. It presents four objects satisfying stated structural hypotheses and violating the desired conclusion, and asks whether a proposed mechanism distinguishes ζ from them. Its unit is a structural property.

The two are logically independent. A route may be blocked by an archive entry while the mechanism underlying it passes the firewall — N5, the natural-boundary obstruction, blocks a route for reasons having nothing to do with the counter-models. Conversely a mechanism may fail the firewall while no archive entry names the route it belongs to; this is the situation for most naive proposals, which is why the firewall is the faster test.

Remark 6.67 (Which to apply first). The firewall, always. It takes four questions and requires no knowledge of the proposed argument's internal structure, only of what it uses. The archive requires identifying which route the argument follows, which can be laborious and can be contested. In our experience the firewall disposes of most proposals in under an hour, and the archive is what one reaches for when a proposal survives it.

Remark 6.68 (Where they meet). Entry N21 is the point of contact: it is an archive entry whose proof *is* a firewall application, applied to a method rather than to a mechanism. It is also, by the classification of §6.11, the only Grade-3 entry in the archive. If a reader wants a single statement summarising what Part I claims to have established negatively, N21 is it: the additive-correlation machine cannot prove RH, and the proof of that is an object.

6.14 The audit protocol

We close Part I by describing how the claims in this book were checked. The description is included because a book making the claims this one makes about honesty should be checkable on that point as well.

6.14.1 The four passes

Pass 1 — statement audit. Every numbered statement was read in isolation, without its proof, and checked for: complete quantification, explicit hypotheses, and a status label consistent with Definition 6.1(E6) and the front-matter key. Statements failing this pass were rewritten before proof-checking, on the principle that a proof of an ambiguous statement is not checkable.

Pass 2 — proof audit. Every proof was checked line by line against its statement, with particular attention to: use of RH or of any statement equivalent to it; use of statements labelled [O] or [E]; and domain and convergence conditions in the analytic arguments. Proofs citing the literature were checked against the cited source, not against a secondary description of it.

Pass 3 — scope audit. Every [N] statement was checked for scope inflation by the following procedure: construct the widest class of arguments the statement plausibly rules out, then find an argument in that class the statement does *not* rule out, and include it in the (E5) clause. Where no such argument could be found, the (E5) clause says so, and this occurred in three entries.

Pass 4 — firewall audit. Every proposed mechanism, including this book's own, was run through the protocol of §2.6 and the resulting table filled in. Rows that could not be completed are marked as incomplete rather than omitted; §4.5 records two such rows.

6.14.2 Citation checking

Citations were checked in three tiers.

Tier 1 — load-bearing. A citation is load-bearing if a claim in this book is false when the cited result is false. These were checked against the published article: journal, volume, year, page range, and the statement of the theorem as published. The list includes [DMV06], [RT20], [Con99], [CL00], [KS03], [Han11], [OR85] and [Has58].

Tier 2 — contextual. Citations supporting historical or orientational claims were checked for existence and for approximate accuracy of the attribution.

Tier 3 — flagged. Entries whose bibliographic details we could not confirm to Tier-1 standard carry a `% [VERIFY]` comment in the bibliography source. A reader compiling the bibliography can extract the full list mechanically.

> One citation correction is recorded explicitly because the error is widespread: the Diamond–Montgomery–Vorhauer paper appeared in *Mathematische Annalen* **334** (2006) 1–36, and is frequently cited to *Inventiones Mathematicae* **165**. See §2.4.

6.14.3 Numerical checking

Every [E] claim carries the five components listed in §5.14: algorithm, parameters, precision, condition number, and stability range. Numerical results were recomputed in a second implementation with a different arithmetic backend where feasible. Where a numerical claim could not be independently recomputed, it is marked and its use restricted.

6.14.4 What the audit does not guarantee

The audit was carried out by the authors and by the readers named in the acknowledgements. It is not a substitute for refereeing, and we do not present it as one. Its purpose is narrower: to make the specific failure modes of §5.3 unlikely, and to leave a record that can itself be checked. An error surviving four passes is possible, and the checklist R1–R10 of §5.10 is offered as the fifth.

What this chapter does not establish

1. It does not establish that the twenty-three obstructions are exhaustive. They are the obstructions we encountered; §6.1 states the selection bias explicitly.
2. It does not establish that the routes obstructed are the routes most likely to succeed. An obstruction to a family of arguments is evidence about that family only.
3. It does not contain complete proofs of most entries. Nineteen of the twenty-three carry proof sketches with forward references; the forward references are to chapters written by the same authors under the same protocol, and a reader unwilling to accept that should treat the sketches as sketches.
4. It does not establish either of the two open problems of §6.8, and does not claim progress toward them beyond the statement of their relation, which is deferred to Part VIII.
5. It does not establish that equality-type positivity *can* work. Corollary 6.52 shows that gapped positivity cannot; the complement of a ruled-out class is not thereby shown to be non-empty.

6. It does not establish RH, and no combination of the entries in this archive bears on the truth of RH in either direction.
7. The audit protocol of §6.14 does not establish that the book is free of error; it establishes what was checked and how.

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